

INAUGURAL - DISSERTATION
submitted
to the
Combined Faculty of Mathematics, Engineering and Natural Sciences
of
Ruprecht-Karls-University
Heidelberg
for the Degree of Doctor of Natural Sciences

Put forward by
Sriram Chinthalagiri Venkata, M.Sc.

from Tirupati, India

Day of the oral exam: July 20, 2026

Bruhat-Tits buildings, harmonic cocycles and an Eichler-Shimura map in the function field setting

Supervisor: Prof. Dr. Gebhard Böckle

Abstract

Let K be a global function field, ∞ a fixed place of K , K_∞ the corresponding completion and $\Gamma \subset \mathrm{GL}_r(K)$ be a principal congruence subgroup. In this thesis we establish an equivariant homotopy equivalence between subregions of the Bruhat-Tits building for $\mathrm{GL}_r(K_\infty)$, which are unstable with respect to Γ , and the Tits buildings for $\mathrm{GL}_r(K)$, for $r \geq 2$ an integer. In rank 2 this is due to Serre and in higher ranks we adapt the proof strategy of a similar result of Grayson-Quillen in [Gra82]. The role of simplices admitting non-trivial Harder-Narasimhan filtrations resp. semistable simplices in [Gra82] is played by Γ -unstable resp. Γ -stable simplices in our case. Using this we generalize a result of Teitelbaum in [Tei91] for $r = 2$, about reconstruction of Γ -invariant harmonic cocycles from their values on the Γ -stable edges, to higher ranks by combining a novel technique of constructing harmonic cocycles on Bruhat-Tits buildings from the Steinberg module using modular symbols à la Ash-Rudolph [AR79]. All this is covered in Part I, at the end of which we consider the simplifications occurring in some of the results in the case $K = \mathbb{F}_q(t)$ and ∞ being the usual place at infinity, by generalizing to higher ranks Serre's ideas in [Ser80] regarding the action of $\mathrm{GL}_2(\mathbb{F}_q[t])$ on the Bruhat-Tits tree. In Part II we consider a geometric application of the results in Part I namely the construction of prototype of an Eichler-Shimura map in higher rank motivated from the work of Böckle in [Boe02].

Zusammenfassung

Sei K ein globaler Funktionenkörper, ∞ eine fixierte Stelle von K , K_∞ die entsprechende Kompletzierung und $\Gamma \subset \mathrm{GL}_r(K)$ eine Hauptkongruenzuntergruppe. In dieser Dissertation etablieren wir eine äquivariante Homotopieäquivalenz zwischen Teilbereichen des Bruhat-Tits-Gebäudes für $\mathrm{GL}_r(K_\infty)$, die in Bezug auf Γ instabil sind, und den Tits-Gebäuden für $\mathrm{GL}_r(K)$ für eine ganze Zahl $r \geq 2$. Im Rang 2 ist dies auf Serre zurückzuführen, und für höhere Ränge adaptieren wir die Beweisstrategie eines ähnlichen Resultats von Grayson-Quillen in [Gra82]. Die Rolle der Simplizes, die nicht-triviale Harder-Narasimhan-Filtrationen bzw. semistabile Simplizes in [Gra82] zulassen, wird in unserem Fall von Γ -instabilen bzw. Γ -stabilen Simplizes eingenommen. Hiermit verallgemeinern wir ein Resultat von Teitelbaum in [Tei91] für $r = 2$, welches die Rekonstruktion von Γ -invarianten harmonischen Kozyklen aus ihren Werten auf den Γ -stabilen Kanten betrifft, auf höhere Ränge. Dies geschieht durch die Kombination einer neuartigen Technik zur Konstruktion harmonischer Kozyklen auf Bruhat-Tits-Gebäuden aus dem Steinberg-Modul unter Verwendung von Modulsymbolen à la Ash-Rudolph [AR79]. All dies wird in Teil I behandelt. Am Ende von Teil I betrachten wir die Vereinfachungen, die in einigen Ergebnissen im Fall $K = \mathbb{F}_q(t)$ und ∞ als dem üblichen Punkt im Unendlichen auftreten, indem wir Serres Ideen aus [Ser80] für $r = 2$ bezüglich der Operation von $\mathrm{GL}_2(\mathbb{F}_q[t])$ auf dem Bruhat-Tits-Baum auf höhere Ränge verallgemeinern. In Teil II betrachten wir eine geometrische Anwendung der Resultate aus Teil I, nämlich die Konstruktion eines Prototyps einer Eichler-Shimura-Abbildung für höheren Rang, motiviert durch die Arbeit von Böckle in [Boe02].

Contents

Abstract	v
Zusammenfassung	vii
Contents	ix
1 Introduction	1
1.1 The rank 2 case	2
1.1.1 Classical Eichler-Shimura isomorphism	2
1.1.2 Work of Serre, Teitelbaum and Böckle	5
1.2 Higher rank case	9
1.2.1 A homotopy equivalence due to Grayson-Quillen	11
1.2.2 Main results of the thesis - a synopsis	13
Acknowledgements	17
I An equivariant homotopy equivalence and generalization of a result of Teitelbaum	21
Summary of chapters - Part I	23
2 Some equivariant properties of G-simplicial complexes	25
2.1 Preliminaries on $(G-)$ simplicial complexes	25
2.1.1 Geometric realization, $(G-)$ CW complexes and equivariant Whitehead's theorem	26
2.2 Homotopy properties of G -simplicial complexes	28
2.2.1 Simplicial complexes on G -posets	29
2.2.2 Some results on G -equivariant homotopies	32

3	Bruhat-Tits building and harmonic cocycles	35
3.1	Bruhat-Tits buildings	35
3.1.1	$\mathcal{BT}_\bullet(W)$ as a building of vector bundles.	36
3.1.2	Labeling the vertices of the building.	38
3.1.3	Geometric realization $ \mathcal{BT}_\bullet $	39
3.2	Stabilizers of simplices	42
3.2.1	Congruence subgroups.	42
3.2.2	Description of stabilizers	43
3.3	Apartments, Sectors and Harmonic cocycles	45
3.3.1	Apartments and sectors of $\mathcal{BT}_\bullet(W)$	45
3.3.2	Harmonic cocycles	49
4	Harmonic cocycles via Ash-Rudolph symbols	51
4.1	Tits building and the Ash-Rudolph symbols	51
4.2	Construction of a map $\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow C_{\text{har}}(V)^\Gamma$	54
5	An equivariant homotopy equivalence à la Grayson-Quillen	61
5.1	Contractible subcomplexes of the Γ -unstable complex	61
5.2	An equivariant homotopy $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}} \rightarrow \mathcal{T}_\bullet(W)$	65
5.3	Relating the various (un)stable complexes	67
6	On a result of Teitelbaum	73
6.1	Ash-Rudolph symbol and Γ -stable chambers in an apartment	73
6.1.1	Projected cube of length N , $\mathcal{PC}_N$	74
6.1.2	Boundary of a projected cube in $\mathcal{A}_\bullet(\underline{w})$	75
6.1.3	Describing the Γ -unstable region in $\mathcal{A}_\bullet(\underline{w})$ outside certain bounded regions	79
6.1.4	The Γ -stable cycle in $\mathcal{A}_\bullet(\underline{w})$ and the Ash-Rudolph symbol $[[\underline{w}]]$	81
6.2	Harmonic cocycles and its values on the Γ -stable part	83
7	The $\mathbb{F}_q[t]$-case	87
7.1	Motivation: the case of Bruhat-Tits tree	87
7.2	A fundamental domain for $\text{GL}_r(\mathbb{F}_q[t])$	89
7.3	Numbering the chambers and $\Gamma(I)$ invariant harmonic cocycles	98

II	An Eichler-Shimura map in higher ranks	105
	Summary of chapters - part II	107
8	Arithmetic quotient of Drinfeld period domain and Drinfeld moduli space	109
8.1	Arithmetic quotient of Drinfeld period domain	109
8.1.1	Rigid analytic geometry - a quick tour	109
8.1.2	Drinfeld period domain	112
8.1.3	Rigid analytic structure on arithmetic quotient	116
8.2	Brief recap of the theory of Drinfeld modules	117
8.2.1	Drinfeld modules and their moduli spaces	118
8.2.2	Drinfeld modules over rigid spaces	121
9	τ-sheaves, crystals and A-motives	127
9.1	Algebraic theory	127
9.2	Analytic theory	131
9.2.1	Extension by zero and direct image with compact support	134
9.2.2	Rigid τ sheaves with algebraic coefficients	135
9.3	Comparison	137
9.4	Uniformizable A -modules	139
10	An Eichler-Shimura map	145
10.1	Preparations	145
10.2	Pink's Satake compactification and a Stein cover of the analytification	147
10.3	Construction of the map	149
10.4	Some contemplations	155
10.4.1	Relation to Gräf's work	155
10.4.2	Future directions	157
	Bibliography	159

Chapter 1

Introduction

*In algebraic number theory, one starts out with $\mathbb{Q}$, completes it to obtain the real numbers $\mathbb{R}$ and then adjoins i to obtain the complex numbers $\mathbb{C}$, and then stops. But for function fields, when one completes, one ends up with a local field which not only has extensions of arbitrarily large degree, but also can have infinitely many distinct extensions of fixed degree. As such, the algebraic closure of this field is huge. In it, so to speak, number theory has “**room to roam**”. Whereas in classical theory, objects may play many different roles all at once, for function fields, these different roles are played by **different** players. (Emphasis added)*

- David Goss [Gos96]

The Bruhat-Tits building is a versatile combinatorial object which also codifies data of arithmetic as well as geometric nature as part of its structure. In this dissertation we focus on such buildings associated with general linear groups over local fields in positive characteristic and ascertain homotopy properties of some of its subcomplexes of arithmetic relevance in the function field setting, inspired by similar results in [Ser80; Gra82]. We also provide two applications of this, first to harmonic cocycles on the building generalizing some results of Teitelbaum [Tei91] and second for the construction of an Eichler-Shimura map in higher ranks emulating a similar map of Böckle in [Boe02]. Our objective in this chapter is to give the motivations for this work and hence focus our attention more on how results over function fields mirror analogous results from the classical setting. In the same vein, most of the notions discussed below in the rank 2 case of function fields has an analogue in the higher rank discussed in the main text of the thesis and we refer the reader there for more details.

1.1 The rank 2 case

Theories in function field arithmetic, more often than not, have a counterpart in the arithmetic over number fields and are motivated from them. But once important results are established the theory takes a life of its own, so to speak, revealing the intricacies inherent in arithmetic over function fields, involving techniques from geometry, combinatorics and much more. This is most evident in the efforts to adapt the classical Eichler-Shimura isomorphism to the function field setting in Böckle's habilitation work [Boe02]. As this is a prime inspiration for the present thesis, it would be worth looking at briefly on how a characteristic p version of an Eichler-Shimura isomorphism was proven. But let us first review the classical setting.

1.1.1 Classical Eichler-Shimura isomorphism

We mainly follow [Hid93; Del71] for the discussion in this subsection. Recall that the upper half plane $\mathbb{H}$ is defined as

$$\{z = x + iy \in \mathbb{C} \mid y > 0\}. \quad (1.1)$$

The special linear group $\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ via Möbius transformations. Of particular importance for number theoretic reasons is the action of $\mathrm{SL}_2(\mathbb{Z})$ and its finite index subgroups, called *congruence subgroups* Γ , which contain the *principal congruence subgroup* $\Gamma(N) := \mathrm{Ker}(\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}))$ for some integer $N > 1$.

Given a congruence subgroup Γ , it turns out that the quotient $Y_\Gamma := \Gamma \backslash \mathbb{H}$ has a canonical structure of Riemann surface and the set

$$X_\Gamma := Y_\Gamma \sqcup \Gamma \backslash \mathbb{P}^1(\mathbb{Q})$$

has the structure of a compact Riemann surface such that $Y_\Gamma \subset X_\Gamma$ is an open immersion. The finite set of points $\Gamma \backslash \mathbb{P}^1(\mathbb{Q})$ are called the *cusps* for Γ .

For a torsion free Γ , the quotient $\mathbb{H} \rightarrow Y_\Gamma$ is a universal covering space with the automorphism group Γ . This hints at a relation between cohomologies of Γ and Y_Γ . In fact there is a canonical isomorphism given as follows. Let M a left $R[\Gamma]$ -module for a commutative ring R and $\mathcal{M}$ denote the sheaf on Y_Γ given locally as sections for the projection $\Gamma \backslash (\mathbb{H} \times M) \rightarrow Y_\Gamma$ (Γ acts diagonally on $\mathbb{H} \times M$). Then

$$H^i(\Gamma, M) \simeq H_{sh}^i(Y_\Gamma, \mathcal{M})$$

for $i \geq 0$ and “ sh ” denotes the sheaf cohomology. Further if one defines the image in RHS of the compactly supported sheaf cohomology by

$$H_{sh,!}^i(Y_\Gamma, \mathcal{M}) := \mathrm{Im}(H_{sh,c}^i(Y_\Gamma, \mathcal{M}) \rightarrow H_{sh}^i(Y_\Gamma, \mathcal{M})), \quad i \geq 0$$

then it is isomorphic to the *parabolic cohomology*

$$H_{par}^i(\Gamma, M).$$

See [Hid93; DH05]. $H_{par}^1(\Gamma, M)$ can be described as the kernel of the restriction map on cohomology

$$H^1(\Gamma, M) \rightarrow \bigoplus_{s \in \Gamma \backslash \mathbb{P}^1(\mathbb{Q})} H^1(\Gamma_s, M)$$

where Γ_s denotes the stabilizer of the cusps.

The above cohomology groups are of arithmetic relevance not only in their own right but also due to their deep relation to the space of (classical) *cusp forms* on $\mathbb{H}$ for suitable Γ and weight, established by Eichler and Shimura in [Eic57; Shi71]. Let $n \geq 0$ be an integer and Γ be a torsion free congruence subgroup (for instance $\Gamma_1(N)$ for $N \geq 4$) and denote by $S_{n+2}^{cl}(\Gamma)$ the space of weight $(n+2)$ cusp forms for Γ . Moreover consider the tautological action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{Z}^2$ inducing a $\mathbb{Z}[\Gamma]$ -module $\mathrm{Sym}^n(\mathbb{Z}^2)$ and denote by $\mathcal{S}ym^n(\mathbb{Z}^2)$ the induced sheaf on Y_Γ .

Then there is an isomorphism of $\mathbb{R}$ -vector spaces

$$S_{n+2}^{cl}(\Gamma) \simeq H_{par}^1(\Gamma, \mathrm{Sym}^n(\mathbb{Z}^2)) \otimes_{\mathbb{Z}} \mathbb{R}. \quad (1.2)$$

In [Del71], Deligne considered the version of this isomorphism after a change of coefficients by $\mathbb{C}$, which combining with the earlier relation between $H_{sh,!}^1$ and H_{par}^1 and the fact that $\mathcal{S}ym^n(\mathbb{Z}^2) \otimes_{\mathbb{Z}} \mathbb{R} \simeq \mathrm{Sym}^n(R^1 f_* \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{R}$ for $f : E_\Gamma \rightarrow Y_\Gamma$ denoting the *universal elliptic curve* on Y_Γ , gives rise to an isomorphism which is compatible with the action of Hecke operators

$$S_{n+2}^{cl}(\Gamma) \oplus \overline{S_{n+2}^{cl}(\Gamma)} \simeq_* H_{par}^1(\Gamma, \mathrm{Sym}^n(\mathbb{Z}^2)) \otimes_{\mathbb{Z}} \mathbb{C} \simeq_{**} H_{sh,!}^1(Y_\Gamma, \mathrm{Sym}^n(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}, \quad (1.3)$$

where $\overline{S_{n+2}^{cl}(\Gamma)}$ denotes the space of anti-holomorphic cusp forms. Deligne famously used this combined with Eichler-Shimura relations as well as his proof of the Weil conjectures, to prove Ramanujan's conjecture on τ -function.

It is worth noting that there are at least three proofs of the Eichler-Shimura isomorphism involving different techniques. In [Shi71], 1.2 is proven by first constructing a canonical injective map between the required spaces and surjectivity is shown by checking that the dimensions match. In [Hid93] the aforementioned surjectivity is established by constructing the map in 1.3 using the de Rham realization of $H_{sh}^1(Y_\Gamma, \mathrm{Sym}^n(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}$ and showing that the image is precisely $H_{sh,!}^1(Y_\Gamma, \mathrm{Sym}^n(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}$. Crucial to the proof here is the description of $\mathbb{H}$ as the symmetric space

$$\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R}). \quad (1.4)$$

for the special orthogonal group $\mathrm{SO}_2(\mathbb{R})$ which is maximal compact subgroup of $\mathrm{SL}_2(\mathbb{R})$. Another elegant proof using de Rham cohomology is given in [BN81]. Among the three proofs, it is worth noting that, [BN81] prove the isomorphism

$$S_{n+2}^{cl}(\Gamma) \oplus \overline{S_{n+2}^{cl}(\Gamma)} \simeq H_{sh,!}^1(Y_\Gamma, \mathrm{Sym}^n(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C} \quad (1.5)$$

without passing through the parabolic cohomology group, realizing the above as a Hodge decomposition for degenerating coefficients by regarding the RHS as a de Rham cohomology group.

Notation

Before proceeding, let us set up some basic notation and terminology in the function field setting that follows, which also applies for the rest of the write up.

- Let $p > 0$ be a prime number and $q = p^m$ for some integer $m \geq 1$.
- Let C denote a smooth, projective and geometrically connected curve over $\mathbb{F}_q$.
- Fix a closed point ∞ of C and set $A := \mathcal{O}_C(C \setminus \{\infty\})$. This is a Dedekind domain whose fraction field we denote by K , where the latter also coincides with the function field of C .
- The local ring $R := \mathcal{O}_{C,\infty}$ is principal ideal domain and we fix a choice of uniformizer $\varpi \in R$ for the maximal ideal of R .
- Let K_∞ denote the completion of K at ∞ and $\mathcal{O}_\infty \subset K_\infty$ be the corresponding ring of integers. $\mathcal{O}_\infty$ can also be described as the completion $\widehat{R} := \lim_n R/(\varpi^n)$.
- Let $\mathbb{C}_\infty$ denote the completion of an algebraic closure of K_∞ , which is again algebraically closed by Krasner's lemma.
- Let $r \geq 2$ be an integer and W denote a K -vector space of dimension r . Furthermore $P \subset W$ be a rank r projective A -submodule.
- The automorphism group of the set $\{0, \dots, r-1\}$ will be denoted by $\mathcal{S}_r$.

For the reader more comfortable with the number field setting, it would be beneficial to keep the following rough analogy in mind

$$A \longleftrightarrow \mathbb{Z}, K \longleftrightarrow \mathbb{Q}, K_\infty \longleftrightarrow \mathbb{R} \text{ and } \mathbb{C}_\infty \longleftrightarrow \mathbb{C}$$

1.1.2 Work of Serre, Teitelbaum and Böckle

We now discuss the corresponding results from the previous section in the function field setting. We follow the treatment in the $r = 2$ case from [Ser80; Tei91; Boe02] and this will serve as a motivation for the main results of our thesis in higher ranks later. The reader is referred to the main text of the dissertation for more details on most of the technicalities introduced in this subsection. For simplicity assume that $P = A^{\oplus 2}$ and $W = K^{\oplus 2}$. Here a *congruence subgroup* will mean a finite index subgroup of $\mathrm{GL}_2(A)$ containing the *principal congruence subgroup* $\Gamma(I) := \mathrm{Ker}(\mathrm{GL}_2(A) \rightarrow \mathrm{GL}_2(A/I))$ for some nonzero proper ideal $I \subset A$. Such a subgroup is said to be *p' -torsion free* if the torsion elements have order a power of p . For instance every principal congruence subgroup is p' -torsion free. In the rest of this section we restrict ourselves to such congruence subgroups.

The upper half plane $\mathbb{H}$ admits the following two distinct analogues in the function field setting:-

- (i) In view of the description $\mathbb{H}^{\pm} = \mathbb{C} \setminus \mathbb{R}$, one has *Drinfeld's upper half plane*

$$\Omega^2$$

a rigid analytic K_{∞} -space of dimension 1, whose $\mathbb{C}_{\infty}$ points are given by $\Omega := \Omega^2(\mathbb{C}_{\infty}) = \mathbb{C}_{\infty} \setminus K_{\infty}$ (8.1.7).

- (ii) *Bruhat-Tits tree*

$$\mathcal{BT}_{\bullet}$$

which is an infinite tree with valency $q^{\deg(\infty)} + 1$. Its vertices $\mathcal{BT}_0$ are the homothety classes $\langle L \rangle$ of R -lattices L of rank 2 and $\langle L_0 \rangle, \langle L_1 \rangle$ are said to form an edge, if after choosing suitable representatives one has $L_0 \supsetneq L_1 \supsetneq \varpi L_0$. In fact $\mathcal{BT}_0$ is in bijection with $\mathrm{GL}_2(K_{\infty})/K_{\infty}^{\times} \cdot \mathrm{GL}_2(O_{\infty})$ where $\mathrm{GL}_2(O_{\infty}) \subset \mathrm{GL}_2(K_{\infty})$ is a maximal (open) compact subgroup (3.1.1, 3.1.12). This is similar description of $\mathbb{H}$ equivalent to 1.4, given by $\mathbb{H}^{\pm} = \mathrm{GL}_2^+(\mathbb{R})/\mathbb{R}^{>0} \cdot \mathrm{SO}_2(\mathbb{R})$ where $\mathrm{GL}_2^+(\mathbb{R})$ consists of all invertible matrices over $\mathbb{R}$ with positive determinant.

$\mathrm{GL}_2(K_{\infty})$ acts via Möbius transformations on Ω and it also acts canonically on $\mathcal{BT}_{\bullet}$. The stabilizer of a vertex/edge with respect to a congruence subgroup Γ is finite, hence a p -group for a p' -torsion free Γ . There is a $\mathrm{GL}_2(K_{\infty})$ -equivariant map due to the work of Goldman-Iwahori [GI63], called the *reduction map* (8.1.10)

$$\Omega \xrightarrow{\rho} |\mathcal{BT}_{\bullet}|,$$

where $|\cdot|$ denotes the geometric realization regarding $\mathcal{BT}_{\bullet}$ as a simplicial complex. It has the property that the preimage under ρ of star shaped regions with rational radius around vertices in $\mathcal{BT}_0$

defines affinoid K_∞ -subspaces of Ω^2 (8.1.11). For a congruence subgroup Γ this fact allows one to define a rigid analytic K_∞ -structure on the quotient $\Gamma \backslash \Omega^2$. In [Boe02, Chap.4] the construction of a compactification $\overline{\Gamma \backslash \Omega^2}$ is described by “adding” the finitely many points of $\Gamma \backslash \mathbb{P}^1(K)$, also called *cusps for Γ* here. In [Ser80], a parallel description for $\Gamma \backslash \mathcal{BT}_\bullet$ is discussed, as a graph (not necessarily a tree anymore) on finitely many vertices of $\mathcal{BT}_0$ attached to finitely many disjoint (infinite) half lines $Y_c \subset \mathcal{BT}_\bullet$ for $c \in \Gamma \backslash \mathbb{P}^1(K)$. See Figure 1.1 for some examples. Consequently one can “compactify” $\Gamma \backslash |\mathcal{BT}_\bullet|$ to $\overline{\Gamma \backslash |\mathcal{BT}_\bullet|}$, by contracting the infinite half lines to a single edge and ρ induces a map

$$\overline{\Gamma \backslash \Omega^2} \xrightarrow{\bar{\rho}_\Gamma} \overline{\Gamma \backslash |\mathcal{BT}_\bullet|}.$$

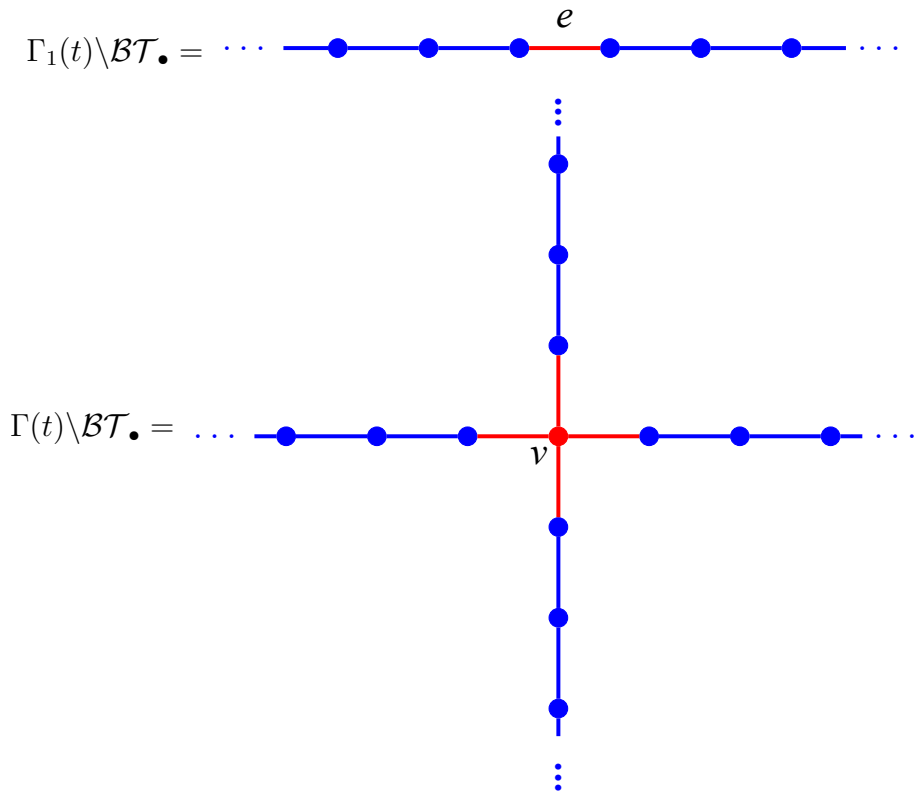


Figure 1.1. Graphs of quotient of Bruhat-Tits tree in the $A = \mathbb{F}_q[t]$ case with respect to

$$\Gamma_1(t) \text{ resp. } \Gamma(t) := \left\{ \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(A) \mid a \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \text{ resp. } \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{t} \right\}.$$

The second figure represents the $q = 3$ case but the general case is similar with $q + 1$ infinite lines. Red resp. blue vertices/edges represent Γ -stable resp. unstable simplices.

The results from [Ser80] clarify the bijection between the infinite half lines of $\Gamma \backslash \mathcal{BT}_\bullet$ and the

finite coset space $\Gamma \backslash \mathbb{P}^1(K)$ in a precise manner as follows. ¹Call a vertex/edge s of $\mathcal{BT}_\bullet$ as Γ -*unstable* if the stabilizer Γ_s is non-trivial, else call it Γ -*stable*. Denote by $\mathcal{BT}_\bullet^{\Gamma\text{-un}}$ the simplicial subcomplex of $\mathcal{BT}_\bullet$ consisting of Γ -unstable vertices and edges. Results by Serre in [Ser80] helps one to construct an explicit equivariant homotopy equivalence

$$\mathcal{BT}_\bullet^{\Gamma\text{-un}} \sim \mathbb{P}^1(K) \quad (1.6)$$

In particular, there is an isomorphism of Γ -modules of (reduced) simplicial homologies

$$\tilde{H}_0(\mathcal{BT}_\bullet^{\Gamma\text{-un}}, \mathbb{Z}) \simeq \tilde{H}_0(\mathbb{P}^1(K), \mathbb{Z}) := \text{Ker}(\mathbb{Z}[\mathbb{P}^1(K)] \xrightarrow{\text{aug}} \mathbb{Z}),$$

where $\text{aug}(\sum n_x[x]) = \sum n_x$. The $\mathbb{Z}[\text{GL}_2(K)]$ -module $\tilde{H}_0(\mathbb{P}^1(K), \mathbb{Z})$ is called the *Steinberg module* and will be denoted in this section by St . We observe that $\mathcal{BT}_\bullet^{\Gamma\text{-st}} := \mathcal{BT}_\bullet \setminus \mathcal{BT}_\bullet^{\Gamma\text{-un}}$ (“st” here abbreviates stable and should not be confused with Steinberg from the previous line) is not necessarily a simplicial complex (see Figure 1.1). Nevertheless one may still declare its homology as $H_\bullet(\mathcal{BT}_\bullet^{\Gamma\text{-st}}, \mathbb{Z}) := H_\bullet(\mathcal{BT}_\bullet, \mathcal{BT}_\bullet^{\Gamma\text{-un}}, \mathbb{Z})$ (cf. 5.3). Using a long exact sequence in homology, the Γ -module $\tilde{H}_0(\mathcal{BT}_\bullet^{\Gamma\text{-un}}, \mathbb{Z})$ is also equivariantly isomorphic to $\text{St}^I := H_1(\mathcal{BT}_\bullet^{\Gamma\text{-st}}, \mathbb{Z})$. Consequently we have an isomorphism of Γ -modules

$$\text{St}^I \simeq \text{St}. \quad (1.7)$$

Teitelbaum([Tei91]) used this to generalize 1.2 and we briefly indicate it here. Since the upper half plane $\mathbb{H}$ has two avatars in the function field setting, the space of classical cusp forms also admits two versions, one on the Drinfeld’s upper half plane and one on $\mathcal{BT}_\bullet$. On Ω , Goss ([Gos80]) introduced the notion of *Drinfeld cusp forms of weight k for Γ* , denoted $S_k(\Gamma)$, akin to their classical counterparts, albeit the uniformizers at cusps are now defined as the inverse of Drinfeld’s exponential function for certain rank 1 projective A -modules. In [Tei91], using the reduction map ρ Teitelbaum found that suitable “currents” on $\mathcal{BT}_\bullet$ which are invariant under Γ are the right analogue of cusp forms and are called *harmonic cocycles* (see 3.3.2) which are invariant under Γ taking values in a $\mathbb{C}_\infty[\Gamma]$ -module $V_n := \text{Sym}^n(\mathbb{C}_\infty^2)^*$, $(\)^*$ denoting the $\mathbb{C}_\infty$ -dual and the space of such maps is denoted by $C_{\text{har}}(V_n)^\Gamma$. The precise result of Teitelbaum is an isomorphism of $\mathbb{C}_\infty$ -vector spaces

$$S_{n+2}(\Gamma) \xrightarrow[\sim]{\text{Res}_\Gamma} C_{\text{har}}(V_n)^\Gamma. \quad (1.8)$$

The inverse to Res_Γ is provided by constructing, for a harmonic cocycle in $C_{\text{har}}(V_n)^\Gamma$, a measure on $\mathbb{P}_{K_\infty}^1$ and integrating a “Poisson kernel” with respect to this measure gives a Drinfeld cusp form.

¹We will only consider principal congruence subgroups in the rest of this section to simplify the exposition and set $\Gamma := \Gamma(I)$ and also in view of our main results in this thesis, though most results in rank 2 also generalizes to other p' -torsion free congruence subgroups.

Using an ingenious argument involving *sources of edges of* $\mathcal{BT}_1$ in the set of all Γ -stable edges, he also proved an isomorphism

$$C_{har}(V_n)^\Gamma \xrightarrow{\text{Te}_\Gamma} \text{St}_\Gamma^I \otimes V_n. \quad (1.9)$$

Then using 1.7 one obtains an isomorphism

$$S_{n+2}(\Gamma) \simeq \text{St}_\Gamma \otimes V_n,$$

which mirrors 1.3(*) in light of the isomorphism

$$\text{St}_{\Gamma(I)} \otimes V_n \simeq H_1(\Gamma, \{\Gamma_s\}_{s \in \Gamma \backslash \mathbb{P}^1(K)}, V_n) := \text{Ker} \left(\bigoplus_{s \in \Gamma \backslash \mathbb{P}^1(K)} (V_n)_{\Gamma_s} \rightarrow (V_n)_\Gamma \right).$$

Concerning the function field version of 1.3(**), it should be noted that the analogue of elliptic curves are the *Drinfeld A -modules* which admit uniformization by A -lattices in $\mathbb{C}_\infty$ of rank 2. Such objects were introduced by Drinfeld in his landmark work [Dri74] (cf. 8.2). Similar to elliptic curves, Drinfeld A -modules also admit level structures and the analytic moduli space parametrizing this combined data has connected components of the form $\Gamma \backslash \Omega^2$. Consequently there exists a universal Drinfeld A -module over $\Gamma \backslash \Omega^2$ and one should expect to take this into account in order to generalize 1.3(**). This was made precise by Böckle in his seminal habilitation work [Boe02]. Using the theory of function field crystals developed in [BP09], he proved an isomorphism of A -modules

$$ES_2^{(n)} : (\text{St}_\Gamma^I \otimes \tilde{V}_n)^* \xrightarrow{\sim} (\underline{\mathcal{S}}_{n+2}(\Gamma)^{\mathfrak{D}_{A-an}})^\tau \quad (1.10)$$

where

- $\tilde{V}_n := \text{Sym}^n(A^{\oplus 2})^*$ has a tautological $A[\Gamma]$ -module structure and $()^*$ denotes the A -dual. In particular $V_n = \tilde{V}_n \otimes_A \mathbb{C}_\infty$ (cf. 10.1.1),
- $\underline{\mathcal{S}}_{n+2}(\Gamma)$ is an appropriate crystal on K_∞ with coefficients in A that can be described as first cohomology on $\overline{\Gamma \backslash \Omega^2}$ of the n -th symmetric power τ -sheaf with A -coefficients arising from the universal Drinfeld A -module on $\Gamma \backslash \Omega^2$ (cf. 10.3.110.3.2),
- $\mathfrak{D}_A := \text{Sp}(K_\infty \langle T \rangle_{\mathbb{F}_q[T]} \otimes A)$ with respect to a finite flat morphism $\mathbb{F}_q[T] \rightarrow A$ and $(.)^{\mathfrak{D}_{A-an}}$ corresponds to the change of coefficients from A to $\mathcal{O}(\mathfrak{D}_A)$ (9.3.1).

Combining this with the aforementioned results of Teitelbaum one obtains an isomorphism of finite dimensional $\mathbb{C}_\infty$ -vector spaces

$$(S_{n+2}(\Gamma))^* \xrightarrow{\sim} (\underline{\mathcal{S}}_{n+2}(\Gamma)^{\mathfrak{D}_{A-an}})^\tau \otimes_A \mathbb{C}_\infty. \quad (1.11)$$

In op.cit. an appropriate action by Hecke operators on both sides are described and the isomorphism is shown to be equivariant with respect to such operators. Note that unlike 1.5 there is no doubling of the space of cusp forms here.

The map $ES_2^{(n)}$ is constructed using the Čech cohomological description of RHS for which one takes into consideration the finite admissible cover of $\overline{\Gamma \backslash \Omega^2}$ obtained via $\bar{\rho}_\Gamma$ corresponding to vertices and edges of $\Gamma \backslash \mathcal{BT}^{\Gamma\text{-st}}$ as well as the cusps for $\Gamma(I)$ and the map turns out to be injective. To show surjectivity, one matches the $\mathbb{F}_q$ -dimensions on both sides of the map after a tensor product $\otimes_A A/\mathfrak{n}$ for all nonzero proper ideals $\mathfrak{n} \subset A$. The dimension of the LHS involves an alternating sum of number of vertices and edges in $\Gamma \backslash \mathcal{BT}_\bullet^{\Gamma\text{-st}}$ and that of the RHS can also be explicitly computed and follows from a formula of Pink for Euler-Poincare characteristic of constructible étale sheaves in [Pin00] after noting that $(\underline{\mathcal{S}}_{n+2}(\Gamma)^{\mathfrak{D}_A - an})^\tau \otimes_A A/\mathfrak{n}$ relates to the étale cohomology of a suitable constructible sheaf on the projective curve over K_∞ associated to $\overline{\Gamma \backslash \Omega^2}$.

Remark 1.1.1. As mentioned at the end of the previous subsection, [BN81] prove 1.5 without passing through parabolic cohomology group, by realizing the cohomology of the local system as a certain de Rham cohomology group on the compactification of the arithmetic quotient of $\mathbb{H}$ and obtaining the classical Eichler-Shimura isomorphism as a Hodge decomposition for degenerating coefficients. It is natural to wonder whether a similar strategy exists in the function field setting and if 1.11 can be proven analogously.

In view of the importance of the de Rham cohomology in the classical case, it should be pointed out that by the work of Anderson, Deligne, Gekeler and Yu one can attach to Drinfeld A -modules over schemes, a locally free sheaf also called a *de Rham cohomology* which mirrors many properties of its counterpart from the classical setting. See [Gek89; Gek90a; Gek90b] for more details. In particular, the universal Drinfeld module on the Drinfeld moduli space admits a de Rham cohomology. In a joint work with Oğuz Gezmiş [GV25], we extend this locally free sheaf to the compactified moduli space and show that it can be used to give a canonical description of *nearly holomorphic Drinfeld modular forms* inspired by a similar result for classical nearly holomorphic forms in [Urb14, Prop.2.2.3].

1.2 Higher rank case

As alluded to in the preceding discussions and evident there, there is a close analogy between the setting of elliptic curves and the theory of Drinfeld modules uniformized by A -lattices of rank 2 in $\mathbb{C}_\infty$. But unlike the classical setting a distinguishing feature of the function field setting is that one can also make sense of Drinfeld modules of ranks higher than 2, primarily due to the extension

$\mathbb{C}_\infty$ over K_∞ being infinite, thereby admitting a rich theory by allowing us to generalize notions and techniques from rank 2. For instance, in the past decade Basson, Breuer and Pink [BBP24] and Gekeler [Gek25b] have independently developed a theory of higher rank Drinfeld modular forms. Though a higher rank Drinfeld module can be regarded as an analogue of abelian variety, there is no concrete correspondence in the classical setting.

Extrapolating the discussion in 1.1.2 to higher ranks deem the following constructions relevant. Let $r \geq 2$ and consider *principal congruence subgroups* of the form $\Gamma_P(I) := \text{Ker}(\text{Aut}_A(P) \rightarrow \text{Aut}_A(P/IP))$ for nonzero proper ideals $I \subsetneq A$. For simplicity of exposition, we will abbreviate $\Gamma_P(I)$ by Γ . Let $W = K^r$ and regard Γ as a subset of $\text{GL}_r(K)$. The analytic moduli space of Drinfeld A -modules of rank r with level I -structure admit connected components of the form $\Gamma \backslash \Omega^r$ (8.2.13) where Ω^r is the *Drinfeld period domain of dimension $r-1$* over K_∞ (8.1.7). Consequently there is a *universal* Drinfeld A -module over $\Gamma \backslash \Omega^r$. From the work of Pink [Pin12], one can form a compactification $\overline{\Gamma \backslash \Omega^r}$ and analogous to before, construct an A -crystal $\underline{\mathcal{S}}_{n+r}(\Gamma)$ over K_∞ with coefficients in A for $n \geq 0$ (10.3.1). On the other hand, from [BBP24; Gek25b; deS01], one can consider the space of Drinfeld cusp forms weight $n+r$ for Γ , denoted $S_{n+r}(\Gamma)$, over Ω^r and also harmonic cocycles admit generalization to functions on pointed chambers of the *Bruhat-Tits building* $\mathcal{BT}_\bullet(W)$ (3.1.1), denoted by $C_{\text{har}}(V_n)^\Gamma$ (3.3.10) where $V_n := \text{Sym}^n(W^*)^*$ has a² canonical left $K[\Gamma]$ action. Motivated by the residue map of Teitelbaum, Gräf constructed in the $r=3$ and $A = \mathbb{F}_q[t]$ case a map $\text{Res}_\Gamma : S_{n+r}(\Gamma) \rightarrow C_{\text{har}}(V_n)^\Gamma \otimes_K \mathbb{C}_\infty$ which is conjectured to be an isomorphism ([Grä21]). Taking into account the higher rank version of the Steinberg module $\text{St}(W)$, we display our expectations based on 1.11 in the following diagram

$$\begin{array}{ccc}
 S_{n+r}(\Gamma) & \xrightarrow{\text{Res}} & C_{\text{har}}(V_n)^\Gamma \otimes_K \mathbb{C}_\infty \\
 & & \\
 C_{\text{har}}(V_n)^\Gamma & \xrightarrow[\text{?}]{\text{Te}_\Gamma} & \text{St}(W) \otimes_\Gamma V_n \\
 & & \\
 & & (\text{St}(W) \otimes_\Gamma V_n)^* \xrightarrow[\text{?}]{\text{ES}_{r,\Gamma}^{(n)}} (\underline{\mathcal{S}}_{n+r}(\Gamma)^{\mathfrak{D}_{A-an}^\tau} \otimes_A K
 \end{array} \tag{1.12}$$

In this thesis, we aim at the construction of the lower two maps Te_Γ and $\text{ES}_{r,\Gamma}^{(n)}$ in the above diagram and our primary focus will be on generalizing Teitelbaum's results exploring properties of the middle map. For the construction of a prospective Te_Γ as above, one expects to pass through an isomorphism between $\text{St}(W)$ and the *stable Steinberg module* $\text{St}^I(P)$ (5.3.5), similar to Serre's

²For a principal congruence subgroup Γ , the $K[\Gamma]$ module V_n agrees with Gräf's $V_n(K)$ described in [Grä21, pg.18]. See 10.1.1 for the precise definition of the space of coefficients V_n used in Part II(cf. Lemma 10.3.6).

result 1.7. We show the latter isomorphism by relying on the work of Grayson and Quillen in [Gra82] where a certain generalization of 1.7 is proven. Before giving an overview of our main results let us briefly go through the technical heart of [Gra82] as the ideas there form the cornerstone of our dissertation. It should be noted that the notion of semistability and Harder-Narasimhan filtrations discussed briefly in the next subsection do not have a direct bearing on the main content of the thesis and are mentioned only so that the exposition is self contained.

1.2.1 A homotopy equivalence due to Grayson-Quillen

Quillen obtained a certain generalization of Serre's isomorphism 1.7 which was sufficient and instrumental in proving the finite generation of K -groups of C . Quillen's ideas were elaborated with much clarity and detail by Grayson in [Gra82] and we will succinctly discuss it, as we adapt the techniques there to provide a more precise generalization of 1.7 which turns out to be crucial in our quest to understand a higher rank Eichler-Shimura isomorphism.

Henceforth in the text, until otherwise stated, r is an integer that is at least 2 and $\mathcal{BT}_\bullet(W)$ denotes the Bruhat-Tits building of rank $r - 1$ consisting of classes of R -lattices of rank r inside W (3.1.1). In [Ser80; Gra82] a precedent to conclude interesting properties of congruence subgroups is the interpretation of the Bruhat-Tits building $\mathcal{BT}_\bullet(W)$ as the building $\mathcal{X}_\bullet(P)$ of classes of rank r vector bundles on C whose restriction to $\mathrm{Spec}(A) \subset C$ is P . Here two vector bundles which are twists of each other with respect to the ample line bundle $\mathcal{O}(\infty)$ are said to lie in the same class (3.1.6). In this viewpoint, the notion of Γ -stability/unstability of simplices for a congruence subgroup $\Gamma \subset \mathrm{Aut}_A(P)$ admit a more refined analogue of *(slope) semistability/non semistability* in $\mathcal{X}_\bullet(P)$. Let us briefly recall their definitions.

Recall that the **degree** of a vector bundle $\mathcal{F}$ on C is the degree of the line bundle $\det(\mathcal{F})$. One defines the slope of $\mathcal{F}$, denoted $\mu(\mathcal{F})$, as

$$\mu(\mathcal{F}) := \frac{\deg(\mathcal{F})}{\mathrm{rank} \mathcal{F}}.$$

Also $\mathcal{E} \hookrightarrow \mathcal{F}$ is said to be a **(vector) subbundle** if the quotient $\mathcal{F}/\mathcal{E}$ is again a vector bundle on C .

Definition 1.2.1. A vector bundle $\mathcal{F}$ on C is said to be **semistable** if for all non-zero subbundles $\mathcal{E} \subsetneq \mathcal{F}$, we have $\mu(\mathcal{E}) \leq \mu(\mathcal{F})$.

The above formulation of semistability is equivalent to stating that, for non-zero quotient bundles $\mathcal{F} \twoheadrightarrow \mathcal{H}$, one has $\mu(\mathcal{F}) \leq \mu(\mathcal{H})$.

The following result, originally done over algebraically closed fields in [HN74, Lemma 1.3.7] also works over arbitrary fields

Proposition 1.2.2. ([Gra82, Prop.3.3]) A vector bundle $\mathcal{F}$ on C has a unique flag of subbundles $0 = \mathcal{F}_0 \subsetneq \mathcal{F}_1 \subsetneq \dots \subsetneq \mathcal{F}_n = \mathcal{F}$ such that the following conditions are satisfied:-

- (i) $\mathcal{F}_i/\mathcal{F}_{i-1}$ is semistable for all i ;
- (ii) $\mu(\mathcal{F}_i/\mathcal{F}_{i-1}) > \mu(\mathcal{F}_{i+1}/\mathcal{F}_i)$.

Since the flag in the proposition consists of subbundles, it is determined by the flag of subspaces $(0 = W_0 \subsetneq W_1 \subsetneq \dots \subsetneq W_n = W)$ where $W_i := \eta^*(\mathcal{F}_i) \forall i$, and $\eta : \text{Spec}(K) \rightarrow C$ is the generic point of C .

Definition 1.2.3. The flag of subspaces $(0 = W_0 \subsetneq W_1 \subsetneq \dots \subsetneq W_n = W)$ is called the **Harder-Narasimhan filtration** of $\mathcal{F}$, which we will denote here by $\text{HN}(\mathcal{F})$.

In particular $\text{HN}(\mathcal{F})$ is a simplex of the *Tits building* $\mathcal{T}_\bullet(W)$ where the latter is defined as the poset of nonzero proper K -subspaces of W and it has rank $r - 2$ regarded as simplicial complex. Using the observation that $\text{HN}(\cdot)$ is independent of the twist by $\mathcal{O}(\infty)$, one can define for $\sigma \in \mathcal{T}_\bullet(W)$ a simplicial subcomplex $\mathcal{X}_\bullet(P)_\sigma \subset \mathcal{X}_\bullet(P)$ based on the classes of vector bundles $\mathcal{F}$ such that $\sigma \subset \text{HN}(\mathcal{F})$. Consequently define the **unstable subcomplex** as

$$\mathcal{X}_\bullet(P)^{un} := \bigcup_{\substack{\sigma \in \mathcal{T}_\bullet(W) \\ \text{simplex}}} \mathcal{X}_\bullet(P)_\sigma.$$

Theorem 1.2.4. ([Gra82, Thm.4.1],[Gra82, Cor.4.2])

- (i) For each simplex σ of the Tits building $\mathcal{T}_\bullet(W)$, $\mathcal{X}^\bullet(P)_\sigma$ is contractible.
- (ii) There is a natural Γ_P -equivariant homotopy equivalence

$$\mathcal{X}^\bullet(P)^{un} \simeq \mathcal{T}_\bullet(W).$$

We briefly indicate the main idea that goes into the proof of (i). Suppose σ is given by $(0 \subsetneq W_1 \subsetneq W_2 \dots \subsetneq W_n = W)$, then define the flag $\bar{\sigma} := (0 \subsetneq W_2/W_1 \dots \subsetneq W/W_1) \in \mathcal{T}_\bullet(W/W_1)$. The canonical quotient $\mathcal{F} \mapsto \mathcal{F}/\mathcal{F} \cap W_1$ induces a map of simplicial complexes

$$\alpha : \mathcal{X}_\bullet(P)_\sigma \rightarrow \mathcal{X}_\bullet(P/P \cap W_1)_{\bar{\sigma}}.$$

By using properties on concatenation of Harder-Narasimhan filtrations [Gra82, Cor.3.4], one can construct a section of α

$$\beta : \mathcal{X}_\bullet(P/P \cap W_1)_{\bar{\sigma}} \rightarrow \mathcal{X}_\bullet(P)_\sigma$$

as well as self maps

$$G_n : \mathcal{X}_\bullet(P)_\sigma \rightarrow \mathcal{X}_\bullet(P)_\sigma, \quad n \in \mathbb{Z}$$

such that G_n is homotopic to G_{n+1} on the geometric realizations and they further satisfy the following relation for $\langle \mathcal{F} \rangle \in \mathcal{X}_\bullet(P)_\sigma$

$$G_n(\langle \mathcal{F} \rangle) := \begin{cases} \langle \mathcal{F} \rangle; & n \ll 0 \\ G_n \beta \alpha(\langle \mathcal{F} \rangle); & n \gg 0 \end{cases}.$$

A straightforward application of Whitehead's lemma (cf. 2.1.8) then concludes the proof of (i). (ii) follows from (i) using a general argument for showing homotopy equivalence between simplicial complexes given in [Gra82, Lemma 1.9]. One can also refer to Proposition 2.2.19 below where this result is updated to the equivariant setting.

From (ii) we can conclude an isomorphism between the top (reduced) simplicial homologies

$$\tilde{H}_{r-2}(\mathcal{X}^\bullet(P)^{un}, \mathbb{Z}) \simeq \tilde{H}_{r-2}(\mathcal{T}_\bullet(W), \mathbb{Z}).$$

As alluded to before, the right side is defined to be the *Steinberg module*, denoted here by $\text{St}(W)$. If one declares a simplex in the complement $\mathcal{X}_\bullet(P) \setminus \mathcal{X}_\bullet(P)^{un}$ to be a *semistable simplex* (so that a semistable vertex is precisely the class of a semistable bundle), then the relative homology $H_\bullet(\mathcal{X}_\bullet(P)^{ss}, \mathbb{Z}) := H_\bullet(\mathcal{X}_\bullet(P), \mathcal{X}_\bullet(P)^{un}, \mathbb{Z})$ can be described as the homology of a complex on semistable simplices. From long exact sequence in homology, we obtain

$$H_{r-1}(\mathcal{X}_\bullet(P)^{ss}, \mathbb{Z}) \simeq \text{St}(W). \quad (1.13)$$

Remark 1.2.5. Using [LRS93, pg. 237, (4.3)], which is an elaboration of a remark of Seshadri in [Ses82], one can show that every semistable vertex of $\mathcal{X}_\bullet(P)$ is $\Gamma_P(I)$ -stable for a non-zero proper ideal $I \subset A$. In the same vein, one can also prove that a semistable simplex is $\Gamma_P(I)$ -stable if $\deg(I) > \deg(\infty)(r-1)$. Though we do not attempt to provide more details regarding this observation, in this light 1.13 may be regarded as a higher rank analogue of 1.7.

1.2.2 Main results of the thesis - a synopsis

The contents of the main text are split into two parts. The main thrust of the first part is to understand the relation between Γ harmonic cocycles and their values on the Γ -stable part of the Bruhat-Tits building, for $\Gamma \subset \text{GL}(W)$ a principal congruence subgroup. Crucial for this are, a technique for constructing harmonic cocycles using generators of the Steinberg module and the generalization of 1.7 by adapting the proof strategy of Grayson-Quillen explained earlier. We are able to show that the

higher rank version of Teitelbaum's restriction map 1.9 is surjective in general and an isomorphism in the $\mathbb{F}_q[t]$ case. In part two, after setting up sufficient prerequisites on Drinfeld moduli space and function field crystals, using results from part I we construct a Eichler-Shimura map similar to 1.10.

Part I

Chapter 2 contains preparatory material on simplicial complexes which serve as a prerequisite for the rest of the thesis. Of particular importance are the results from [Gra82, §1] regarding homotopy equivalences of simplicial complexes, updated in this chapter to their equivariant counterparts following [TW91]. In chapter 3 we collect the various equivalent definitions of Bruhat-Tits building of rank $r - 1$ that will be utilized by us later and also recap some well known and some easily deducible properties of stabilizers in congruence subgroups of simplices in the building. We also setup the terminology regarding apartments $\mathcal{A}_\bullet(\underline{w})$ and sectors inside the Bruhat-Tits building, for an ordered K -basis $\underline{w}$ of W . We conclude the chapter by defining the space of harmonic cocycles $C_{har}(V)$ for a L -vector space V , where $L \supset K$ is a field.

In the work of Ash et al [AR79; AGM12], a set of generators of the Steinberg module $\text{St}(W)$ is explicitly described for each ordered K -basis $\underline{w}$ as above and denoted by $[[\underline{w}]]$. Using these generators, called *Ash-Rudolph symbols* in chapter 4, we construct a canonical harmonic cocycle which vanishes outside the apartment $\mathcal{A}_\bullet(\underline{w})$ thereby giving a map $\text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow C_{har}(V)$. Moreover if V is a left $L[\Gamma]$ -module for a congruence subgroup Γ , by an “averaging” argument as well as analyzing the growth of stabilizers of Γ in apartments, one obtains a map of L -vector spaces (4.2.8)

$$\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow C_{har}(V)^\Gamma.$$

In chapter 5 we extend and generalize the homotopy equivalences 1.6 and Theorem 1.2.4(ii) adapting the latter's proof technique, to an $\text{Aut}_A(P)$ equivariant homotopy equivalence (5.2.2)

$$\mathcal{BT}_\bullet(W)^{\Gamma_P(I)\text{-un}} \sim \mathcal{T}_\bullet(W).$$

A long exact sequence in homology gives an $\text{Aut}_A(P)$ -equivariant isomorphism

$$\text{GQ}_{\Gamma_P(I)} : \text{St}^I(P) \simeq \text{St}(W)$$

and is a generalization of 1.7.

The map obtained by restricting harmonic cocycles to the Γ -stable part, as in 1.9, can be defined in a straightforward manner in higher ranks and combined with the maps AR and GQ it gives rise

to a triangle

$$\begin{array}{ccc}
 & C_{har}(V)^{\Gamma_P(I)} & \\
 \swarrow \text{Te}_{\Gamma_P(I)} & & \nwarrow \text{AR}_{\Gamma_P(I)} \\
 \text{St}^I(P) \otimes_{\Gamma_P(I)} V & \xleftarrow[\sim]{\text{GQ}_{\Gamma_P(I)}^{-1} \otimes 1} & \text{St}(W) \otimes_{\Gamma_P(I)} V
 \end{array}$$

In chapter 6 we show that this is a commutative triangle, see 6.2.3. The crux of the argument requires us to describe explicitly the preimage under $\text{GQ}_{\Gamma_P(I)}$ of the Ash-Rudolph symbols and this is done in Theorem 6.1.12. In the concluding chapter 7 of the first part, we show that the maps AR and Te are isomorphisms in the $\mathbb{F}_q[t]$ case. This is achieved by generalizing to $\mathcal{BT}_\bullet(W)$, the description of Bruhat-Tits tree by Serre in [Ser80, Ch.2, §2.4.1], using the fact that the fundamental domain of $\text{GL}_r(\mathbb{F}_q[t])$ action on the building is the standard sector. The latter fact follows from the work of Soulé in [Sou79] but in order to be self contained we also provide a technically alternate proof of it in Theorem 7.2.1.

This brief discussion on the first part of the thesis can be summarized as follows.

Theorem 1.2.6. (i) (Proposition 4.2.8) Let $\Gamma \subset \text{GL}(W)$ be a congruence subgroup and V be a left $L[\Gamma]$ -module for a field $L \supset K$. Then there exists a canonical map of L -vector spaces

$$\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow C_{har}(V)^\Gamma,$$

defined as follows:- for any ordered K -basis $\underline{w} = (w_1, \dots, w_r)$ of W and $v \in V$, the pure tensor $[[w_1, \dots, w_r]] \otimes v$ is sent to $\sum_{\gamma \in \Gamma} \gamma f_{\underline{w}}^v \gamma^{-1}$, where $f_{\underline{w}}^v$ is the harmonic cocycle which vanishes outside the apartment $\mathcal{A}_\bullet(\underline{w})$ associated with $\underline{w}$ and has the value v on the standard pointed chamber of $\mathcal{A}_\bullet(\underline{w})$.

- (ii) (Theorem 5.2.2) There is a natural $\text{Aut}_A(P)$ -equivariant homotopy equivalence between the geometric realizations $|\mathcal{BT}_\bullet(W)^{\Gamma_P(I)\text{-un}}|$ and $|\mathcal{T}_\bullet(W)|$.
- (iii) (Proposition 6.2.3) AR_Γ is a section of $(\text{GQ}_{\Gamma_P(I)} \otimes 1_V) \circ \text{Te}_{\Gamma_P(I)}$. In particular, $\text{Te}_{\Gamma_P(I)}$ is surjective.
- (iv) (Corollary 7.3.8) When $A = \mathbb{F}_q[t]$, $\text{Te}_{\Gamma_P(I)}$ as well as $\text{AR}_{\Gamma_P(I)}$ are isomorphisms.

Part II

Here we emulate the construction in [Boe02] of the map $ES_2^{(n)}$ (1.10) for higher ranks. Setting up the jargon to define the appropriate crystal on the compactified Drinfeld moduli space for such a

map is taken up in chapters 8 and 9. In the former, the relation between algebraic and analytic Drinfeld moduli space is clarified following [Dri74; Boe02] while the latter is a recapitulation of results on function field crystals given in [BH07; Boe02; BP09]. Using Pink's Satake compactification of the algebraic Drinfeld moduli space, we construct the analogue of the map $ES_2^{(n)}$ in Theorem 10.3.11. Combining this with the PhD work of Gräf [Grä21], we draw a parallel in higher rank of the diagram 1.12 and conclude by speculating on some future directions.

Acknowledgements

“Every book is a quotation; and every house is a quotation out of all forests and mines and stone quarries; and every man is a quotation from all his ancestors.”

Ralph Waldo Emerson [Eme76]

First and foremost I would like to thank my advisor Gebhard Böckle for his patience in guiding me as a PhD student and for his insights and expertise leading to the main results of the thesis. I am equally grateful for his sagacious advice on structuring the thesis coherently and bringing it to fruition. I am deeply indebted to him for the countless weekly meetings we had over the years, which apart from being the genesis of the present work, were also a prominent source of learning experience for me on how to think about the theory and problems of function field arithmetic, and also other areas of math, which I could not have known by reading any book. I am also grateful to him for our joint work [BV26] which has been reproduced in this thesis and for letting me incorporate in the dissertation some ideas which are purely his. In those instances I was merely a note taker such as for the result of chapter 4 on constructing harmonic cocycles from Ash-Rudolph symbols, as well as the description of standard sector in Bruhat-Tits building as the fundamental domain for the $GL_r(\mathbb{F}_q[t])$ action in section 7.2. I am also intellectually indebted to Daniel Grayson and Daniel Quillen for their work [Gra82], whose ideas form the heart of the thesis.

I am thankful to Avner Ash for pointing out the appropriate reference on generators of Steinberg module from his works, cited here. I would like to also express my gratitude to Mihran Papikian for refereeing my thesis, for pointing out corrections as well as his practical remarks towards making the presentation better and for inviting me to give an online talk on the thesis in the Algebra and Number Theory seminar at Penn State and helpful conversations regarding the same. In this regard I would like to thank my colleagues Dahli, Alberto, Alireza, Andrea, Miriam and Pietro as well as my brother Srikar for patiently listening to a rehearsal of my talk and suggesting constructive changes. For fruitful conversations regarding the thesis I am also thankful to Ariane Mezard and Peter Gräf.

I am grateful to Oğuz Gezmiş for collaborating with me on our paper [GV25], which not only broadened my understanding of function field arithmetic but his experience also helped me grow as a writer of mathematics and thanks to him for letting me reuse parts of this joint work in the thesis.

Special thanks to the math department at Heidelberg and also CRC 326 GAUS for organizing the most interesting and enlightening workshops, seminars and colloquia over the last four years. In particular I am grateful to the past and present members of AG Böckle for the countless hours we spent co-organizing and learning through seminars and in this regard I would like to mention Gebhard, Judith, Oğuz, Barinder, Alireza, Andrea, Peter, Paola, Julian, Jakob, Theresa and Giacomo. Also thanks to Judith Ludwig for giving me the opportunity to teach in her lecture course on “Representation theory of p -adic groups” and to her and Junyan for helping me formalize an interesting bit of math in the Lean workshop on “Formalizing algebraic geometry” held in November 2025.

I also acknowledge the support of Deutsche Forschungsgemeinschaft (DFG) through CRC-TRR 326 ‘Geometry and Arithmetic of Uniformized Structures’, project number 444845124, during the whole duration of my PhD. I am also grateful to Department of Science and Technology, Government of India for funding my undergraduate and postgraduate studies at CMI through the KVPY fellowship.

I would also like to acknowledge the use of Gemini (Google) for technical assistance with LaTeX typesetting, formatting adjustments and for the code as well as refining the clarity of most figures.

My interest in mathematics developed while I was a student in India and I would like to express my sincere gratitude to all the teachers of that land who have been a perpetual source of inspiration for me. Especially I would like to thank teachers in all the schools I attended, as well as professors at my alma mater, CMI, for inculcating and nurturing in me the spirit of learning which has brought me this far. I am particularly grateful to Pramath Sastry for introducing me to rigid analytic geometry and to Sukhendu Mehrotra for guiding me for a master’s thesis in arithmetic geometry.

This work could not have been completed without the constant motivation from my family. I am indebted to my brother for presenting me with the math books that got this journey started and encouraging me to pursue a doctorate abroad; to my father for his persistence in my joining CMI and being a guiding figure; and last but not least to my mother for her unwavering counsel and support through all the ebbs and flows.

To the memory of my grandfather Shri. V. S. Dhruva Rao and to all my teachers.

Part I

An equivariant homotopy equivalence and generalization of a result of Teitelbaum

Summary of chapters - Part I

Chapter 2 sets up the terminology regarding simplicial complexes, their geometric realization as CW complexes and focuses attention on their equivariant properties. Significant for our concerns later are the equivariant version of Whitehead's theorem 2.1.8 and also equivariant counterparts of the discussions from [Gra82, §1] covered in Section 2.2.

Chapter 3 introduces the simplicial complex fundamental for the thesis namely the Bruhat-Tits building $\mathcal{BT}_\bullet(W)$ of rank $r - 1$ along with its other incarnations and we also describe its geometric realization via Goldman-Iwahori in 3.1.3. In section 3.2 the description of the stabilizers in congruence subgroups of simplices of $\mathcal{BT}_\bullet(W)$ is given. We also setup the terminology regarding apartments $\mathcal{A}_\bullet(\underline{w})$ and sectors $\mathcal{S}_\bullet(\underline{w}, \sigma)$ inside the Bruhat-Tits building, for an ordered K -basis $\underline{w}$ of W and $\sigma \in \mathcal{S}_r$. Among other things, we note here an interesting interplay between two permutations $\tau_{\underline{w}}$ and $\sigma_{\underline{w}}^{x-nbd}$ associated to chambers in $\mathcal{A}_\bullet(\underline{w})$, see Proposition 3.3.7. In section 3.3.2 harmonic cocycles $C_{har}(V)$ for a L -vector space where $L \supset K$ is a field are defined.

Chapter 4 contains the first main result of the thesis. We give here a construction of harmonic cocycles on $\mathcal{BT}_\bullet(W)$ invariant under a congruence subgroup from the generators of Steinberg module, which we call Ash-Rudolph symbols. The terminology and properties of the latter are developed first in section 4.1. In section 4.2 the aforementioned construction is given in three steps, where the first two steps show that a certain construction of a harmonic cocycle on an apartment corresponding to a Ash-Rudolph symbol is well defined and the last step involves an analysis of the size of simplex stabilizers in congruence subgroups to construct the required harmonic cocycle invariant under congruence subgroups.

Chapter 5 is the technical heart of the thesis where a equivariant homotopy equivalence between Γ -unstable region of the Bruhat-Tits building and the rational Tits boundary is shown for a principal congruence subgroup $\Gamma \subset \mathrm{GL}(W)$. We emulate the proof technique of a similar result in [Gra82]. The analogue of [Gra82, Prop.4.1] is taken up in section 5.1 and that of the homotopy equivalence [Gra82, Cor.4.2] in section 5.2. The concluding section 5.3 discusses the compatibility of the above homotopy equivalence while changing Γ .

Chapter 6 presents an application of the results from previous chapters by obtaining a partial generalization of a result of Teitelbaum [Tei91] showing that harmonic cocycles invariant under suitable congruence subgroups Γ are uniquely determined by their values on the Γ -stable edges of the Bruhat-Tits tree. In section 6.1 we analyze the cycle corresponding to the Ash-Rudolph symbol in the homotopy equivalence of the last chapter and use this in section 6.2 to show that, Teitelbaum's restriction map on harmonic cocycles to the Γ -stable part of the building, for principal congruence subgroups Γ , is a surjective map.

Chapter 7 concludes by showing that in the $\mathbb{F}_q[t]$ -case, the restriction map of Teitelbaum from before is also injective and hence an isomorphism. This is motivated from an idea implicit in Serre's description of the fundamental domain of Bruhat-Tits tree for the $\mathrm{GL}_2(\mathbb{F}_q[t])$ -action and briefly discussed in section 7.1. Following this in higher ranks, in section 7.2 we describe the fundamental domain of Bruhat-Tits building by $\mathrm{GL}_r(\mathbb{F}_q[t])$ -action as the standard sector. After analyzing the arrangement of chambers in the standard sector using cones of simplices, in section 7.3 we obtain the required isomorphism.

Chapter 2

Some equivariant properties of G-simplicial complexes

2.1 Preliminaries on (G-)simplicial complexes

We collect here important definitions and results on simplicial complexes which will be required by us. We mainly follow [Gra82, §1] and for more details refer the reader there.

Recall that a *simplicial complex* is a pair (V, X) , where V is called the set of vertices and X is a collection of finite subsets of V such that if $Y \in X$ and $Z \subset Y$ then $Z \in X$. A set $Y \in X$ of cardinality $d+1$, for $d \geq 0$, is called a *d-simplex*. In particular, vertices are *0-simplices*. We denote the collection of all *d*-simplices of the pair (V, X) by X_d . Since $X_0 = V$ we will often denote the pair (V, X) simply by X . Moreover, a simplicial complex X is said to be a rank n , if $X_n \neq \emptyset$ and $X_i = \emptyset \forall i > n$. If $X_d \neq \emptyset$ for all $d \in \mathbb{N}$, then X is said to have ∞ -rank. A map between simplicial complexes $f : X \rightarrow Y$ is said to be a *simplicial map*, if it is a map of sets $X_0 \rightarrow Y_0$ such that the image of a simplex is again a simplex.

By axiom of choice, we can choose a partial ordering on X_0 such that each simplex becomes totally ordered. This motivates the following definition

Definition 2.1.1. An **ordered simplicial complex** is a simplicial complex X together with a partial ordering on X_0 with respect to which each simplex is totally ordered.

Example 2.1.2. Let $d \geq 0$ be an integer. The *standard d-simplex*, denoted Δ_d , is the ordered simplicial complex with vertices $\{0, \dots, d\}$, together with the order inherited from $\mathbb{N}$, and its simplices are all finite subsets of $\{0, \dots, d\}$. It has rank d .

2.1.1 Geometric realization, $(G-)$ CW complexes and equivariant Whitehead's theorem

Assume for simplicity that $X = (X_i)$ is a rank n simplicial complex for a non-negative integer n . One can attach a canonical topological space associated to X inside $\mathbb{R}^n$, denoted by $|X|$, as follows. Let $I := [0, 1]$ be the closed unit interval. Then define

$$|X| := \{(t_v) \in \prod_{v \in X_0} I \mid \{v \in X_0 \mid t_v \neq 0\} \text{ is a simplex and } \sum_{v \in X_0} t_v = 1\}.$$

In particular, we see that $|\Delta_d| = \{(t_i)_{i=0}^{d-1} \in \mathbb{R}^{d+1} \mid \sum_i t_i = 1\}$, which is the standard geometric d -simplex. Consequently for each simplex $\sigma \in X_d$ which is simplicial complex in its own right, $|\sigma|$ is homeomorphic to $|\Delta_d|$. So declare the topology on $|X|$ as the weak topology with respect to these geometric simplices in the sense that, $M \subset |X|$ is closed if and only if $M \cap |\sigma| \subset |\sigma|$ is closed for all $\sigma \in \cup_{i \geq 0} X_i$.

A G -simplicial complex is a simplicial complex X equipped with a (left) action of G such that each $g \in G$ acts via simplicial maps on X . The consequent G action on $|X|$ which is continuous. Later on, it will be beneficial for us to also regard the geometric realization $|X|$ as a G -CW complex. We recall the required definition. Call a topological space $\mathcal{Y}$ a G -space, if G acts left continuously on $\mathcal{Y}$.

Definition 2.1.3. ([Mis03, §2], [Hat02, Ch.0, Page 5], [MPC96])

- (i) A topological space $\mathcal{Y}$ is said to be a **CW complex of rank n** if it admits a filtration of closed subsets $\mathcal{Y}_0 \subset \mathcal{Y}_1 \subset \dots \subset \mathcal{Y}_n = \mathcal{Y}$ such that the following are satisfied:-
 - (a) $\mathcal{Y}_0 \subset \mathcal{Y}$ is a discrete topological space;
 - (b) Let $\mathbb{D}^d := \{(x_i)_{i=1}^d \in \mathbb{R}^d \mid \sum_i x_i^2 \leq 1\}$ denote the d -dimensional unit disc and $\mathbb{S}^{d-1} := \partial \mathbb{D}^d$ denote the corresponding $(d-1)$ -dimensional unit sphere. Then there exists a discrete topological space P_d for all $0 \leq d \leq n$ and maps $f_d : \mathbb{S}^{d-1} \times P_d \rightarrow \mathcal{Y}_{d-1}$ and $\hat{f}_d : \mathbb{D}^d \times P_d \rightarrow \mathcal{Y}_d$ such that we have a pushout diagram

$$\begin{array}{ccc} \mathbb{S}^{d-1} \times P_d & \xrightarrow{f_d} & \mathcal{Y}_{d-1} \\ \downarrow & & \downarrow \\ \mathbb{D}^d \times P_d & \xrightarrow{\hat{f}_d} & \mathcal{Y}_d \end{array} \quad \lrcorner$$

where the left vertical arrow is given by natural inclusion on the first coordinate and identity on the second coordinate, and the right vertical arrow is inclusion.

- (ii) Moreover let $\mathcal{Y}$ be a G -space. Then $\mathcal{Y}$ is said to be a **G-CW complex** if $\mathcal{Y}$ is a CW complex as above, with the further condition that each $\mathcal{Y}_i$ is a closed G -space of $\mathcal{Y}$, each P_d is a G -set and all the maps f_d and $\widehat{f}_d$ are G -equivariant (with G -acting trivially on $\mathbb{S}^{d-1}$ and $\mathbb{D}^d$).

Lemma 2.1.4. Let X be a G -simplicial complex of rank n . Then $|X|$ is a G -CW complex. Moreover, for any two G -simplicial complexes X, Y , any G -simplicial map $X \rightarrow Y$ induces a canonical continuous G -map of G -CW-complexes $|X| \rightarrow |Y|$.

Proof. We first consider the case $G = \{1\}$. In the notation of the definition above, we set $\mathcal{Y} = |X|$, $\mathcal{Y}_d = \bigcup_{\sigma \in X_d} |\sigma|$ and $P_d = X_d$. We claim that $|X|$ becomes a CW complex with respect to these $\mathcal{Y}_i$'s and P_d 's. 2.1.3(i)(a) is clearly true since $|X|$ is equipped with the weak topology. For a simplex $\sigma \in X_d$ denote by $\partial\sigma := \{\tau \mid \tau \subset \sigma \text{ and } \tau \in X_{d-1}\}$. It is clear that there is a natural pushout diagram

$$\begin{array}{ccc} \mathbb{S}^{d-1} \times \{\sigma\} & \xrightarrow{f_{d,\sigma}} & \bigcup_{\tau \in \partial\sigma} |\tau| \\ \downarrow & & \downarrow \\ \mathbb{D}^d \times \{\sigma\} & \xrightarrow{\widehat{f}_{d,\sigma}} & |\sigma| \end{array}$$

where $f_{d,\sigma}$ (resp. $\widehat{f}_{d,\sigma}$) is the natural homotopy between $\bigcup_{\tau \in \partial\sigma} |\tau|$ (resp. $|\sigma|$) and boundary of the unit disc (resp. the unit disc) with barycenter as the center and drawn around the vertices of σ . Considering $\bigcup_{\sigma \in X_d} f_{d,\sigma}$ and $\bigcup_{\sigma \in X_d} \widehat{f}_{d,\sigma}$, we see that 2.1.3(i)(b) also holds for $|X|$, and hence $|X|$ is a CW complex.

It is clear from the construction of $\mathcal{Y}_i$'s, P_d 's and $f_d, \widehat{f}_d$, that all the necessary conditions of 2.1.3(ii) are also satisfied by them so that $|X|$ is a G -CW complex, concluding the proof. $\square$

G -CW complexes have very nice G -equivariant homotopy properties. A fundamental and very useful result is the equivariant version of Whitehead's theorem. Before stating it, let us recap the following.

- Definition 2.1.5.** (i) Let $\mathcal{Y}, \mathcal{Z}$ be two G -spaces and $f, g : \mathcal{Y} \rightarrow \mathcal{Z}$ be two continuous maps of G -spaces. Then f and g are said to be **G -homotopic** to each other, if there is a homotopy $F : \mathcal{Y} \times [0, 1] \rightarrow \mathcal{Z}$ between f and g such that F is also G -equivariant (with G acting trivially on $[0, 1]$).
- (ii) A G -equivariant map $f : \mathcal{Y} \rightarrow \mathcal{Z}$ which is a homotopy equivalence is said to be a **G -equivariant homotopy equivalence**. Moreover, if there exists a G -map $f' : \mathcal{Z} \rightarrow \mathcal{Y}$ such that $f' \circ f$ is G -homotopic to $Id_{\mathcal{Y}}$ and $f \circ f'$ is G -homotopic to $Id_{\mathcal{Z}}$, then f is said to be a **G -homotopy equivalence**.

(iii) A G -space $\mathcal{Y}$ is said to be G -**contractible**, if $f : \mathcal{Y} \rightarrow \{*\}$ is a G -homotopy equivalence.

Below, for a G -space $\mathcal{Y}$ and a subgroup $H < G$, we set $\mathcal{Y}^H := \{y \in \mathcal{Y} \mid gy = y \forall g \in H\}$.

Remark 2.1.6. A necessary condition for a G -space $\mathcal{Y}$ to be G -contractible is that $\mathcal{Y}^G \neq \emptyset$. Indeed, from (ii) above, we see that the existence of a G -map $f' : \{*\} \rightarrow \mathcal{Y}$ implies that the image of $*$ is a G -invariant point.

Theorem 2.1.7. (Equivariant Whitehead's theorem I, [Mis03, Thm.2.2], [MPC96, Cor.3.3]) Let $\mathcal{Y}, \mathcal{Z}$ be G -CW complexes and $f : \mathcal{Y} \rightarrow \mathcal{Z}$ a G -map between them. If for all subgroups $H < G$, for all $y_0 \in \mathcal{Y}^H$ and for all $i \geq 0$, the induced map on homotopy groups $\pi_i(\mathcal{Y}^H, y_0) \rightarrow \pi_i(\mathcal{Z}^H, f(y_0))$ is an isomorphism, then f is a G -homotopy equivalence.

Applying the theorem to the constant map $\mathcal{Y} \rightarrow \{*\}$, we get as an immediate consequence,

Corollary 2.1.8. (Equivariant Whitehead's theorem II) Let $\mathcal{Y}$ be a G -space and suppose that for all subgroups $H < G$, $\mathcal{Y}^H \neq \emptyset$. If for all base points $y_0 \in \mathcal{Y}^H$ and all $i \geq 0$, the homotopy groups $\pi_i(\mathcal{Y}^H, y_0) = \{1\}$, then $\mathcal{Y}$ is G -contractible.

2.2 Homotopy properties of G -simplicial complexes

Definition 2.2.1. ([Gra82, Defn.1.2]) Let X, Y be G -simplicial complexes such that X is ordered. Two G -simplicial maps $f, g : X \rightarrow Y$ are said to be **adjacent**, if for each simplex σ of X and for each $x \in \sigma$, the set

$$\{f(x') \mid x' \in \sigma \text{ and } x' \leq x\} \cup \{g(x') \mid x' \in \sigma \text{ and } x \leq x'\}$$

is a simplex in Y .

Remark 2.2.2. [Gra82, Defn.1.2] has a typographical error rendering the Corollary [Gra82, p. 1.3] wrong (which is the Proposition 2.2.5 below). This has been corrected above.

Definition 2.2.3. Given ordered G -simplicial complexes X, Y , the **product simplicial complex**, denoted by $X \times Y$, is the simplicial complex whose vertex set is $X_0 \times Y_0$, and $\{(x_1, y_1), \dots, (x_i, y_i)\}$ is a simplex iff $x_1 \leq \dots \leq x_i$ and $y_1 \leq \dots \leq y_i$, and moreover $\{x_1, \dots, x_i\}$ as well as $\{y_1, \dots, y_i\}$ are simplices. It is again a G -simplicial complex with respect to the diagonal G -action.

Lemma 2.2.4. ([Gra82, Lemma 1.1]) The natural map $|X \times Y| \rightarrow |X| \times |Y|$ is a G -homeomorphism.

Proof. For $G = \{1\}$, the proof is in op.cit. For general G , the same proof follows. $\square$

The notion of adjacent maps captures the idea of homotopy and on the level of simplicial complexes, it is explained by the following

Proposition 2.2.5. Suppose X is an ordered G -simplicial complex and Y is a G -simplicial complex. Then adjacent simplicial maps between X to Y are G -homotopic, that is their geometric realizations are G -homotopic.

Proof. Let $f, g : X \rightarrow Y$ be two adjacent G -simplicial maps. Recall the 1-simplex $\Delta_1 := \{0, 1\}$. Adjacency is precisely the condition required to ensure that the following map is G -simplicial (with G acting trivially on Δ_1)

$$F : X \times \Delta_1 \rightarrow Y$$

sending

$$(x, i) \mapsto \begin{cases} f(x) & ; i = 0 \\ g(x) & ; i = 1. \end{cases}.$$

Lemma 2.2.4 finishes the proof since the geometric realization $|\Delta_1|$ is the closed unit interval $[0, 1]$. $\square$

2.2.1 Simplicial complexes on G -posets

Let G be a group. In the following, by a G -poset we mean a partially ordered set equipped with a (left) G -action which is compatible with the partial order. Most results from this subsection are generalized from their non-equivariant counterparts, which can be recovered by simply choosing $G = 1$ (or by choosing a trivial action of G).

Example 2.2.6. 1. Let $\mathcal{T}$ be a G -poset. Then one obtains a G -simplicial complex, whose vertex set is $\mathcal{T}$ and whose simplices are finite, totally ordered subsets of $\mathcal{T}$. We will denote the associated simplicial complex by $\mathcal{T}_\bullet$.

2. Let $\mathbb{E}$ be a G -poset and suppose that it carries a $\mathbb{Z}$ action commuting with the G -action. The $\mathbb{Z}$ -action is said to be **cofinal**, if for any $x, x' \in \mathbb{E}$, there exists $n \in \mathbb{Z}$ s.t. $x + n \geq x'$. For such an action of $\mathbb{Z}$ on $\mathbb{E}$, we denote by $\langle \mathbb{E} \rangle$ the G -simplicial complex, whose vertex set consists of $\mathbb{Z}$ -orbits in $\mathbb{E}$, and a simplex is defined to be any finite nonempty set of vertices whose union is a chain in $\mathbb{E}$. For $t \in \mathbb{E}$, the corresponding $\mathbb{Z}$ -orbit is denoted by $\langle t \rangle \in \langle \mathbb{E} \rangle$.
3. The free abelian group $\mathbb{Z}^r$ is a G -poset with the trivial G -action and a cofinal $\mathbb{Z}$ -action defined as follows:- we declare $(x_1, \dots, x_r) \leq (y_1, \dots, y_r)$, for $x_i, y_j \in \mathbb{Z}$, if and only if $x_i \leq y_i$ for all $1 \leq i \leq r$. The $\mathbb{Z}$ -action is given by acting $n \in \mathbb{Z}$ on $(x_1, \dots, x_r)$ to give $(x_1 + n, \dots, x_r + n)$.

The corresponding simplicial complex $\langle \mathbb{Z}^r \rangle$ is called the **Coxeter complex of type $\widetilde{A}_{r-1}$** . We will denote it by $\mathcal{A}_\bullet$ omitting the dependence on r when it is clear from the context.

Lemma 2.2.7. ([Gra82, Lemma 1.5]) Let $\mathbb{E}_1, \mathbb{E}_2$ be two G -posets with a cofinal $\mathbb{Z}$ -action on them and $f : \mathbb{E}_1 \rightarrow \mathbb{E}_2$ be a G -map satisfying

- (i) for $t \leq t'$ in $\mathbb{E}_1$, $f(t) \leq f(t')$;
- (ii) for all $t \in \mathbb{E}_1$, $f(t+1) = f(t) + 1$.

Then the map $\langle f \rangle : \langle \mathbb{E}_1 \rangle \rightarrow \langle \mathbb{E}_2 \rangle$, defined by sending $\langle x \rangle \mapsto \langle f(x) \rangle$, is a G -simplicial map.

Proof. The proof of [Gra82, Lemma 1.5] is the statement of the lemma with $G = 1$. The proof for general G is also identical. $\square$

We would like to extend the Definition 2.2.3 to the situation of posets with cofinal $\mathbb{Z}$ -action, so as to obtain a result similar to Proposition 2.2.5. For this, we need the following definition

Definition 2.2.8. Let $\mathbb{E}$ be a G -poset with a cofinal $\mathbb{Z}$ -action. A map $\varepsilon : \mathbb{E} \rightarrow \mathbb{Z}$ is said to be an **augmentation map**, if ε satisfies the two conditions (i), (ii) of Lemma 2.2.7.

Also in this case, set $\mathbb{E}_0 := \{t \in \mathbb{E} \mid \varepsilon(t) = 0\}$.

Remark 2.2.9. (i) Note that above G -equivariance of ε is not assumed.

- (ii) Given an augmentation map $\varepsilon : \mathbb{E} \rightarrow \mathbb{Z}$, it is easy to observe that $\mathbb{E}_0$ is a simplicial complex with $\{x_0, \dots, x_d\}$ being a d -simplex if and only if $x_0 < \dots < x_d < x_0 + 1$ ([Gra82, Defn.1.6]). $\mathbb{E}_0$ is isomorphic to $\langle \mathbb{E} \rangle$, but $\mathbb{E}_0$ being a subset of $\mathbb{E}$, is also ordered.

Definition 2.2.10. Let $\mathbb{E}_1, \mathbb{E}_2$ be two G -posets with a cofinal $\mathbb{Z}$ -action and $\varepsilon : \mathbb{E}_1 \rightarrow \mathbb{Z}$ be an augmentation map. Then two G -maps $f, g : \mathbb{E}_1 \rightarrow \mathbb{E}_2$ are said to be **adjacent** (with respect to ε) if they satisfy the following:-

- (i) for $t \in \mathbb{E}_1$, $f(t) \leq g(t)$;
- (ii) for $t < t'$ in $\mathbb{E}_1$ with $\varepsilon(t) < \varepsilon(t')$, $g(t) \leq f(t')$.

We can now state the analogue of Proposition 2.2.5 in this situation

Proposition 2.2.11. ([Gra82, Lemma 1.8]) Suppose $\mathbb{E}_1, \mathbb{E}_2$ are as above and if $f, g : \mathbb{E}_1 \rightarrow \mathbb{E}_2$ are adjacent G -maps, the $\langle f \rangle, \langle g \rangle : \langle \mathbb{E}_1 \rangle \rightarrow \langle \mathbb{E}_2 \rangle$ are G -homotopic to each other, on the geometric realizations.

Homological complex associated to a G -poset

Here we set up the terminology regarding the (simplicial) homological complex of a G -poset $\mathcal{T}$ by explicitly describing such a complex. This will be helpful to us in later chapters where we will consider the maps induced on the homology groups, from homotopy equivalences between posets.

Definition 2.2.12. Let $\mathcal{T}$ be a G -poset and $\mathcal{T}_\bullet$ be the associated simplicial complex. Suppose that the rank of $\mathcal{T}_\bullet$ is finite, say d . Then we denote by $(C_i(\mathcal{T}_\bullet, \mathbb{Z}), \partial_i)$ the homological complex which is defined as:-

$$(i) \ C_i(\mathcal{T}_\bullet, \mathbb{Z}) = 0 \text{ for all } i > d \text{ and } i < -1;$$

$$(ii) \ C_{-1}(\mathcal{T}_\bullet, \mathbb{Z}) = \mathbb{Z};$$

$$(iii) \text{ For } 0 \leq i \leq d,$$

$$C_i(\mathcal{T}_\bullet, \mathbb{Z}) = \bigoplus_{\substack{\{x_0, \dots, x_i\} \in \mathcal{T}_i \\ x_0 < \dots < x_i}} \mathbb{Z}[x_0, \dots, x_i].$$

(iv) The differentials are given as:-

$$(a) \text{ For } 1 \leq i \leq d,$$

$$\partial_i : C_i(\mathcal{T}_\bullet, \mathbb{Z}) \rightarrow C_{i-1}(\mathcal{T}_\bullet, \mathbb{Z})$$

sends

$$[x_0, \dots, x_i] \mapsto \sum_{0 \leq j \leq i} (-1)^j [x_0, \dots, \widehat{x_j}, \dots, x_i].$$

$$(b) \ \partial_0 : C_0(\mathcal{T}_\bullet, \mathbb{Z}) = \bigoplus_{x \in \mathcal{T}_0} \mathbb{Z}[x] \rightarrow \mathbb{Z} \text{ sends } [x] \text{ to } 1.$$

Remark 2.2.13. (i) Given a simplex $s \in \mathcal{T}_i$, we will mean by $[x]_{x \in s}$ the basis symbol $[x_0, \dots, x_i]$ where $s = \{x_0, \dots, x_i\}$ and $x_0 < \dots < x_i$.

(ii) Let $\mathcal{T}'$ be another poset and $f : \mathcal{T} \rightarrow \mathcal{T}'$ be a map respecting the partial order. This induces a map of homological complexes $C_\bullet(f) : C_\bullet(\mathcal{T}_\bullet, \mathbb{Z}) \rightarrow C_\bullet(\mathcal{T}'_\bullet, \mathbb{Z})$ as follows:- given an i -simplex $s = \{x_0, \dots, x_i\}$, if $f(s) = \{f(x_0), \dots, f(x_i)\}$ is also an i -simplex in $\mathcal{T}'_\bullet$ (that is, $f(x_j) \neq f(x_k)$ for all $j \neq k$), then $C_i(f)$ maps $[s]$ to $[f(s)] \in C_i(\mathcal{T}', \mathbb{Z})$, else it sends $[s]$ to 0.

2.2.2 Some results on G -equivariant homotopies

We conclude this section by proving a G -equivariant version of [Gra82, Lemma 1.9]. This requires us to also restate [Qui78, Cor.1.8] in terms of G -equivariant homotopy, which we do so following [TW91, Thm.1].

Checking homotopy properties of maps between posets can be easy. For example, if $f, g : \mathcal{U} \rightarrow \mathcal{T}$ is a map of posets such that $f(u) \leq g(u)$, $\forall u \in \mathcal{U}$, then $|f|$ and $|g|$ are homotopic to each other ([Qui78, §1.3]). One enhances it to a G -homotopy, in the case of G -posets.

Lemma 2.2.14. (cf. [TW91, Prop.1.1]) Let $\mathcal{U}, \mathcal{T}$ be G -posets and suppose $f, g : \mathcal{U} \rightarrow \mathcal{T}$ be two maps of G -posets such that $f(u) \leq g(u)$, $\forall u \in \mathcal{U}$. Then $|f|$ and $|g|$ are G -homotopic to each other.

Proof. Note that for each simplex σ of $\mathcal{U}$ and for each $u \in \sigma$, the set $\{f(u') \mid u' \in \sigma \text{ and } u' \leq u\} \cup \{g(u') \mid u' \in \sigma \text{ and } u \leq u'\}$ is a totally ordered subset of $\mathcal{T}$ by hypothesis and hence a simplex in $\mathcal{T}$. Therefore f, g are adjacent maps(2.2.3) and hence one can conclude by applying Proposition 2.2.5. $\square$

We get two consequences of this lemma, one by setting f to be the constant map and the other from $f = id$.

Corollary 2.2.15. Let $\mathcal{T}$ be a G -poset and suppose that it contains a smallest(or largest) element, which is moreover fixed under G . Then $\mathcal{T}$ is G -contractible.

Corollary 2.2.16. ([TW91, Cor.1.2]) Let $\mathcal{T}$ be a G -poset together with a map $\phi : \mathcal{T} \rightarrow \mathcal{T}$ such that $\phi(x) \geq x$ for all $x \in \mathcal{T}$. The $\phi : \mathcal{T} \rightarrow Im(\phi)$ is a G -homotopy equivalence.

Given a G -poset $\mathcal{T}$, for $\sigma \in \mathcal{T}$, we denote by $\mathcal{T}_{\leq \sigma}$ (resp. $\mathcal{T}_{\geq \sigma}$) the set

$$\{t \in \mathcal{T} \mid t \leq \sigma\} \text{ (resp. } \{t \in \mathcal{T} \mid t \geq \sigma\}).$$

For any G -set X and $x \in X$, the stabilizer subgroup $G_x \subset G$ is defined as the subgroup of all $g \in G$ such that $gx = x$. We have the following G -equivariant version of [Qui78, Prop.1.6]

Proposition 2.2.17. ([TW91, Thm.1]) Let $\mathcal{U}, \mathcal{T}$ be two G -posets and $\phi : \mathcal{U} \rightarrow \mathcal{T}$ a map of G -posets such that, for all $\sigma \in \mathcal{T}$ $\phi^{-1}(\mathcal{T}_{\leq \sigma})$ is G_σ -contractible, that is, its geometric realization is contractible. Then ϕ is a G -homotopy equivalence on their geometric realizations. Similar statement holds true for $\mathcal{T}_{\geq \sigma}$.

We would like to also rewrite [Qui78, Cor.1.8] in G -equivariant terms. In the following, given a G -poset $\mathcal{T}$, by a G -closed subset S of $\mathcal{T}$, we mean a G -subposet that is closed under *specialization*, that is, $x' \leq x \in S \Rightarrow x' \in S$.

Proposition 2.2.18. (cf. [Qui78, Prop.1.7]) Let $\mathcal{U}, \mathcal{T}$ be two G -posets and $\mathcal{Z} \subset \mathcal{U} \times \mathcal{T}$ be a G -closed subset. Let $p_1 : \mathcal{Z} \rightarrow \mathcal{U}$ be the first projection. If the fiber $\mathcal{Z}_u := \{t \in \mathcal{T} \mid (u, t) \in \mathcal{Z}\}$, is G_u -contractible for all $u \in \mathcal{U}$, then p_1 is a G -homotopy equivalence on the geometric realizations.

Proof. We apply Proposition 2.2.17 to the map p_1 , which is a map of G -posets. Consider the two maps of posets

$$f : \mathcal{Z}_u \rightarrow p_1^{-1}(\mathcal{U}_{\geq u}), \quad g : p_1^{-1}(\mathcal{U}_{\geq u}) \rightarrow \mathcal{Z}_u,$$

given by $f(t) := (u, t)$ and $g(u', t) := t$. Note that g is well defined since $(u', t) \in p_1^{-1}(\mathcal{U}_{\geq u}) \Rightarrow (u, t) \leq (u', t)$, and since $\mathcal{Z}$ is closed, $(u, t) \in \mathcal{Z}_u$. f, g are G_u -equivariant, and $g \circ f(t) = t$ and $f \circ g(u', t) \leq (u', t)$. Hence by Lemma 2.2.14, f and g are G_u -homotopy equivalences and hence by hypothesis on $\mathcal{Z}_u$, $p_1^{-1}(\mathcal{U}_{\geq u})$ is G_u -contractible. Applying Proposition 2.2.17 finishes the proof. $\square$

Proposition 2.2.19. ([Gra82, Lemma 1.9]) Let X be a G -simplicial complex, $\mathcal{T}$ a G -poset and for each $\sigma \in \mathcal{T}$, $X_\sigma \subset X$ be a subcomplex. Suppose the following conditions holds true:-

- (i) $\sigma \leq \tau \Rightarrow X_\sigma \supset X_\tau$;
- (ii) every simplex of X is a simplex of X_σ for some $\sigma \in \mathcal{T}$, that is $X = \bigcup_{\sigma \in \mathcal{T}} X_\sigma$;
- (iii) each X_σ is contractible;
- (iv) for each simplex γ in X , the poset $\mathcal{T}_\gamma := \{\sigma \in \mathcal{T} \mid \gamma \text{ is a simplex of } X_\sigma\}$ is G_γ -contractible.

Then there is a natural G -equivariant homotopy equivalence between $|X|$ and $|\mathcal{T}|$. Moreover, if (iii) is strengthened so that each X_σ being G_σ -contractible, then $|X|$ and $|\mathcal{T}|$ are naturally G -homotopy equivalent.

Proof. Since a G -simplicial complex and its barycentric subdivision are naturally G -homotopy equivalent to each other, so without loss of generality, we can replace X and X_σ 's by their barycentric subdivision. Hence we may assume X is a poset and each of the X_σ 's is a closed subset of X , and condition (iv) can be reformulated as the poset $\mathcal{T}_x := \{\sigma \in \mathcal{T} \mid x \in X_\sigma\}$ being G_x -contractible, for all $x \in X$. Consequently, we can form the closed G -subposet $Z \subset X \times \mathcal{T}$, defined by

$$Z := \{(x, \sigma) \mid x \in X_\sigma\}.$$

The projection $p_1 : Z \rightarrow X$ is a G -equivariant map and its fiber at $x \in X$ is precisely $\mathcal{T}_x$, which by hypothesis is G_x -contractible. Hence by Proposition 2.2.18, p_1 is a G -homotopy equivalence. Also since the fiber of the second projection $p_2 : Z \rightarrow \mathcal{T}$ at $\sigma \in \mathcal{T}$ is X_σ , which is contractible, from

the non-equivariant version of Proposition 2.2.18(which is [Qui78, Prop.1.7]), we know that p_2 is a G -equivariant homotopy equivalence. If we denote by p'_1 the G -homotopy inverse of p_1 obtained by the same proposition, then $X \xrightarrow{p'_1} Z \xrightarrow{p_2} \mathcal{T}$ induces a G -equivariant homotopy equivalence between X and $\mathcal{T}$.

Moreover, when each X_σ is also G_σ -contractible, then p_2 defines a G -homotopy equivalence as well, and hence in this case $|X|$ and $|\mathcal{T}|$ become G -homotopy equivalent. $\square$

Remark 2.2.20. Note that, in all of the “ G -equivariant” results stated in this section, one obtains a “non-equivariant” version by choosing $G = 1$. Later on, this is what we will mean by a non-equivariant version of any of the above results in this section.

Chapter 3

Bruhat-Tits building and harmonic cocycles

3.1 Bruhat-Tits buildings

Let $r \geq 2$ be an integer. In this section, we recap the basic facts about the Bruhat-Tits building of rank $r - 1$ for $\mathrm{GL}_r(R)$. Recall that $R \subset K$ is the discrete valuation ring corresponding to ∞ . We first describe it as a simplicial complex on homothety classes of R -lattices in W of rank r . In a later subsection we provide an alternate description of this building as classes of rank r vector bundles on the curve C whose restriction to $\mathrm{Spec}(A)$ is a fixed projective A -module, following [Gra82, §4] and also [Ser80, Chap.II] in rank 2 case. We will see there how these two descriptions are equivalent to the standard description of Bruhat-Tits building for GL_{r,K_∞} .

Recall the notation from Chapter 1, where W denotes a fixed K -vector space of dimension r and $P \subset W$ is a rank r projective A -submodule. Set $\Gamma_P := \mathrm{Aut}_A(P)$, which is a subgroup of $\mathrm{GL}(W)$.

Also recall Example 2.2.6(2) for the following.

Definition 3.1.1. Define the poset

$$\mathbb{BT}(W) := \{L \subset W \text{ an } R\text{-submodule of rank } r\},$$

equipped with subset inclusion as the partial order. $\mathbb{Z}$ acts cofinally on it via

$$n + L := \varpi^{-n}L; \quad n \in \mathbb{Z}, L \in \mathbb{BT}(W).$$

Then

$$\mathcal{BT}_\bullet(W) := \langle \mathbb{BT}(W) \rangle.$$

As before, we let $\mathcal{BT}_i(W)$ denote its set of i simplices. The complex $\mathcal{BT}_\bullet(W)$ is called the *Bruhat-Tits building of rank $r - 1$* .

- Remark 3.1.2.** (i) The set of vertices $\mathcal{BT}_0(W)$ consists of the homothety classes of R -lattices $L \subset W$ of rank r , denoted by $\langle L \rangle$. Similarly, $\{\langle L_0 \rangle, \dots, \langle L_d \rangle\} \in \mathcal{BT}_d(W)$ is a d -simplex, if after reordering the L_i 's if necessary, $L_0 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$.
- (ii) Note that there is a canonical action of $\mathrm{GL}(W)$ acts on $\mathcal{BT}_\bullet(W)$. In particular, there is an induced action of Γ_P on $\mathcal{BT}_\bullet(W)$.
- (iii) We denote by $\widehat{\mathcal{BT}}_i(W)$ as the set of pointed i -simplices, that is the set of pairs (x, s) for an i -simplex s and $x \in s$. Note that the pointed simplex (x, s) canonically defines a ordering tuple on the vertices of s , which we will denote by $\vec{s}_x$.

3.1.1 $\mathcal{BT}_\bullet(W)$ as a building of vector bundles.

Let $\widetilde{W}$ denote the constant sheaf on C associated to W . Fix the ample line bundle $\mathcal{O}(1) := \mathcal{O}(\infty)$ (dual of the vanishing ideal at ∞ , which is a line bundle). Recall the uniformizer ϖ of the maximal ideal of R .

We will be concerned with vector bundles whose sections over $\mathrm{Spec}(A)$ are P and whose stalk at ∞ varies over rank r lattices, as explained below.

Lemma 3.1.3. (cf. [Wie05, §2.8]) Let L be an R lattice of rank r inside W . Then there exists a unique rank r vector bundle E over C such that

$$E|_{\mathrm{Spec}(A)} = \widetilde{P}, \quad E_{C, \infty} = L$$

Proof. Let $i : \mathrm{Spec}(A) = C - \infty \rightarrow C$ and $j : \mathrm{Spec}(\mathcal{O}_{C, \infty}) = \{\eta, \infty\} \rightarrow C$ be the canonical morphisms. Consider the sheaf $E := i_*(\widetilde{P}) \cap j_*(\widetilde{L}) \subset \widetilde{W}$. Then E is a coherent sheaf which is locally free, for all its stalks are free modules over the corresponding local rings. Thus we have a vector bundle satisfying the hypothesis of the lemma. Now let $E, E' \subset \widetilde{W}$ be two vector bundles in $\mathbb{E}(P)$, so that their stalks at ∞ coincide. Then $E/E \cap E' \simeq (E + E')/E'$, with the R.H.S. having all its stalks zero. Hence $E \subset E'$. By symmetry of the argument, we get $E = E'$. Hence the uniqueness of such a vector bundle. $\square$

Definition 3.1.4. Let $L \subset W$ be an $\mathcal{O}_\infty$ -module inside W of rank r , and similarly $P \subset W$ be a rank r projective A -submodule. Denote by $\mathcal{E}(P, L)$ the unique rank r vector bundle on C satisfying the hypothesis Lemma 3.1.3.

Remark 3.1.5. (i) It is easy to deduce from the proof above that $H^0(C, \mathcal{E}(P, L)) = P \cap L$, where the intersection is inside W .

- (ii) It is instructive to point out that the association $(P, L) \mapsto \mathcal{E}(P, L)$, from the category of pairs (P, L) satisfying the hypothesis of Lemma 3.1.3 to the category of rank r vector bundles on C , commutes with dualization and tensor product. That is, $\text{Hom}_{\mathcal{O}_C}(\mathcal{E}(P, L), \mathcal{O}_C) = \mathcal{E}(\text{Hom}_A(P, A), \text{Hom}_R(L, R))$ and $\mathcal{E}(P_1, L_1) \otimes_{\mathcal{O}_C} \mathcal{E}(P_2, L_2) = \mathcal{E}(P_1 \otimes_A P_2, L_1 \otimes_R L_2)$.

Definition 3.1.6. Define the poset

$$\mathbb{E}(P) := \{\mathcal{E}(P, L) \mid L \subset W \text{ a rank } r \text{ } R\text{-submodule}\}.$$

It admits a cofinal $\mathbb{Z}$ -action given by

$$n + \mathcal{E}(P, L) := \mathcal{E}(P, L)(n) := \mathcal{E}(P, L) \otimes \mathcal{O}(1)^{\otimes n}; \quad n \in \mathbb{Z}, \quad \mathcal{E}(P, L) \in \mathbb{E}(P).$$

Define the simplicial complex

$$\mathcal{X}_\bullet(P) := \langle \mathbb{E}(P) \rangle$$

and let $\mathcal{X}_d(P)$ denote the set of d -simplices of $\mathcal{X}_\bullet(P)$ for $d \geq 0$. The top dimensional simplices, in $\mathcal{X}_{r-1}(P)$, are called *chambers*.

Remark 3.1.7. Note that the set of vertices $\mathcal{X}_0(P)$ consists of $\mathbb{Z}$ orbit of vector bundles $\mathcal{E}(P, L)$ and $(\langle \mathcal{E}_0 \rangle, \dots, \langle \mathcal{E}_d \rangle) \in \mathcal{X}_d(P)$ forms a d -simplex if after choosing appropriate representatives $\mathcal{E}_0, \dots, \mathcal{E}_d$ we have $\mathcal{E}_0 \supsetneq \dots \supsetneq \mathcal{E}_d \supsetneq \mathcal{E}_0(-1)$.

There is a canonical left action of Γ_P on $\mathcal{X}_\bullet(P)$, given by $g \cdot \mathcal{E}(P, L) := \mathcal{E}(P, gL)$ for $g \in \Gamma_P$.

A priori it is not clear whether the geometry of the building $\mathcal{X}_\bullet(P)$ is independent of P . The following lemma shows that $\mathcal{X}_\bullet(P)$ is in fact isomorphic to the Bruhat-Tits building $\mathcal{BT}_\bullet(W)$, as simplicial complexes.

Lemma 3.1.8. The map $\mathbb{B}(W) \rightarrow \mathbb{E}(P)$ sending $L \mapsto \mathcal{E}(P, L)$ is bijective and $\mathbb{Z}$ -equivariant. In particular, this induces an isomorphism of simplicial complexes $\mathcal{BT}_\bullet(W) \simeq \mathcal{X}_\bullet(P)$ that is Γ_P -equivariant.

Proof. It is enough to observe that $\mathcal{E}(P, L)(-1) = \mathcal{E}(P, \varpi L)$. From this and Lemma 3.1.3, we obtain the required isomorphism of simplicial complexes. Also Γ_P -equivariance is clear. $\square$

Consequently notions involving the simplicial structure on $\mathcal{X}_\bullet(P)$ have an equivalent notion in $\mathcal{BT}_\bullet(W)$ and vice versa.

Remark 3.1.9. Let $\widehat{W} := W \otimes_K K_\infty$. In many references(eg. [Boe02],[Dri74],[DH87]), the rank $r - 1$ Bruhat-Tits building, $\mathcal{BT}_\bullet(\widehat{W})$, is defined as the simplicial complex whose

- (1) *Vertices* are given by homothety classes, $\langle L \rangle$, of rank r $\widehat{R}$ -lattices $L \subset \widehat{W}$.
(2) *d-simplices* are given by $\{\langle L_1 \rangle, \dots, \langle L_d \rangle\}$, satisfying $L_0 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$.

The 1-1 correspondence

$$\{R\text{-lattices inside } W\} \xrightarrow{1-1} \{\widehat{R}\text{-lattices inside } \widehat{W}\}$$

$$L \mapsto L \otimes_R \widehat{R},$$

establishes an isomorphism of simplicial complexes $\mathcal{BT}^\bullet(W) \rightarrow \mathcal{BT}_\bullet(\widehat{W})$ which is also $GL_r(K)$ equivariant.

Let us fix a K -basis $\{e_1, \dots, e_r\}$ of W so that $W \simeq K^r$. Based on the remark above we can describe a canonical one-one correspondence between simplices of $\mathcal{BT}_\bullet(W)$ and quotients of $GL_r(K_\infty)$ by certain open compact subgroups. We provide such a description only in the case of vertices and chambers with a later application in mind.

Definition 3.1.10. Define the *Iwahori subgroup* $\mathcal{IW}_r$ of $GL_r(K_\infty)$ to be the inverse image of the standard Borel subgroup of $GL_r(\widehat{R}/(\varpi))$ under the natural map $GL_r(\widehat{R}) \rightarrow GL_r(\widehat{R}/(\varpi))$.

Remark 3.1.11. More explicitly, one can also describe the Iwahori subgroup $\mathcal{IW}_r$ as the subgroup of matrices $(a_{ij})_{1 \leq i, j \leq r} \in GL_r(\widehat{R})$ such that $v_\infty(a_{ij}) \geq 1$ for all $i > j$.

Proposition 3.1.12. There exist following canonical 1-1 correspondences of sets

$$(i) \mathcal{BT}_0(W) \leftrightarrow GL_r(K_\infty)/K_\infty^\times \cdot GL_r(\widehat{R});$$

$$(ii) \widehat{\mathcal{BT}}_{r-1}(W) \leftrightarrow GL_r(K_\infty)/K_\infty^\times \cdot \mathcal{IW}_r.$$

Proof. By the previous remark, it suffices to show the bijections for vertices and chambers in $\mathcal{BT}_\bullet(\widehat{W})$.

(i) Note that the natural transitive action of $GL_r(K_\infty)$ has the stabilizer subgroup $K_\infty^\times GL_r(\widehat{R})$ for the standard vertex v_0 , which is the class of the lattice $\widehat{R}e_1 + \dots + \widehat{R}e_r$. The bijection then follows.

(ii) This is covered in [ST97, Lemma 7]. □

3.1.2 Labeling the vertices of the building.

Later we will consider the homological chain complex of the Bruhat-Tits building, for which purpose it is important to consider the orientation on the simplices. With this in view, we define a

labeling on the vertices of $\mathcal{X}_\bullet(P)$ generalizing the notions of *even* and *odd* vertices of the Bruhat-Tits tree as in [Ser80, Ch.2, Prop.5].

For the following lemma, we fix a basis $\{e_1, \dots, e_r\}$ of W and consider the lattice $L_0 := Re_1 + \dots + Re_r \subset W$.

Lemma 3.1.13. Let $\mathcal{E}(P, L) \in \mathbb{E}(P)$ be a vector bundle for an R -lattice $L \subset W$. Then $\det(\mathcal{E}(P, L)) = \mathcal{L}_0(n)$ for $\mathcal{L}_0 = \mathcal{E}(\det(P), \det L_0)$ and some $n \in \mathbb{Z}$.

Proof. By Remark 3.1.5, we have that $\det(\mathcal{E}(P, L)) = \mathcal{E}(\det(P), \det(L))$. But by elementary divisors theorem, $\det(L) = \varpi^n \det(L_0)$ for some integer $n \in \mathbb{Z}$ and hence the lemma. $\square$

Combining the above with the following fact for a vector bundle $\mathcal{E} \in \mathbb{E}(P)$ that, $\det(\mathcal{E}(n)) = \det(\mathcal{E})(rn)$, gives the following

Definition 3.1.14. (i) The set of *types* is defined as

$$Types := \{\mathcal{L}_i := \mathcal{L}_0(-i) = \mathcal{E}(\det(P), \varpi^i R) \mid 0 \leq i \leq r-1\}.$$

A vertex $\langle \mathcal{E} \rangle$ is said to be of *type* i , if i is the unique integer in $[0, r-1]$ such that $\langle \mathcal{E} \rangle = \langle \mathcal{L}_i \rangle$.

In this case we will also denote $type(\langle \mathcal{E} \rangle) := i$.

(ii) Let s be an i -simplex in $\mathcal{X}_i(P)$. Then denote by

$$x(s) := \{type(x) \mid x \in s\}.$$

Remark 3.1.15. (i) By the proof of Proposition 3.2.4, if $\gamma \in \Gamma_P$, then there exists a K -basis $\{f_i\}_{i=1}^r$ s.t. $L = \sum_{1 \leq j \leq r} Rf_j$ and $\gamma L = \sum_{1 \leq j \leq r} \varpi^{a_j} Rf_j$ with $\sum_{1 \leq j \leq r} a_j = 0$. In particular, we see that $\det(L) = \det(\gamma(L))$ or γL has the same type as that of L .

(ii) By Lemma 3.1.8, we can, and will, define the type of a vertex $x \in \mathcal{BT}_\bullet(W)$ as the type of the corresponding vertex in $\mathcal{X}_\bullet(P)$. Similarly we also generalize the notation regarding $x(s)$ for $s \in \mathcal{BT}_i(W)$.

3.1.3 Geometric realization $|\mathcal{BT}_\bullet|$.

The geometric realization of any simplicial complex was defined in §1.1.1. Here we will make this explicit for the Bruhat-Tits building $\mathcal{BT}_\bullet$, by relating $|\mathcal{BT}_\bullet|$ to a metric space $\mathcal{N}$, the space of *dilation classes of norms on K_∞^r* , hence inducing a metric on the former. This is achieved by the work of Goldman-Iwahori and we follow its treatment as described in [DH87, Chap.3] and the reader is referred there for more details. For simplicity, we will denote by $q_\infty := q^{deg(\infty)}$ the cardinality of the residue field of $\mathcal{O}_\infty$.

Definition 3.1.16. (i) Let V be a finite dimensional vector space over K_∞ . A *norm* $\alpha : V \rightarrow \mathbb{R}$ is a function satisfying the following three properties:-

- given $v \in V$, $\alpha(v) \geq 0$ and $\alpha(v) = 0$ if and only if $v = 0$;
- $\alpha(av) = |a|_\infty \alpha(v)$, for $a \in K_\infty, v \in V$;
- $\alpha(v + v') \leq \max\{\alpha(v), \alpha(v')\}$, for $v, v' \in V$.

Two norms α, α' as above are said to lie in the same *dilation class* if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $\alpha' = \lambda\alpha$. Denote the dilation class of a norm α by $[\alpha]$.

(ii) Denote by $(\mathcal{N}, d)$ the metric space, whose points consist of dilation classes $[\alpha]$ of norms $\alpha : K_\infty^r \rightarrow \mathbb{R}$ and the metric is defined by

$$d([\alpha], [\alpha']) := \log_{q_\infty} \left(\sup_{x \neq 0} \frac{\alpha(x)}{\alpha'(x)} \cdot \sup_{x \neq 0} \frac{\alpha'(x)}{\alpha(x)} \right)$$

for two dilation $[\alpha], [\alpha'] \in \mathcal{N}$. Note that the metric is independent of the representatives of the dilation class chosen.

Before stating the relation of $\mathcal{N}$ with $|\mathcal{BT}_\bullet|$, we make precise some terminologies on the latter. For the following, recall the definition of geometric realization of a simplicial complex from the beginning of §1.1.1.

Definition 3.1.17. Let $y \in |\mathcal{BT}_\bullet|$ and $\{t_{v_0}, \dots, t_{v_d}\}$ be the set of real numbers between 0 and 1 associated to y , consisting of a d -simplex $s = \{v_0, \dots, v_d\} \in \mathcal{BT}_d$ and satisfying $\sum_{0 \leq i \leq d} t_{v_i} = 1$. Then $\{v_0, \dots, v_d\}$ is called the *facet* containing y and the set $\{t_{v_0}, \dots, t_{v_d}\}$ will be called the *barycentric coordinates* of y .

Proposition 3.1.18. There is a canonical isomorphism of topological spaces

$$\theta : |\mathcal{BT}_\bullet| \rightarrow \mathcal{N}.$$

Proof. We will only describe here the map θ . For a proof that it is an isomorphism, the reader is referred to either the original article of Goldman-Iwahori [GI63] or to [DH87, Chap.3, Thm.2.2].

Given an O_∞ lattice L of rank r inside K_∞^r , define the norm $\alpha_L : K_\infty^r \rightarrow \mathbb{R}$ given by $x \mapsto \inf\{1/a \in K_\infty | ax \in L\}$. Note that choosing a lattice L' homothetic to L , gives rise to a norm $\alpha_{L'}$ in the same dilation class as α_L . Hence this gives a well defined map from $\theta : \mathcal{BT}_0 \rightarrow \mathcal{N}$ sending $\langle L \rangle \mapsto [\alpha_L]$. θ can be extended to a map on the geometric realization of $\mathcal{BT}_\bullet$ as follows. Let $y \in |\mathcal{BT}_\bullet|$ be a point and consider its barycentric coordinates $(t_{v_0}, \dots, t_{v_d})$ with respect to its

facet $\{v_0, \dots, v_d\}$. Suppose v_i corresponds to lattices L_i , for $0 \leq i \leq d$, so that $L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$. Then set

$$\theta(y) := \max_{0 \leq i \leq d} \{q_\infty^{t_{v_i} + \dots + t_{v_d}} \cdot \alpha_{L_i}\}.$$

□

We end this subsection by stating some calculations on distances in $|\mathcal{BT}_\bullet|$ that will be useful to us later. By abuse of notation, we also denote by d the induced metric on $|\mathcal{BT}_\bullet|$ from $\mathcal{N}$. Let $L \subset K_\infty^r$ be an $\mathcal{O}_\infty$ lattice of rank r and α a norm on K_∞^r . Then by [DH87, Chap.3, Ex.3.3], the following formula holds in $\mathcal{N}$

$$d(\alpha, \alpha_L) = \log_{q_\infty} \left(\sup_{x \in L \setminus \varpi L} \alpha(x) \right) - \log_{q_\infty} \left(\inf_{x \in L \setminus \varpi L} \alpha(x) \right). \quad (3.1)$$

Adapting this formula to $|\mathcal{BT}_\bullet|$ we have

Proposition 3.1.19. (i) Let $y \in |\mathcal{BT}_\bullet|$ and $\{t_{v_0}, \dots, t_{v_d}\}$ be its barycentric coordinates with respect to its facet $\{v_0, \dots, v_d\}$. Then $d(y, v_i) = 1 - t_{v_i}$.

(ii) Let L, U be two R -lattices such that $L \supset U \supset \varpi^r L$ and $\varpi L \not\supset U \not\supset \varpi^{r-1} L$. Then $d(\langle L \rangle, \langle U \rangle) = r$.

Proof. (i) For $0 \leq i \leq d$, let L_i be $\mathcal{O}_\infty$ lattices corresponding to v_i such that there is a descending chain $L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$. Also let $\alpha := \theta(y)$ be as in the proof of Proposition 3.1.18. Then we need to show $d(\alpha, \alpha_{L_i}) = 1 - t_{v_i}$. By rescaling the lattices L_i and ordering them, if necessary, it suffices to assume $i = 0$. We claim the following:-

$$(a) \log_{q_\infty} \left(\sup_{x \in L_0 \setminus \varpi L_0} \alpha(x) \right) = 2 - t_{v_0};$$

$$(b) \log_{q_\infty} \left(\inf_{x \in L_0 \setminus \varpi L_0} \alpha(x) \right) = 1.$$

The above claim together with 3.1 finishes the proof that $d(\alpha, \alpha_{L_0}) = 1 - t_{v_0}$. It is not difficult to observe that, for any $x \in K_\infty^r$, $\alpha_{L_0}(x) \leq \alpha_{L_1}(x) \leq \dots \leq \alpha_{L_d} \leq q_\infty \cdot \alpha_{L_0}$. Moreover, for any $x \in L_0 \setminus \varpi L_0$ $\alpha_{L_0}(x) = 1$. Also for each $i \geq 1$, there exists $x_i \in L_0 \setminus \varpi L_0$ such that $\alpha_{L_j}(x_i) = 1$ for $0 \leq j < i$ and $\alpha_{L_k}(x_i) = q_\infty$ for $k \geq i$. Note that $\log_{q_\infty} \left(\sup_{x \in L_0 \setminus \varpi L_0} \alpha(x) \right) = \sup_{x \in L_0 \setminus \varpi L_0} \left(\sup_{0 \leq i \leq d} (t_{v_i} + \dots + t_{v_d} + \log_{q_\infty}(\alpha_{L_i}(x))) \right)$. The expression $\sup_{0 \leq i \leq d} (t_{v_i} + \dots + t_{v_d} + \log_{q_\infty}(\alpha_{L_i}(x)))$ attains the value $t_{v_l} + \dots + t_{v_d} + 1$, for $x = x_l$ and $l \geq 1$. Thus the LHS of (a) is $\geq t_{v_1} + \dots + t_{v_d} + 1 = 2 - t_{v_0}$. On the other hand, by above observations, for any $x \in L_0 \setminus \varpi L_0$, $\log_{q_\infty}(\alpha_{L_0}(x)) = 0$ and $0 \leq \log_{q_\infty}(\alpha_{L_i}(x)) \leq 1$. Hence

$\log_{q_\infty} \left(\sup_{x \in L_0 \setminus \varpi L_0} \alpha(x) \right) \leq t_{v_1} + \dots + t_{v_d} + 1 = 2 - t_{v_0}$, hence showing (a). The claim (b) is shown similarly.

(ii) This is the content of [DH87, Chap.3, Prop.3.4]. □

Remark 3.1.20. For two vertices $\langle L \rangle, \langle U \rangle$ in $\mathcal{BT}_\bullet$, it is not hard to show that $d(\langle L \rangle, \langle U \rangle)$ can also be interpreted as the length of the path $x_0 := \langle L \rangle, x_1 := \langle L_1 \rangle, \dots, x_n := \langle U \rangle$ of minimal length between $\langle L \rangle$ and $\langle U \rangle$, such that $\langle L_i \rangle$ and $\langle L_{i+1} \rangle$ form an edge in $\mathcal{BT}_\bullet$. We will only require this fact below for $n = 1$, that is, for two vertices $v, w \in \mathcal{BT}_0$, $d(v, w) = 1$ if and only if $\{v, w\}$ is an edge.

3.2 Stabilizers of simplices

3.2.1 Congruence subgroups.

Here we follow the treatment from [Boe02, §3.4]. Note that the notion of congruence subgroup is developed there in the rank 2 case but many results easily generalize to higher rank.

Definition 3.2.1. A subgroup $\Gamma \leq GL(W)$ is called a **congruence subgroup** if there is a non zero ideal $I \subset A$ and such that Γ contains $\Gamma(I) := \text{Ker}(GL_r(A) \rightarrow GL_r(A/I))$, as a finite index subgroup. Here $GL_r(A)$ is the group of A -automorphisms of $\sum_{1 \leq i \leq r} Ae_i$, for a fixed K -basis $\{e_i\}_{i=1}^r$ of W .

Recall that two subgroups $G', G'' \leq G$ are said to be *commensurable* if $G' \cap G''$ is a finite index subgroup inside both G' and G'' .

Lemma 3.2.2. Let $\Gamma', \Gamma'' \subset GL(W)$ be two congruence subgroups. Then Γ and Γ' are commensurable.

Proof. Suppose Γ' contains $\Gamma(I)$ as a finite index subgroup and similarly $\Gamma'' \supset \Gamma(J)$. Then both Γ', Γ'' contain $\Gamma(IJ)$ as a common finite index subgroup. Consequently, also $\Gamma' \cap \Gamma''$ is a finite index subgroup of both Γ' and Γ'' . □

We also give below an equivalent definition of a congruence subgroup which is the definition for an arithmetic subgroup in [Gek86, p. V.2].

Lemma 3.2.3. Let $\Gamma \leq GL(W)$ be a subgroup. Then Γ is a congruence subgroup if and only if there exist a projective A -module $P \subset W$ of rank r and an ideal $I \subset A$ such that $\Gamma_P(I) := \text{Ker}(\text{Aut}_A(P) \rightarrow \text{Aut}_A(P/IP)) \subset \Gamma \subset \Gamma_P := \text{Aut}_A(P)$.

Proof. The proof of [Boe02, Prop. 3.14] verbatim holds true for arbitrary ranks as well. $\square$

Also we have the analogue of [Boe02, Prop. 3.16], which says that an arithmetic group stabilizes a simplex if and only if it stabilizes all the vertices of the simplex.

Proposition 3.2.4. Let Γ be a congruence group, with $\gamma \in \Gamma \setminus \{1\}$ and $v \in \mathcal{BT}_\bullet(W)$ a vertex. Then $\det(\Gamma) \subset \mathbb{F}_q^\times$ is a finite group and $d(v, \gamma v) \neq 1$.

Proof. Since Γ contains $\Gamma(I)$ as a finite index subgroup, for some non-zero proper ideal $I \subset A$ and $\det(\Gamma(I)) = \{1\}$, we have that $\det(\Gamma) \subset K^\times$ is a finite subgroup. But the only finite subgroups of $K^\times$ are that of $\mathbb{F}_q^\times \subset K^\times$, hence the first statement.

As in [Boe02, Prop. 3.16], we follow the argument leading to corollary in [Ser80, Pg. 75], to prove the next statement. Suppose v is represented by a lattice L . Let $\{e_i\}_{i=1}^r$ be an R -basis spanning L such that $\{\pi^{a_i} e_i\}_{i=1}^r$ spans the lattice γL , with $a_i \leq a_{i+1}$. So $\gamma = \gamma_1 \gamma_2$, where γ_1 is the diagonal matrix with entries π^{a_i} and γ_2 is a matrix in $GL_r(\mathcal{O}_\infty)$. Since the determinant of γ is a unit, we have that $\sum a_i = 0$.

On the other hand, by Proposition 3.1.19(ii) and Remark 3.1.20, $d(L, \gamma L) = a_r - a_1$. Clearly, $a_r - a_1 \neq 1$ since $\sum a_i = 0$. $\square$

Consequently we have

Corollary 3.2.5. Let $v \in \mathcal{BT}_\bullet(W)$ and $L \subset W$ be an R -lattice representing v . Then for any congruence subgroup $\Gamma \subset GL(W)$, $\Gamma_L = \Gamma_v$.

Proof. Clearly $\Gamma_L \subset \Gamma_v$. Conversely given $\gamma \in \Gamma_v$, $\gamma L = c.L$ for some $c \in K^\times$. Since $\det(\gamma) \in \mathbb{F}_q^\times$, we have that $c \in R^\times$ hence $\gamma L = L$. $\square$

Corollary 3.2.6. Let $s = (x_0, \dots, x_d)$ be a simplex in $\mathcal{BT}^\bullet(W)$ and Γ be as above. If $\gamma \in \Gamma$ stabilizes s , then it stabilizes all the vertices of s . In particular, $\Gamma_s = \bigcap_{0 \leq i \leq d} \Gamma_{x_i}$.

Proof. By 3.2.4, $\gamma(x_i) \in s$ and is not equal to any of x_j for $j \neq i$. Hence $\gamma(x_i) = x_i$ for all i . $\square$

3.2.2 Description of stabilizers

Let Γ be a congruence subgroup. By the Lemma 3.2.3, we may assume $\Gamma \subset \Gamma_P$. Here we would like to study the finiteness of the stabilizer subgroup Γ_e , for e any simplex in the Bruhat-Tits building, and when possible would also like to describe explicitly the stabilizer. Recall that the stabilizer subgroup Γ_e in Γ , is the subgroup of all $\gamma \in \Gamma$ such that $\gamma e = e$. By Lemma 3.1.8, without loss of generality, we consider the avatar $\mathcal{X}_\bullet(P)$ of the Bruhat-Tits building. We have the following result

Proposition 3.2.7. (cf. [Gra82, §5]) Γ_e is a finite group.

Proof. From the Corollary 3.2.6, any $\gamma \in \Gamma_e$ fixes all the vertices of e as well, and hence it is enough to assume e is a vertex. We assume that e is the class of a vector bundle $\mathcal{E} \in \mathbb{E}(P)$.

Clearly the stabilizer $(\Gamma_P)_e = \text{Aut}(\mathcal{E})$. But this is a finite group since $\text{Aut}(\mathcal{E}) \subset \text{End}(\mathcal{E}) \simeq H^0(C, \mathcal{E}^\vee \otimes \mathcal{E})$, and the latter is a finite dimensional $\mathbb{F}_q$ vector space and hence is a finite set. Since $\Gamma \subset \Gamma_P$, Γ_e is also a finite set. □

When the arithmetic group Γ is a principal congruence subgroup $\Gamma_P(I)$ for some ideal non-zero proper ideal $I \subset A$, then we can explicitly describe the stabilizer group. For any subgroup $G \subset \text{GL}(W)$ and $L \subset W$ an R -lattice of maximal rank, we denote by G_L the subgroup of G fixing L .

Proposition 3.2.8. Let L be a nR -lattice in W of rank r and $I \subset A$ be an ideal as above.

(i) Then

$$H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)) = \text{End}_A(P)_L.$$

(ii) Let $\mathcal{I} := \mathcal{E}(I, R)$ and for any vector bundle $\mathcal{F}$ on C and $n \in \mathbb{Z}$, denote by $\mathcal{F}(I^n) := \mathcal{F} \otimes_{\mathcal{O}_C} \mathcal{I}^{\otimes n}$. Then for an element

$$\gamma \in H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I)) \subset H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)) = \text{End}_A(P)_L,$$

we have $\gamma P \subset IP$ and that γ is nilpotent.

Proof. (i) From the Remark 3.1.5(i,ii), it is clear that

$$H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)) = \text{End}_A(P) \cap \text{End}_R(L),$$

where the intersection on the right side is inside $\text{End}_K(W)$. Clearly $\text{End}_A(P) \cap \text{End}_R(L) = \text{End}_A(P)_L$.

(ii) Again by Remark 3.1.5(ii),

$$\mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I) = \mathcal{E}((P^\vee) \otimes_A IP, L^\vee \otimes_R L) = \mathcal{E}(\text{Hom}_A(P, IP), \text{End}_R(L)).$$

Hence by part (i) of the same remark, $H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I)) = \text{Hom}_A(P, IP) \cap \text{End}_R(L)$, and hence we can conclude that $\gamma P \subset IP$. Also it is clear that for any $n \geq 0$, $\gamma^n \in \text{Hom}_A(P, I^n P) \cap \text{End}_R(L) = H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I^n))$. By [Liu02, Ch.7, Prop.4.4], the line bundle $\mathcal{I}^\vee$ is ample (since $\deg(\mathcal{I}) < 0$). Hence by Serre vanishing theorem ([Liu02, Ch.5, Prop.3.6]), $H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I^n)) = 0$ for all $n \gg 0$. Hence γ is nilpotent. □

Corollary 3.2.9. $\Gamma_P(I)_L = 1 + H^0(C, \mathcal{E}(P, L)^\vee \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I))$, where the sum on the right is inside $\text{End}_K(W)$.

Proof. By Proposition 3.2.8, the inclusion $\supset$ is clear. For the other direction, it is enough to observe that for any $\gamma \in \Gamma_P(I)$, $\gamma - 1 \in \text{Hom}_A(P, IP)_L$. $\square$

3.3 Apartments, Sectors and Harmonic cocycles

Before defining harmonic cocycles on Bruhat-Tits building, it is beneficial to study certain Coxeter subcomplexes of $\mathcal{BT}_\bullet(W)$ called *apartments*. For our purposes it suffices to use an adhoc and simple description of these apartments involving the Coxeter complex of type $\widetilde{A}_{r-1}$ introduced in 2.2.6, for which we follow [KY25, §3.2.1]. For the more general definition of apartments, the reader is referred to the relevant portions of [AB10]. We follow the convention that $\{w_1, \dots, w_r\}$ etc. denote an unordered basis of W and $(w_1, \dots, w_r)$ etc. denote an ordered basis of W and we also use the notation $\underline{w}$ etc. for the latter.

3.3.1 Apartments and sectors of $\mathcal{BT}_\bullet(W)$

Recall the Coxeter complex $\mathcal{A}_\bullet$ of type $\widetilde{A}_{r-1}$ introduced in 2.2.6; it is the simplicial complex $\langle \mathbb{Z}^r \rangle$, associated with the poset $\mathbb{Z}^r$ equipped with a cofinal $\mathbb{Z}$ -action defined by translations of the vector $(1, \dots, 1)$. Hence there is an induced simplicial complex structure on the quotient abelian group $\mathbb{Z}^r / \mathbb{Z}(1, \dots, 1)$. Henceforth by $\mathcal{A}_\bullet$ we mean this simplicial complex. It is clear that $s \subset \mathbb{Z}^r / \mathbb{Z}(1, \dots, 1)$ is a simplex of $\mathcal{A}_\bullet$ if and only if under the quotient $\pi : \mathbb{Z}^r \rightarrow \mathbb{Z}^r / \mathbb{Z}(1, \dots, 1)$, $\pi^{-1}(s)$ is a totally ordered subset. It is not hard to observe that $\mathcal{A}_\bullet$ is a rank $(r - 1)$ simplicial complex, which in fact a chamber complex.

Fix an ordered K -basis $\underline{w} = (w_0, \dots, w_{r-1})$ of W and consider the natural map of $\mathbb{Z}$ -posets

$$\mathbb{Z}^r \rightarrow \mathbb{B}(W),$$

defined by sending $(n_1, \dots, n_r) \mapsto R\text{-span of } \{\varpi^{n_i} w_i\}_{0 \leq i \leq r-1}$. This factors through a natural injective map of simplicial complexes, $i_{\underline{w}} : \mathcal{A}_\bullet \rightarrow \mathcal{BT}_\bullet(W)$.

Definition 3.3.1. (i) The *apartment on $\underline{w}$* , denoted $\mathcal{A}_\bullet(\underline{w})$, is the image of $i_{\underline{w}}$ together with its inherited simplicial structure from $\mathcal{A}_\bullet$. The class of $(n_0, \dots, n_{r-1})$ will be denoted by $(\underline{w}, [n_0, \dots, n_{r-1}])$ and when $\underline{w}$ is clear from the context, only by $[n_0, \dots, n_{r-1}]$.

- (ii) Let $\sigma \in \mathcal{S}_r$, where $\mathcal{S}_r$ is the symmetric group on the r elements $\{0, \dots, r-1\}$. A *sector* $\mathcal{S}_\bullet(\underline{w}, \sigma)$ in $\mathcal{A}(\underline{w})$ is the simplicial subcomplex on the vertex set

$$\{[\underline{w}, [n_0, \dots, n_{r-1}]] \mid n_{\sigma(0)} \leq n_{\sigma(1)} \leq \dots \leq n_{\sigma(r-1)}\}.$$

We refer to the sector corresponding to the identity permutation as the *standard sector with respect to* $(\underline{w})$ and denote it by $\mathcal{S}_\bullet(\underline{w})$.

Remark 3.3.2. One can also consider apartments and sectors in other descriptions of the Bruhat-Tits buildings given in this chapter namely $\mathcal{BT}_\bullet(\widehat{W})$ and $\mathcal{X}_\bullet(P)$.

The following lemma shows that $\mathcal{A}_\bullet(\underline{w})$ is a disjoint union of sectors $\mathcal{S}_\bullet(\underline{w}, \sigma)$, so that each chamber is contained in a unique sector.

Lemma 3.3.3. With notation as above, let C be a chamber inside an apartment $\mathcal{A}_\bullet(\underline{w})$. Then there is a unique $\tau \in \mathcal{S}_r$ such that $C \in \mathcal{S}(\underline{w}, \tau)$.

Proof. Suppose $C = \{\langle L_0 \rangle, \dots, \langle L_{r-1} \rangle\}$ satisfying $L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_{r-1} \supsetneq \varpi L_0$. Let L_{r-1} be given by $[\varpi^{n_0} w_0, \varpi^{n_1} w_1, \dots, \varpi^{n_{r-1}} w_{r-1}]$ for integers $n_0, \dots, n_{r-1} \in \mathbb{Z}$. Moreover let $\tau' \in \mathcal{S}_r$ be *some* permutation, so that $n_{\tau'(0)} = \dots = n_{\tau'(t_0)} < n_{\tau'(t_0+1)} = \dots < n_{\tau'(t_{l-1}+1)} = \dots = n_{\tau'(t_l)}$, with $t_l = r-1$, for $t_0 < \dots < t_l$ being $l+1$ distinct elements in $\{0, \dots, r-1\}$. If $l = r-1$, τ' determines a unique sector in $\mathcal{A}_\bullet(\underline{w})$ by definition. Otherwise, we update the permutation $\{\tau'(0), \dots, \tau'(r-1)\}$ of $\{0, \dots, r-1\}$ as follows. Since C is a chamber, L_{r-2}/L_{r-1} is a 1-dimensional $R/(\varpi)$ -vector space, hence there exists unique $\mathbf{m}_0 \in \{0, \dots, r-1\}$ such that $\varpi(L_{r-2} \cap Kw_{\tau'(\mathbf{m}_0)}) = L_{r-1} \cap Kw_{\tau'(\mathbf{m}_0)}$ and $L_{r-2} \cap Kw_j = L_{r-1} \cap Kw_j$, $\forall j \neq \tau'(\mathbf{m}_0)$. There is a unique $d \in \{0, \dots, l-1\}$ s.t. $t_d + 1 \leq \mathbf{m}_0 \leq t_{d+1}$, so that $n_{\tau'(t_d+1)} = n_{\tau'(\mathbf{m}_0)} = n_{\tau'(t_{d+1})}$. We update the

$$\{\tau'(0), \dots, \tau'(t_d), \dots, \tau'(\mathbf{m}_0), \dots, \tau'(t_{d+1}), \dots, \tau'(r-1)\}$$

to

$$\{\tau'(0), \dots, \tau'(\mathbf{m}_0), \tau'(t_d), \dots, \tau'(\mathbf{m}_0 - 1), \tau'(\mathbf{m}_1 + 1), \dots, \tau'(t_{d+1}), \dots, \tau'(r-1)\}.$$

Continuing thus, by going from $i = 0$ to $r-1$, we arrive at the permutation τ of $\{0, \dots, r-1\}$, which determines the unique sector containing C . It is clear from the proof that τ is independent of the τ' we begin with. $\square$

Definition 3.3.4. For a chamber C inside the apartment $\mathcal{A}_\bullet(\underline{w})$, denote by $\tau_{\underline{w}}^{sec}(C)$ the permutation determined by Lemma 3.3.3.

Let $\vec{C}_x := (C, x)$ be a pointed chamber for a vertex $x \in C$. We will also denote a pointed chamber $\vec{C}_x$ by $(\langle L_0 \rangle, \dots, \langle L_{r-1} \rangle)$, where R -lattices (or suitable representatives thereof) $\{L_0, \dots, L_{r-1}\}$ represent the chamber C such that $L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_{r-1} \supsetneq \varpi L_0$ and $\langle L_0 \rangle$ represents the vertex x . For the following definition, analogous to Lemma 3.3.3, we define $\mathbf{m}_j$ for $0 \leq j \leq r-1$. For any $1 \leq i \leq r-1$, L_{i-1}/L_i is a 1-dimensional $R/(\varpi)$ -vector space. Hence $\mathbf{m}_j$ be the unique integer in $\{0, \dots, r-1\}$ such that $\varpi(L_{r-j-1} \cap K w_{\mathbf{m}_j}) = L_{r-j} \cap K w_{\mathbf{m}_j}$ and $L_{r-j-1} \cap K w_i = L_{r-j} \cap K w_i$, $\forall i \neq \mathbf{m}_j$, where we set $L_r := \varpi L_0$.

Definition 3.3.5. Let C and x be as above.

- (i) We will denote by $\sigma_{\underline{w}}^{x-nbd}(C)$ the permutation of $\{0, \dots, r-1\}$ given by $(0, \dots, r-1) \mapsto (\mathbf{m}_0, \dots, \mathbf{m}_{r-1})$. Then define

$$\text{sgn}_{\underline{w}}(\vec{C}_x) := \text{sgn}(\sigma_{\underline{w}}^{x-nbd}(C)).$$

- (ii) Let $\vec{C}_x, \vec{C}_{x'} \in \widehat{\mathcal{BT}}_{r-1}(W)$ be two pointed chambers, over the same chamber $C = \{\langle L_0 \rangle, \dots, \langle L_{r-1} \rangle\}$, not necessarily with a given choice of an apartment containing them. Suppose $x = \langle L_0 \rangle$ and $x' = \langle L_f \rangle$ for some $0 \leq f \leq r-1$. Then define

$$\text{sgn}(C, x, x') := \text{sgn}(\rho),$$

where ρ is the permutation $(0, \dots, r-1) \mapsto (f, \dots, r-1, 0, \dots, f-1)$.

Remark 3.3.6. (i) The superscript “ x - nbd ” is motivated from the fact that there are exactly $r!$ chambers in $\mathcal{A}_{\bullet}(\underline{w})$ adjoining the vertex x (due to the fact that the link of a vertex is the Tits building for $\text{GL}_r(\mathbb{F}_q)$) and this is reflected in the bijection between such chambers and the symmetric group $\mathcal{S}_r$, given by the map $C \mapsto \sigma_{\underline{w}}^{x-nbd}(C)$.

- (ii) Note that by sgn , we mean the sign homomorphism $\mathcal{S}_r \rightarrow \{\pm 1\}$.
- (iii) Given two pointed chambers $\vec{C}_x, \vec{C}_{x'}$, it is not hard to see that $\text{sgn}_{\underline{w}}(\vec{C}_x) \cdot (\text{sgn}_{\underline{w}}(\vec{C}_{x'}))^{-1}$ is equal to $\text{sgn}(C, x, x')$ and independent of the apartment $\mathcal{A}_{\bullet}(\underline{w})$ containing C .
- (iv) When r is odd, it is easy to observe that $\text{sgn}(C, x, x') = 1$ is independent of C, x, x' .

We can describe the following relation between the two permutations introduced in this section as follows.

Proposition 3.3.7. Let C resp. $x \in C$ be a chamber resp. vertex in an apartment $\mathcal{A}_{\bullet}(\underline{w})$ as above. Suppose that $C := ([n_0^{(0)}, \dots, n_{r-1}^{(0)}], \dots, [n_0^{(r-1)}, \dots, n_{r-1}^{(r-1)}])$ for $x := [n_0^{(0)}, \dots, n_{r-1}^{(0)}]$. For simplicity, denote by $\tau := \tau_{\underline{w}}^{sec}(C)$ and $\sigma := \sigma_{\underline{w}}^{x-nbd}(C)$. Then the following holds true

$$0 \leq i < i' \leq r-1 \text{ and } n_{\tau(i)}^{(0)} = n_{\tau(i')}^{(0)} \Rightarrow \sigma^{-1}\tau(i) < \sigma^{-1}\tau(i').$$

Proof. Consider the action of the symmetric group on $\mathcal{A}_\bullet(\underline{w})$ defined by letting a permutation $\theta \in \mathcal{S}_r$ act on vertices as follows

$$[n_0, \dots, n_{r-1}] \mapsto [n_{\theta(0)}, \dots, n_{\theta(r-1)}].$$

Hence τ is the unique permutation such that $\tau(C)$ lies in the standard sector of $\mathcal{A}_\bullet(\underline{w})$.

We may assume without loss of generality that the R -lattices $L_i := \langle \varpi^{n_0^{(i)}} w_0, \dots, \varpi^{n_{r-1}^{(i)}} w_{r-1} \rangle_R$, for $0 \leq i \leq r-1$, satisfy

$$L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_{r-1} \supsetneq \varpi L_0.$$

Then σ is the unique permutation such that for all $0 \leq j \leq r-1$,

$$L_{r-j} \cap Kw_{\sigma(j)} = \varpi L_{r-j-1} \cap Kw_{\sigma(j)},$$

where we set $L_r := \varpi L_0$.

From the definition of σ and τ it is not hard to observe that

$$\sigma_{\underline{w}}^{\tau x - nbd}(\tau C) = \tau^{-1} \sigma_{\underline{w}}^{x - nbd}(C) (= \tau^{-1} \sigma) \quad - (*)$$

If C is in the standard sector of $\mathcal{A}_\bullet(\underline{w})$ so that $\tau = 1$, it is easy to see that the statement of the proposition holds true. For an arbitrary chamber C in the apartment, since $\tau(C)$ lies in the standard sector, we have

$$0 \leq i < i' \leq r-1 \text{ and } n_{\tau(i)}^{(0)} = n_{\tau(i')}^{(0)} \Rightarrow (\sigma_{\underline{w}}^{\tau x - nbd}(\tau C))^{-1}(i) < (\sigma_{\underline{w}}^{\tau x - nbd}(\tau C))^{-1}(i').$$

Simplifying it using $(*)$ finishes the proof. $\square$

Remark 3.3.8. The above relation reappears in Chapter 6 for the case $\tau = 1$, when describing the fundamental domain of $\mathrm{GL}_r(\mathbb{F}_q[t])$ action on $\mathcal{BT}_\bullet(W)$. See Definition 7.2.2.

We conclude this subsection with a short lemma, which shows that given a pointed panel in an apartment, the two pointed chambers containing it have opposite signs. This will be used later to define a harmonic cocycle on an apartment.

Lemma 3.3.9. Let $\vec{e}_x \in \widehat{\mathcal{BT}^{r-2}}(W)$ be a pointed panel inside the apartment $\mathcal{A}_\bullet(\underline{w})$. Then precisely two pointed chambers of the apartment contain $\vec{e}_x$. Moreover these pointed chambers have opposite signs.

Proof. Suppose the pointed panel is given by $\vec{e}_x := (\langle L_0 \rangle, \dots, \langle L_{i-1} \rangle, \langle L_{i+1} \rangle, \dots, \langle L_{r-1} \rangle)$ with $x = \langle L_0 \rangle$, such that $\dim_{R/(\varpi)}(L_{i+1}/L_{i-1}) = 2$. Then there exists w_k, w_l , $1 \leq l \neq k \leq r$, so that

$L_{i-1} \cap K.w_j = R\varpi^{n_j}w_j$ and $L_{i+1} \cap K.w_j = R\varpi^{n_j-1}w_j$, for $j \in \{k, l\}$ and $n_k, n_l \in \mathbb{Z}$ some integers. Hence there are only two choices for a lattice $L_i \in \mathcal{A}_\bullet(\underline{w})$, between L_{i-1}, L_{i+1} , namely $L'_i := L_{i-1} + R\varpi^{n_k-1}w_k$ and $L''_i := L_{i-1} + R\varpi^{n_l-1}w_l$, so that $(\overrightarrow{C_1})_x := (\langle L_0 \rangle, \dots, \langle L_{i-1} \rangle, \langle L'_i \rangle, \langle L_{i+1} \rangle, \dots, \langle L_{r-1} \rangle)$ and $(\overrightarrow{C_2})_x := (\langle L_0 \rangle, \dots, \langle L_{i-1} \rangle, \langle L''_i \rangle, \langle L_{i+1} \rangle, \dots, \langle L_{r-1} \rangle)$ are the two chambers containing $\vec{e}_x$ in the apartment $\mathcal{A}_\bullet(\underline{w})$.

Now it is clear from the construction of $(\overrightarrow{C_1})_x, (\overrightarrow{C_2})_x$, that $\text{sgn}_{\underline{w}}((\overrightarrow{C_1})_x)$ and $\text{sgn}_{\underline{w}}((\overrightarrow{C_2})_x)$ differ by sign of the transposition

$$(1, \dots, k, k+1, \dots, l-1, l, \dots, r) \mapsto (1, \dots, l, k+1, \dots, l-1, k, \dots, r),$$

hence the last statement. □

3.3.2 Harmonic cocycles

Now we are ready to formulate the definition of a harmonic cocycle on the Bruhat-Tits building.

Definition 3.3.10. Below let L be any field containing K .

1. Let V be a finite dimensional L -vector space. A *harmonic cocycle with values in V* is a function $f : \widehat{\mathcal{BT}_{r-1}}(W) \rightarrow V$, satisfying the following conditions:-

- (a) For any chamber $C \in \mathcal{BT}_{r-1}(W)$ and two vertices $x, x' \in C$,

$$f(\vec{C}_x) = \text{sgn}(C, x, x')f(\vec{C}_{x'}),$$

- (b) For any pointed panel $\vec{e}_x \in \widehat{\mathcal{BT}_{r-2}}(W)$, one has

$$\sum_{\substack{\vec{C}_x \in \widehat{\mathcal{BT}_{r-1}}(W) \\ e \subset C}} f(\vec{C}_x) = 0.$$

We denote the L -vector space of harmonic cocycles by $C_{\text{har}}(V)$.

2. Let $\Gamma \subset \text{GL}(W)$ be a subgroup and in addition to the above setting, suppose now V is also a left $L[\Gamma]$ -module. Then a Γ *harmonic cocycle with values in V* , is a harmonic cocycle $f : \widehat{\mathcal{BT}_{r-1}}(W) \rightarrow V$ such that for all $\gamma \in \Gamma$ and $\vec{C}_x \in \widehat{\mathcal{BT}_{r-1}}(W)$, $f(\gamma\vec{C}_x) = \gamma f(\vec{C}_x)$, that is f is Γ -equivariant.

Remark 3.3.11. (i) Note that [deS01, p. 3.1], gives a more general version of a harmonic cocycle, over the set of pointed simplices of *any* rank, and it agrees with the above definition in the case of pointed chambers.

- (ii) By Remark 3.3.6(iv), the condition 1(a) is vacuous when r is odd.
- (iii) Note that there is no harm in defining more generally harmonic cocycles taking values in an arbitrary abelian group but with our applications in mind, we restrict ourselves to the above situation.

Chapter 4

Harmonic cocycles via Ash-Rudolph symbols

As the title suggests this chapter develops a novel method of constructing harmonic cocycles from the generators of the Steinberg module $\text{St}(W)$, thereby giving a map $\text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow C_{\text{har}}(V)$. In the case V is a Γ -module as in the previous chapter for a congruence subgroup Γ , further analysis on the growth of stabilizers of Γ in apartments of Bruhat-Tits buildings allows one to enhance the map to a homomorphism $\text{St}(W) \otimes_{\Gamma} V \rightarrow C_{\text{har}}(V)^{\Gamma}$.

4.1 Tits building and the Ash-Rudolph symbols

We recall first the definition of the Tits building associated to W .

Definition 4.1.1. The *Tits building for W* , denoted by $\mathcal{T}_{\bullet}(W)$, is the simplicial complex based on the poset of *proper, non-zero K -subspaces of W* .

In particular, the set of vertices $\mathcal{T}_0(W)$ consists of proper non-zero subspaces of W and d -simplices $\mathcal{T}_d(W)$ consists of proper chains of $(d + 1)$ -many such subspaces. We will denote a vertex in $\mathcal{T}_{\bullet}(W)$ by the subspace representing it. Also the group $\text{GL}(W)$ has canonical action on $\mathcal{T}_{\bullet}(W)$ via simplicial automorphisms.

An important result for Tits building is the *Solomon-Tits theorem*

Theorem 4.1.2. ([Qui73, §2]) If $r \geq 2$, then $\mathcal{T}_{\bullet}(W)$ has the homotopy type of a bouquet of $(r - 2)$ -spheres, that is, it is a wedge sum of a family of $\mathbb{S}^{r-2}$.

Consequently, the (reduced) homology is non vanishing only in the top dimension. Also the action of $\text{GL}(W)$ on $\mathcal{T}_{\bullet}(W)$ induces an action of $\text{GL}(W)$ on the homology.

Definition 4.1.3. The $\mathbb{Z}[\mathrm{GL}(W)]$ -module $\mathrm{St}(W) := \tilde{H}_{r-2}(|\mathcal{T}_\bullet(W)|, \mathbb{Z})$ is called the *Steinberg module for W*

Thanks to the work of A. Ash et al in [AR79; AGM12], one has an explicit description of $\mathrm{St}(W)$ via a set of generators and relations, with a natural action of $\mathrm{GL}(W)$ on the set of generators. In the rest of this subsection, we discuss this construction.

Ash-Rudolph symbols/relations. Let $d \geq 1$ be an integer and recall the standard d -simplex Δ_d on $I_d := \{0, \dots, d\}$. Let $\Delta_d^{(1)}$ denote its first barycentric subdivision, so that its vertices are the non-empty subsets of I_d and a simplex is a chain of subsets.

Define the *boundary* of $\Delta_d^{(1)}$, denoted $\partial\Delta_d^{(1)}$, as the subcomplex given by removing the subset I_d from $\Delta_d^{(1)}$. It is easy to observe using ideas in the proof of Lemma 2.1.4 that $\partial\Delta_d^{(1)}$ is naturally homotopic to the $(d-1)$ -sphere $\mathbb{S}^{d-1}$. Consequently $\tilde{H}_{d-1}(\partial\Delta_d^{(1)}, \mathbb{Z}) \simeq \mathbb{Z}$. We can explicitly describe a generator for the homology, regarding $\partial\Delta_d^{(1)}$ as a poset and using its associated homological complex 2.2.12.

Lemma 4.1.4. The cycle $\xi_d := \sum_{\sigma \in \mathcal{S}_{d+1}} \mathrm{sgn}(\sigma) [\{\sigma(0)\}, \{\sigma(0), \sigma(1)\}, \dots, \{\sigma(0), \dots, \sigma(d-1)\}] \in C_{d-1}(\partial\Delta_d^{(1)}, \mathbb{Z})$ generates the homology group $\tilde{H}_{d-1}(\partial\Delta_d^{(1)}, \mathbb{Z})$.

Proof. Recall from 2.2.12 that, since $\partial\Delta_d^{(1)}$ has rank $(d-1)$ as a simplicial complex, the homological complex $C_i(\partial\Delta_d^{(1)})$ vanishes for $i > d-1$ and $H_{d-1}(\partial\Delta_d^{(1)}, \mathbb{Z}) = \mathrm{Ker}(\partial_{d-1})$. A standard computation shows that $\partial_{d-1}(\xi_d) = 0$. Let ξ be a generator of $\mathrm{Ker}(\partial_{d-1})$. Then there is an integer $n \in \mathbb{Z}$ such that $n\xi = \xi_d$. From the description of ξ_d , clearly $n = \pm 1$, hence ξ_d also generates the top homology group. $\square$

Definition 4.1.5. The homology class ξ_d generating $\tilde{H}_{d-1}(\partial\Delta_d^{(1)}, \mathbb{Z})$ as a $\mathbb{Z}$ -module, will be called the *fundamental d -class*.

Now we return to the setting from the beginning of this subsection.

Definition 4.1.6. Let $w_0, \dots, w_{r-1} \in W \setminus \{0\}$ be given. Then the *Ash-Rudolph symbol on (the ordered tuple) $(w_0, \dots, w_{r-1})$* , denoted $[[w_0, \dots, w_{r-1}]]$, is the homology class in $\tilde{H}_{r-2}(|\mathcal{T}_\bullet(W)|, \mathbb{Z})$, given as the image of ξ_{r-1} under the map on homologies, induced by the simplicial map $\partial\Delta_{r-1}^{(1)} \rightarrow \mathcal{T}_\bullet(W)$ sending $J \mapsto K$ -span of $\{w_j\}_{j \in J}$,

$$\tilde{H}_{r-2}(\partial\Delta_{r-1}^{(1)}, \mathbb{Z}) \rightarrow \mathrm{St}(W) = \tilde{H}_{r-2}(|\mathcal{T}^\bullet(W)|, \mathbb{Z})$$

induced by , for any proper subset $J \subsetneq I_{r-1}$. $[[w_0, \dots, w_{r-1}]]$ will be called a *Ash-Rudolph symbol*.

Remark 4.1.7. (i) From Lemma 4.1.4 we see that a canonical class in $C_{r-2}(\mathcal{T}_\bullet(W), \mathbb{Z})$ representing $[[w_0, \dots, w_{r-1}]]$ is the cycle $\sum_{\sigma \in \mathcal{S}_r} \text{sgn}(\sigma) [Kw_{\sigma(0)}, Kw_{\sigma(0)} + Kw_{\sigma(1)}, \dots, Kw_{\sigma(0)} + \dots + Kw_{\sigma(r-2)}]$.

(ii) In [AR79; AGM12], $[[w_0, \dots, w_{r-1}]]$ as above is called a modular symbol, motivated by the classical theory of such symbols.

The following lemma records many properties of the Ash-Rudolph symbols.

Lemma 4.1.8. ([AR79, Prop.2.2]) For $i \in \{0, \dots, r\}$, let w_i denote any non-zero element of W . The Ash-Rudolph symbol $[[\cdot]]$ satisfies the following properties:-

(i) It is anti-symmetric, that is

$$[[\dots, w_i, \dots, w_j, \dots]] = -[[\dots, w_j, \dots, w_i, \dots]];$$

(ii) It is homogeneous of degree 1, that is, for any $a \in K \setminus \{0\}$,

$$[[aw_0, \dots, w_{r-1}]] = [[w_0, \dots, w_{r-1}]];$$

(iii) $[[w_0, \dots, w_{r-1}]] = 0$ if $\{w_0, \dots, w_{r-1}\}$ does not span W ;

(iv) $\sum_{i=0}^r (-1)^i [[w_0, \dots, \widehat{w}_i, \dots, w_r]] = 0$ for any set of $(r+1)$ -elements $w_0, \dots, w_r \in W$.

These will be called the *Ash-Rudolph relations*.

In [AGM12], it is shown that the Ash-Rudolph symbols generate $\text{St}(W)$ with the Ash-Rudolph relations. To formulate it, as in op.cit., we introduce the following definition.

Definition 4.1.9. The *Sharbly complex*, denoted $Sh_*(W)$, is the (homological) complex of $\mathbb{Z}[\text{GL}(W)]$ -modules such that $Sh_k(W)$ is the quotient of the free abelian group generated by the symbols $[w_0, \dots, w_{r+k-1}]$, where w_i is a non-zero vector in W for $0 \leq i \leq r+k-1$, modulo the submodule generated by the following relations

- $[w_{\sigma(0)}, \dots, w_{\sigma(r+k-1)}] - \text{sgn}(\sigma)[w_0, \dots, w_{r+k-1}]$, for permutations $\sigma \in \text{Sym}(r+k)$;
- $[w_0, \dots, w_{r+k-1}]$, if $w_0, \dots, w_{r+k-1}$ do not span W ;
- $[w_0, \dots, w_{r+k-1}] - [aw_0, \dots, w_{r+k-1}]$, $a \in K$.

For any $\gamma \in \text{GL}(W)$, define $\gamma[w_0, \dots, w_{r+k-1}] := [\gamma w_0, \dots, \gamma w_{r+k-1}]$.

The differential $\partial_k : Sh_k(W) \rightarrow Sh_{k-1}(W)$ is given by the map

$$[w_0, \dots, w_{r+k-1}] \mapsto \sum_{i=0}^{r+k-1} (-1)^i [w_0, \dots, \widehat{w}_i, \dots, w_{r+k-1}].$$

It is clear from the second relation above that $Sh_k = 0$ for $k < 0$. Also it follows from Lemma 4.1.8 that the natural map $\alpha : Sh_0(W) \rightarrow \text{St}(W)$ given by $[w_0, \dots, w_{r-1}] \mapsto [[w_0, \dots, w_{r-1}]]$ is a well defined map of $\mathbb{Z}[\text{GL}(W)]$ -modules.

Theorem 4.1.10. ([AGM12, Thm.5]) $Sh_*(W) \xrightarrow{\alpha} \text{St}(W) \rightarrow 0$ is an exact sequence of $\mathbb{Z}[\text{GL}(W)]$ -modules.

In particular, $\alpha : Sh_0(W) \rightarrow \text{St}(W)$ is a surjective map of $\mathbb{Z}[\text{GL}(W)]$ -modules.

4.2 Construction of a map $\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow \mathbf{C}_{har}(V)^\Gamma$

Throughout this subsection, we assume $\Gamma \subset \text{GL}_r(K)$ is a *congruence subgroup* such that it contains the principal congruence subgroup $\Gamma_P(I)$ as a finite index subgroup, for a proper ideal $I \subset A$. Also as in 3.3.10, we require L be a field containing K and V a finite dimensional L -vector space equipped which is also a left $L[\Gamma]$ -module. We shall construct a L -linear map

$$\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow \mathbf{C}_{har}(V)^\Gamma,$$

achieved through the following three steps:-

1. Firstly we construct a map of abelian groups

$$\mathbb{Z}[W^r] \otimes_{\mathbb{Z}} V \rightarrow \mathbf{C}_{har}(V),$$

defined by sending a pure tensor of the form $[w_0, \dots, w_{r-1}] \otimes v$ to a harmonic cocycle $f_{\underline{w}}^v$, which is assigned $+v$ or $-v$ for pointed chambers in $\mathcal{A}_\bullet(\underline{w})$ suitably (here $\underline{w} = (w_0, \dots, w_{r-1})$), and is zero outside this apartment. We then extend it L -linearly to obtain the said map.

Then we show that the map factorizes via the quotient $\mathbb{Z}[W^r] \otimes_{\mathbb{Z}} V \twoheadrightarrow Sh_0 \otimes_{\mathbb{Z}} V$, to give a map $Sh_0 \otimes_{\mathbb{Z}} V \rightarrow \mathbf{C}_{har}(V)$.

2. Furthermore, considering any $(r+1)$ -tuple of vectors $(w_0, \dots, w_r)$ in W and analyzing the intersections of apartments determined by K -basis inside the tuple, we can further show that the above map factorizes via $\alpha \otimes 1$, to give us a map $\text{AR} : \text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow \mathbf{C}_{har}(V)$.

3. In the final step, we construct the sought after map $\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow C_{\text{har}}(V)^\Gamma$ from AR , by “averaging over Γ ” in the following sense:- for any $f_{\underline{w}}^v \in C_{\text{har}}(V)$ as in Step (1), we would like to form $f_{\underline{w}, \Gamma}^v := [(\vec{C}_x \in \widehat{\mathcal{BT}^{r-1}}(W)) \mapsto \sum_{\gamma \in \Gamma} \gamma f_{\underline{w}}^v(\gamma^{-1} \vec{C}_x)]$. For this to be well defined, one has to show that for any pointed chamber $\vec{C}_x$, the intersection $\Gamma \vec{C}_x \cap \mathcal{A}_\bullet(\underline{w})$ is finite. This is achieved by applying the theorem of Riemann-Roch, to show that there are only finitely many chambers in $\mathcal{A}_\bullet(\underline{w})$ of any bounded order.

Step 1. Let $\underline{w} = (w_0, \dots, w_{r-1}) \in W^r$ and $v \in V$ be arbitrary elements. Define the function $f_{\underline{w}}^v \in \text{Maps}(\widehat{\mathcal{BT}^{r-1}}(W), V)$ to be 0, if $\underline{w}$ does not span W as a K -vector space. Otherwise, define it as (recall 3.3.5(i))

$$f_{\underline{w}}^v(\vec{C}_x) := \begin{cases} \text{sgn}_{\underline{w}}(\vec{C}_x)v; & \text{if } \vec{C}_x \in \mathcal{A}(\underline{w}); \\ 0; & \text{otherwise,} \end{cases} \quad (4.1)$$

for $\vec{C}_x \in \widehat{\mathcal{BT}^{r-1}}(W)$.

Lemma 4.2.1. $f_{\underline{w}}^v$ is a harmonic cocycle with values in V .

Proof. Suffices to assume that $\underline{w}$ is a K -basis of W . We need to verify that $f_{\underline{w}}^v$ satisfies the conditions 3.3.10(1.a,1.b). Condition 3.3.10(1.a) follows from Remark 3.3.6(2) and conditions 3.3.10(1.b) follows from Lemma 3.3.9. $\square$

By linear extension we get a map of L -vector spaces $\mathbb{Z}[W^r] \otimes_{\mathbb{Z}} V \rightarrow C_{\text{har}}(V)$. The following lemma shows that the above map factorizes via $\mathbb{Z}[W^r] \otimes_{\mathbb{Z}} V \rightarrow Sh_0(W) \otimes_{\mathbb{Z}} V$.

Lemma 4.2.2. With the notation as above, the following properties hold true:-

- (i) if $\underline{w}$ does not span W as a K -vector space, then $f_{\underline{w}}^v = 0$,
- (ii) $f_{\sigma \underline{w}}^v = \text{sgn}(\sigma) f_{\underline{w}}^v$, where $\sigma \in \mathcal{S}(r)$ $\sigma \underline{w} = (w_{\sigma(0)}, \dots, w_{\sigma(r-1)})$,
- (iii) given $a \in K^\times$, $f_{(a,1,\dots,1).\underline{w}}^v = f_{\underline{w}}^v$ where $(a, 1, \dots, 1).\underline{w} := (aw_0, \dots, w_{r-1})$.

Proof. (i) holds by definition of $f_{\underline{w}}^v$, (iii) follows from the definition of sign of pointed chamber inside an apartment. Assume $\underline{w}$ is a K -basis of W and let $\vec{C}_x$ be a pointed chamber inside $\mathcal{A}_\bullet(\underline{w})$. (ii) follows from the observation that $\text{sgn}_{\underline{w}}(\vec{C}_x)$ is determined by the permutation $(w_0, \dots, w_{r-1}) \mapsto (w_{\mathbf{m}_0}, \dots, w_{\mathbf{m}_{r-1}})(\mathbf{m}_i$'s as in Definition 3.3.5) and $\text{sgn}_{\sigma \underline{w}}(\vec{C}_x)$ is the sign of $(w_{\sigma(0)}, \dots, w_{\sigma(r-1)}) \mapsto (w_{\mathbf{m}_0}, \dots, w_{\mathbf{m}_{r-1}})$. $\square$

By Definition 4.1.9 and the previous lemma, we obtain a map $\text{AR}' : Sh_0(W) \otimes_{\mathbb{Z}} V \rightarrow C_{har}(V)$, sending a pure tensor $[w_0, \dots, w_{r-1}] \otimes v$ to $f_{\underline{w}}^v$.

Step 2. To show that AR' factorizes via $\alpha \otimes 1 : Sh_0 \otimes_{\mathbb{Z}} V \rightarrow \text{St}(W) \otimes_{\mathbb{Z}} V$, we need to prove that any $(r+1)$ -tuple $(w_0, \dots, w_r)$ of elements in W satisfy

$$\sum_{i=0}^r (-1)^i f_{\underline{w}^{(i)}}^v = 0,$$

where $\underline{w}^{(i)} := (w_0, \dots, \widehat{w_i}, \dots, w_r)$. By Lemma 4.2.2, there is an integer $0 < m \leq r$ such that without loss of generality, we may assume that $w_0 + \dots + w_m = 0$ holds true and that $\{w_0, \dots, \widehat{w_m}, \dots, w_r\}$ is a K -linearly independent set of vectors in W . Choose the minimal such m . In particular $\underline{w}^{(i)}$ does not span W for $i > m$ and hence $f_{\underline{w}^{(i)}}^v = 0$ in such cases. Consequently the relation required to be proved by us becomes

$$\sum_{i=0}^m (-1)^i f_{\underline{w}^{(i)}}^v = 0.$$

In order to prove this, we need to analyze the intersection of the apartments $\mathcal{A}_{\bullet}(\underline{w}^{(i)})$ for $0 \leq i \leq m$, which we do next.

Proposition 4.2.3. Denote $\mathcal{A}_{\bullet}^{(i)} := \mathcal{A}(\underline{w}^{(i)})$ for $0 \leq i \leq m$. Then for any chamber $C \in \bigcup_{0 \leq i \leq m} \mathcal{A}_{r-1}^{(i)}$, there are exactly two indices $0 \leq i_C \neq j_C \leq m$ such that C lies inside $\mathcal{A}_{r-1}^{(i_C)} \cap \mathcal{A}_{r-1}^{(j_C)}$.

Proof. Let $0 \leq i \neq j \neq k \leq m$ be three distinct indices. We analyze the intersection $\mathcal{A}_{\bullet}^{(i)} \cap \mathcal{A}_{\bullet}^{(j)}$ and $\mathcal{A}_{\bullet}^{(i)} \cap \mathcal{A}_{\bullet}^{(j)} \cap \mathcal{A}_{\bullet}^{(k)}$ in two separate cases.

Case I. Vertices in $\mathcal{A}_0^{(i)} \cap \mathcal{A}_0^{(j)}$.

Let $v \in \mathcal{A}_0^{(i)} \cap \mathcal{A}_0^{(j)}$ be a vertex so that there exists tuple of integers $(n_0, \dots, n_{i-1}, n_{i+1}, \dots, n_r)$, $(m_0, \dots, m_{j-1}, m_{j+1}, \dots, m_r)$ such that

$$v = \langle \varpi^{n_0} w_0, \dots, \varpi^{n_{i-1}} w_{i-1}, \varpi^{n_{i+1}} w_{i+1}, \dots, \varpi^{n_r} w_r \rangle_R = \langle \varpi^{m_0} w_0, \dots, \varpi^{m_{j-1}} w_{j-1}, \varpi^{m_{j+1}} w_{j+1}, \dots, \varpi^{m_r} w_r \rangle_R,$$

where $\langle s_1, \dots, s_r \rangle_R$ denotes the R -lattice generated by $s_1, \dots, s_r$. From the hypothesis $w_0 + \dots + w_m = 0$, it is not hard to conclude the following relations

$$\begin{cases} m_i = n_j; \\ m_k = n_k \quad \forall 0 \leq k \neq i, j \leq r; \\ m_i \geq \max_{0 \leq k \neq i, j \leq m} \{m_k\}; \\ n_j \geq \max_{0 \leq k \neq i, j \leq m} \{n_k\} \end{cases} \quad (4.2)$$

Moreover we can also observe that the above conditions are also sufficient to determine a common vertex of $\mathcal{A}_0^{(i)}$ and $\mathcal{A}_0^{(j)}$. In particular, 4.2 implies that $\mathcal{A}_\bullet^{(i)} \cap \mathcal{A}_\bullet^{(j)}$ is the union of $\frac{(r-1)!}{(m-1)!}$ -sectors of the form $\mathcal{S}_\bullet(\underline{w}^{(i)}, \sigma)$, for suitable permutations $\sigma \in \mathcal{S}_r$.

Case 2. Vertices in $\mathcal{A}_0^{(i)} \cap \mathcal{A}_0^{(j)} \cap \mathcal{A}_0^{(k)}$.

In addition to the equality of lattices considered above, we also need add an extra condition of them coinciding with $[\varpi^{l_0} w_0, \dots, \varpi^{l_{k-1}} w_{k-1}, \varpi^{l_{k+1}} w_{k+1}, \dots, \varpi^{l_r} w_r]_R$, for a tuple of integers $(l_0, \dots, l_{k-1}, l_{k+1}, \dots, l_r)$. Applying Eq.4.2 twice, we obtain $m_i = m_k \geq \max_{1 \leq z \leq m} \{m_z\}$ and similar equations hold for n 's and l 's. In particular, no chamber can lie inside $\mathcal{A}_\bullet^{(i)} \cap \mathcal{A}_\bullet^{(j)} \cap \mathcal{A}_\bullet^{(k)}$.

From these two cases, we can conclude that for a chamber C inside $\bigcup_{0 \leq i \leq m} \mathcal{A}_\bullet^{(i)}$, there is a unique pair of integers (i_C, j_C) with $1 \leq i_C \neq j_C \leq m$ such that C is contained inside precisely $\mathcal{A}_\bullet^{(i_C)}$ and $\mathcal{A}_\bullet^{(j_C)}$. □

As above, let C is a chamber inside the union $\bigcup_{0 \leq i \leq m} \mathcal{A}_\bullet^{(i)}$ so that by previous proposition, it is contained inside $\mathcal{A}_\bullet^{(i_C)} \cup \mathcal{A}_\bullet^{(j_C)}$. As a consequence of the proof above, we obtain

Corollary 4.2.4.

$$(-1)^{j_C+1-i_C} \operatorname{sgn}_{\underline{w}^{(i_C)}}(\vec{C}_x) = \operatorname{sgn}_{\underline{w}^{(j_C)}}(\vec{C}_x).$$

Proof. For simplicity in this proof, we denote $i := i_C, j := j_C$, and without loss of generality, assume that $i < j$. Recall from Definition 3.3.5, that $\operatorname{sgn}_{\underline{w}^{(i)}}(\vec{C}_x)$ (resp. $\operatorname{sgn}_{\underline{w}^{(j)}}(\vec{C}_x)$) is determined with respect to the ordered basis $\underline{w}^{(i)} = (w_0, \dots, w_{i-1}, w_{i+1}, \dots, w_r)$ (resp. $\underline{w}^{(j)} = (w_0, \dots, w_{j-1}, w_{j+1}, \dots, w_r)$). Applying the cyclic permutation $\sigma = (i+1, \dots, j)$ to the ordered basis $\underline{w}^{(i)}$, we obtain

$$\sigma(\underline{w}^{(i)}) = (w_0, \dots, w_{i-1}, w_j, w_{i+1}, \dots, w_{j-1}, w_{j+1}, \dots, w_r).$$

From 3.3.5(i) and 4.2, it is clear that $\operatorname{sgn}_{\sigma(\underline{w}^{(i)})}(\vec{C}_x) = \operatorname{sgn}_{\underline{w}^{(j)}}(\vec{C}_x)$. Since $\operatorname{sgn}(\sigma) = (-1)^{j-i+1}$, from the proof of Lemma 4.2.2, we also have $\operatorname{sgn}_{\sigma(\underline{w}^{(i)})}(\vec{C}_x) = (-1)^{j-i+1} \operatorname{sgn}_{\underline{w}^{(i)}}(\vec{C}_x)$, finishing the proof. □

Corollary 4.2.5. With the notation as above, for $v \in V$ we have

$$\sum_{i=0}^m (-1)^i f_{\underline{w}^{(i)}}^v = 0.$$

Proof. By definition, the sum above is zero on pointed chambers not in the union $\bigcup_{0 \leq i \leq m} \mathcal{A}_\bullet^{(i)}$, so it suffices to consider C inside $\bigcup_{0 \leq i \leq m} \mathcal{A}_\bullet^{(i)}$ and show that $\sum_{i=0}^m (-1)^i f_{\underline{w}^{(i)}}^v(\vec{C}_x) = 0$ for all $x \in C$. By

Proposition 4.2.3, $\vec{C}_x$ is contained in exactly two apartments, namely $\mathcal{A}_\bullet^{(i_C)}$ and $\mathcal{A}_\bullet^{(j_C)}$. So we are reduced to showing,

$$(-1)^{i_C} f_{\underline{w}^{(i_C)}}^v(\vec{C}_x) + (-1)^{j_C} f_{\underline{w}^{(j_C)}}^v(\vec{C}_x) = 0.$$

But since $f_{\underline{w}^{(i_C)}}^v(\vec{C}_x) = \text{sgn}_{\underline{w}^{(i_C)}}(\vec{C}_x)v$ and $f_{\underline{w}^{(j_C)}}^v(\vec{C}_x) = \text{sgn}_{\underline{w}^{(j_C)}}(\vec{C}_x)v$, from Corollary 4.2.4 we obtain the desired result. $\square$

In particular, the map $\text{AR}' : Sh_0(W) \otimes_{\mathbb{Z}} V \rightarrow C_{har}(V)$ factorizes via $\alpha \otimes 1$, inducing a map of L -vector spaces $\text{AR} : \text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow C_{har}(V)$.

Step 3. Let Γ be a congruence subgroup inside $\text{GL}(W)$ containing $\Gamma_P(I)$ as a finite index subgroup, for some non-zero proper ideal $I \subset A$. As mentioned before, in this last step we would like to “average $f_{\underline{w}^{(i)}}^v$ over Γ ” to obtain a map from $\text{St}(W) \otimes_{\Gamma} V \rightarrow C_{har}(V)^{\Gamma}$. For the sake of this being well defined, we need the following lemma.

Lemma 4.2.6. Let $\underline{w} = (w_0, \dots, w_{r-1})$ be a K -basis of W . Denote by $[n_1, \dots, n_{r-1}]$ the vertex in $\mathcal{A}_0(\underline{w})$ corresponding to the R -lattice generated by $w_0, \varpi^{n_1}w_1, \dots, \varpi^{n_{r-1}}w_{r-1}$. Then

$$\#\Gamma_{[n_1, \dots, n_{r-1}]} \rightarrow \infty \text{ as } \left| \sum_{i=1}^{r-1} n_i \right| \rightarrow \infty.$$

In particular, given an integer $N \in \mathbb{N}$, there are only finitely many vertices $[n_1, \dots, n_{r-1}]$ with $\#\Gamma_{[n_1, \dots, n_{r-1}]} \leq N$.

Proof. It suffices to show the statement of the lemma for $\Gamma_P(I)$, for Γ contains $\Gamma_P(I)$ as a finite index subgroup. Also from Corollaries 3.2.5 and 3.2.9, it suffices to analyze the dimension of the space of global sections $H^0(C, \mathcal{E}(P, L)^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I))$, for $L = \langle w_0, \varpi^{n_1}w_1, \dots, \varpi^{n_{r-1}}w_{r-1} \rangle_R$. Moreover, the cokernel of $\mathcal{E}(P, L)^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)(I) \hookrightarrow \mathcal{E}(P, L)^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}(P, L)$, is a skyscraper sheaf with $\mathbb{F}_q$ dimension that is independent of the n_i 's, and hence we can restrict ourselves to estimating the dimension of $H^0(C, \mathcal{E}(P, L)^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}(P, L))$. For convenience, we set $n_0 := 0$ in the rest of the proof.

Following Remark 3.1.5(ii), let us denote by $\mathcal{E}^{(n_i)} := \mathcal{E}(P, L) \cap \widetilde{K}w_i = \mathcal{E}(P \cap Kw_i, R\varpi^{n_i}w_i)$, for $0 \leq i \leq r-1$, and similarly the quotient bundle $\mathcal{E}_{(n_i)} := \mathcal{E}(P, L) / (\mathcal{E}(P, L) \cap \sum_{\substack{0 \leq j \leq r \\ j \neq i}} Kw_j) = \mathcal{E}(P / (P \cap \sum_{\substack{0 \leq j \leq r \\ j \neq i}} Kw_j), R\varpi^{n_i}w_i)$. Consequently, we obtain an injective map of vector bundles (for $0 \leq i \neq j \leq r-1$)

$$(\mathcal{E}_{(n_i)})^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}^{(n_j)} \hookrightarrow \mathcal{E}(P, L)^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}(P, L).$$

Recall the terminology in §1.5 that $\mathcal{O}(1) = \mathcal{O}(\infty)$ is an ample line bundle and hence by Remark 3.1.5(ii), we get $(\mathcal{E}_{(n_i)})^{\vee} \otimes_{\mathcal{O}_C} \mathcal{E}^{(n_j)} = \mathcal{E}((P / (P \cap \sum_{\substack{0 \leq k \leq r \\ k \neq i}} Kw_k))^{\vee} \otimes_A (P \cap Kw_j), (R\varpi^{n_i}w_i)^{\vee} \otimes_R$

$R\varpi^{n_j} w_j) = (\mathcal{E}_{(0)})^\vee \otimes_{\mathcal{O}_C} \mathcal{E}^{(0)}(n_i - n_j)$. By the Riemann-Roch theorem ([Liu02, Ch.7, Thm.3.17]) and Serre vanishing theorem ([Liu02, Ch.5, Prop.3.6]), we have, for $n_i - n_j \gg 0$, $H^0(C, (\mathcal{E}_{(n_i)})^\vee \otimes_{\mathcal{O}_C} \mathcal{E}^{(n_j)}) = c_{ij}(n_i - n_j) + d_{ij}$, for some constants $c_{ij}, d_{ij} \in \mathbb{Q}$ independent of the n_i 's. Now $\max_{0 \leq i < j \leq r-1} |n_i - n_j| \rightarrow \infty$, if $|\sum_{1 \leq i \leq r-1} n_i| \rightarrow \infty$. This establishes the lemma. $\square$

As an immediate consequence we obtain

Corollary 4.2.7. Let C be any chamber of $\mathcal{BT}_\bullet(W)$. Then the apartment $\mathcal{A}_\bullet(\underline{w})$, for a K -basis $\underline{w} = (w_0, \dots, w_{r-1})$, contains only finitely many chambers in the orbit ΓC .

Proof. It suffices to show that, for any vertex $x \in C$, $\mathcal{A}_\bullet(\underline{w}) \cap \Gamma x$ is a finite set. For all vertices $x' \in \Gamma x$, the size of its stabilizer $\#\Gamma_{x'}$ is the same, say N . Hence by Lemma 4.2.6, there are finitely many vertices in $\mathcal{A}_\bullet(\underline{w})$ with cardinality at most N . $\square$

The previous corollary allows us to define the following Γ -invariant harmonic cocycle with values in V , given $v \in V$ and $\underline{w}$ a K -basis,

$$(f_{\underline{w}}^v)_\Gamma := [(\vec{C}_x \in \widehat{\mathcal{BT}^{r-1}}(W)) \mapsto \sum_{\gamma \in \Gamma} \gamma f_{\underline{w}}^v(\gamma^{-1} \vec{C}_x)]. \quad (4.3)$$

The sum on the right is well defined by Corollary 4.2.7, and consequently the harmonicity and Γ -invariance are also clear. Therefore we obtain a map

$$\tilde{\theta} : \text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow C_{\text{har}}(V)^\Gamma,$$

which on the pure tensors is defined by

$$[[w_0, \dots, w_{r-1}]] \otimes v \mapsto (f_{\underline{w}}^v)_\Gamma.$$

Proposition 4.2.8. AR factors via $\text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow \text{St}(W) \otimes_\Gamma V$. We denote by $\text{AR}_\Gamma : \text{St}(W) \otimes_\Gamma V \rightarrow C_{\text{har}}(V)^\Gamma$ the resulting map.

Proof. Let $\delta \in \Gamma$ be an arbitrary element. We have $\text{AR}([[\delta w_1, \dots, \delta w_r]] \otimes \delta v) = [(\vec{C}_x \in \widehat{\mathcal{BT}^{r-1}}(W)) \mapsto \sum_{\gamma \in \Gamma} \gamma f_{\delta \underline{w}}^{\delta v}(\gamma^{-1} \vec{C}_x)]$. Now

$$\gamma^{-1} \vec{C}_x \in \mathcal{A}_\bullet(\delta \underline{w}) \iff (\gamma \delta)^{-1} \vec{C}_x = \delta^{-1} \gamma^{-1} \vec{C}_x \in \mathcal{A}_\bullet(\underline{w}).$$

Hence in such a case, $\gamma f_{\delta \underline{w}}^{\delta v}(\gamma^{-1} \vec{C}_x) = \gamma(\delta v) = (\gamma \delta)v = \gamma \delta f_{\underline{w}}^v((\gamma \delta)^{-1} \vec{C}_x)$. Consequently,

$$\sum_{\gamma \in \Gamma} \gamma f_{\delta \underline{w}}^{\delta v}(\gamma^{-1} \vec{C}_x) = \sum_{\gamma \in \Gamma} \gamma \delta f_{\underline{w}}^v((\gamma \delta)^{-1} \vec{C}_x) = \sum_{\gamma' \in \Gamma} \gamma' f_{\underline{w}}^v((\gamma')^{-1} \vec{C}_x),$$

or equivalently, $\text{AR}([[\delta w_1, \dots, \delta w_r]] \otimes \delta v) = \text{AR}([w_1, \dots, w_r] \otimes v)$. Hence as claimed, AR factors via $\text{St}(W) \otimes_{\mathbb{Z}} V \rightarrow \text{St}(W) \otimes_\Gamma V$. $\square$

It will be shown in Proposition 6.2.3, AR_Γ is an injective map when Γ is a principal congruence subgroup, and moreover in the case $A = \mathbb{F}_q[t]$ it is an isomorphism (Corollary 7.3.8).

Chapter 5

An equivariant homotopy equivalence à la Grayson-Quillen

The results of this chapter have been covered in [BV26]. Here we will show that certain subcomplex(es) of the Bruhat-Tits building are homotopic to the Tits building $|\mathcal{T}_\bullet(W)|$. But as is clear from the conclusion of the previous chapter, the action of a congruence subgroup on the Tits building plays an important role in relating its top homology, that is the Steinberg module, to invariant harmonic cocycles. Hence it would be prudent to incorporate the congruence subgroup actions in the homotopies to be discussed, so that the results from Chapter 2 are significant here.

Recall the $\mathbb{Z}$ -poset $\mathbb{B}(W)$ from 3.1.1, which consists of all R -lattices of rank r inside W , with $n \in \mathbb{Z}$ acting on $L \in \mathbb{B}(W)$ via $n + L := \varpi^{-n}L$. Also we had denoted in Definition 3.1.4 the unique vector bundle on C , such that its sections over $\text{Spec}(A)$ are P and its stalk at ∞ is an R -lattice $L \subset W$, by $\mathcal{E}(P, L)$. The congruence subgroups we will be concerned with in this section are the principal ones, namely we let $I \subset A$ be a non-zero proper ideal and set $\Gamma := \Gamma_P(I)$ for simplicity, so that $\Gamma \trianglelefteq \Gamma_P$.

5.1 Contractible subcomplexes of the Γ -unstable complex

Let $0 \subsetneq W_1 \subsetneq W$ be a proper subspace and we denote the corresponding vertex in $\mathcal{T}_\bullet(W)$ by σ and we define the following subposet of $\mathbb{B}(W)$,

$$\mathbb{B}(W)_\sigma := \{L \in \mathbb{B}(W) \mid \exists g \in \Gamma_L \setminus \{1\} \text{ s.t. } g|_{W_1} = 1|_{W_1}\}$$

Clearly the $\mathbb{Z}$ -action on $\mathbb{B}(W)$ induces a cofinal $\mathbb{Z}$ -action on $\mathbb{B}(W)_\sigma$. We form the simplicial complex

$$\mathcal{BT}_\bullet(W)_\sigma,$$

whose vertices are the $\mathbb{Z}$ -orbits of elements of $\mathbb{B}(W)_\sigma$, and the $\mathbb{Z}$ -orbits of lattices $L_0, \dots, L_d \in \mathbb{B}(W)_\sigma$ such that $L_0 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$ form a d -simplex s if and only if $\Gamma_s = \bigcap_{0 \leq i \leq d} \Gamma_{L_i} \neq \{1\}$ and there exists $g \in \Gamma_s \setminus \{1\}$ such that $g|_{W_1} = 1|_{W_1}$. The $\mathbb{Z}$ -orbit in $\mathcal{BT}_\bullet(W)_\sigma$ of a lattice $L \in \mathbb{B}(W)_\sigma$ will be denoted by $[L]$.

Remark 5.1.1. Note that the complex $\mathcal{BT}_\bullet(W)_\sigma$ is not the same as $\langle \mathbb{B}(W)_\sigma \rangle$. The latter is a possibly larger complex.

The main results of this section are motivated from [Gra82, Thm.4.1 and Cor.4.2]. Even the proof is structured around the corresponding results in op.cit., only changing as necessitated by our setting.

Theorem 5.1.2. (cf. [Gra82, Thm.4.1]) $|\mathcal{BT}_\bullet(W)_\sigma|$ is contractible.

Proof. Denote by $P_1 := P \cap W_1$ and let

$$\alpha : \mathbb{B}(W)_\sigma \rightarrow \mathbb{B}(W/W_1)$$

be the map sending

$$L \mapsto L/(L \cap W_1).$$

We denote by $[\alpha] : \mathcal{BT}_\bullet(W)_\sigma \rightarrow \mathcal{BT}_\bullet(W/W_1)$ the induced simplicial map.

To ease the notation below, for any lattice $L' \subset W/W_1$ (resp. $L_1 \subset W_1$), we denote by $\mathcal{E}_{L'}$ (resp. $\mathcal{E}_{L_1}$) the vector bundle $\mathcal{E}(P/(P \cap W_1), L')$ (resp. $\mathcal{E}(P_1, L_1)$). Since $P_1 \subset P$ is the quotient of a saturated projective A -submodule, P/P_1 is also a projective A -module. We fix a splitting $P \simeq P_1 \oplus P/P_1$, consequently also fixing a splitting of $W \simeq W_1 \oplus W/W_1$. Also fix an R -lattice $\tilde{L} \subset W_1$ of maximal rank, so that we can form the vector bundle $\tilde{\mathcal{E}} := \mathcal{E}_{\tilde{L}}$.

Now define an augmentation map (see 2.2.8)

$$\epsilon : \mathbb{B}(W/W_1) \rightarrow \mathbb{Z}$$

by

$$L' \mapsto \min \{m \mid H^0(C, \mathcal{E}_{L'}^\vee \otimes \tilde{\mathcal{E}}(I + m\infty)) \neq \{0\}\} + 1.$$

The convention regarding the twist $(I + m\infty)$ is as in Proposition 3.2.8(ii). This allows us to define

$$\beta : \mathbb{B}(W/W_1) \rightarrow \mathbb{B}(W)$$

given by

$$L' \mapsto \varpi^{-\epsilon(L')} \tilde{L} \oplus L',$$

where the direct sum above is with respect to the splitting of W fixed above.

Claim 1. $\beta(L') \in \mathbb{B}(W)_\sigma$. Moreover, given a d -simplex $s = \{\langle L'_0 \rangle, \dots, \langle L'_d \rangle\}$ in $\mathcal{BT}_\bullet(W/W_1)$, $\beta(s) := \{[\beta(L'_0)], \dots, [\beta(L'_d)]\}$ is a d -simplex in $\mathcal{BT}_\bullet(W)_\sigma$.

Proof. From Corollary 3.2.9, we know that for any R -lattice $L \subset W$ of maximal R -rank

$$\Gamma_L = 1 + H^0(C, \mathcal{E}^\vee(P, L) \otimes \mathcal{E}(P, L)(I)).$$

If we set $\mathcal{E}'_L := \mathcal{E}_{L/(L \cap W_1)}$ and $\mathcal{E}_{L,1} := \mathcal{E}_{L \cap W_1}$, then we have a subsheaf

$$(\mathcal{E}'_L)^\vee \otimes \mathcal{E}_{L,1}(I) \hookrightarrow \mathcal{E}^\vee(P, L) \otimes \mathcal{E}(P, L)(I).$$

First we show that $\Gamma_{\beta(L')} \neq 1$, for which it is sufficient to show that $H^0(C, (\mathcal{E}'_{\beta(L')})^\vee \otimes \mathcal{E}_{\beta(L'),1}(I)) \neq 0$. Now $\mathcal{E}'_{\beta(L')} = \mathcal{E}_{L'}$ and $\mathcal{E}_{\beta(L'),1} = \tilde{\mathcal{E}}(\epsilon(L')\infty)$. Hence by definition of ϵ , we have the non-vanishing of $H^0(C, \mathcal{E}_{L'}^\vee \otimes \tilde{\mathcal{E}}(I + \epsilon(L')\infty))$, and consequently also the non-vanishing of $H^0(C, \mathcal{E}^\vee(P, L) \otimes \mathcal{E}(P, L)(I))$. Also any element $g' \in H^0(C, \mathcal{E}_{L'}^\vee \otimes \tilde{\mathcal{E}}(I + \epsilon(L')\infty))$, acts as zero on W_1 , so that $g = 1 + g'$ acts as the identity on W_1 . This proves the first part of the claim 1.

For the second part, consider the d -simplex in $\mathcal{BT}_\bullet(W/W_1)$ determined by the flag $L'_0 \supsetneq \dots \supsetneq L'_d \supsetneq \varpi L'_0$. To show that $(\beta(L'_0) \supsetneq \dots \supsetneq \beta(L'_d) \supsetneq \varpi \beta(L'_0))$ determines a simplex in $\mathcal{BT}_\bullet(W)_\sigma$, it suffices to obtain a non-zero element in the intersection of the space of sections

$$\bigcap_{0 \leq i \leq d} H^0(C, \mathcal{E}_{L'_i}^\vee \otimes \tilde{\mathcal{E}}(I + \epsilon(L'_i)\infty)), \quad (5.1)$$

as in the proof of the first part. The intersection above takes place inside $\text{End}(W)$.

Note that we have the following relations

$$\mathcal{E}_{L'_0}^\vee \subset \dots \subset \mathcal{E}_{L'_d}^\vee \text{ and } \epsilon(L'_0) - 1 \leq \epsilon(L'_d) \leq \dots \leq \epsilon(L'_0).$$

The second set of inequalities result from the definition of the augmentation map. Consequently the intersection in 5.1 contains $H^0(C, \mathcal{E}_{L'_0}^\vee \otimes \tilde{\mathcal{E}}(I + (\epsilon(L'_0) - 1)\infty))$. From the definition of ϵ clearly the latter space is a non-zero $\mathbb{F}_q$ -vector space. Hence this concludes the proof of latter half of the claim. $\square$

We denote the induced simplicial map from β by $[\beta] : \mathcal{BT}_\bullet(W/W_1) \rightarrow \mathcal{BT}(W)_\sigma$.

Now define a new map

$$\hat{\epsilon} : \mathbb{B}(W)_\sigma \rightarrow \mathbb{Z}$$

as the composition $\mathbb{B}(W)_\sigma \xrightarrow{\alpha} \mathbb{B}(W/W_1) \xrightarrow{\epsilon} \mathbb{Z}$. Since α is order preserving and $\mathbb{Z}$ -equivariant, $\hat{\epsilon}$ is again an augmentation map.

Continuing the line of thought from the proof of [Gra82, Thm.4.1], we introduce for any integer $n \in \mathbb{Z}$, a map of posets

$$G_n : \mathbb{B}(W)_\sigma \rightarrow \mathbb{B}(W),$$

given by

$$L \mapsto L + \varpi^{-n-\hat{\epsilon}(L)} \tilde{L}.$$

Claim 2. For any simplex $\gamma \in \mathcal{BT}_\bullet(W)$ and any $n \in \mathbb{Z}$, there is an inclusion of subgroups

$$\{g \in \Gamma_\gamma \mid g|_{W_1} = 1_{W_1}\} \subset \{g \in \Gamma_{G_n(\gamma)} \mid g|_{W_1} = 1_{W_1}\}.$$

Consequently, G_n has target inside $\mathbb{B}(W)_\sigma$ and induces a simplicial map denoted by $[G_n] : \mathcal{BT}_\bullet(W)_\sigma \rightarrow \mathcal{BT}_\bullet(W)_\sigma$.

Proof. Let γ be determined by the flag of lattices $L_0 \supsetneq \dots \supsetneq L_d \supsetneq \varpi L_0$ and $g \in \Gamma_\gamma \setminus \{1\}$ be given such that $g|_{W_1} = 1_{W_1}$. Since $K\tilde{L} = W_1$, g also stabilizes each of $G_n(L_i)$ for $0 \leq i \leq d$. Hence the required inclusion of subgroups. $\square$

From the Definition 2.2.10, we can check that G_n and G_{n+1} are adjacent maps with respect to $\hat{\epsilon}$. Indeed, condition 2.2.10(i) follows since for any L , $G_n(L) \subset G_{n+1}(L)$. Suppose $L \subsetneq M$ are two R -lattices such that $\hat{\epsilon}(L) < \hat{\epsilon}(M)$. Then $G_{n+1}(L) = L + \varpi^{-n-1-\hat{\epsilon}(L)} \tilde{L} \subset M + \varpi^{-n-\hat{\epsilon}(M)} \tilde{L}$, hence condition 2.2.10(ii) is also satisfied. Proposition 2.2.11 shows then that the geometric realizations of $\langle G_n \rangle$ and $\langle G_{n+1} \rangle$ are homotopic to each other. But we will need a stronger homotopy property, which can be obtained by following the proof of the same result as follows.

Claim 3. $||[G_n]||$ and $||[G_{n+1}]||$ are homotopic to each other.

Proof. Let $(\mathbb{B}(W)_\sigma^{un})_0$ be the simplicial subcomplex of $\mathbb{B}(W)_\sigma$, whose vertices are $\hat{\epsilon}^{-1}(0)$ and a chain $(L_0 \subset \dots \subset L_d)$ is defined to be a d -simplex, if $\gamma = \{[L_0], \dots, [L_d]\}$ is a d -simplex in $\mathcal{BT}_\bullet(W)_\sigma$ and $\exists g \in \Gamma_\gamma \setminus \{1\}$ such that $g|_{W_1} = 1_{W_1}$. Similar to Remark 2.2.9, we obtain the natural isomorphism $(\mathbb{B}(W)_\sigma^{un})_0 \simeq \mathcal{BT}_\bullet(W)_\sigma$. Denote the composition of $[G_n]$ resp. $[G_{n+1}]$ with this isomorphism by $[G_n]_0$ resp. $[G_{n+1}]_0$. For a d -simplex γ in $(\mathbb{B}(W)_\sigma^{un})_0$ as above, $\hat{\epsilon}(L_d) = 0 > -1 = \hat{\epsilon}(\varpi L_0)$. Since G_n and G_{n+1} are adjacent, for any $0 \leq i \leq d$, the flag

$$G_{n+1}(L_0) \supset \dots \supset G_{n+1}(L_i) \subset G_n(L_i) \subset \dots \subset G_n(L_d)$$

determines a simplex $\tilde{\gamma}$ in $(\mathbb{B}(W)_\sigma^{un})_0$, as $G_n(L_d) \supset G_{n+1}(\varpi L_0)$ and also by Claim 2, if $g \in \Gamma_\gamma \setminus \{1\}$ satisfies $g|_{W_1} = 1_{W_1}$, then $g \in \Gamma_{G_i(L_j)} \setminus \{1\}$ for $i = n, n+1$ and for all $0 \leq j \leq d$, and hence $g \in \Gamma_{\tilde{\gamma}} \setminus \{1\}$ and it acts as identity on W_1 . Consequently applying Proposition 2.2.5, we conclude that $[G_n]_0$ and $[G_{n+1}]_0$ are adjacent to each other, or their geometric realization's are homotopic to each other. In particular, $||[G_n]||$ and $||[G_{n+1}]||$ are homotopic to each other. $\square$

Now the rest of the proof of the theorem is exactly as in [Gra82, Thm.4.1]. For any R -lattice $L \subset W$ of rank r , we have

$$G_n(L) := \begin{cases} L; & n \ll 0 \\ G_n \beta \alpha(L); & n \gg 0 \end{cases}.$$

We can now finish the proof using the (non-equivariant) Whitehead's theorem (Theorem 2.1.8). Let $f : Z \rightarrow |\mathcal{BT}_\bullet(W)_\sigma|$ be a continuous map from a compact topological space Z . Then its image is in the span of finitely many vertices of $\mathcal{BT}_\bullet(W)_\sigma$. Hence by above, we have

$$[G_n] \circ f = \begin{cases} f; & n \ll 0 \\ [G_n] \circ [\beta] \circ [\alpha] \circ f; & n \gg 0 \end{cases}.$$

Consequently, $|f|$ is homotopic to $[[G_n]| \circ |[\beta]| \circ |[\alpha]| \circ |f|]$. Since $|[\alpha]| \circ |f|$ is null-homotopic (since the Bruhat-Tits building is contractible [Gra82, Thm.2.1]), $[[G_n]| \circ |[\beta]| \circ |[\alpha]| \circ |f|$ is null-homotopic, and we can conclude that $|f|$ is null-homotopic. By Whitehead's theorem, $|\mathcal{BT}_\bullet(W)_\sigma|$ is contractible. $\square$

5.2 An equivariant homotopy $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}} \rightarrow \mathcal{T}_\bullet(W)$

Define the Γ -unstable subcomplex of $\mathcal{BT}_\bullet(W)$ denoted $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ as follows. A simplex $\gamma = \{\langle L_0 \rangle, \dots, \langle L_d \rangle\}$ is in $\mathcal{BT}_\bullet(W)^{\text{un}}$ if and only if $\Gamma_\gamma \neq \{1\}$.

Analogous to [Gra82, Cor.4.2], we would like to use the above theorem together with Theorem 2.2.19, to conclude a Γ_P -equivariant homotopy equivalence between $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ and $\mathcal{T}_\bullet(W)$. The strategy to construct the sought after homotopy equivalence is to use Theorem 2.2.19. In the notation of that theorem, we apply it as follows. Set $X := \mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ and $\mathcal{T} := \mathcal{T}_\bullet(W)$. For $\sigma \in \mathcal{T}$, define $X_\sigma := \mathcal{BT}_\bullet(W)_\sigma$.

In this setting 2.2.19(i) is obvious and 2.2.19(iii) follows from Theorem 5.1.2. Also 2.2.19(ii) follows from the following standard fact from group representation theory over positive characteristic fields.

Lemma 5.2.1. (cf. [KY21, Lem.38]) Let G be a finite p -group acting on a finite dimensional K -vector space V . Then $V^G \neq \{0\}$.

Proof. We do induction on the size of G . Suppose $G = \langle g \rangle$ is a cyclic group generated by g . Then $g^{p^n} = 1$ for some $n \in \mathbb{Z}$. Consequently, the characteristic polynomial of g is of the form $X^{p^n} - 1$, so the only eigenvalue for the action of g on V is 1. Hence $V^{\langle g \rangle} \neq \{0\}$.

Now let G be an arbitrary p -group. It is well known fact that the centralizer $C_G \neq \{1\}$, so pick a $g \in C_G$. Then $\langle g \rangle \subset G$ is a normal subgroup and $G/\langle g \rangle$ acts on $V^{\langle g \rangle}$. By induction hypothesis, we have $V^G = (V^{\langle g \rangle})^{G/\langle g \rangle} \neq \{0\}$. $\square$

Then 2.2.19(ii) follows in our setting since given any simplex γ in $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$, $\Gamma_\gamma \neq \{1\}$. Hence $\{0\} \subsetneq W^{\Gamma_\gamma} \subsetneq W$ determines a vertex $\sigma \in \mathcal{T}_\bullet(W)$, so that γ is a simplex of $\mathcal{BT}_\bullet(W)_\sigma$.

Finally one needs to show that given a simplex $\gamma \in \mathcal{BT}_\bullet(W)^{un}$, the poset

$$\mathcal{T}_\bullet(W)_\gamma := \{\sigma \in \mathcal{T}_\bullet(W) \mid \gamma \text{ is a simplex of } \mathcal{BT}_\bullet(W)_\sigma\}$$

is $(\Gamma_P)_\gamma$ -contractible. We will achieve this by considering the $(\Gamma_P)_\gamma$ -map of posets

$$\mathcal{T}_\bullet(W)_\gamma \xrightarrow{\phi} \mathcal{T}_\bullet(W)_\gamma$$

sending

$$\sigma = (0 \subsetneq W' \subsetneq W) \mapsto (0 \subsetneq W' + W^{\Gamma_\gamma} \subsetneq W),$$

which is $(\Gamma_P)_\gamma$ since $\Gamma_\gamma \trianglelefteq (\Gamma_P)_\gamma$ is a normal subgroup. That $W' + W^{\Gamma_\gamma}$ is a proper subspace of W follows from the fact that if $1 \neq g \in \Gamma_\gamma$ stabilizes W' , then it also stabilizes $W' + W^{\Gamma_\gamma}$ and hence the latter must not be the whole vector space W . Consequently, since $\phi(\sigma) \geq \sigma$ for all $\sigma \in \mathcal{T}_\bullet(W)_\gamma$, by Corollary 2.2.16, $\mathcal{T}_\bullet(W)_\gamma$ is $(\Gamma_P)_\gamma$ -homotopy equivalent to the poset $Im(\phi)$. But $Im(\phi)$ has the smallest element, namely the vertex corresponding to W^{Γ_γ} , which is also $(\Gamma_P)_\gamma$ -invariant. Hence by Corollary 2.2.15 $Im(\phi)$ is $(\Gamma_P)_\gamma$ -contractible, or $\mathcal{T}_\bullet(W)_\gamma$ is also $(\Gamma_P)_\gamma$ -contractible.

To summarize the discussion in this section, we have

Theorem 5.2.2. There is a natural Γ_P -equivariant homotopy equivalence between $|\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}|$ and $|\mathcal{T}_\bullet(W)|$.

Remark 5.2.3. (i) Since $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ and $\mathcal{T}_\bullet(W)$ are simplicial complexes, it follows that their corresponding (reduced)simplicial homology groups of the geometric realizations are also naturally isomorphic. In particular, there is an isomorphism

$$\tilde{H}_i(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \simeq \tilde{H}_i(\mathcal{T}_\bullet(W), \mathbb{Z}),$$

for all $i \geq 0$, that is also Γ_P equivariant (the former is contained in any standard text on algebraic topology, eg. [Mun84, Chap.2, §19], and the latter follows from the Γ_P -equivariance of the homotopy equivalence).

(ii) It is worth highlighting that when $r = 2$ the homotopy equivalence in Theorem 5.2.2 is not a Γ_P homotopy equivalence(that is, there do not exist a Γ_P -equivariant homotopy inverse

$|\mathcal{T}_\bullet(W)| \rightarrow |\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}|$. See Definition 2.1.5(ii)) and we expect the same to hold true in higher ranks as well. Indeed to see this fact in $r = 2$ case, recall that $\mathcal{T}_\bullet(W) = \mathbb{P}(W)$ and note by (i) that an inverse Γ_P -equivariant homotopy inverse $|\mathbb{P}(W)| \rightarrow |\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}|$ would result in a Γ_P equivariant map $g : C_0(\mathbb{P}(W), \mathbb{Z}) \rightarrow C_0(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$. In particular, for each $c \in \mathbb{P}(W)$, $g([c]) = \sum_{x \in \mathcal{BT}_0(W)^{\Gamma\text{-un}}} n_x [x]$, where $n_x \in \mathbb{Z}$ are almost all zero. The stabilizer $(\Gamma_P)_c$ has infinite cardinality (see [Ser80, Chap.2, §2.5]) whereas the (finitely many) vertices occuring in the sum $g([c])$ have finite Γ_P stabilizers. Hence the Γ_P -equivariance of g means that $g \equiv 0$. But this is a contradiction since $C_\bullet(\mathbb{P}(W), \mathbb{Z})$ is not contractible.

5.3 Relating the various (un)stable complexes

In this section, we will be interested in understanding the compatibility of the isomorphisms in Theorem 5.2.2 when Γ runs over $\Gamma_P(I)$ for I any proper non-zero ideal of A . In particular, we will show a compatible family of isomorphisms between the homology of the various $\Gamma_P(I)$ -stable complexes (defined below) with the Steinberg module. Let $\Gamma := \Gamma_P(I)$ as in the previous section. First we will explicitly describe a simplicial homological complex for the building $\mathcal{BT}_\bullet(W)$ which consequently also defines it for $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$. Let us fix a K -basis $\{e_1, \dots, e_r\}$ of W and by $\text{type}(x)$ for a vertex $x \in \mathcal{BT}_0(W)$ we mean the type with respect to this fixed basis; see Remark 3.1.15.

Definition 5.3.1. We denote by $(C_i(\mathcal{BT}_\bullet(W), \mathbb{Z}), \partial_i)$ the homological complex which is defined as follows:

- (i) $C_i(\mathcal{BT}_\bullet(W), \mathbb{Z}) = 0$ for all $i > r - 1$ and $i < -1$;
- (ii) $C_{-1}(\mathcal{BT}_\bullet(W), \mathbb{Z}) = \mathbb{Z}$
- (iii) For $0 \leq i \leq r - 1$,

$$C_i(\mathcal{BT}_\bullet(W), \mathbb{Z}) := \bigoplus_{\substack{\{x_0, \dots, x_i\} \in \mathcal{BT}_i(W) \\ \text{type}(x_0) < \dots < \text{type}(x_i)}} \mathbb{Z}[x_0, \dots, x_i],$$

where for a simplex $s = \{x_0, \dots, x_i\} \in \mathcal{BT}_i(W)$, we denote by $[\vec{s}] = [x_0, \dots, x_i]$ the corresponding basis symbol in $C_i(\mathcal{BT}_\bullet(W), \mathbb{Z})$. Similarly we denote by $[\vec{s}_{x_j}] := [x_j, x_{j+1}, \dots, x_{j-1}] := (-1)^{ij} [\vec{s}]$. Note that $(-1)^{ij}$ is the sign of the permutation $(0, 1, \dots, i) \rightarrow (j, j+1, \dots, i, 0, \dots, j-1)$.

- (iv) For $1 \leq i \leq r - 1$, the differentials are given by

$$\partial_i : C_i(\mathcal{BT}_\bullet(W), \mathbb{Z}) \rightarrow C_{i-1}(\mathcal{BT}_\bullet(W), \mathbb{Z})$$

sending

$$[x_0, \dots, x_i] \mapsto \sum_{0 \leq j \leq i} (-1)^j [x_0, \dots, \widehat{x}_j, \dots, x_i].$$

∂_0 sends $[x]$ to 1, for a vertex $x \in \mathcal{BT}_0(W)$.

Remark 5.3.2. (i) The homology of the complex $(C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z}), \partial_i)$ is the reduced simplicial homology of the space $|\mathcal{BT}_\bullet(W)|$, so we will denote it by $\tilde{H}_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})$.

(ii) As already alluded to before, $|\mathcal{BT}_\bullet(W)|$ is contractible, that is, $\tilde{H}_i(\mathcal{BT}_\bullet(W), \mathbb{Z}) = 0$ for all $i \neq 0$.

(iii) It is clear that the complex $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})$ is equipped with the obvious $\mathrm{GL}(W)$ -action making it a complex of $\mathbb{Z}[\mathrm{GL}(W)]$ -modules.

Denote by $C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-un}}, \mathbb{Z} \subset C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})$ the subcomplex involving only the Γ unstable simplices and the corresponding homology will be denoted by $\tilde{H}_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-un}}, \mathbb{Z}$.

Definition 5.3.3. Define

$$C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}} := \mathrm{coker}(C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-un}}, \mathbb{Z} \hookrightarrow C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})).$$

The homology of the complex $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}}$ will be denoted by $H_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$.

Remark 5.3.4. The set of stable simplices does not form a simplicial complex. Therefore, a priori it is not clear whether $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}}$ can be defined as the complex associated to a subset of simplices of $\mathcal{BT}_\bullet(W)$. To clarify this, let $\mathcal{BT}_d(W)^{\Gamma\text{-st}} := \mathcal{BT}_d(W) \setminus \mathcal{BT}_d(W)^{\Gamma\text{-un}}$ for $d \geq 0$, so that $\mathcal{BT}_d(W)^{\Gamma\text{-st}}$ consists of those d -simplices with trivial Γ -stabilizer. Let $C_d(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$ be the free $\mathbb{Z}$ -module on generators $[s]$ for $s \in \mathcal{BT}_d(W)^{\Gamma\text{-st}}$, and define the differential $\partial_d : C_d(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z} \rightarrow C_{d-1}(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$ by

$$[v_0, \dots, v_d]_{\mathrm{st}} \mapsto \sum_{i=0}^d (-1)^i [v_0, \dots, \widehat{v}_i, \dots, v_d]_{\mathrm{st}},$$

where for a simplex $s = \{t_0, \dots, t_j\}$ the symbol $[s]_{\mathrm{st}} = 0$ if s is Γ -unstable, and otherwise $[s]_{\mathrm{st}}$ is the symbol $[s]$ defined above. Then as the reader may easily verify, this defines a complex $C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$ and one has a canonical isomorphism $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}} \rightarrow C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$.

Since $\Gamma \trianglelefteq \Gamma_P$ is a normal subgroup, the action of Γ_P on $\mathcal{BT}_\bullet(W)$ induces a natural action of Γ_P on $C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z}$. Consequently we have a short exact sequence of Γ_P -modules

$$0 \rightarrow C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-un}}, \mathbb{Z} \rightarrow C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z}) \rightarrow C_\bullet(\mathcal{BT}_\bullet(W))^{\Gamma\text{-st}}, \mathbb{Z} \rightarrow 0,$$

giving rise to a long exact sequence of Γ_P -modules

$$\dots \rightarrow \tilde{H}_i(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \rightarrow \tilde{H}_i(\mathcal{BT}_\bullet(W), \mathbb{Z}) \rightarrow H_i(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \rightarrow \tilde{H}_{i-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \rightarrow \dots \quad (5.2)$$

Because the Bruhat-Tits building $\mathcal{BT}_\bullet(W)$ is contractible, the above long exact sequence together with Remark 5.2.3 yields isomorphisms of Γ_P -modules

$$H_{i-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \stackrel{(5.2)}{\simeq} \tilde{H}_{i-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \stackrel{\text{Thm. 5.2.2}}{\simeq} \tilde{H}_{i-2}(\mathcal{T}_\bullet(W), \mathbb{Z}) \quad (5.3)$$

for all $i \geq 2$, and the vanishing $H_0(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) = 0$. Moreover as recalled in Theorem 4.1.2, one also has $\tilde{H}_i(\mathcal{T}_\bullet(W), \mathbb{Z}) = 0$ unless $i = r - 2$ and $\tilde{H}_{r-2}(\mathcal{T}_\bullet(W), \mathbb{Z}) = \text{St}(W)$. Hence the augmented complex

$$0 \rightarrow H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \rightarrow C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}} \quad (5.4)$$

is acyclic. This motivates the following definition:

Definition 5.3.5. The Steinberg module for level I is defined as

$$\text{St}^I(P) := H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}),$$

and the isomorphism $\text{St}^I(P) \simeq \text{St}(W)$ of Γ_P -modules from (5.3) is denoted by GQ_Γ .

Lemma 5.3.6. Each set $\mathcal{BT}_d(W)^{\Gamma\text{-st}}$ is a finite union of free Γ -orbits.

Proof. We rely on the proof of [BKW13, Prop.13.8]: Recall the K -basis $\{e_i\}_{i=1}^r$ of W from before and consider the setting $S := \{\infty\}$, $\mathcal{G} := (\text{GL}_r)_K$ in the notation of op.cit., so that $\mathcal{G}(K) = \text{GL}(W)$ with respect to $\{e_i\}_{i=1}^r$ and for $\text{GL}_r^0(A) := \text{Im}(\text{GL}_r(A) \rightarrow \text{Aut}(\mathcal{BT}_\bullet(W)))$, the series

$$\sum_{C \in \text{GL}_r(A) \backslash \mathcal{BT}_{r-1}(W)} \frac{1}{|\text{GL}_r^0(A)_C|}$$

converges by [BKW13, Prop.13.8]. The kernel of $\text{GL}_r(A) \rightarrow \text{Aut}(\mathcal{BT}_\bullet(W))$ is a finite group by Proposition 3.2.7, hence the series

$$\sum_{C \in \text{GL}_r(A) \backslash \mathcal{BT}_{r-1}(W)} \frac{1}{|\text{GL}_r(A)_C|}$$

also converges. In particular, given $M \in \mathbb{N}$ there are finitely many $\text{GL}_r(A)$ orbit of simplices with the order of the associated $\text{GL}_r(A)$ stabilizer bounded above by M . Since Γ and $\text{GL}_r(A)$ are commensurable from Lemma 3.2.2, we can conclude that there are at most finitely Γ orbits of Γ -stable simplices. $\square$

It follows from Lemma 5.3.6 and Remark 5.3.4 that $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}}$ is a complex of free and finitely generated $\mathbb{Z}[\Gamma]$ -modules, and from the acyclicity of (5.4) we deduce:

Corollary 5.3.7. The Γ_P -module $\text{St}(W)$ is projective and finitely generated as a $\mathbb{Z}[\Gamma]$ -module.

Our next result concerns the compatibility of the maps $\text{GQ}_{\Gamma_P(I)}$ as I runs through nonzero proper ideals of A :

Theorem 5.3.8. Let $J \subset I \subset A$ be non-zero proper ideals and set $\Gamma' := \Gamma_P(J)$, so that $\Gamma' \trianglelefteq \Gamma \trianglelefteq \Gamma_P$. Then there is a canonical Γ_P -equivariant homomorphism of abelian groups $i_{J,I} : \text{St}^J(P) \rightarrow \text{St}^I(P)$ such that the following diagram commutes

$$\begin{array}{ccc} \text{St}^J(P) & \xrightarrow{\text{GQ}_{\Gamma'}} & \text{St}(W) \\ \downarrow i_{J,I} & & \uparrow \text{GQ}_{\Gamma} \\ \text{St}^I(P) & & \end{array} \quad (5.5)$$

Proof. Note that any simplex that is Γ' -unstable is also Γ -unstable. Hence we have an embedding of complexes $C_\bullet(\mathcal{BT}_\bullet(W)^{\Gamma'\text{-un}}, \mathbb{Z}) \hookrightarrow C_\bullet(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$, giving rise to the quotient of the corresponding stable complexes $C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma'\text{-st}} \twoheadrightarrow C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}}$. This yields natural Γ_P -equivariant maps $\pi_{J,I} : \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma'\text{-un}}, \mathbb{Z}) \rightarrow \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$ as well as $i_{J,I} : \text{St}^J(P) \rightarrow \text{St}^I(P)$ such that the following diagram of Γ_P -modules commutes

$$\begin{array}{ccc} \text{St}^J(P) & \longrightarrow & \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma'\text{-un}}, \mathbb{Z}) \\ \downarrow i_{J,I} & & \downarrow \pi_{J,I} \\ \text{St}^I(P) & \longrightarrow & \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \end{array} \quad (5.6)$$

Therefore it will suffice to show the commutativity of

$$\begin{array}{ccc} \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma'\text{-un}}, \mathbb{Z}) & \xrightarrow{\alpha_J} & \text{St}(W) \\ \downarrow \pi_{J,I} & & \uparrow \alpha_I \\ \tilde{H}_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) & & \end{array} \quad (5.7)$$

We briefly recall how α_J and α_I arises out of the homotopy equivalence in 5.2.2, which in turn is invoked from the Theorem 2.2.19. For $* \in \{\Gamma', \Gamma\}$, let $\mathcal{BT}_\bullet(W)_1^{*\text{-un}}$ denote the first barycentric subdivision of $\mathcal{BT}_\bullet(W)^{*\text{-un}}$ and $Z_* \subset \mathcal{BT}_\bullet(W)_1^{*\text{-un}} \times \mathcal{T}_\bullet(W)$ be the closed subposet such that $(\gamma, \sigma) \in Z_*$ if and only if $\gamma \in \mathcal{BT}_\bullet(W)_\sigma$ (note that here the $\mathcal{BT}_\bullet(W)_\sigma$ are defined with respect

to the corresponding $*$. The reader is referred to section 5.1 for details on the definition of the subcomplex $\mathcal{BT}_\bullet(W)_\sigma$. Then there is a correspondence of the homological complexes

$$\begin{array}{ccc}
 & C_\bullet(Z_*, \mathbb{Z}) & \\
 p_{1,*} \swarrow & & \searrow p_{2,*} \\
 C_\bullet(\mathcal{BT}_\bullet(W)^{*-un}, \mathbb{Z}) & \xrightarrow{\sim} & C_\bullet(\mathcal{BT}_\bullet(W)_1^{*-un}, \mathbb{Z}) \\
 & & C_\bullet(\mathcal{T}_\bullet(W), \mathbb{Z})
 \end{array} \quad (5.8)$$

where $p_{1,*}$ resp. $p_{2,*}$ is the first resp. second projection. It turns out, as argued in the proof of 5.2.2, that $p_{1,*}$ is a Γ_P -homotopy equivalence and consequently this induces α_J resp. α_I for $*$ = Γ' resp. Γ . Consequently it is clear that we need to show only that the following two diagrams commute, which is clear from construction

$$\begin{array}{ccc}
 C_\bullet(\mathcal{BT}_\bullet(W)_1^{\Gamma'-un}, \mathbb{Z}) & \xleftarrow{p_{1,\Gamma'}} & C_\bullet(Z_{\Gamma'}, \mathbb{Z}) \\
 \downarrow & & \downarrow \\
 C_\bullet(\mathcal{BT}_\bullet(W)_1^{\Gamma-un}, \mathbb{Z}) & \xleftarrow{p_{1,\Gamma}} & C_\bullet(Z_\Gamma, \mathbb{Z})
 \end{array} \quad \text{and} \quad \begin{array}{ccc}
 C_\bullet(Z_{\Gamma'}, \mathbb{Z}) & \xrightarrow{p_{2,\Gamma'}} & C_\bullet(\mathcal{T}_\bullet(W), \mathbb{Z}) \\
 \downarrow & & \uparrow \\
 C_\bullet(Z_\Gamma, \mathbb{Z}) & \xrightarrow{p_{2,\Gamma}} & C_\bullet(\mathcal{T}_\bullet(W), \mathbb{Z})
 \end{array}$$

Here the right vertical map in the first diagram (or the vertical map in the second) is the canonical embedding, similar to that of the left vertical embedding. Consequently we get the commutativity of diagram (5.5). $\square$

Remark 5.3.9. (i) Choosing J to be in the descending chain of ideals $\{I^n\}_{n \geq 1}$, we get a sequence of simplicial complexes, $\dots \subset \mathcal{BT}_\bullet(W)^{\Gamma_P(I^{n+1})-un} \subset \mathcal{BT}_\bullet(W)^{\Gamma_P(I^n)-un} \subset \dots$, (and also a corresponding increasing sequence $\dots \subset \mathcal{BT}_\bullet(W)^{\Gamma_P(I^n)-st} \subset \mathcal{BT}_\bullet(W)^{\Gamma_P(I^{n+1})-st} \subset \dots$). The results of this section shows that the corresponding (top dimensional) simplicial homology is compatibly Γ_P -equivariantly isomorphic to the Steinberg module $\text{St}(W)$.

(ii) A different family of subcomplexes of $\mathcal{BT}_\bullet(W)$ homotopy equivalent to the Tits building is also considered in [KY21, §11.2, 11.3] using the theory of Harder-Narasimhan polygons, and following more closely [Gra82].

Chapter 6

On a result of Teitelbaum

Let us continue to denote $\Gamma_P(I)$ by Γ as in the last chapter. In his seminal work [Tei91], Teitelbaum shows among other things that when $r = 2$, a Γ -invariant harmonic cocycle on the Bruhat-Tits tree can be reconstructed uniquely from its values on the Γ -stable part of the tree. More precisely, he shows that the restriction map, $C_{har}(V)^\Gamma \rightarrow \text{St}^I(P) \otimes_\Gamma V$, for V a finite dimensional K_∞ -vector space is an isomorphism of K_∞ -vector spaces. We will consider the generalization of this result to higher rank Bruhat-Tits building here. As in 3.3.10, we let L be a field containing K and V a finite dimensional L -vector space equipped with a left $L[\Gamma]$ -module structure as well.

6.1 Ash-Rudolph symbol and Γ -stable chambers in an apartment

Motivated by [AR79, §3, Rmk.3], giving an explicit description of a representative of the image of the Ash-Rudolph symbol under the “boundary” map

$$H_{n-2}(T_n, \mathbb{Z}) \simeq H_{n-2}(\partial \overline{X}, \mathbb{Z}) \simeq H_{n-1}(\overline{X}, \partial \overline{X}, \mathbb{Z}),$$

($X := \text{SL}_n(\mathbb{R})/\text{SO}_n(\mathbb{R})$ is the symmetric space and $\overline{X}$ is its Borel-Serre bordification), we also try to give an explicit representative of the image of the Ash-Rudolph symbol in our setting, namely under

$$\text{GQ}_\Gamma^{-1} : \text{St}(W) \simeq \text{St}^I(P).$$

We claim that a cycle in $\text{St}^I(P)$ representing the image of the Ash-Rudolph symbol $[[w_0, \dots, w_{r-1}]]$ is a sum of all Γ -stable chambers (with suitable orientation) inside the apartment $\mathcal{A}_\bullet(w_0, \dots, w_{r-1})^1$.

¹Note that by Lemma 4.2.6, there are finitely many Γ -stable chambers inside the apartment $\mathcal{A}_\bullet(w_0, \dots, w_{r-1})$

For simplicity, we denote the ordered basis $(w_0, \dots, w_{r-1})$ by $\underline{w}$ and the corresponding apartment by $\mathcal{A}_\bullet(\underline{w})$.

A key tool useful for the proof is the projection of the r -dimensional cube in $\mathbb{Z}^r$ to the apartment $\mathcal{A}_\bullet(\underline{w})$. Appropriately interpreting this projected cube as a $(r-1)$ -chain, its boundary turns out to be a $(r-2)$ -cycle in the *unstable* complex, $C_\bullet(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$ and one of the primary concluding steps in the main proof of this section involves showing that this cycle maps to the Ash-Rudolph symbol $[[\underline{w}]]$ under the homotopy equivalence in 5.2.2.

6.1.1 Projected cube of length N , $\mathcal{PC}_N$.

Recall the definition of the simplicial complexes $\mathcal{A}_\bullet$, $\mathcal{A}_\bullet(\underline{w})$ from Example 2.2.6(iii) and Definition 3.3.1.

Definition 6.1.1. (i) Let $N > 0$ be a positive integer. The *projected cube of length N* , denoted $\mathcal{PC}_N$, is defined to be the image in $\mathcal{A}_\bullet$ of the following subset of $\mathbb{Z}^r$ under the projection $\mathbb{Z}^r \rightarrow \mathcal{A}_\bullet$:-

$$\mathcal{C}_N := \{(n_0, \dots, n_{r-1}) \in \mathbb{Z}^r \mid 0 \leq n_i \leq N \ \forall 0 \leq i \leq r-1\}.$$

We call $\mathcal{C}_N$ the *cube of length N* in $\mathbb{Z}^r$.

(ii) Denote by $\mathcal{PC}_N(\underline{w}) \subset \mathcal{A}_\bullet(\underline{w})$ the image of $\mathcal{PC}_N$ under the isomorphism $\mathcal{A}_\bullet \simeq \mathcal{A}_\bullet(\underline{w})$. We will refer to it as the *projected cube of length N* in the apartment $\mathcal{A}_\bullet(\underline{w})$.

We record here some easily deducible properties of the projected cube $\mathcal{PC}_N$

Lemma 6.1.2. The following hold true:-

1. There is a canonical simplicial structure on $\mathcal{PC}_N$ so that it becomes a simplicial subcomplex of $\mathcal{A}_\bullet$. The resulting complex will be denoted by $(\mathcal{PC}_N)_\bullet$.
2. Under the projection $\mathbb{Z}^r \rightarrow \mathcal{A}_\bullet$, the subposet $(\mathcal{C}_N^{\text{top}})_\bullet$ of $\mathbb{Z}^r$ based on the vertices

$$\{(n_0, \dots, n_{r-1}) \in \mathcal{C}_N \mid \exists j \text{ s.t. } n_j = N\}$$

of $\mathbb{Z}^r$ is isomorphic to $(\mathcal{PC}_N)_\bullet$.

3. Let finitely many points $y_1, \dots, y_m \in \mathcal{A}_\bullet$ be given. Then for $N \gg 0$, $\mathcal{PC}_N$ contains $\{y_1, \dots, y_m\}$.

Identical statements as above also hold true for $\mathcal{A}_\bullet(\underline{w})$ and $\mathcal{PC}_N(\underline{w})$. The simplicial structure on the latter will be denoted by $\mathcal{PC}_N(\underline{w})_\bullet$.

Proof. 1. For an integer $0 \leq d \leq r - 1$, we declare the set of d -simplices of $(\mathcal{PC}_N)_\bullet$ to be the set $\mathcal{A}_d \cap \mathcal{PC}_N$. To see that these simplices induce a simplicial structure on $\mathcal{PC}_N$, one needs to show that, if s is a d -simplex of $\mathcal{PC}_N$ and $e \subset s$ then $e \in \mathcal{A}_l \cap \mathcal{PC}_N$, where $l + 1$ is the cardinality of e . But this is obvious from the definition of $\mathcal{PC}_N$.

Henceforth, for an integer $0 \leq d \leq r - 1$, we denote by $(\mathcal{PC}_N)_d$ the set of d -simplices of $\mathcal{PC}_N$ described in the above proof.

2. First note that for any integer $0 \leq d \leq r - 1$, the canonical map $(\mathcal{C}_N^{\text{top}})_d \rightarrow (\mathcal{PC}_N)_d$ is injective. Indeed it suffices to see this for $d = 0$ and this follows since no two vertices of $\mathcal{C}_N^{\text{top}}$ can differ by an integral multiple of $(1, \dots, 1)$.

To show the surjectivity of the above map on d -simplices, we will explicitly exhibit a preimage of a d -simplex $s := \{[n_0^{(0)}, \dots, n_{r-1}^{(0)}], \dots, [n_0^{(d)}, \dots, n_{r-1}^{(d)}]\} \in (\mathcal{PC}_N)_d$ as follows. Choosing appropriate representative of s after scaling $(n_0^{(0)}, \dots, n_{r-1}^{(0)})$ etc. by an element in $(n, \dots, n)$ for some $n \in \mathbb{N}$, we can assume without loss of generality that $(n_0^{(0)} + 1, \dots, n_{r-1}^{(0)} + 1) \leq (N, \dots, N)$ and moreover that there exists $0 \leq j \leq r - 1$ satisfying $n_j^{(0)} + 1 = N$. Consequently there exists an integer $0 \leq i \leq d$ such that $(n_0^{(i)}, \dots, n_{r-1}^{(i)}) < (N, \dots, N)$ and for some index $0 \leq j' \leq r - 1$ (not necessarily equal to j) $n_{j'}^{(i+1)} = N$. Then $(n_0^{(i+1)}, \dots, n_{r-1}^{(i+1)}) \leq (n_0^{(i+2)}, \dots, n_{r-1}^{(i+2)}) \leq \dots \leq (n_0^{(i)} + 1, \dots, n_{r-1}^{(i)} + 1)$ is a d -simplex in $(\mathcal{C}_N^{\text{top}})$ and corresponds to s .

3. Choose preimages $\tilde{y}_i \in \mathbb{Z}^r$ of y_i for $1 \leq i \leq m$. It is clear that for $N \gg 0$, $\mathcal{C}_N$ contains all the $\tilde{y}_i$'s. Hence we are done. □

6.1.2 Boundary of a projected cube in $\mathcal{A}_\bullet(\underline{w})$

Recall the convention regarding the symbol $[\vec{C}]$ from Definition 5.3.1. In what follows we construct a natural $(r - 1)$ -chain on the $\mathcal{BT}_\bullet(W)$.

Definition 6.1.3. Let $N > 0$ be a positive integer. Then define

$$[\mathcal{PC}_N(\underline{w})] := \sum_{C \in (\mathcal{PC}_N)_{r-1}} \text{sgn}_{\underline{w}}(\vec{C})[\vec{C}] \in C_{r-1}(\mathcal{BT}_\bullet(W), \mathbb{Z}).$$

Lemma 6.1.4. The image of the chain $[\mathcal{PC}_N(\underline{w})]$ under the quotient

$$C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z}) \rightarrow C_\bullet(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}}$$

(cf. 5.3.3) is independent of N , for $N \gg 0$.

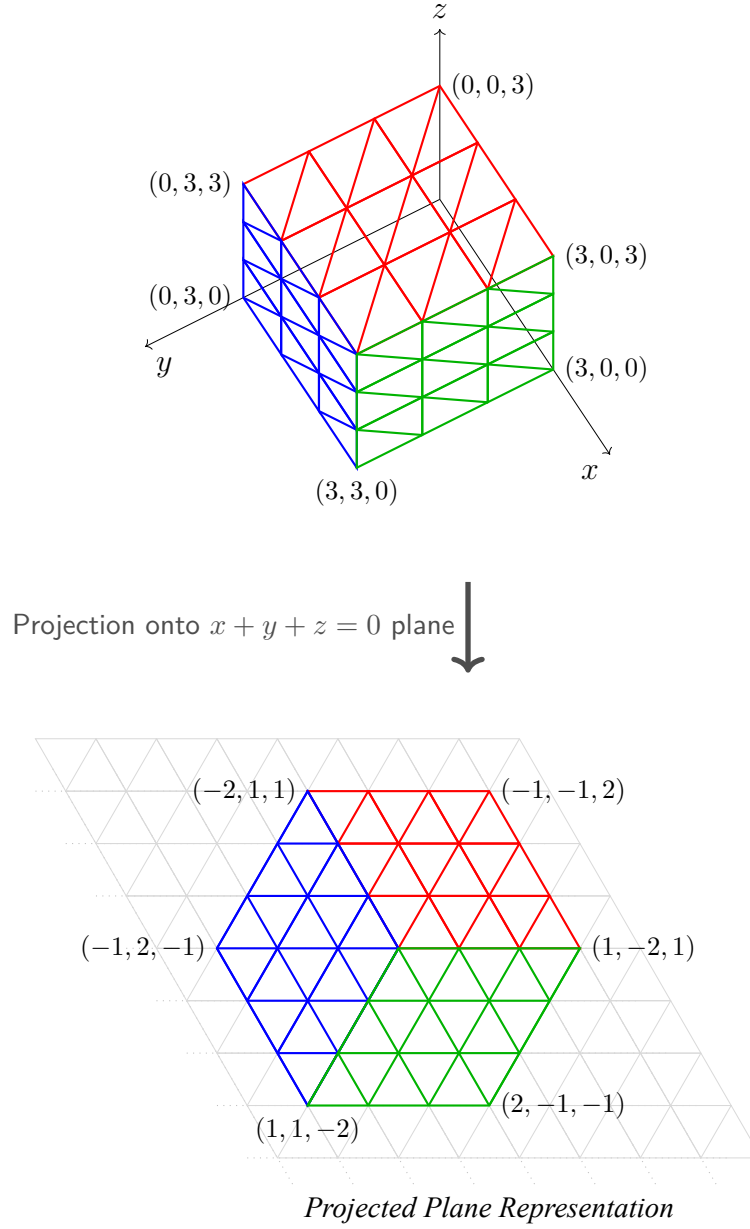


Figure 6.1. The top resp. bottom figure is a representation in Euclidean space of $\mathcal{C}_3^{top}$ resp. projected cube of length 3 $\mathcal{PC}_3$, when $r = 3$.

For the proof, recall that $\mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}} \cap \mathcal{A}_{r-1}(\underline{w})$ is finite by Lemma 4.2.6.

Proof. Define

$$\text{StC}_{\Gamma}(\underline{w}) := \sum_{C \in \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}} \cap \mathcal{A}_{r-1}(\underline{w})} \text{sgn}_{\underline{w}}(\vec{C}) [\vec{C}]_{st} \in C_{r-1}(\mathcal{BT}_{\bullet}(W)^{\Gamma\text{-st}}, \mathbb{Z}),$$

which is a $(r-1)$ -chain consisting of a sum of Γ -stable chambers in the apartment $\mathcal{A}_{\bullet}(\underline{w})$. It suffices to show that the image of $[\mathcal{PC}_N]$, for $N \gg 0$, is $\text{StC}_{\Gamma}(\underline{w})$. By Lemma 6.1.2(iii), for all $N \gg 0$, the projected cube of length N contains all the chambers C in $\mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}} \cap \mathcal{A}_{r-1}(\underline{w})$. Hence $\mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}} \cap \mathcal{A}_{r-1}(\underline{w}) \subset (\mathcal{PC}_N)_{r-1}$ and since for any chamber C of $(\mathcal{PC}_N)_{r-1}$ which is not Γ -stable, $[\vec{C}]_{st} = 0$, we are done. $\square$

We will show below that $\text{StC}_{\Gamma}(\underline{w})$ defined in the above proof is a $(r-1)$ -cycle in $C_{r-1}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}}$. Hence a posteriori it would be appropriate to call it the Γ -stable cycle in $\mathcal{A}_{\bullet}(\underline{w})$.

The boundary of $[\mathcal{PC}_N(\underline{w})]$.

Let $N > 1$ be an integer. Here by boundary we mean the image of the chain $[\mathcal{PC}_N(\underline{w})]$ under the differential $C_{r-1}(\mathcal{BT}_{\bullet}(W), \mathbb{Z}) \xrightarrow{\partial_{r-1}} C_{r-2}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})$. We will give a description of this boundary in $C_{r-2}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})$ which suffices for our purpose later.

Let us write $\partial_{r-1}([\mathcal{PC}_N(\underline{w})]) = \sum_{e \in \mathcal{BT}_{r-2}(W)} m_e [\vec{e}]$, where $m_e \in \mathbb{Z}$. By definition m_e vanishes for almost all $\vec{e}$ and in fact, it vanishes for e outside $\mathcal{PC}_N(\underline{w})$. We will use the phrase “ e occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ ” to mean those panels e such that $m_e \neq 0$. Similarly for a panel $e \subset \mathcal{PC}_N(\underline{w})$ we say “ e lies in the region $n_i = N_i, n_j = N_j, \dots$ ” if the panel corresponding to e inside $\mathcal{C}_N^{\text{top}}$, via the isomorphism in 6.1.2(2), lies in the said region.

Proposition 6.1.5. A panel $e \in \mathcal{PC}_N(\underline{w})_{r-2}$ occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ if and only if there exists i such that $0 \leq i \leq r-1$ and e lies in the region $n_i = 0$.

Proof. Suppose e occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$. Then by Lemma 3.3.9 we know that the one of the two chambers of $\mathcal{A}_{\bullet}(\underline{w})$ containing e is not contained in $\mathcal{PC}_N(\underline{w})$. Denote by

$$\tilde{e} := \{(n_0^{(0)}, \dots, n_{r-1}^{(0)}), (n_0^{(1)}, \dots, n_{r-1}^{(1)}), \dots, (n_0^{(r-2)}, \dots, n_{r-1}^{(r-2)})\}$$

the panel corresponding to e in $\mathcal{C}_N^{\text{top}}$ (see the proof of Lemma 6.1.2(ii) for the construction of $\tilde{e}$). Then observe from the proof of Lemma 3.3.9 that there exists two indices $0 \leq i \neq j \leq r-1$ such that $n_i^{(s)}$ resp. $n_j^{(s)}$ are equal to some integers say, N_i resp. N_j for all $0 \leq s \leq r-2$. By definition of $\mathcal{C}_N^{\text{top}}$ we can assume without loss of generality that $N_j = N$. If $N_i > 0$, then it is easy to see that

the two chambers containing $\tilde{e}$ are contained in C_N^{top} . Hence $N_i = 0$. This finishes the proof of the forward direction.

Conversely, since any panel e lying in the region $n_i = 0$ is contained in precisely one chamber of $\mathcal{PC}_N(\underline{w})$, it necessarily occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$. $\square$

Example 6.1.6. Consider the panel

$$e_0 := \{[0, 0, \dots, 0, N], [0, 0, \dots, 1, N], \dots, [0, 1, \dots, 1, N]\} \in \mathcal{A}_{r-2}(\underline{w}).$$

From previous proposition, e_0 occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$. As observed in the above proof, there is precisely one chamber of $\mathcal{PC}_N(\underline{w})$ containing $\vec{e}_0$ which is clearly

$$C_0 := \{[0, 0, \dots, 0, N], [0, 0, \dots, 1, N], \dots, [0, 1, \dots, 1, N], [1, 1, \dots, 1, N]\}.$$

We note that for $x_{r-1} := [1, \dots, 1, N]$, $\text{sgn}_{\underline{w}}((\vec{C}_0)_{x_{r-1}}) = 1$. This follows from choosing the following representative lattices of the vertices of C_0 and arranging in the descending order

$$\begin{aligned} \langle \varpi w_0, \dots, \varpi w_{r-2}, \varpi^N w_{r-1} \rangle_R &\supsetneq \langle \varpi w_0, \dots, \varpi w_{r-2}, \varpi^{N+1} w_{r-1} \rangle_R \supsetneq \dots \supsetneq \\ &\langle \varpi^1 w_0, \varpi^2 w_1, \dots, \varpi^2 w_{r-2}, \varpi^{N+1} w_{r-1} \rangle_R \end{aligned}$$

and checking that $\sigma_{\underline{w}}^{x_{r-1}-nbd}(C_0) = 1$ (3.3.5(i)).

Given a permutation $\sigma \in \mathcal{S}_r$, its action on vertices $[n_0, \dots, n_{r-1}] \in \mathcal{A}_0(\underline{w})$ is given by sending it to $[n_{\sigma(0)}, \dots, n_{\sigma(r-1)}]$. Then it is not hard to see that

$$\sigma e_0 = \{\sigma[0, 0, \dots, 0, N], \sigma[0, 0, \dots, 1, N], \dots, \sigma[0, 1, \dots, 1, N]\}$$

also occurs in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ and the unique chamber of $\mathcal{PC}_N(\underline{w})$ containing σe_0 is σC_0 .

Proposition 6.1.7. (i) Let $\partial_{r-1}([\mathcal{PC}_N(\underline{w})]) = \sum_{e \in \mathcal{BT}_{r-2}(W)} m_e[\vec{e}]$ and $\sigma \in \mathcal{S}_r$ and $x_0 := [0, 1, \dots, 1, N]$. Then $m_{e_0}[\vec{e}_0] = [(\vec{e}_0)_{x_0}]$ and for $\vec{e} = \sigma \vec{e}_0$, $m_e = \text{sgn}(\sigma)m_{e_0}$.

(ii) Let e' be a panel occurring in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ distinct from the panels σe_0 for any $\sigma \in \mathcal{S}_r$. Then e' does not contain a vertex of the form $[0, \dots, N, \dots, 0]$ all of whose coordinates are zero except one.

Proof. (i) First assume $\sigma = 1$ and set $x_{r-1} := [1, \dots, 1, N]$. Then by previous remark the only chamber C_0 of $\mathcal{PC}_N(\underline{w})$ containing e_0 satisfies $\text{sgn}_{\underline{w}}((\vec{C}_0)_{x_{r-1}}) = 1$ and hence the summand corresponding to $(\vec{C}_0)_{x_{r-1}}$ appearing in $[\mathcal{PC}_N(\underline{w})]$ is $[(\vec{C}_0)_{x_{r-1}}]$ (note that $\text{sgn}_{\underline{w}}((\vec{C}_0)_x)[(\vec{C}_0)_x]$ is independent of the vertex x). Hence $\partial_{r-1}((\vec{C}_0)_{x_1}) = [(\vec{e}_0)_{x_0}] + \dots$, and in particular $m_{e_0}[\vec{e}_0] = [(\vec{e}_0)_{x_1}]$. Analogously the only chamber containing σe_0 in $\mathcal{PC}_N(\underline{w})$ is σC_0 and $\text{sgn}_{\underline{w}}((\sigma \vec{C}_0)_{\sigma x_0}) = \text{sgn}(\sigma)$. Hence $m_{\sigma e_0} = \text{sgn}(\sigma)m_{e_0}$.

- (ii) Let $e' = \{(n_1^{(0)}, \dots, n_r^{(0)}), (n_1^{(1)}, \dots, n_r^{(1)}), \dots, (n_1^{(r-2)}, \dots, n_r^{(r-2)})\}$ be an oriented panel occurring in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$. Suppose e' contains a vertex of the form $[0, \dots, N, \dots, 0]$ with N at the l -th coordinate. Then it is not difficult to observe that by virtue of e' being a panel in $\mathcal{PC}_N(\underline{w})$, the coordinates of the other vertices of e' satisfy $n_t^{(s)} = 0$ or 1 for all $0 \leq s \leq r-2$ and $0 \leq t \neq l \leq r-1$. Consequently it is of the form $\sigma \vec{e}_0$ for some $\sigma \in \mathcal{S}_r$.

□

Remark 6.1.8. (i) Continuing the notation from before, if we set

$$x_0 := [0, \dots, 0, N], x_1 := [0, \dots, 1, N], \dots, x_{r-2} := [0, 1, \dots, 1, N],$$

then the basis symbol $[\vec{e}_{0x_0}]$ is given explicitly by $[x_0, x_1, \dots, x_{r-2}]$. Similarly $[\overrightarrow{\sigma(e_0)_{\sigma x_0}}]$ is given $[\sigma x_0, \dots, \sigma x_{r-2}]$. (5.3.1(iii))

- (ii) The above proposition can be summarized as saying that

$$\partial_{r-1}([\mathcal{PC}_N(\underline{w})]) = \sum_{\theta \in \mathcal{S}_r} \text{sgn}(\theta) [\overrightarrow{\theta(e_0)_{\theta x_1}}] + \sum_{\substack{e' \neq \theta e_0 \\ \theta \in \mathcal{S}_r}} m_{e'} [\vec{e}']$$

and none of the panels e' appearing in the second sum contain a vertex of the form $[0, \dots, N, \dots, 0]$.

6.1.3 Describing the Γ -unstable region in $\mathcal{A}_\bullet(\underline{w})$ outside certain bounded regions

For a vertex $[n_0, \dots, n_{r-1}] \in \mathcal{A}_\bullet(\underline{w})$ representing the class of R -lattice $\langle \varpi^{n_0} w_0, \dots, \varpi^{n_{r-1}} w_{r-1} \rangle_R$, we will find sufficient conditions on the integers n_i 's that will enable us to find the vertices $\sigma \in \mathcal{T}_\bullet(\underline{w})$ such that $[n_0, \dots, n_{r-1}] \in \mathcal{BT}_\bullet(W)_\sigma$ (recall the simplicial subcomplex $\mathcal{BT}_\bullet(W)_\sigma \subset \mathcal{BT}_\bullet(W)$ from 5.1). Let us denote by $\mathcal{E}_\underline{n}$ be the vector bundle $\mathcal{E}(P, [n_0, \dots, n_{r-1}])$ and let the vertex in $\mathcal{T}_0(W)$ corresponding to Kw_i , for $0 \leq i \leq r-1$, be denoted by σ_i . Also by the sum $\sum_{i \in I} \sigma_i$, for a proper subset $I \subsetneq \{0, \dots, r-1\}$, we mean the vertex corresponding to $\sum_{i \in I} Kw_i$ in $\mathcal{T}_0(W)$.

Let us consider the elements of $\text{End}(W)$ in a matrix form with respect to the basis $(w_0, \dots, w_{r-1})$. Then for $0 \leq i \neq j \leq r-1$, define

$$\mathbf{M}_{i,j} := \{(a_{kl})_{0 \leq k, l \leq r-1} \in \text{End}(W) \mid a_{kl} = 0 \forall (k, l) \neq (i, j)\}.$$

The lemma below is easy to observe

Lemma 6.1.9. Let i, j be integers such that $0 \leq i \neq j \leq r-1$. Then any $g \in \mathbf{M}_{i,j}$ maps the codimension 1 subspace $\sigma_0 + \dots + \widehat{\sigma_j} + \dots + \sigma_{r-1}$ to zero, that is $g|_{\sigma_0 + \dots + \widehat{\sigma_j} + \dots + \sigma_{r-1}} = 0$.

□

Consequently, it is clear and stated in the corollary below, that if we can find a non-zero element of $H^0(C, \mathcal{E}_{\underline{n}}^\vee \otimes \mathcal{E}_{\underline{n}}(I)) \cap M_{i,j}$ (intersection inside $\text{End}(W)$), then $[n_1, \dots, n_r] \in \mathcal{BT}_\bullet(W)_{\sigma_1 + \dots + \widehat{\sigma_j} + \dots + \sigma_r}$. In this direction, we can use Riemann-Roch theorem, analogous to its use in Lemma 4.2.6, to determine sufficient conditions depending on (i, j) , for n_i 's so that the intersection above furnishes a nonzero element. Note that under the isomorphism $W^\vee \otimes_K W \simeq \text{End}(W)$, the K -subspace $M_{i,j}$ on the right corresponds to $(Kw_j)^\vee \otimes_K (Kw_i)$ on the left, where the embedding $(Kw_j)^\vee \hookrightarrow W^\vee$ is induced by the quotient $W \twoheadrightarrow W / (\sum_{l \neq j} Kw_l) \simeq Kw_j$. Hence we have

Lemma 6.1.10. The image of the following injection upon applying the global section functor H^0

$$(\mathcal{E}_{\underline{n}} / (\mathcal{E}_{\underline{n}} \cap \sum_{l \neq j} Kw_l))^\vee \otimes_{\mathcal{O}_C} (\mathcal{E}_{\underline{n}} \cap Kw_i)(I) \hookrightarrow (\mathcal{E}_{\underline{n}})^\vee \otimes_{\mathcal{O}_C} \mathcal{E}_{\underline{n}}(I)$$

is contained in the intersection $H^0(C, \mathcal{E}_{\underline{n}}^\vee \otimes \mathcal{E}_{\underline{n}}(I)) \cap M_{i,j}$. $\square$

Notice that the sheaf on the left in the above injection is a line bundle which can be described as $\mathcal{E}((P / (P \cap \sum_{l \neq j} Kw_l))^\vee \otimes_A (P \cap Kw_i) \otimes_A I, \varpi^{n_i - n_j}(Rw_j)^\vee \otimes_R (Rw_i))$. Hence for an integer $N \gg 0$ and for $n_i - n_j \leq -N$, by the Riemann-Roch theorem, the line bundle above admits non-trivial global sections. Consequently, this observation together with the previous two lemmas give

Corollary 6.1.11. There exists an integer $N \gg 0$ such that for any $0 \leq i \neq j \leq r - 1$, if a simplex $\{[n_0^{(0)}, \dots, n_{r-1}^{(0)}], \dots, [n_0^{(d)}, \dots, n_{r-1}^{(d)}]\}$ lies in the region $n_i - n_j \leq -N$, then it is a simplex in $\mathcal{BT}_\bullet(W)_{\sigma_0 + \dots + \widehat{\sigma_j} + \dots + \sigma_{r-1}}$.

Proof. Let us first assume $d = 0$ and $L = [n_0, \dots, n_{r-1}]$ be a vertex of $\mathcal{A}_\bullet(\underline{w})$ such that $n_i - n_j \leq -N$ for some $0 \leq i \neq j \leq r - 1$. Then by Corollary 3.2.9, $\Gamma_L = 1 + H^0(C, \mathcal{E}_{\underline{n}}^\vee \otimes_{\mathcal{O}_C} \mathcal{E}_{\underline{n}})$. From the discussion above we have found a non-trivial element in the intersection $H^0(C, \mathcal{E}_{\underline{n}}^\vee \otimes_{\mathcal{O}_C} \mathcal{E}_{\underline{n}}) \cap M_{i,j}$, say g . By 6.1.9, $\gamma := 1 + g$ acts as identity on $\sigma_0 + \dots + \widehat{\sigma_j} + \dots + \sigma_{r-1}$.

More generally, let $s := ([n_0^{(0)}, \dots, n_{r-1}^{(0)}], \dots, [n_0^{(d)}, \dots, n_{r-1}^{(d)}])$ be a simplex in $\mathcal{A}_d(\underline{w})$. Then the set of line bundles, for $0 \leq u \leq d$, $(\mathcal{E}_{\underline{n}^{(u)}} / (\mathcal{E}_{\underline{n}^{(u)}} \cap \sum_{l \neq j} Kw_l))^\vee \otimes_{\mathcal{O}_C} (\mathcal{E}_{\underline{n}^{(u)}} \cap Kw_i)(I) = \mathcal{E}((P / (P \cap \sum_{l \neq j} Kw_l))^\vee \otimes_A (P \cap Kw_i) \otimes_A I, \varpi^{n_i^{(u)} - n_j^{(u)}}(Rw_j)^\vee \otimes_R (Rw_i))$, form a chain, hence also do their global sections form a chain inside $\text{End } W$. Using the Riemann-Roch theorem analogous to its use in the previous paragraph, for $N \gg 0$, the set of global sections given by $H^0(C, \mathcal{E}((P / (P \cap \sum_{l \neq j} Kw_l))^\vee \otimes_A (P \cap Kw_i) \otimes_A I, \varpi^{n_i^{(u)} - n_j^{(u)}}(Rw_j)^\vee \otimes_R (Rw_i)))$ form a nontrivial chain of subspaces for $n_i^{(u)} - n_j^{(u)} \leq -N$ and consequently furnish a non-trivial element in the stabilizer Γ_s stabilizing $\sigma_0 + \dots + \widehat{\sigma_j} + \dots + \sigma_{r-1}$. $\square$

6.1.4 The Γ -stable cycle in $\mathcal{A}_\bullet(\underline{w})$ and the Ash-Rudolph symbol $[[\underline{w}]]$.

We can now state and prove the following main result of this section. Recall $\text{StC}_\Gamma(\underline{w})$ from the proof of Lemma 6.1.13.

Theorem 6.1.12. $\text{StC}_\Gamma(\underline{w})$ is a $(r-1)$ -cycle. Its associated class in $H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z})$ is mapped to $[[\underline{w}]]$ under the isomorphism $\text{GQ}_\Gamma : \text{St}^I(P) = H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \simeq \text{St}(W)$.

Recall from section 5.3 that $\text{GQ}_\Gamma = \alpha \circ \beta$, where $\alpha : H_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \simeq \text{St}(W)$ is the isomorphism induced from 5.2.2 and $\beta : H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \simeq H_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$ is an results from the connecting homomorphism of the long exact sequence in homology

$$\begin{aligned} \dots \rightarrow H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \rightarrow H_{r-1}(\mathcal{BT}_\bullet(W), \mathbb{Z}) \rightarrow H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z}) \\ \xrightarrow{\beta} H_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \dots \end{aligned}$$

In particular, β arises from an application of the Snake lemma to the following commutative diagram of short exact sequences of abelian groups

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) & \longrightarrow & C_{r-1}(\mathcal{BT}_\bullet(W), \mathbb{Z}) & \longrightarrow & C_{r-1}(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}} \longrightarrow 0 \\ & & \downarrow \partial_{r-1}^{\Gamma\text{-un}} & & \downarrow \partial_{r-1} & & \downarrow \partial_{r-1}^{\Gamma\text{-st}} \\ 0 & \longrightarrow & C_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) & \longrightarrow & C_{r-2}(\mathcal{BT}_\bullet(W), \mathbb{Z}) & \longrightarrow & C_{r-2}(\mathcal{BT}_\bullet(W), \mathbb{Z})^{\Gamma\text{-st}} \longrightarrow 0 \end{array}$$

Lemma 6.1.13. For $N \gg 0$, the boundary of the chain corresponding to the projected cube of length N in $\mathcal{A}_\bullet(\underline{w})$, $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$, is contained in $C_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$.

In particular, $\text{StC}_\Gamma(\underline{w})$ is a $(r-1)$ -cycle and, by abuse of notation, denoting the classes of $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ resp. $\text{StC}_\Gamma(\underline{w})$ in $H_{r-1}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-st}}, \mathbb{Z})$ resp. $H_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z})$ also by the same notation, we have $\beta(\text{StC}_\Gamma(\underline{w})) = \partial_{r-1}([\mathcal{PC}_N(\underline{w})])$.

Proof. From Proposition 6.1.5 and Corollary 6.1.11 we know that any panel occuring in $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ is Γ -unstable. The second part of the lemma follows from an application of Snake lemma described above. $\square$

Proof of Theorem 6.1.12. From the previous lemma it suffices to show that the image of $\partial_{r-1}([\mathcal{PC}_N(\underline{w})])$ under

$$\alpha : H_{r-2}(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \simeq \text{St}(W)$$

is $[[\underline{w}]]$. Recall that α arises out of the homotopy equivalence in 5.2.2, which in turn is invoked from the Theorem 2.2.19. Below it is necessary to explicitly work through the proof details of

that theorem and it would be prudent to refer to 2.2.12 and 2.2.13 for the terminology regarding the homological complex of posets to be used here. Let $\mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}}$ denote the first barycentric subdivision of $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ and $Z_\bullet \subset \mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}} \times \mathcal{T}_\bullet(W)$ be the closed subset such that $(\gamma, \sigma) \in Z_\bullet$ if and only if $\gamma \in \mathcal{BT}_\bullet(W)_\sigma$. Then there is a correspondence of the homological complexes

$$\begin{array}{ccc}
 & C_\bullet(Z_\bullet, \mathbb{Z}) & \\
 p_1 \swarrow & & \searrow p_2 \\
 C_\bullet(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) & \xrightarrow{\sim} C_\bullet(\mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}}, \mathbb{Z}) & C_\bullet(\mathcal{T}_\bullet(W), \mathbb{Z})
 \end{array} \tag{6.1}$$

where p_1 resp. p_2 is the first resp. second projection. As shown in the proof of 5.2.2 that p_1 is a Γ_P -homotopy equivalence, consequently this induces α .

Let us denote the homotopy equivalence between homological complexes of $\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}$ and $\mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}}$ by

$$B_\bullet : C_\bullet(\mathcal{BT}_\bullet(W)^{\Gamma\text{-un}}, \mathbb{Z}) \rightarrow C_\bullet(\mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}}, \mathbb{Z})$$

which is explicitly given by

$$B_d : [\vec{s}_{v_0}] := [v_0, \dots, v_d] \mapsto \sum_{\tau \in \mathcal{S}_{d+1}} \text{sgn}(\tau) [\{v_{\tau(0)}\}, \{v_{\tau(0)}, v_{\tau(1)}\}, \dots, \{v_{\tau(0)}, \dots, v_{\tau(d)}\}], \quad 0 \leq d \leq r-1.$$

See [Lö23, Prop.3.4.1] for a proof. We need to apply B_{r-2} to $\partial_{r-1}(\mathcal{PC}_N(\underline{w}))$. To simplify, write

$$\partial_{r-1}([\mathcal{PC}_N(\underline{w})]) = \Sigma_1 + \Sigma_2$$

for $\Sigma_1 := \sum_{\theta \in \mathcal{S}_r} \text{sgn}(\theta) [\theta(\vec{e}_0)_{\theta x_1}]$ and $\Sigma_2 := \sum_{\substack{\vec{e}' \neq \theta \vec{e}_0 \\ \theta \in \mathcal{S}_r}} n_{\vec{e}'} [\vec{e}']$. Recall the notation $e_0 = \{x_0, \dots, x_{r-2}\}$

from Remark 6.1.8(i). Then we have

$$\begin{aligned}
 B_{r-2}(\Sigma_1) &= \sum_{\theta \in \mathcal{S}_r} \text{sgn}(\theta) [\{\theta(x_0)\}, \{\theta(x_0), \theta(x_1)\}, \dots, \{\theta(x_0), \dots, \theta(x_{r-2})\}] + \\
 &\sum_{\substack{\tau \in \mathcal{S}_{r-1} \\ \tau \neq 1}} \text{sgn}(\tau) \sum_{\theta \in \mathcal{S}_r} \text{sgn}(\theta) [\{\theta(x_{\tau(0)})\}, \{\theta(x_{\tau(0)}), \theta(x_{\tau(1)})\}, \dots, \{\theta(x_{\tau(0)}), \dots, \theta(x_{\tau(r-2)})\}].
 \end{aligned}$$

Let us further denote the former sum by Σ'_1 and the latter by Σ''_1 . Also writing $\Sigma'_2 := B_{r-2}(\Sigma_2)$, we have

$$B_{r-2}(\partial_{r-1}(\mathcal{PC}_N(\underline{w}))) := \Sigma'_1 + \Sigma''_1 + \Sigma'_2 \in C_\bullet(\mathcal{BT}_\bullet(W)_1^{\Gamma\text{-un}}, \mathbb{Z}).$$

Our task is now to find appropriate pre-images of Σ'_1, Σ''_1 and Σ'_2 in $C_\bullet(Z_\bullet, \mathbb{Z})$ (via p_1). First consider Σ'_1 . By Corollary 6.1.11 we have that $\{x_0, \dots, x_i\} \in \mathcal{BT}_\bullet(W)_{\sigma_0+\dots+\sigma_i}$ for $0 \leq i \leq r-2$. This provides the following preimage of Σ'_1

$$\tilde{\Sigma}'_1 := \sum_{\theta \in \mathcal{S}_r} \text{sgn}(\theta) [(\{\theta x_0\}, \sigma_{\theta(0)}), (\{\theta x_0, \theta x_1\}, \sigma_{\theta(0)} + \sigma_{\theta(1)}), \dots, (\{\theta x_0, \dots, \theta x_{r-2}\}, \sigma_{\theta(0)} + \dots + \sigma_{\theta(r-2)})].$$

Clearly the image of $\tilde{\Sigma}'_1$ under p_2 is precisely the Ash-Rudolph symbol $[[w]]$ by 4.1.7(i). To conclude the proof, it suffices to find pre-images $\tilde{\Sigma}''_1$ resp. $\tilde{\Sigma}'_2$ of Σ''_1 resp. Σ'_2 which map to the zero under p_2 . Let $[\vec{s}]_1 := m_s[s_0, \dots, s_{r-2}]$ be a symbol appearing in either Σ''_1 or Σ'_2 for $m_s \in \{\pm 1\}$. Then combining Proposition 6.1.5 and Corollary 6.1.11, there exists an integer, which we denote by $N_{[\vec{s}]_1}$, between 0 and $r-1$ such that all the simplices $s_0, \dots, s_{r-2}$ lie in $\mathcal{BT}_\bullet(W)_{\sigma_{N_{[\vec{s}]_1}}}$. Consequently one can form

$$\tilde{\Sigma}''_1 := \sum_{\substack{[\vec{s}]_1 = m_s[s_0, \dots, s_{r-2}] \\ \text{appears in } \Sigma''_1}} m_s [(s_0, \sigma_{N_{[\vec{s}]_1}}), (s_1, \sigma_{N_{[\vec{s}]_1}}), \dots, (s_{r-2}, \sigma_{N_{[\vec{s}]_1}})],$$

and similarly

$$\tilde{\Sigma}'_2 := \sum_{\substack{[\vec{s}]_1 = m_s[s_0, \dots, s_{r-2}] \\ \text{appears in } \Sigma'_2}} m_s [(s_0, \sigma_{N_{[\vec{s}]_1}}), (s_1, \sigma_{N_{[\vec{s}]_1}}), \dots, (s_{r-2}, \sigma_{N_{[\vec{s}]_1}})].$$

Clearly both $\tilde{\Sigma}''_1$ and $\tilde{\Sigma}'_2$ are mapped to zero under $p_2 : C_{r-2}(Z_\bullet, \mathbb{Z}) \rightarrow C_{r-2}(\mathcal{T}_\bullet(W), \mathbb{Z})$ (see 2.2.13(ii)). $\square$

6.2 Harmonic cocycles and its values on the Γ -stable part

As already alluded to before, in [Tei91] Teitelbaum shows in the rank 2 case that a $\Gamma(I)$ -invariant harmonic cocycle with values in a K -vector space V is uniquely determined by its values on the $\Gamma(I)$ -stable part. More precisely, let $\Gamma(I) := \text{Ker}(\text{GL}_2(A) \rightarrow \text{GL}_2(A/I))$ be the principal congruence subgroup and $\text{St} := \text{St}^I(A^{\oplus 2})$. Then in the course of the proof of [Tei91, Prop.21], the following map

$$C_{\text{har}}(V)^{\Gamma(I)} \xrightarrow{\text{Res}} C_1(\mathcal{BT}_\bullet, \mathbb{Z})^{\Gamma(I)\text{-st}} \otimes_{\Gamma(I)} V,$$

is shown to have image inside $\text{St} \otimes_{\Gamma(I)} V$ and is an isomorphism onto it. Our aim in this subsection is to obtain a (partial) generalization of this isomorphism to the case of higher rank buildings. For the proof of the isomorphism, Teitelbaum first shows surjectivity by attaching to any edge of the Bruhat-Tits tree $\mathcal{BT}_\bullet$ a canonical *source* of $\Gamma(I)$ -stable edges and thereby constructing a harmonic

cocycle given a cycle in $\text{St} \otimes_{\Gamma} V$ cf. [Tei91, Lem.20]. Due to harmonicity, given a harmonic cocycle it can be reconstructed from its values on the $\Gamma(I)$ -stable edges, thus also showing the injectivity.

Now let r be an integer at least 2 and $\Gamma := \Gamma_P(I)$ be as in the previous subsections. Let V be a finite dimensional L -vector space equipped with a left action of Γ , for L a field containing K

We can generalize Teitelbaum's restriction map as follows.

Definition 6.2.1. Define

$$\text{Te}_{\Gamma} : C_{\text{har}}(V)^{\Gamma} \rightarrow C_{r-1}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}} \otimes_{\Gamma} V$$

to be

$$f \mapsto \sum_{C \in \Gamma \backslash \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes f(\vec{C}).$$

Note that it is well defined due to finiteness of $\Gamma \backslash \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}$ (see Lemma 5.3.6) as well as Γ -invariance of f .

Recall that $C_{\bullet}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}}$ is a complex of $\mathbb{Z}[\Gamma]$ -projective modules. In particular, $\text{St}^I(P) \otimes_{\Gamma} V \rightarrow C_{r-1}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}} \otimes_{\Gamma} V$ is injective. Therefore we identify $\text{St}^I(P) \otimes_{\Gamma} V$ with its image in $C_{r-1}(\mathcal{BT}_{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}} \otimes_{\Gamma} V$.

Lemma 6.2.2. Te_{Γ} has image inside $\text{St}^I(P) \otimes_{\Gamma} V$.

Proof. This follows from the definition of harmonicity. It suffices to show that for a Γ -invariant harmonic cocycle f , $\sum_{\vec{C} \in \Gamma \backslash \mathcal{BT}^{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes f(\vec{C})$ vanishes under the differential

$$C_{r-1}(\mathcal{BT}^{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}} \otimes_{\Gamma} V \xrightarrow{\partial_{r-1} \otimes 1} C_{r-2}(\mathcal{BT}^{\bullet}(W), \mathbb{Z})^{\Gamma\text{-st}} \otimes_{\Gamma} V$$

. We have

$$\partial_{r-1} \otimes 1 \left(\sum_{C \in \Gamma \backslash \mathcal{BT}^{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes f(\vec{C}) \right) = \sum_{\substack{\vec{C} = (x_0, \dots, x_{r-1}) \\ \in \Gamma \backslash \mathcal{BT}^{r-1}(W)^{\Gamma\text{-st}}}} \sum_{0 \leq i \leq r-1} (-1)^i [(x_0, \dots, \hat{x}_i, \dots, x_{r-1})]_{st} \otimes f(\vec{C}).$$

Note that if $(x_0, \dots, \hat{x}_i, \dots, x_{r-1})$ is a Γ -stable panel then the convention from 5.3.1(iii) is that

$$(-1)^i [(x_0, \dots, \hat{x}_i, \dots, x_{r-1})] \otimes f(\vec{C}) = (-1)[(x_{i-1}, \hat{x}_i, \dots, x_{i-2})] \otimes f(\vec{C}_{x_{i-1}}),$$

where we denote for $j \in \mathbb{Z}$, $x_j := x_{j'}$ for $j' := j \bmod r$ (in particular, $x_{-1} = x_{r-1}$, $x_{-2} = x_{r-2}$ and so on). With this observation we can rewrite the last sum above as

$$- \sum_{\substack{\vec{C} = (x_0, \dots, x_{r-1}) \\ \in \Gamma \backslash \mathcal{BT}^{r-1}(W)^{\Gamma\text{-st}}}} \sum_{0 \leq i \leq r-1} [(x_{i-1}, \hat{x}_i, \dots, x_{i-1})]_{st} \otimes f(\vec{C}_{x_{i-1}})$$

and exchanging the inner and outer sums gives

$$- \sum_{\substack{e \in \Gamma \setminus \mathcal{BT}_{r-2}(W)^{\Gamma\text{-st}} \\ x \in e}} [\vec{e}_x]_{st} \otimes \left(\sum_{\substack{C \in \Gamma \setminus \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}} \\ C \supset e}} f(\vec{C}_x) \right).$$

Clearly the last expression vanishes by harmonicity (see Definition 3.3.10(1)). $\square$

By abuse of notation, we denote the resulting map $C_{har}(V)^\Gamma \rightarrow \text{St}^I(P) \otimes_\Gamma V$ also by Te_Γ .

The discussions up till in this chapter can be neatly combined into a triangular diagram,

$$\begin{array}{ccc} & C_{har}(V)^\Gamma & \\ \swarrow \text{Te}_\Gamma & & \nwarrow \text{AR}_\Gamma \\ \text{St}^I(P) \otimes_\Gamma V & \xrightarrow[\text{GQ}_\Gamma \otimes 1]{\sim} & \text{St}(W) \otimes_\Gamma V \end{array} \quad (6.2)$$

It is natural to ask whether this is a commutative diagram.

Proposition 6.2.3. The diagram 6.2 is commutative, that is

$$\text{Te}_\Gamma \circ \text{AR}_\Gamma = \text{GQ}_\Gamma^{-1} \otimes 1.$$

Proof. By Theorem 4.1.10, the Ash-Rudolph symbols generate the Steinberg module $\text{St}(W)$. Hence it suffices to show for a (ordered) K -basis $\underline{w} = (w_1, \dots, w_r)$ of W and $v \in V$, that we have

$$\text{Te}_\Gamma \circ \text{AR}_\Gamma([\underline{w}] \otimes v) = \text{GQ}_\Gamma^{-1}([\underline{w}]) \otimes v.$$

The right side was shown to be $\text{StC}_\Gamma(\underline{w}) \otimes v$ in Theorem 6.1.12. So we need to only compute the LHS. Recall the harmonic cocycle $(f_{\underline{w}}^v)_\Gamma$ from 4.3. By definition, $\text{AR}_\Gamma([\underline{w}] \otimes v) = (f_{\underline{w}}^v)_\Gamma$, so that

$$\begin{aligned} \text{Te}_\Gamma((f_{\underline{w}}^v)_\Gamma) &= \sum_{C \in \Gamma \setminus \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes (f_{\underline{w}}^v)_\Gamma(\vec{C}) = \sum_{C \in \Gamma \setminus \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes \left(\sum_{\gamma \in \Gamma} \gamma f_{\underline{w}}^v(\gamma^{-1} \vec{C}) \right) \\ &= \sum_{\gamma \in \Gamma} \sum_{C \in \Gamma \setminus \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} [\gamma^{-1} \vec{C}]_{st} \otimes f_{\underline{w}}^v(\gamma^{-1} \vec{C}) = \sum_{C \in \mathcal{A}_{r-1}(\underline{w}) \cap \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} [\vec{C}]_{st} \otimes f_{\underline{w}}^v(\vec{C}). \end{aligned}$$

In the second line, the first equality follows since the computation happens inside $\text{St}^I(P) \otimes_\Gamma V$ and the second equality is evident from the definition of the harmonic cocycle $f_{\underline{w}}^v$ (cf. 4.1), which is given by $f_{\underline{w}}^v(\vec{C}) = \text{sgn}_{\underline{w}}(\vec{C})v$ for $C \in \mathcal{A}(\underline{w})$ and 0 otherwise. Hence the last expression above equals

$$\sum_{C \in \mathcal{A}(\underline{w}) \cap \mathcal{BT}_{r-1}(W)^{\Gamma\text{-st}}} \text{sgn}_{\underline{w}}(\vec{C}) [\vec{C}]_{st} \otimes v = \text{StC}(\underline{w}) \otimes v.$$

$\square$

Due to GQ_Γ being an isomorphism, we also have that

Corollary 6.2.4. The restriction map $\mathrm{Te}_\Gamma : C_{har}^r(V)^\Gamma \rightarrow \mathrm{St}^I(P) \otimes_\Gamma V$ is surjective and $\mathrm{AR}_\Gamma : \mathrm{St}(W) \otimes_\Gamma V \rightarrow C_{har}(V)^\Gamma$ is injective. $\square$

In the next chapter we will show the above maps to be isomorphisms in the case $A = \mathbb{F}_q[t]$.

Chapter 7

The $\mathbb{F}_q[t]$ -case

In this chapter we restrict ourselves to the situation of the curve $C = \mathbb{P}_{\mathbb{F}_q}^1$ with coordinate t and ∞ being the place corresponding to $(1/t)$. Thus $A = \mathbb{F}_q[t]$ and $K = \mathbb{F}_q(t)$. Let $r \geq 2$ be an integer and $W = K^r$. In this case it is considerably easier to analyse the action of a principal congruence subgroup $\Gamma \leq \mathrm{GL}_r(\mathbb{F}_q[t])$ on the Bruhat-Tits building of rank $r - 1$ and consequently deduce properties of Γ -invariant harmonic cocycles on them. In particular, in this case we will show that the Teitelbaum restriction map Te_Γ defined in the preceding chapter is an isomorphism.

The argument for showing this will involve first obtaining the standard sector $\mathcal{S}_\bullet$ as a fundamental domain for the action of $\mathrm{GL}_r(\mathbb{F}_q[t])$ on $\mathcal{BT}_\bullet := \mathcal{BT}_\bullet(W)$. Then an induction argument based on a certain indexing by non-negative integers for chambers in $\mathcal{S}_\bullet$ establishes the result. For analyzing the growth of these indices associated to chambers as the latter “move towards the boundary”, we define and use the convex cone of a vertex, analogous to its use in [Mü15] in the $r = 3$ case. But first we provide a prototype of this argument from the tree case implicit in [Ser80].

7.1 Motivation: the case of Bruhat-Tits tree

Let $r = 2$ so that $W = K^2$ is a two dimensional K -vector space with respect to a basis $\{e_0, e_1\}$. The building $\mathcal{BT}_\bullet = \mathcal{BT}_\bullet(W)$ is then a tree. In [Ser80, Ch.2, §2.4.1], Serre records the following facts regarding the action of the group $\mathrm{GL}_2(\mathbb{F}_q[t])$ on $\mathcal{BT}_\bullet$:-

- (1) A fundamental domain for the action of $\mathrm{GL}_2(\mathbb{F}_q[t])$ on $\mathcal{BT}_\bullet$ is the standard sector $\mathcal{S}_\bullet(e_0, e_1)$, which in this case is the canonical simplicial complex on the set of vertices $\{\langle e_0, \varpi^n e_1 \rangle_R\}_{n \in \mathbb{N}_{\geq 0}}$ (see the statement of Theorem 7.2.1 below for the definition of a fundamental domain in this context).

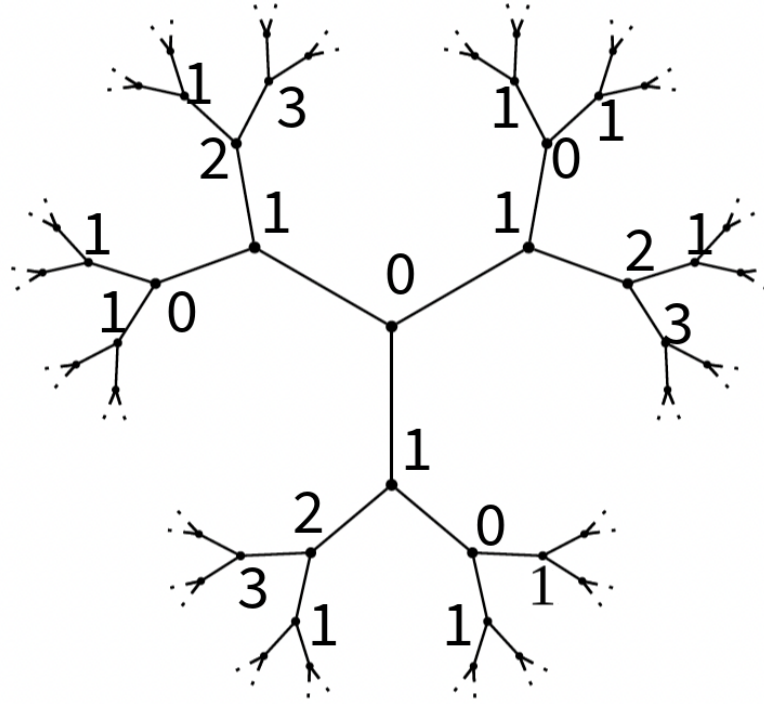


Figure 7.1. The Bruhat-Tits tree for $K = \mathbb{F}_2(t)$, $A = \mathbb{F}_2[t]$.

- (2) Consequently one can define a labeling on vertices of $\mathcal{BT}_\bullet$ by calling x of type n , if x lies in the $\mathrm{GL}_2(\mathbb{F}_q[t])$ orbit of the vertex associated to $\langle e_0, \varpi^n e_1 \rangle_R$. The tree $\mathcal{BT}_\bullet$ then has the following description:- Any type 0 vertex has all its $(q + 1)$ -neighboring vertices of type 1, and for $i \geq 1$ any vertex of type i has q neighboring vertices of type $i - 1$ and one neighboring vertex of type $i + 1$. A pictorial description when $q = 2$ is given in Figure 7.1.

- (3) Moreover any edge of type $(0, 1)$ is $\Gamma(I)$ -stable for $I \neq 0$ a non-zero proper ideal of $\mathbb{F}_q[t]$.

The above description of the Bruhat-Tits tree enables one to give a simple, alternate proof of the injectivity of restriction map $\mathrm{Te}_{\Gamma(I)} : C_{\mathrm{har}}(V)^{\Gamma(I)} \rightarrow \mathrm{St}^s(A^{\oplus 2})_{\Gamma(I)} \otimes V$, for a V a $\mathbb{Z}[\Gamma(I)]$ -module and finite dimensional over a field containing K (in fact, only the former condition on V is necessary for what follows) from last chapter, via an induction argument as follows:- since any edge of type $(0, 1)$ is $\Gamma(I)$ -stable, assume that a harmonic cocycle f vanishes on all edges of type $(0, 1)$. By induction hypothesis, suppose for an integer k , f vanishes on all edges of type $(i, i + 1)$ for $0 \leq i \leq k$. Let $\vec{e} = (v, w)$ be an edge of type $(k + 1, k + 2)$. From (2) above, there exist q edges $\vec{e}_1, \dots, \vec{e}_q$ meeting $\vec{e}$ at v such that each $\vec{e}_i$ is of type $(k, k + 1)$. By induction hypothesis f vanishes on all edges $\vec{e}_i$ and hence also on $\vec{e}$. We can conclude that $f \equiv 0$.

Remark 7.1.1. In [Cak18], the theory of harmonic cocycles in the case $A = \mathbb{F}_q[t]$ and $r = 2$ is

discussed more elaborately, using Serre's description of the quotient tree described above. The reader is referred there for more details.

In the rest of the chapter, we generalize the facts (1), (2) and (3) to higher ranks so that an argument as above can be analogously carried out to conclude that $\mathrm{Te}_{\Gamma(I)}$ is injective, and hence an isomorphism by Corollary 6.2.4, for $r \geq 2$.

7.2 A fundamental domain for $\mathrm{GL}_r(\mathbb{F}_q[t])$

Let $\underline{e} = (e_0, \dots, e_{r-1})$ denote the standard basis of $W = K^r$. Recall from Definition 3.3.1(ii), the standard sector $\mathcal{S}_\bullet(\underline{e})$ with respect to $\underline{e}$. In what follows we abbreviate $\mathcal{S}_\bullet(\underline{e})$ by $\mathcal{S}_\bullet$. The aim of this section is to show the following theorem

Theorem 7.2.1. $\mathcal{S}_\bullet$ is a fundamental domain for the action of $\mathrm{GL}_r(\mathbb{F}_q[t])$ on $\mathcal{BT}_\bullet$, that is

- (i) Any simplex of $\mathcal{BT}_\bullet$ lies in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of a simplex in $\mathcal{S}_\bullet$;
- (ii) The $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of two distinct simplices s, s' in $\mathcal{S}_\bullet$ are disjoint.

The rest of the section will give a proof of this result. The main idea of the proof can be summarized as follows:-

For (i) Using the fact that $\mathcal{BT}_\bullet$ is a chamber complex, it suffices to show that any *chamber* is in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of a chamber in $\mathcal{S}_\bullet$. In fact, it is enough if we can show a similar statement for *pointed chambers*. Showing the latter is especially convenient since pointed chambers of $\mathcal{BT}_\bullet$ admit a description as the quotient $\mathrm{GL}_r(K_\infty)/K_\infty^\times \cdot \mathcal{IW}_r$ (see 3.1.12), where the completion of K at ∞ in the current setting is given by $K_\infty = \mathbb{F}_q((1/t))$ and $\mathcal{IW}_r \subset \mathrm{GL}_r(K_\infty)$ is the Iwahori subgroup. Hence to conclude (i), we will show that each class of double coset $\mathrm{GL}_r(\mathbb{F}_q[t]) \backslash \mathrm{GL}_r(K_\infty)/K_\infty^\times \cdot \mathcal{IW}_r$ has a representative which corresponds to a pointed chamber in $\mathcal{S}_\bullet$.

For (ii) We will show that no two distinct *vertices* of the standard sector are $\mathrm{GL}_r(\mathbb{F}_q[t])$ translates of each other.

Proof of 7.2.1(i)

Note that the ring of integers $\mathcal{O}_\infty \subset K_\infty$ is given by $\mathbb{F}_q[[1/t]]$ and we fix the choice of an uniformizer to be $\varpi := 1/t$. For the rest of this chapter, we will identify $\mathcal{BT}_\bullet(W)$ with $\mathcal{BT}_\bullet(W \otimes_K K_\infty)$ under the simplicial isomorphism described in Remark 3.1.9, and work with the

latter description of Bruhat-Tits buildings as classes of O_∞ lattices. We denote it also by $\mathcal{BT}_\bullet$ and similarly $\mathcal{S}_\bullet$ will mean the standard sector corresponding to the basis $\{e_0, \dots, e_{r-1}\}$ of $W \otimes_K K_\infty$. Hence given integers $m_0, \dots, m_{r-1} \in \mathbb{Z}$, we will denote by $[m_0, \dots, m_{r-1}]$ the class of O_∞ lattice $\langle \varpi^{m_0} e_0, \dots, \varpi^{m_{r-1}} e_{r-1} \rangle_{O_\infty}$. We will denote by v_∞ the additive valuation on K_∞ with respect to ϖ , given by $v_\infty : K_\infty \rightarrow \mathbb{Z} \cup \{\infty\}$ sending a non-zero element $\sum_{n \geq n_0} a_n (1/t)^n$, with $a_{n_0} \neq 0$, to n_0 and 0 to ∞ .

Definition 7.2.2. (i) Denote by PC_r the subset of monomial matrices $(a_{ij})_{0 \leq i, j \leq r-1}$ in $\mathrm{GL}_r(K_\infty)$ equipped with a permutation $\sigma \in \mathcal{S}_r$ and an r -tuple of integers $(n_0, \dots, n_{r-1})$ satisfying the following conditions

1. $a_{i\sigma(i)} = \varpi^{n_i}$, and for all $0 \leq i \leq r-1$ and $j \neq \sigma(i)$, we have $a_{ij} = 0$;
2. $n_0 \leq \dots \leq n_{r-1}$;
3. $i < i'$ and $n_i = n_{i'} \Rightarrow \sigma(i) < \sigma(i')$.

(ii) Let $g \in \mathrm{GL}_r(K_\infty)$. Then we denote by $R_i(g)$ the i -th row of g and by $C_j(g)$ the j -th column of g , for $0 \leq i, j \leq r-1$.

(iii) The matrix representation of g in $\mathrm{GL}_r(K_\infty)$ will be denoted by $(a_{ij}(g))$. We define the minimal valuation of the i -th row as

$$v_i(g) := \min_{0 \leq j \leq r-1} (v_\infty(a_{ij}(g))).$$

We also introduce the notation, for each $0 \leq i \leq r-1$,

$$j_i(g) := \min\{0 \leq j \leq r-1 \mid v_\infty(a_{ij}(g)) = v_i(g)\}.$$

Remark 7.2.3. (i) Note that every element of PC_r is determined by a pair $(\sigma, \underline{n})$, where $\underline{n} = (n_0, \dots, n_{r-1})$ is a tuple of r integers satisfying the conditions 7.2.2(i)(1,2,3) described above, and vice versa.

(ii) PC is meant to be an abbreviation of “pointed chambers” and motivated from the next proposition.

We will denote by $\widehat{\mathcal{S}}_{r-1}$ the pointed chambers in $\widehat{\mathcal{BT}}_{r-1}$ contained in the standard sector $\mathcal{S}_\bullet$. We will denote by $PC_r/\varpi^\mathbb{Z}$ the image of PC_r in $\mathrm{GL}_r(K_\infty)/K_\infty^\times$.

Proposition 7.2.4. The bijection in Proposition 3.1.12(ii) induces a 1-1 correspondence between $\widehat{\mathcal{S}}_{r-1}$ and $PC_r/\varpi^\mathbb{Z}$.

Proof. Recall that the bijection in 3.1.12(ii) is defined by sending the class $[g]$ of an element $g \in \mathrm{GL}_r(K_\infty)$ to the pointed chamber (gx_0, gC_0) where x_0 is the standard vertex $[0, \dots, 0]$ and C_0 is the standard chamber given by $\{[0, 0, \dots, 0], [0, 0, \dots, 1], \dots, [0, 1, \dots, 1]\}$. Suppose first that g be associated with a permutation $\sigma \in \mathcal{S}_r$ and a tuple of integers $(n_0, \dots, n_{r-1})$, as in the Definition 7.2.2. Note that

$$g[0, \dots, 0] = \text{class of } \langle \varpi^{n_{\sigma^{-1}(0)}} e_{\sigma^{-1}(0)}, \dots, \varpi^{n_{\sigma^{-1}(r-1)}} e_{\sigma^{-1}(r-1)} \rangle_{O_\infty}$$

$$g[0, \dots, 1] = \text{class of } \langle \varpi^{n_{\sigma^{-1}(0)}} e_{\sigma^{-1}(0)}, \dots, \varpi^{n_{\sigma^{-1}(r-1)}+1} e_{\sigma^{-1}(r-1)} \rangle_{O_\infty}$$

$$\vdots$$

$$g[0, 1, \dots, 1] = \text{class of } \langle \varpi^{n_{\sigma^{-1}(0)}} e_{\sigma^{-1}(0)}, \varpi^{n_{\sigma^{-1}(1)}+1} e_{\sigma^{-1}(1)}, \dots, \varpi^{n_{\sigma^{-1}(r-1)}+1} e_{\sigma^{-1}(r-1)} \rangle_{O_\infty}.$$

In particular, $g[0, \dots, 0] = [n_0, \dots, n_{r-1}]$, so that by 7.2.2(i)(2) it lies in $\mathcal{S}_0$. By 7.2.2(i)(1,3) it is not hard to check that gx for $x \in C \setminus \{x_0\}$, also lies in $\mathcal{S}_0$, hence the pointed chamber (gx_0, gC_0) lies in $\widehat{\mathcal{S}}_{r-1}$.

Any g as above as well as λg for $\lambda \in K_\infty^\times$, induce the same pointed chamber. In particular this induces a map of sets $\phi : PC_r / \varpi^\mathbb{Z} \rightarrow \widehat{\mathcal{S}}_{r-1}$.

To show that this map is a bijection, we can provide an inverse as follows. Let a pointed chamber $\vec{C}_x \in \widehat{\mathcal{S}}_{r-1}$ be given and suppose $x = [n_0, \dots, n_{r-1}]$ so that we have $n_0 \leq \dots \leq n_{r-1}$. Using Remark 3.3.6(ii) and the fact that $\tau_{\underline{e}}^{sec}(C) = 1$ (see Definition 3.3.4), we obtain 7.2.2(3) if we set $\sigma := \sigma_{\underline{e}}^{x-nbd}(C)^{-1}$. More precisely, recalling the definition of $\sigma_{\underline{e}}^{x-nbd}(C)$ from 3.3.5, σ is the inverse of the permutation $(0, \dots, r-1) \mapsto (\mathbf{m}_0, \dots, \mathbf{m}_{r-1})$ where the $\mathbf{m}_i$'s are given by:- write C as descending chain of lattices $L_0 \supsetneq L_1 \supsetneq \dots \supsetneq L_{r-1} \supsetneq \varpi L_0$ and let $\mathbf{m}_j$ be the unique integer in $\{0, \dots, r-1\}$ such that $\varpi(L_{r-j-1} \cap Ke_{\mathbf{m}_j}) = L_{r-j} \cap Ke_{\mathbf{m}_j}$ and $L_{r-j-1} \cap Ke_i = L_{r-j} \cap Ke_i$, $\forall i \neq \mathbf{m}_j$, where we set $L_r := \varpi L_0$.

This defines a map $\psi : \widehat{\mathcal{S}}_{r-1} \rightarrow PC_r / \varpi^\mathbb{Z}$ given by sending $\vec{C}_x$ to the element in $PC_r / \varpi^\mathbb{Z}$ associated with the pair $(\sigma, (n_0, \dots, n_{r-1}))$.

The two maps ϕ, ψ are inverses to each other. We will show only $\psi \circ \phi = 1$, the other identity $\phi \circ \psi = 1$ is proven using a similar argument. Let $g \in PC_r$ be given, associated to a pair $(\sigma, \underline{n})$. From the description of the vertices $g[0, \dots, 0], g[0, \dots, 1], \dots, g[0, 1, \dots, 1]$ given above, it is not hard to see that, setting $\vec{C}_x := \phi(g)$, one has $\sigma_{\underline{e}}^{x-nbd}(C) = \sigma^{-1}$. In particular, we obtain $\psi \circ \phi = 1$. $\square$

As a consequence, for the proof of (i), it suffices to show that the class of any $g \in \mathrm{GL}_r(K_\infty)$ in the double coset $\mathrm{GL}_r(\mathbb{F}_q[t]) \backslash \mathrm{GL}_r(K_\infty) / \mathcal{W}_r$ can be represented by a matrix in PC_r . This will be achieved by an induction argument on the rows of g . First we record few auxilliary definitions and observations that will be used in the induction.

Definition 7.2.5. (i) Define the following matrices in the Iwahori subgroup $\mathcal{IW}_r \subset \mathrm{GL}_r(K_\infty)$:-

$$u_{ij}(a) = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & a & \\ & & & & \ddots \\ & & & & & 1 \end{bmatrix}, \quad 0 \leq i < j \leq r-1, \quad a \in \mathcal{O}_\infty,$$

(i, j)-th entry

$$l_{ij}(a) = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & a & & \\ & & & 1 & \\ & & & & \ddots \\ (i, j)\text{-th entry} & & & & & 1 \end{bmatrix}, \quad 0 \leq j < i \leq r-1, \quad a \in \varpi\mathcal{O}_\infty$$

(i, j)-th entry

$$d_i(a) = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & a & & \\ & & & \ddots & \\ (i, i)\text{-th entry} & & & & & 1 \end{bmatrix}, \quad 0 \leq i \leq r-1, \quad a \in \mathcal{O}_\infty^\times.$$

(i, i)-th entry

(ii) Similarly define $u_{ij}(a)$, $l_{ij}(a)$ and $d_i(b)$ for $a \in \mathbb{F}_q[t]$ and $b \in \mathbb{F}_q^\times$.

(iii) We will identify symmetric group with its image under the canonical embedding $s : \mathcal{S}_r \hookrightarrow \mathrm{GL}_r(\mathbb{F}_q[t])$ sending any $\sigma \in \mathcal{S}_r$ to the permutation matrix $s(\sigma) := (a_{ij})_{0 \leq i, j \leq r-1}$ given by $a_{i, \sigma(i)} = 1$ for all i , $0 \leq i \leq r-1$ and 0 otherwise. In particular, *left multiplication by* σ on a matrix $g \in \mathrm{GL}_r(K_\infty)$ will mean multiplying g on the left by $s(\sigma)$.

Remark 7.2.6. Let $g \in \mathrm{GL}_r(K_\infty)$.

(i) For $a \in \mathcal{O}_\infty$ and $0 \leq i < j \leq r-1$, the right multiplication of g by $u_{ij}(a)$ is equivalent to transforming $g = (\dots, C_i(g), \dots, C_j(g), \dots)$ to $(\dots, C_i(g), \dots, a.C_i(g) + C_j(g), \dots)$. Similarly right multiplication of g by $d_i(a)$ resp. $l_{ij}(b)$ for $a \in \mathcal{O}_\infty^\times$ resp. $b \in \varpi\mathcal{O}_\infty$ is equivalent to the transformation $(\dots, a.C_i(g), \dots)$ resp. $(\dots, C_i(g) + b.C_j(g), \dots, C_j(g), \dots)$.

- (ii) Analogously we can define left multiplication by elements of $\mathrm{GL}_r(\mathbb{F}_q[t])$ as follows. For $a \in \mathbb{F}_q[t]$ and $0 \leq i < j \leq r-1$, the left multiplication of g by $u_{ij}(a)$ is equivalent to transforming $g = \begin{pmatrix} R_i(g) \\ \vdots \\ R_j(g) \end{pmatrix}$ to $\begin{pmatrix} R_i(g)+a \cdot R_j(g) \\ \vdots \\ R_j(g) \end{pmatrix}$. Similarly right multiplication of g by $d_i(a)$ resp. $l_{ij}(b)$ for $a \in \mathbb{F}_q^\times$ resp. $b \in \varpi \mathbb{F}_q[t]$ is the transformation $\begin{pmatrix} \vdots \\ a \cdot R_i(g) \\ \vdots \end{pmatrix}$ resp. $\begin{pmatrix} R_i(g) \\ \vdots \\ a \cdot R_i(g) + R_j(g) \end{pmatrix}$.
- (iii) Given $\sigma \in \mathcal{S}_r$, left multiplication of $g = \begin{pmatrix} R_0(g) \\ \vdots \\ R_{r-1}(g) \end{pmatrix}$ by σ amounts to a permutation of the rows by σ , giving $\begin{pmatrix} R_{\sigma(0)} \\ \vdots \\ R_{\sigma(r-1)} \end{pmatrix}$.

Lemma 7.2.7. Let $g \in \mathrm{GL}_r(K_\infty)$. Then as g' ranges over representatives of the class $[g]$ in the double coset $\mathrm{GL}_r(\mathbb{F}_q[t]) \backslash \mathrm{GL}_r(K_\infty) / \mathcal{IW}_r$, $\sum_{0 \leq i \leq r-1} v_i(g')$ is bounded above.

That is, there exists a constant $C \in \mathbb{R}$ depending only on g such that, for all g' obtained from g by right multiplication by $\mathcal{IW}_r$ resp. left multiplication by $\mathrm{GL}_r(\mathbb{F}_q[t])$, one has $\sum_{0 \leq i \leq r-1} v_i(g') \leq C$.

Proof. Suppose $g' \in \mathrm{GL}_r(K_\infty)$ is a representative of the class $[g]$, then $v_\infty(\det(g)) = v_\infty(\det(g'))$ since the determinant of matrices in $\mathcal{IW}_r$ resp. $\mathrm{GL}_r(\mathbb{F}_q[t])$ lies in $\mathcal{O}_\infty^\times$ resp. $\mathbb{F}_q^\times$. In particular, $v_\infty(\det(g'))$ is constant, say $C_g \in \mathbb{Z}$, for all representatives g' of $[g]$. We claim that C_g is the required upper bound. Let $g' = (a'_{ij})_{0 \leq i, j \leq r-1}$, so that $\det(g') = \sum_{\sigma \in \mathcal{S}_r} \mathrm{sgn}(\sigma) \prod_{i=0}^{r-1} a'_{i\sigma(i)}$. By definition,

$$\sum_{0 \leq i \leq r-1} v_i(g') \leq v_\infty\left(\prod_{i=0}^{r-1} a'_{i\sigma(i)}\right) = C_g. \text{ Together with the property of } v_\infty, \text{ this establishes the claim.}$$

□

The following proposition concludes the proof of (i)

Proposition 7.2.8. The injection of sets $PC_r \hookrightarrow \mathrm{GL}_r(K_\infty)$ induces a surjection

$$PC_r \twoheadrightarrow \mathrm{GL}_r(\mathbb{F}_q[t]) \backslash \mathrm{GL}_r(K_\infty) / \mathcal{IW}_r.$$

Proof. We need to show that given an element in $\mathrm{GL}_r(K_\infty)$ it can be transformed to an element in PC_r , via suitable multiplication on the right by elements of $\mathcal{IW}_r$ and on the left by elements of $\mathrm{GL}_r(\mathbb{F}_q[t])$. We will achieve this by a descending induction argument on the rows of g , wherein each row will be successively transformed to the form $(0, \dots, \varpi^n, \dots, 0)$ for integers $n \in \mathbb{Z}$. One may also regard the argument below as an algorithm to transform g into a monomial matrix lying in PC_r .

We begin by defining a set of column resp. row operations on matrices of $\text{GL}_r(K_\infty)$ that correspond to right multiplication by $\mathcal{IW}_r$ resp. left multiplication by $\text{GL}_r(\mathbb{F}_q[t])$. Let i, l, k be integers between 0 and $r - 1$ such that $k < l$. Now let g be an element in $\text{GL}_r(K_\infty)$.

C_i. $\mathbf{C}_i(g)$ is the matrix obtained from g so that it has the i -th row of the form $(0, \dots, \varpi^{v_i(g)}, \dots, 0)$ (where the non-zero entry lies in the $j_i(g)$ -th column) obtained via column operations corresponding to right multiplication by appropriate matrices $u_{i'j_i(g)}(a)$, $d_i(b)$ and $l_{j_i(g)i''}(c)$ for $0 \leq i'' < j_i(g) < i' \leq r - 1$ and $a \in \mathcal{O}_\infty$, $b \in \mathcal{O}_\infty^\times$ and $c \in \varpi\mathcal{O}_\infty$.

R_{k,l}. Consider the minimum number of row operations on g corresponding to $u_{kl}(a)$ for $a \in \mathbb{F}_q[t]$, so that the resulting matrix, denoted by $\mathbf{R}_{k,l}(g) = (a_{te})_{0 \leq t, e \leq r-1}$, satisfy $v_\infty(a_{kj_l}(g)) > v_\infty(a_{lj_l}(g))$.

Remark 7.2.9. The following are immediate to observe:-

- (i) The tuples of integers $(v_e(g))_{e=0}^{r-1}$ as well as $(j_e(g))_{e=0}^{r-1}$ are invariant under the operation $\mathbf{C}_i$.
- (ii) Analogously, the tuples $(v_t(g))_{t \neq k}$ as well as $(j_t(g))_{t \neq k}$ are invariant under the operation $\mathbf{R}_{k,l}$.

Moreover, we have the inequality $v_k(g) \leq v_k(\mathbf{R}_{kl}(g))$, and hence in particular $\sum_{t=0}^{r-1} v_t(g) \leq$

$$\sum_{t=0}^{r-1} v_t(\mathbf{R}_{kl}(g)).$$

The induction hypothesis $I(k)$, for k between 0 and $r - 1$, is that there is a representative of the class of g in $\text{GL}_r(\mathbb{F}_q[t]) \setminus \text{GL}_r(K_\infty) / \mathcal{IW}_r$ such that it is a matrix of the form

$$\begin{bmatrix} a_{00} & \dots & a_{0(r-1)} \\ \vdots & \ddots & \vdots \\ a_{(k-1)0} & \dots & a_{(k-1)(r-1)} \\ 0 & \varpi^{n_k} \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots \varpi^{n_{r-1}} & 0 \end{bmatrix} \quad (7.1)$$

where the row $R_i(g)$ is of the form $(0, \dots, \varpi^{n_i}, \dots, 0)$ with the non-zero entry in the j_i -th column, for $k \leq i \leq r - 1$, and $j_d \neq j_l$ for all $k \leq d \neq l \leq r - 1$. Moreover we would also like to include the following conditions on the integers n_i 's as part of the hypothesis:-

$$(*_k) \quad n_k \leq n_{k+1} \leq \dots \leq n_{r-1};$$

$$(**_k) \quad \text{for } k \leq i < i' \leq r - 1, \text{ if } n_i = n_{i'}, \text{ then } j_i < j_{i'}.$$

In particular, if $I(0)$ holds true then we have transformed g into an element in PC_r associated to the pair $(\underline{n}, \sigma)$ for $\underline{n} := (n_0, \dots, n_{r-1})$ and σ given by the permutation $(0, \dots, r-1) \mapsto (j_0, \dots, j_{r-1})$.

$I(r-1)$ holds. This is achieved by an application of the operation $\mathbf{C}_{r-1}$ on g , the form of the resulting matrix being

$$\begin{bmatrix} a_{11}(g) & \dots & a_{1(r-1)}(g) \\ \vdots & \ddots & \vdots \\ a_{(r-2)1}(g) & \dots & a_{(r-2)(r-1)}(g) \\ 0 \dots & \varpi^{n_{r-1}} & \dots 0 \end{bmatrix}$$

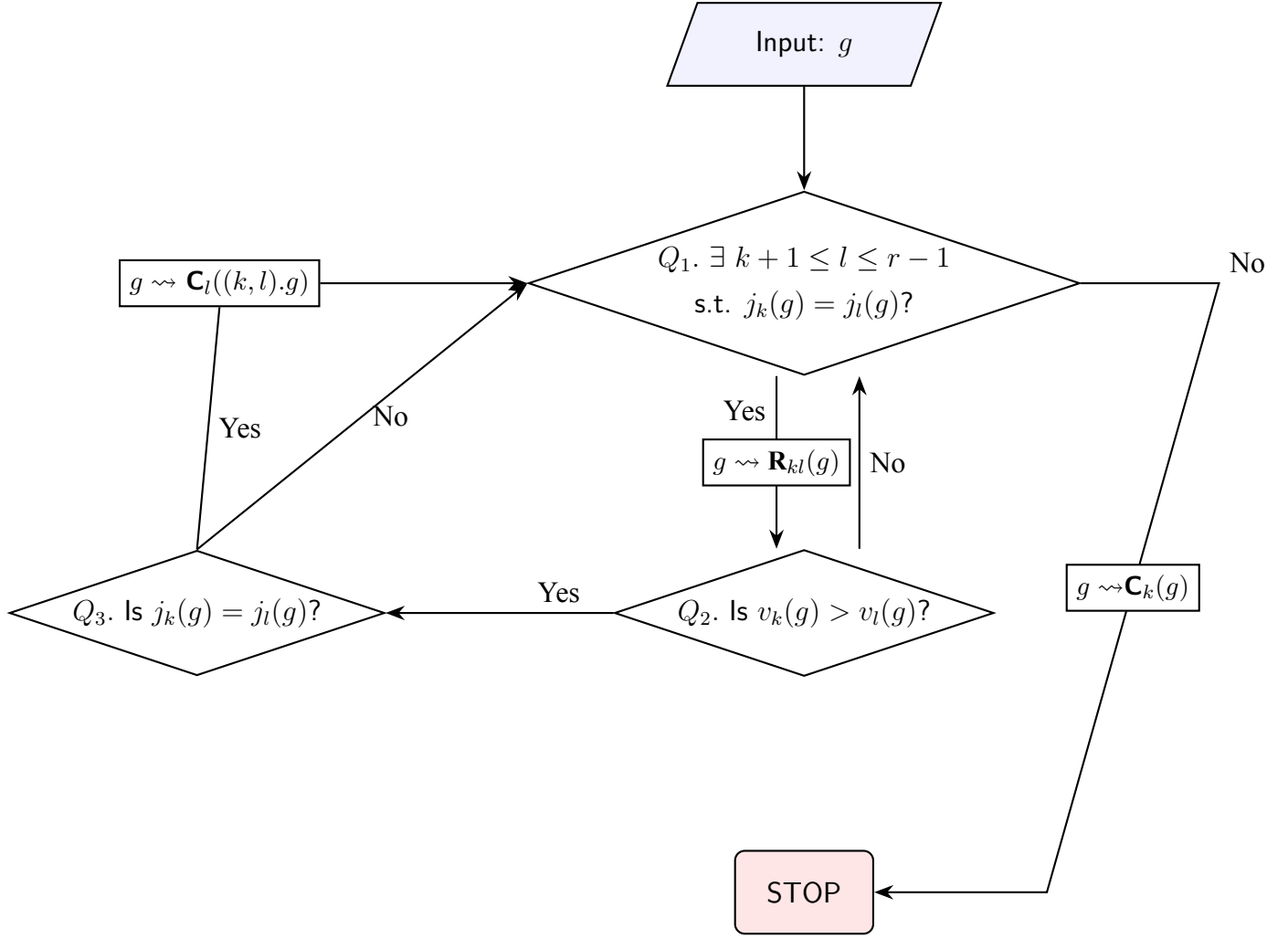
$I(k+1)$ holds $\Rightarrow I(k)$ holds By induction hypothesis we may suppose that g is a matrix of the form

$$\begin{bmatrix} a_{00} & \dots & a_{0(r-1)} \\ \vdots & \ddots & \vdots \\ a_{k0} & \dots & a_{k(r-1)} \\ 0 & \varpi^{n_{k+1}} \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots \varpi^{n_{r-1}} & 0 \end{bmatrix}$$

where the row $R_i(g)$ is of the form $(0, \dots, \varpi^{n_i}, \dots, 0)$ with the non-zero entry in the j_i -th column, for $k+1 \leq i \leq r-1$, and $j_d \neq j_l$ for all $k+1 \leq d \neq l \leq r-1$. Moreover it satisfies $*_{k+1}, **_{k+1}$.

The aim of this step is to transform this matrix via suitable column and row operations corresponding to $\mathbf{C}_l$ and $\mathbf{R}_{kl}$, for $k+1 \leq l \leq r-1$, to a matrix described in 7.1 satisfying $*_k, **_k$.

Consider the following algorithm with initial input being g fed into Q_1 .



Note that in the above diagram $(k, l).g$ means the operation of the transposition (k, l) on g , that is the operation swapping rows k and l .

By an *iteration* of the algorithm, we mean the passing of the algorithm between the decision boxes Q_i 's via a single “Yes” or “No” arrow. Denote by $g^{(n)}$ the n -th iterate of the input $g^{(0)} := g$. We note below some easy observations

- (i) If the $(n + 1)$ -st and the $(n + 2)$ -nd iteration are a loop of the form $Q_1 \xrightarrow{\text{Yes}} Q_2 \xrightarrow{\text{No}} Q_1$ then $j_k(g^{(n)}) < j_k(g^{(n+2)})$. In particular such loops of iterations cannot occur infinitely often.
- (ii) The *structure* of the lower $(r - k - 1) \times (r - 1)$ submatrix of each iterate $g^{(n)}$ remains unchanged, that is, the sequence of tuples $(j_i(g^{(n)}))_{i=k+1}^{r-1}$ remains invariant and each row $R_i(g^{(n)})$ has a single non-zero entry which is a power of ϖ , for all $k + 1 \leq i \leq r - 1$ and $n \in \mathbb{N}$.

(iii) The sum $\sum_{i=0}^{r-1} v_i(g^{(n)})$ is a non-decreasing sequence for $n \in \mathbb{N}$. This is obvious from Remark 7.2.9.

(iv) If the algorithm reaches **STOP** after finitely many iterations, then it is clear that acting by a suitable permutation on the left makes the resulting matrix satisfy conditions $*_k, **_k$ and hence $I(k)$ will hold true.

By (iv) above, the ongoing induction step is complete, if we could show that the algorithm indeed stops. Note that if the iteration $Q_2 \xrightarrow{\text{Yes}} Q_3$ occurs infinitely often then $\sum_{i=0}^{r-1} v_i(g^{(n)})$ becomes unbounded as $n \rightarrow \infty$, contradicting Lemma 7.2.7. Consequently, the iterations after a stage N , for some $N \in \mathbb{N}$, consists of loops of the form $Q_1 \xrightarrow{\text{Yes}} Q_2 \xrightarrow{\text{No}} Q_1$. But we already saw in the observation (i) above that this cannot be happen infinitely often. Hence the algorithm stops after a finite number of iterations.

This completes the induction step.

This concludes the proof of the ongoing proposition as well as that of Theorem 7.2.1(i). $\square$

Proof of 7.2.1(ii)

Recall from the discussion below Theorem 7.2.1 that it suffices to show that no two distinct vertices of the standard sector $\mathcal{S}_\bullet$ are $\text{GL}_r(\mathbb{F}_q[t])$ translates of each other. Let $v_1 := [m_0, \dots, m_{r-1}]$ resp. $v_2 := [n_0, \dots, n_{r-1}]$ be two vertices in the class of $\langle \varpi^{m_0} e_0, \dots, \varpi^{m_{r-1}} e_{r-1} \rangle_{\mathcal{O}_\infty}$ resp. $\langle \varpi^{n_0} e_0, \dots, \varpi^{n_{r-1}} e_{r-1} \rangle_{\mathcal{O}_\infty}$ in $\mathcal{S}_0$, which is equivalent to saying that $m_0 \leq \dots \leq m_{r-1}, n_0 \leq \dots \leq n_{r-1}$. Suppose there exists $g \in \text{GL}_r(\mathbb{F}_q[t])$ such that $gv_1 = v_2$. We need to show $v_1 = v_2$.

Write $g = (a_{ij})_{0 \leq i, j \leq r-1}$. Then by virtue of the relation

$$g \langle \varpi^{m_0} e_0, \dots, \varpi^{m_{r-1}} e_{r-1} \rangle_{\mathcal{O}_\infty} = c \cdot \langle \varpi^{n_0} e_0, \dots, \varpi^{n_{r-1}} e_{r-1} \rangle_{\mathcal{O}_\infty},$$

for some $c \in K_\infty^\times$, there exists $a'_{il} \in \mathcal{O}_\infty$, for all $0 \leq i, l \leq r-1$, such that it can be rewritten as

$$\varpi^{m_i} \sum_{j=0}^{r-1} a_{ij} e_j = \sum_{l=0}^{r-1} \varpi^{\tilde{n}_l} a'_{il} e_l, \quad \forall 0 \leq i \leq r-1,$$

where $(\tilde{n}_0, \dots, \tilde{n}_{r-1})$ is a tuple lying in $(n_0, \dots, n_{r-1}) + \mathbb{Z}(1, \dots, 1)$ arising from c . In particular, since $\{e_i\}$ is a basis, we obtain

$$\varpi^{m_i} a_{ij} = \varpi^{\tilde{n}_j} a'_{ij}, \quad 0 \leq i, j \leq r-1.$$

Using the fact that $\det(g) \neq 0$, we can get hold of a $\sigma \in \mathcal{S}_r$ so that the product $\prod_{i=0}^{r-1} a_{i\sigma(i)}$ and hence also $\prod_{i=0}^{r-1} a'_{i\sigma(i)}$ are not equal to 0. Since the elements of $\mathbb{F}_q[t]$ resp. $\mathcal{O}_\infty$ have v_∞ valuation ≤ 0 resp. ≥ 0 , applying v_∞ to the relation above, we obtain a set of inequalities

$$m_i \geq \tilde{n}_{\sigma(i)}, \quad 0 \leq i \leq r-1.$$

Claim:- $m_i \geq \tilde{n}_i$ for all $0 \leq i \leq r-1$.

Proof. We use the fact that the sequence of m_i 's and $\tilde{n}_i$'s are non-decreasing. If $\sigma^{-1}(i) \leq i$, then we're done since $m_i \geq m_{\sigma^{-1}(i)} \geq \tilde{n}_i$. Otherwise $\sigma^{-1}(i) \geq i+1$, in which case it is not hard to see that there is an $j \in \{i+1, \dots, r-1\}$ such that $\sigma^{-1}(j) \leq i$. Then again we're done, as we have $m_i \geq m_{\sigma^{-1}(j)} \geq \tilde{n}_j \geq \tilde{n}_i$. $\square$

Note that since $g^{-1} \in \mathrm{GL}_r(\mathbb{F}_q[t])$, by symmetry of the argument above we also obtain the string of inequalities $m_i \leq \tilde{n}_i$ for all $0 \leq i \leq r-1$. Hence $v_1 = v_2$, establishing also part (ii) of Theorem 7.2.1. $\square$

7.3 Numbering the chambers and $\Gamma(I)$ invariant harmonic cocycles

Now that we have obtained the standard sector $\mathcal{S}_\bullet$ as a fundamental domain for the action of $\mathrm{GL}_r(\mathbb{F}_q[t])$ on the Bruhat-Tits building $\mathcal{BT}_\bullet$, we can carry forward the analogy from the $r = 2$ case to higher ranks, as mentioned in §7.1. In this section we generalize the two ideas described in (2), (3) of 7.1 for $r \geq 2$, assigning a non-negative integer to each chamber of $\mathcal{S}_{r-1}$, and consequently also for all chambers C of $\mathcal{BT}_{r-1}$ denoted by N_C . Our main result in this context will be that any chamber C not in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of the standard chamber $C_0 := \{[0, \dots, 0], [0, \dots, 1], \dots, [0, 1, \dots, 1]\}$ contains a unique panel e such that the q chambers $\{C_i\}_{i=1}^q$ distinct from C and containing e satisfy $N_{C_i} = N_{C_j} < N_C$ for all $1 \leq i, j \leq q$. This result is sufficient to show that the restriction map $\mathrm{Te}_{\Gamma(I)}$ is injective once we also show that the standard chamber C_0 is $\Gamma(I)$ -stable.

For the sake of the main result, we will require a rudimentary analysis of the $\mathrm{GL}_r(\mathbb{F}_q[t])$ stabilizers of the chambers of $\mathcal{S}_\bullet$, as they move towards the “boundary”. For this purpose it is helpful to define the convex cone based at a vertex of $\mathcal{S}_\bullet$, analogous to that given in the $r = 3$ case in [Mü15, Defn.1.16]. Our aim in introducing this is to prove an analogue of [Mü15, Lemma 1.35(i), 1.36(i)].

Also taking cue from the numbering of vertices provided by Serre in the example in [Ser80, Ch.2, §2.4.1], we have the following

Definition 7.3.1. (i) Let $v = [n_0, \dots, n_{r-1}]$ be a vertex in $\mathcal{S}_\bullet$. Define the *convex cone at v* to be the canonical simplicial subcomplex of $\mathcal{S}_\bullet$ based on the set of vertices

$$\{[n'_0, \dots, n'_{r-1}] \in \mathcal{S}_0 \mid n'_i - n_i \leq n'_j - n_j \ \forall \ 0 \leq i < j \leq r-1\}.$$

This can also be described as the translate of the standard sector to the vertex v . We denote the resulting simplicial complex by $\mathcal{S}_\bullet^\geq(v)$. Also associate with v the non-negative integer $N_v := \sum_{0 \leq i < j \leq r-1} (n_j - n_i)$. For a simplex $s \in \mathcal{S}_\bullet$, we define

$$\mathcal{S}_\bullet^\geq(s) := \bigcup_{v \in s} \mathcal{S}_\bullet^\geq(v).$$

(ii) Let $\tilde{s}$ be a simplex of $\mathcal{BT}_\bullet$ and s be the unique simplex in $\mathcal{S}_\bullet$ so that $\tilde{s}$ lies in its $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit (cf. 7.2.1). Then we set $\mathcal{S}_\bullet^\geq(\tilde{s}) := \mathcal{S}_\bullet^\geq(s)$.

Let $n \in \mathbb{N}$ be an integer. If $n \geq 0$, then denote by $\{n\}$ the set of polynomials in $A = \mathbb{F}_q[t]$ of degree at most n . In particular, $\{0\} = \mathbb{F}_q$. For $n < 0$, set $\{n\}$ to be 0. Also let $B_r(\mathbb{F}_q[t])$ denote the *Borel subgroup* of $\mathrm{GL}_r(\mathbb{F}_q[t])$, that is the subgroup of upper triangular matrices in $\mathrm{GL}_r(\mathbb{F}_q[t])$. For a vertex $v \in \mathcal{BT}_0$, denote by $B_r(\mathbb{F}_q[t])_v$ the subgroup $\mathrm{GL}_r(\mathbb{F}_q[t])_v \cap B_r(\mathbb{F}_q[t])$.

Lemma 7.3.2. (cf. [Geb96, Kap.3])

(i) Let $v = [n_0, \dots, n_{r-1}] \in \mathcal{S}_0$ be a vertex. Then the stabilizer subgroup $\mathrm{GL}_r(\mathbb{F}_q[t])_v$ is given by elements γ in

$$\begin{bmatrix} \mathbb{F}_q & \{n_1 - n_0\} \dots & \{n_{r-1} - n_0\} \\ \{n_0 - n_1\} & \mathbb{F}_q \dots & \{n_{r-1} - n_1\} \\ \vdots & \ddots & \vdots \\ \{n_0 - n_{r-1}\} & \dots & \mathbb{F}_q \end{bmatrix},$$

satisfying $\det(\gamma) \in \mathbb{F}_q[t]^* = \mathbb{F}_q^*$. The above set can be explicitly described as

$$\{(a_{ij})_{0 \leq i, j \leq r-1} \in \mathrm{GL}_r(\mathbb{F}_q[t]) \mid \begin{array}{l} \text{(i)} a_{ij} \in \{n_j - n_i\} ; \forall i < j \\ \text{(ii)} a_{ii} \in \mathbb{F}_q^* ; \forall i \\ \text{(iii)} a_{ij} \in \{n_j - n_i\} ; \forall i > j \end{array} \}.$$

(ii) The stabilizer of $[0, \dots, 0]$ in $\mathrm{GL}_r(\mathbb{F}_q[t])$ is $\mathrm{GL}_r(\mathbb{F}_q)$.

(iii) Let v' be another vertex of $\mathcal{S}_\bullet$ such that $v \in \mathcal{S}_\bullet^\geq(v')$. Then $B_r(\mathbb{F}_q[t])_{v'} \subset B_r(\mathbb{F}_q[t])_v$.

Proof. (i) Note that $\mathrm{GL}_r(\mathbb{F}_q[t])_v = \mathrm{GL}_r(\mathcal{O}_\infty)_v \cap \mathrm{GL}_r(\mathbb{F}_q[t])$. It is not hard to see that $\mathrm{GL}_r(\mathcal{O}_\infty)_v$ is the intersection of the set $\{(a_{ij})_{0 \leq i, j \leq r-1} \in \mathbf{M}_r(\mathcal{O}_\infty) \mid a_{ij} \in \varpi^{n_i - n_j} \mathcal{O}_\infty \forall i, j\}$ with $\mathrm{GL}_r(\mathcal{O}_\infty)$. Since $\varpi = \frac{1}{t}$, the description of the $\mathrm{GL}_r(\mathbb{F}_q[t])$ stabilizer follows.

(ii) This is obvious from (i)

(iii) This is also clear from (i) and the definition of $\mathcal{S}_\bullet^\geq(-)$.

□

We will apply this lemma in the context of the following. Recall that $\underline{e}$ is the standard basis and we will use the notation e for a panel in $\mathcal{BT}_\bullet$, and hope this does not cause any confusion.

Proposition 7.3.3. Let C be a chamber of $\mathcal{S}_{r-1}$ not equal to C_0 . Then there exists a panel $e \subset C$ such that the chamber C' adjacent to C in the standard apartment $\mathcal{A}_\bullet := \mathcal{A}_\bullet(\underline{e})$ is also contained in $\mathcal{S}_\bullet$ and the following holds true:-

(i) $v := C \setminus e$ belongs to $\mathcal{S}_\bullet^\geq(e)$;

(ii) for $v' := C' \setminus e$, $N_{v'} < N_v$.

Proof. Choose any vertex v_0 of C and consider the permutation $\sigma := \sigma_{\underline{e}}^{v_0 - nbd}(C)$. Note that choosing a vertex other than v_0 changes σ by a cyclic permutation of length r .

Let k be the maximal non-negative integer between 0 and $r - 1$ such that there exists $i \geq 0$ satisfying

$$(\sigma(i), \dots, \sigma(i + k)) = (r - k - 1, \dots, r - 1).$$

We split the proof into two cases:-

Case I. $k = r - 1$ In this case, σ must be a cyclic permutation of length r . Hence by choosing the vertex v_0 appropriately, we may assume without loss of generality that σ is the identity. If we let $v_0 = [n_0, \dots, n_{r-1}]$, then

$$\vec{C}_{v_0} = (v_0 = [n_0, n_1, \dots, n_{r-1}], v_1 := [n_0, n_1, \dots, n_{r-1} + 1], \dots, v_{r-1} := [n_0, n_1 + 1, \dots, n_{r-1} + 1]).$$

Let i_0 be the minimal index between 0 and $r - 2$ such that $n_{i_0} < n_{i_0+1}$. Such an i_0 exists since $C \neq C_0$. We claim that setting

$$v := v_{r-1-i_0} = [n_0, \dots, n_{i_0}, n_{i_0+1} + 1, \dots, n_{r-1} + 1]$$

satisfies the claims of the proposition. (i) clearly holds since v lies in $\mathcal{S}_\bullet^\geq(v_0)$ and hence also lies in $\mathcal{S}_\bullet^\geq(e)$ (where $e = C \setminus v$).

For (ii), note that v' (as defined in the proposition) is given by

$$[n_0, \dots, n_{i_0} + 1, n_{i_0+1}, n_{i_0+2} + 1, \dots, n_{r-1} + 1],$$

so that $C' := e \cup \{v'\}$ is a chamber in $\mathcal{S}_{r-1}$, by the choice of i_0 . A straightforward computation shows that $N_{v'} < N_v$.

Case 2. $k < r - 1$ Let i be as above, so that $(\sigma(i), \dots, \sigma(i + k)) = (r - k - 1, \dots, r - 1)$. By choosing v_0 suitable, we can assume that $i + k = r - 1$. Hence the pointed chamber $\vec{C}_{v_0}$ can be described as

$$(v_0 = [n_0, \dots, n_{r-1}], v_1 := [n_0, \dots, n_{r-1} + 1], \dots, v_{k+1} := [n_0, \dots, n_{r-k-1} + 1, \dots, n_{r-1} + 1],$$

$$v_{k+2} := [n_0, \dots, n_j + 1, \dots, n_{r-k-1} + 1, \dots, n_{r-1} + 1], \dots),$$

where $j \leq r - k - 2$ due to the maximality of k .

We claim that (i) and (ii) are satisfied for the pair (e, v') corresponding to the choice $v := v_{k+1}$. Indeed (i) follows easily from $v_{k+1} \in \mathcal{S}_\bullet^\geq(v_0)$. v' in this context is easily seen to be given by

$$[n_0, \dots, n_j + 1, \dots, n_{r-k-1}, n_{r-k} + 1, \dots, n_{r-1} + 1],$$

which also lies in $\mathcal{S}_\bullet$ since $j \leq r - k - 2$. Similar to the previous case we also have $N_{v'} < N_v$. $\square$

Corollary 7.3.4. Let C, e and C' be as in 7.3.3. Then $\mathrm{GL}_r(\mathbb{F}_q[t])_C = \mathrm{GL}_r(\mathbb{F}_q[t])_e$ and the q chambers distinct from C containing e lie in the $\mathrm{GL}_r(\mathbb{F}_q[t])_e$ orbit of C' .

Proof. First we show that $\mathrm{GL}_r(\mathbb{F}_q[t])_C = \mathrm{GL}_r(\mathbb{F}_q[t])_e$. In fact we claim that $\mathrm{GL}_r(\mathbb{F}_q[t])_e \subset B_r(\mathbb{F}_q[t])$, so that the required equality of stabilizers follows from Corollary 3.2.6, Lemma 7.3.2(iii) and Proposition 7.3.3(ii). By way of contradiction, suppose $\mathrm{GL}_r(\mathbb{F}_q[t])_e \not\subset B_r(\mathbb{F}_q[t])$. Let $e = \{[n_0^{(0)}, \dots, n_{r-1}^{(0)}], \dots, [n_0^{(r-2)}, \dots, n_{r-1}^{(r-2)}]\}$. Then by Corollary 3.2.6 and Proposition 7.3.3(i) we see that there exists $0 \leq i < j \leq r - 1$ such that $n_i^{(l)} = n_j^{(l)}$ for all $0 \leq l \leq r - 2$. Since C is a chamber lying in $\mathcal{S}_\bullet$ containing e , then it is not hard to see using the argument used in the proof of Lemma 3.3.9 that the other chamber of the standard apartment $\mathcal{A}_\bullet$ containing e must not lie in the standard sector $\mathcal{S}_\bullet$. This contradicts the conclusion of Proposition 7.3.3, hence $\mathrm{GL}_r(\mathbb{F}_q[t])_e \subset B_r(\mathbb{F}_q[t])$.

Let C'' be a chamber distinct from C and C' containing e . Then by Theorem 7.2.1(i) there exists $g \in \mathrm{GL}_r(\mathbb{F}_q[t])$ such that $gC'' \in \mathcal{S}_{r-1}$. In particular, $ge \subset gC''$ lies in $\mathcal{S}_{r-2}$. But by Theorem 7.2.1(ii), since $e \in \mathcal{S}_{r-2}$, $ge = e$ or equivalently $g \in \mathrm{GL}_r(\mathbb{F}_q[t])_e$. Consequently gC'' is either C or C' . Since the $\mathrm{GL}_r(\mathbb{F}_q[t])_e$ orbit of C is a singleton by the former part, $gC'' = C'$. $\square$

The above proposition motivates us to define the following

Definition 7.3.5. Let $\tilde{C}$ be a chamber in $\mathcal{BT}_{r-1}$. If $\tilde{C}$ lies in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of C_0 , then define $N_{\tilde{C}} := 0$. Otherwise, let $\tilde{C}$ lie in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of a chamber $C \in \mathcal{S}_{r-1}$ and define

$$N_{\tilde{C}} := \sum_{v \in C} N_v.$$

Note that $N_{\tilde{C}}$ is always a non-negative integer.

The discussion in this section can be summarized by the following result

Corollary 7.3.6. Let $\tilde{C}$ be a chamber in $\mathcal{BT}_{r-1}$ not lying in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of the standard chamber C_0 . Then there exists a panel $\tilde{e} \subset \tilde{C}$ such that the q chambers $\tilde{C}_i$, for $1 \leq i \leq q$, satisfy $N_{\tilde{C}_i} = N_{\tilde{C}_j} < N_{\tilde{C}}$ for all $1 \leq i, j \leq q$.

Proof. Let $\tilde{C}$ be the translate of a unique chamber $C \in \mathcal{S}_{r-1}$ so that $\tilde{C} = gC$ for some $g \in \mathrm{GL}_r(\mathbb{F}_q[t])$. Moreover let $e \subset C$ be a panel such that the vertex $v := C \setminus e$ satisfies the Proposition 7.3.3 and $C' \in \mathcal{S}_{r-1}$ be the chamber distinct from C containing e . By Proposition 7.3.3(ii), $N_{C'} < N_C$, so that in particular $N_{gC'} < N_{gC}$. By Corollary 7.3.4, the q chambers, say $\{C_i\}_{1 \leq i \leq q}$ with $C_1 := C'$, adjoining e distinct from C all lie in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of C' , or equivalently $N_{C_i} = N_{C'}$ for all $1 \leq i \leq q$. Hence the q chambers $\tilde{C}_i := gC_i$ adjoining $\tilde{e} := ge$ satisfy $N_{\tilde{C}_i} = N_{\tilde{C}_j} < N_{\tilde{C}}$ for all $1 \leq i, j \leq q$. $\square$

We finally have all the necessary generalizations of results from $r = 2$ case to emulate the argument from section 7.1 regarding the uniqueness of a $\Gamma(I)$ -invariant harmonic cocycle from its value on the $\mathrm{GL}_r(\mathbb{F}_q[t])$ -orbit of the standard chamber C_0 .

Theorem 7.3.7. Let $f \in C_{\mathrm{har}}(V)$ be a harmonic cocycle on $\mathcal{BT}_\bullet$ taking values in V . Suppose f vanishes on all (pointed)chambers lying in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of the standard chamber C_0 . Then $f \equiv 0$.

Proof. Note that the vanishing of f on a pointed chamber $\vec{C}_x$ depends only on the underlying chamber C . With this in mind, we proceed by induction on the non-negative integer $N_{\tilde{C}}$ for a chamber $\tilde{C}$. By assumption, f vanishes on all chambers $\tilde{C}$ satisfying $N_{\tilde{C}} = 0$.

Now suppose f vanishes on all chambers $\tilde{C}$ satisfying $N_{\tilde{C}} < k + 1$ for a non negative integer k and let $\tilde{D}$ be a chamber satisfying $N_{\tilde{D}} = k + 1$. By Corollary 7.3.6 there exists a panel $\tilde{e} \subset \tilde{D}$ such that all the q chambers, $\{\tilde{D}_i\}_{i=1}^q$ adjoining $\tilde{D}$ via $\tilde{e}$ and distinct from it, satisfy $N_{\tilde{D}_i} < N_{\tilde{D}} = k + 1$. In particular, f vanishes on all the chambers $\tilde{D}_i$ by hypothesis and hence by harmonicity also vanishes on $\tilde{D}$. This finishes the induction step and also concludes the theorem. $\square$

From Lemma 7.3.2(ii) it is clear that given a non-zero proper ideal $I \subset \mathbb{F}_q[t]$, the standard chamber C_0 is $\Gamma(I)$ -stable for $\Gamma_I := \Gamma_{\mathbb{F}_q[t]^r}(I)$. Since $\Gamma(I) \trianglelefteq \mathrm{GL}_r(\mathbb{F}_q[t])$ is normal, the chambers

in the $\mathrm{GL}_r(\mathbb{F}_q[t])$ orbit of C_0 are also $\Gamma(I)$ -stable. Consequently combining Corollary 6.2.4 and Theorem 7.3.7 we can deduce the following

Corollary 7.3.8. The restriction map $\mathrm{Te}_{\Gamma(I)} : C_{\mathrm{har}}(V)^{\Gamma(I)} \rightarrow \mathrm{St}^I(\mathbb{F}_q[t]^r) \otimes_{\Gamma(I)} V$ as well as $\mathrm{AR}_{\Gamma(I)} : \mathrm{St}(\mathbb{F}_q(t)^r) \otimes_{\Gamma(I)} V \rightarrow C_{\mathrm{har}}(V)^{\Gamma(I)}$ are isomorphisms. $\square$

Part II

An Eichler-Shimura map in higher ranks

Summary of chapters - part II

Chapter 8 records well known results about Drinfeld modules, their moduli spaces and the analytic description of the latter involving arithmetic quotient of Drinfeld's period domain. In section 8.1 after recalling the rigid analytic structure on the Drinfeld period domain Ω^r , we give an admissible cover for it as well as its arithmetic quotient following [Dri74, §6], using the reduction map from Ω^r to the geometric realization of the Bruhat-Tits building. In section 8.2 we briefly recap the necessary theory of Drinfeld modules necessary to relate the arithmetic quotients of Ω^r to the moduli space of Drinfeld modules as mentioned before.

Chapter 9 is similar in spirit to the previous chapter, as being a collection of results from [Boe02; BP09; BH07] about τ -sheaves and crystals. In section 9.1 we recap the (algebraic) theory of τ -sheaves and crystals on schemes and functors between them. Section 9.2 discusses analogous definitions and results pertaining to τ -sheaves and crystals on a rigid analytic space. Important examples of τ -sheaves arise from Drinfeld A -modules over schemes resp. rigid analytic spaces known as A -motives (9.1.6(ii)) resp. rigid A -motives (9.2.12). Section 9.3 compares the results of the previous two sections via rigid analytification. The concluding section 9.4 describes a uniformization of some canonical rigid A -motives on Ω^r and we follow [Boe02, Chap.9, §2] for this.

Chapter 10 constructs a prototype of an Eichler-Shimura map, modeled upon its rank 2 version in [Boe02, Defn.10.7]. Section 10.1 defines the space of coefficients V_n (10.1.1) and with respect to it recollects the main results of part I. In section 10.2 we make some basic observations regarding Pink's Satake compactification of the Drinfeld moduli space. The construction of the Eichler-Shimura map is carried out in section 10.3. Finally in section 10.4 we relate this construction to the residue map of Gräf and speculate on some future directions.

Chapter 8

Arithmetic quotient of Drinfeld period domain and Drinfeld moduli space

Let $r \geq 2$ be an integer. For the construction of the Eichler-Shimura map in a later chapter, it is necessary to understand the relation between the Drinfeld period domain Ω^r and the Bruhat-Tits building of rank $r - 1$, $\mathcal{BT}_\bullet$, in particular their arithmetic quotients. Using the reduction map $\Omega^r(\mathbb{C}_\infty) \rightarrow \mathcal{BT}_\bullet$, motivated by [Dri74, Prop.6.2], we will provide below an admissible covering of Ω^r with the corresponding nerve being $\mathcal{BT}_\bullet$. Then one can provide a rigid analytic structure on arithmetic quotients of Ω^r analogous to [Dri74, §6]. There Drinfeld also realizes the components of the base change of the Drinfeld moduli space M_{I,K_∞}^r as appropriate arithmetic quotients of Ω^r . We will also recap this correspondence following its treatment in [Boe02, §4.1, 4.2].

8.1 Arithmetic quotient of Drinfeld period domain

Before giving the definitions of Drinfeld period domain as well as its arithmetic quotients, it would be prudent to recall some notions from the theory of rigid analytic spaces which we do first.

8.1.1 Rigid analytic geometry - a quick tour

Here we will give a very quick introduction to the notion of Grothendieck topologies and the definition of rigid spaces. As alluded to above, our aim is to tweak the admissible affinoid covering of Ω^r provided in [Dri74, Prop.6.2] to suit our requirements later. Below we will develop the bare minimum of the theory of rigid spaces, starting with the definition of Grothendieck topologies and then describing the definition of a rigid analytic space. We assume the knowledge about affinoid algebras and the associated theory of affinoid spaces. For more details on these, the reader is referred

to relevant portions of any standard textbook on rigid geometry, such as [Bos14; FP04].

Definition 8.1.1. ([Bos14, Defn.1, Chap.5.1]) A *Grothendieck topology* $\mathcal{J}$ consists of a category $\text{Cat}_{\mathcal{J}}$ and a set $\text{Cov}_{\mathcal{J}}$ of families $\{U_i \rightarrow U\}_{i \in I}$ of morphisms in $\text{Cat}_{\mathcal{J}}$, called *coverings*, such that the following conditions hold true:-

- (i) If $U \rightarrow V$ is an isomorphism in $\text{Cat}_{\mathcal{J}}$, then it lies in $\text{Cov}_{\mathcal{J}}$.
- (ii) If $(U_i \rightarrow U)_{i \in I}$ and $(V_{ij} \rightarrow U_i)_{j \in J_i}$ belong to $\text{Cov}_{\mathcal{J}}$, then so does the set consisting of all compositions $(V_{ij} \rightarrow U_i \rightarrow U)_{j \in J_i, i \in I}$ lie in $\text{Cov}_{\mathcal{J}}$.
- (iii) If $(U_i \rightarrow U)_{i \in I}$ is in $\text{Cov}_{\mathcal{J}}$ and $V \rightarrow U$ is a morphism in $\text{Cat}_{\mathcal{J}}$, then the fiber products $V \times_U U_i$ exist in $\text{Cat}_{\mathcal{J}}$ for all $i \in I$ and $(V \times_U U_i \rightarrow V)_{i \in I}$ is a covering of V .

For simplicity, we will call a Grothendieck topology by a G -topology. A set X is said to be a *G-topological space* if there is a G -topology on a subset of the power set of X .

Recall that $\mathbb{C}_{\infty}$ denotes the completion of a fixed algebraic closure of K_{∞} . Note that it is a complete and algebraically closed. We denote by $|\cdot|$ the absolute value on K_{∞} as well as its canonical extension to $\mathbb{C}_{\infty}$. In this subsection we denote by L a field that is either K_{∞} or $\mathbb{C}_{\infty}$.

A pair $(X, \mathcal{O}_X)$ is said to be a *locally G-ringed L-space* if X is a G -topological space and $\mathcal{O}_X$ is a sheaf of L -algebras on X with its stalks being local rings. For definition and properties of sheaves associated to a G -topological space, refer to [Bos14, §5.2].

Example 8.1.2. ([Bos14, Chap.5.1, Defn.4, Prop.5]) Let X be an affinoid L -space. We will use the notion of *affinoid subdomains* below and not use its definition explicitly. The interested reader can have a look at [Bos14, Chap.3.3]. Recall that a cover $\mathcal{U} := \{U_{\alpha}\}_{\alpha \in \Lambda}$ is said to be a refinement of another cover $\mathcal{V} := \{V_{\beta}\}_{\beta \in \Delta}$ of a set X , if there is a map $f : \Delta \rightarrow \Lambda$ such that $U_{f(\beta)} \subset V_{\beta}$ for all $\beta \in \Delta$. Then the *strong G-topology* associated to X is defined as follows.

1. A subset $U \subset X$ is said to be *admissible open* if there is a covering $U = \bigcup_{i \in I} U_i$, for $U_i \subset X$ affinoid subdomains and I not necessarily a finite index set, further satisfying that if $\phi : Z \rightarrow X$ is a map of affinoid L -spaces such that $\phi(Z) \subset U$ then $\{\phi^{-1}(U_i)\}_{i \in I}$ has finite refinement by affinoid subdomains covering Z .
2. A family of admissible opens $\{V_j\}_{j \in J}$ of X is said to be an *admissible covering* of an admissible open V , if $V = \bigcup_{j \in J} V_j$ and for each map of affinoid L -spaces $\psi : Z \rightarrow X$ satisfying $\psi(Z) \subset V$, $\{\psi^{-1}(V_j)\}$ has a finite refinement by affinoid subdomains covering Z . In particular, by [Bos14, Chap.3.3, Prop.13] a finite covering of an affinoid space by affinoid subdomains is an admissible covering.

The pair consisting of admissible open subsets of X as well as their admissible coverings comprise the *strong G -topology* on X . It has the following properties:-

- (G0) $\emptyset$ and X are admissible open;
- (G1) let $\{U_i\}_{i \in I}$ be an admissible covering of an admissible open $U \subset X$. Moreover let $V \subset U$ satisfy that $V \cap U_i$ is a admissible open for all $i \in I$. Then V is admissible open in X ;
- (G2) let $\{U_i\}_{i \in I}$ be a covering of an admissible open $U \subset X$ by admissible opens $U_i \subset X$ for all $i \in I$. If $\{U_i\}_{i \in I}$ admits a refinement that is an admissible covering of U , then so is $\{U_i\}_{i \in I}$ an admissible cover.

Due to Tate's acyclicity theorem([Bos14, Chap.4.3]), one can attach a canonical sheaf of L -algebras to X , denoted $\mathcal{O}_X$ (this is described in [Bos14, Chap.5.2]). The ring of global sections $\mathcal{O}_X(X)$ is precisely the affinoid algebra associated with X . Henceforth, a pair such as $(X, \mathcal{O}_X)$ here will be called an affinoid L -space.

Definition 8.1.3. ([Bos14, Chap.5.3, Defn.4]) A rigid analytic L -space is a locally ringed G -space $(X, \mathcal{O}_X)$ such that

- (i) The G -topology on X satisfies G0, G1 and G2 in Example 8.1.2;
- (ii) X admits an admissible covering $\{X_i\}_{i \in I}$ such that $(X_i, \mathcal{O}_{X_i})$ is an affinoid L -space.

(ii) can also be rephrased as saying that the G -topology on X should be compatible with the strong G -topology on X_i for all $i \in I$.

From Definition 8.1.1(ii) and (G2), we can deduce the following result on admissible coverings

Lemma 8.1.4. Let X be a rigid L -space and $\{U_i\}_{i \in I}$ is an admissible covering of X . Suppose $\{U'_j\}_{j \in J}$ is another covering of X by admissible open subsets U'_j of X such that for each $i \in I$, $\{U'_j \cap U_i\}_{j \in J}$ is an admissible covering of U_i . Then $\{U'_j\}_{j \in J}$ is an admissible covering of X .

Remark 8.1.5. Let X be a rigid K_∞ -space. Then one also has a well defined base change, $X_{\mathbb{C}_\infty}$, to a rigid analytic $\mathbb{C}_\infty$ -space equipped with a canonical map $\varphi : X_{\mathbb{C}_\infty} \rightarrow X$ (see [Bos14, Chap.4.2]). Since $\mathbb{C}_\infty$ is algebraically closed, one can identify the G -topological space associated to $X_{\mathbb{C}_\infty}$ with $X(\mathbb{C}_\infty) := \text{Hom}(\text{Sp}(\mathbb{C}_\infty) \rightarrow X)$, where Hom is considered in the category of rigid spaces. We will say that a subset $U \subset X(\mathbb{C}_\infty)$ defines an *admissible open subset over K_∞* , if it is the pull back along φ of an admissible open subset of X .

Example 8.1.6. *Rigid analytification of schemes of finite type.* Let $\mathbf{SchFt}_L$ denote the category of schemes of finite type over L and $\mathbf{Rig}_L$ denote the category of rigid analytic spaces over L . Then there is functor $(\cdot)^{an} : \mathbf{SchFt}_L \rightarrow \mathbf{Rig}_L$ which has the property that any finite type scheme X over L is equipped with a natural morphism of ringed spaces $X^{an} \rightarrow X$ such that for a rigid analytic space Z over L , any morphism of ringed spaces $Z \rightarrow X$ factors through a morphism of rigid spaces $Z \rightarrow X^{an}$. We only give below two examples of analytification of the affine space and projective space and refer the reader to relevant parts of [FP04; Bos14] for more details.

- (i) Let $n \geq 1$ be an integer and consider the affine n -space $\mathbb{A}_L^n := \text{Spec}(L[X_1, \dots, X_n])$. Choose $c \in L$ such that $|c| > 1$. Define the affinoid algebras

$$T_n^{(i)} := L\langle c^{-i}T_1, \dots, c^{-i}T_n \rangle.$$

The inclusions $T_n^{(0)} \hookleftarrow T_n^{(1)} \hookleftarrow \dots$ induce inclusions of their associated affinoid spaces $\text{Sp}(T_n^{(0)}) \hookrightarrow \text{Sp}(T_n^{(1)}) \hookrightarrow \dots$. The analytification $\mathbb{A}_L^{n,an}$ is defined to be the rigid analytic L -space obtained from gluing the affinoid spaces $\{\text{Sp}(T_n^{(i)})\}_{i \geq 0}$ along these inclusions.

- (ii) Let $r \geq 2$ be an integer and consider the projective space $\mathbb{P}_L^{r-1} = \text{Proj}(L[X_1, \dots, X_r])$. Then $\mathbb{P}_L^{r-1,an}$ can be obtained by gluing the analytification of the affine $(r-1)$ -spaces $\text{Spec}(L[\frac{X_1}{X_i}, \dots, \frac{X_r}{X_i}])$, for $1 \leq i \leq r$. In fact it can be shown that the r affinoid spaces $\text{Sp}(L\langle \frac{X_1}{X_i}, \dots, \frac{X_r}{X_i} \rangle)$ for $1 \leq i \leq r$ form an admissible cover $\mathbb{P}_L^{r-1,an}$.

Note that $\text{GL}_r(L)$ acts as isomorphisms on the projective space $\mathbb{P}_L^{r-1}$, which induces an action on $\mathbb{P}_L^{r-1,an}$ which are rigid analytic isomorphisms.

8.1.2 Drinfeld period domain

Consider the $r-1$ dimensional rigid analytic projective space $\mathbb{P}_{K_\infty}^{r-1,an}$ described in Example 8.1.6. Suppose that it is given by coordinates $X_i; 1 \leq i \leq r$. Note that any linear polynomial $f(X_i) := \sum_{i=1}^r a_i X_i; a_i \in K_\infty$ defines a *linear K_∞ -rational hyperplane* of $\mathbb{P}_{K_\infty}^{r-1,an}$, whose $\mathbb{C}_\infty$ points are exactly the points $(z_1, \dots, z_r)^T \in \mathbb{P}^{r-1}(\mathbb{C}_\infty)$ satisfying f .

Definition 8.1.7. Let $\mathcal{H}$ denote the set of all linear K_∞ rational hyperplanes of $\mathbb{P}^{r-1}$. Then define

$$\Omega^r := \mathbb{P}_{K_\infty}^{r-1,an} \setminus \bigcup_{H \in \mathcal{H}} H.$$

Proposition 8.1.8. Ω^r is an admissible open subset of $\mathbb{P}_{K_\infty}^{r-1,an}$.

Proof. We only provide here a sketch of a proof following the part (C) of the argument in [SS91, Prop.1] and refer the reader there for more details.

From 8.1.2(1), 8.1.3 and Example 8.1.6, it suffices to show the following:-

- (a) there exists affinoid admissible open $\Omega_n \subset \mathbb{P}_{K_\infty}^{r-1, an}$, for $n \in \mathbb{N}$, such that $\Omega^r = \bigcup_{n \in \mathbb{N}} \Omega_n$ and
- (b) given a morphism of rigid K_∞ spaces $f : Y \rightarrow \mathbb{P}_{K_\infty}^{r-1, an}$ such that Y is affinoid and $f(Y) \subset \Omega^r$, f factorizes via Ω_n for some $n \in \mathbb{N}$.

We will consider unimodular normalization of the column vector of coordinates $\underline{z} = (z_1, \dots, z_r)^T \in \mathbb{P}^{r-1}(\mathbb{C}_\infty)$, that is $|z_i| \leq 1$ for all $1 \leq i \leq r$ and $|z_j| = 1$ for some j . Let $H \in \mathcal{H}$ be a hyperplane and denote by $[H] \in O_\infty^r$ a row vector such that one of its coordinate has absolute value 1 and $H(\mathbb{C}_\infty) = \{\underline{z} \in \mathbb{P}^{r-1}(\mathbb{C}_\infty) \mid [H].\underline{z} = 0\}$. Define

$$H(|\varpi^n|) := \{\underline{z} \in \mathbb{P}^{r-1}(\mathbb{C}_\infty) \mid [H].\underline{z} \leq |\varpi^n|\}.$$

Identifying the $\text{Aut}_{K_\infty}(\mathbb{C}_\infty)$ conjugate points, this also defines an affinoid open subdomain of $\mathbb{P}_{K_\infty}^{r-1}$, which we also denote by $H(|\varpi^n|)$ (In fact, such an affinoid domain is an example of a *Weierstrass domain*, see [Bos14, Chap.3.3, Defn.7(i)]). Then set

$$\Omega_n := \mathbb{P}_{K_\infty}^{r-1} \setminus \bigcup_{H \in \mathcal{H}} H(|\varpi^n|).$$

[SS91, Lemma 2, 3] shows that for each n , Ω_n is finite intersection of Laurent domains (see [Bos14, Chap.3.3, Defn.7(ii)] for definition of such domains) of the form $\mathbb{P}_{K_\infty}^{r-1} \setminus H(|\varpi^n|)$, $H \in \mathcal{H}$, hence is itself a Laurent domain, and consequently $\Omega^r = \bigcup_{n \in \mathbb{N}} \Omega_n$ satisfies condition (a). In op.cit. it is also shown that condition (b) is satisfied for this covering. See also [Bas14, §6.2]. $\square$

Now we can use the above covering of Ω^r to describe the action of $\text{GL}_r(K_\infty)$ on it, which coincides with its action on $\Omega^r(\mathbb{C}_\infty)$ given in [BBP24, §1]. Let $\gamma \in \text{GL}_r(K_\infty)$ be given. Since γ induces a rigid analytic isomorphism of $\mathbb{P}_{K_\infty}^{r-1, an}$ stabilizing Ω^r , by condition (b) mentioned in the previous proof, for each $n \in \mathbb{N}$ $\gamma(\Omega_n)$ is an affinoid admissible subset of Ω_{m_n} for some $m_n \in \mathbb{N}$. Running over all $n \in \mathbb{N}$ we see that γ induces a rigid analytic homomorphism $\gamma : \Omega^r \rightarrow \Omega^r$, which is also an isomorphism due to γ^{-1} .

Remark 8.1.9. Suppose that γ as above has the matrix form $(a_{ij})_{1 \leq i, j \leq r} \in \text{GL}_r(K_\infty)$. The function $\Omega_n(\mathbb{C}_\infty) \rightarrow \mathbb{C}_\infty$ sending

$$\underline{z} \mapsto \sum_{1 \leq i \leq r} a_{ri} z_i$$

defines a section of $\mathcal{O}_{\Omega^r}(\Omega_n)$ which is non-vanishing. Such sections can be glued together as n ranges over $\mathbb{N}$ and the resulting section in $\mathcal{O}_{\Omega^r}(\Omega^r)$ will be denoted by $j(\gamma, -)$, which is clearly also non-vanishing. As a function on $\Omega^r(\mathbb{C}_\infty)$ this is precisely the *factor of automorphy* defined in [BBP24, §1].

The reduction map and an admissible cover of Ω^r

For the following recall the metric space of dilation classes of norms on K_∞^r , $\mathcal{N}$, discussed in connection with the geometric realization $|\mathcal{BT}_\bullet|$ in 3.1.3, related by a topological isomorphism θ between them described in Proposition 3.1.18. Observe that there is also a canonical map

$$\xi : \Omega^r(\mathbb{C}_\infty) \rightarrow \mathcal{N}$$

sending

$$\underline{z} \mapsto [x \mapsto |x \cdot \underline{z}|_\infty],$$

where $|\cdot|_\infty$ denotes the canonical extension of the norm in K_∞ and elements of K_∞^r are regarded as row vectors. It is not hard to see that the map is well defined.

Definition 8.1.10. The **reduction map** is defined as $\rho := \theta^{-1} \circ \xi : \Omega^r(\mathbb{C}_\infty) \rightarrow |\mathcal{BT}_\bullet|$.

ρ is $GL_r(K_\infty)$ -equivariant.

Proposition 8.1.11. [Dri74, Prop.6.1(i)] Let $x_1, \dots, x_k \in |\mathcal{BT}_\bullet|$ be points corresponding to the vertices of the underlying building and $c \in \mathbb{Q}$. Let $V^c := \{y \in |\mathcal{BT}_\bullet| \mid \sum_i d(x_i, y) \leq c\}$. Then $U^c := \rho^{-1}V^c$ defines an admissible open affinoid K_∞ -subset of $\mathbb{P}_{K_\infty}^{r-1}$.

Proposition 8.1.12. ([Dri74, Prop.6.2]) Fix a rational number c with $0 < c < 1$. Consider the following set ($\Delta \subset \mathcal{BT}_{k-1}$ a simplex)

$$V_\Delta^c := \{y \in |\mathcal{BT}_\bullet| \mid d(x, y) \leq 1 - \frac{3-c}{4^{k-1}} \forall x \in \Delta \text{ and } \sum_{x \in \Delta} d(x, y) \leq k-1 + \frac{1+c}{4^k}\}$$

Then for $U_\Delta^c := \rho^{-1}V_\Delta^c$, the sets $\{U_\Delta^c\}$ form an admissible covering of Ω^r such that $U_{\Delta g}^c = gU_\Delta^c$ for $g \in GL_r(K_\infty)$. Also the nerve of the covering is the barycentric subdivision of $\mathcal{BT}_\bullet$.

Now we're in a position to state and prove an analogue of the above result [Dri74, Prop. 6.2] of Drinfeld. We adapt the proof strategy here and change the cover slightly, so that the opens in the following cover are indexed by vertices in $\mathcal{BT}_0$ of Bruhat Tits building and nerve of this covering is $\mathcal{BT}_\bullet$.

Proposition 8.1.13. Fix a rational number c with $0 < c < 1$. Define the set

$$V_x'^c := \{y \in |\mathcal{BT}_\bullet| \mid d(x, y) \leq 1 - \frac{3-c}{4^r}\}$$

for $x \in \mathcal{BT}_0$. Then $\{V_x'^c\}_{x \in \mathcal{BT}_0}$ is a covering of the Bruhat Tits building. Furthermore setting $W_x^c := \rho^{-1}V_x'^c$, the sets $\{W_x^c\}$ form an admissible covering of Ω^r such that $W_{xg}^c = gW_x^c$, for $g \in GL_r(K_\infty)$. Also the nerve of the covering is $\mathcal{BT}_\bullet$.

Proof. The notation from Proposition 8.1.12 continues to hold here.

Covering. To show that the sets W_x^c cover Ω^r , as x ranges over $\mathcal{BT}_0$, it is enough to show that $\{V_x'^c\}_{x \in \mathcal{BT}_0}$ is a cover of $|\mathcal{BT}_\bullet|$. Let $y \in |\mathcal{BT}_\bullet|$ be a point and suppose that its barycentric coordinates are $(t_{v_0}, \dots, t_{v_d})$ with respect to its facet $\{v_0, \dots, v_d\}$. In particular, the distance between y and v_i is $1 - t_{v_i}$ by Proposition 3.1.19(i). Since $\sum_{0 \leq i \leq d} t_{v_i} = 1$ and $d \leq r - 1$, we have that $t_{v_i} \geq \frac{3-c}{4^r}$ for some $0 \leq i \leq d$, or equivalently $d(y, v_i) \leq 1 - \frac{3-c}{4^r}$. Hence $y \in V_{v_i}'^c$.

Admissibility. First note that for each $x \in \mathcal{BT}_0$, W_x^c is admissible open in Ω^r by Proposition 8.1.11. Also by Lemma 8.1.4 it suffices to show that for each $\Delta \in \mathcal{BT}_\bullet$, $\{W_x^c \cap U_\Delta^c\}_{x \in \mathcal{BT}_0}$ is an admissible covering of U_Δ^c . Since W_x^c and U_Δ^c are affinoid open in $\mathbb{P}_{K_\infty}^{r-1}$, by [Bos14, Chap.6.3] the intersection $W_x^c \cap U_\Delta^c$ is affinoid open and hence it is enough to show that $\{V_x'^c \cap V_\Delta^c\}_{x \in \mathcal{BT}_0}$ is a finite cover of V_Δ^c . Suppose Δ is a d -simplex $\{v_0, \dots, v_d\}$. Then observe that if y lies in the intersection $V_x'^c \cap V_\Delta^c$, then $d(x, y) < 1, d(v_i, y) < 1 \Rightarrow d(x, v_i) < 2$ for all $0 \leq i \leq d$, which means x lies in a chamber containing Δ , which are finitely many. Hence the covering is finite.

Nerve of the covering. It suffices to show that $\{V_x'^c\}_{x \in \mathcal{BT}_0}$ has the nerve $\mathcal{BT}_\bullet$.

Suppose $\Delta = \{v_0, \dots, v_d\}$ is a d -simplex as before. Let $\Phi = \{v_0, \dots, v_d, \dots, v_{r-1}\}$ be a chamber containing Δ and consider a point $y \in |\Phi|$ with barycentric coordinates $(t_{v_0}, \dots, t_{v_{r-1}})$. Since $d(y, v_i) = 1 - t_{v_i}$, we see that $y \in V_{v_0}'^c \cap \dots \cap V_{v_d}'^c$ if and only if $\frac{3-c}{4^r} \leq t_i$ for $0 \leq i \leq d$ and $\sum_{0 \leq i \leq r-1} t_{v_i} = 1$. Clearly one can find a non-empty set of solutions in $[0, 1]^r$ for such a system of inequalities for $(t_{v_0}, \dots, t_{v_{r-1}})$, hence $V_{v_0}'^c \cap \dots \cap V_{v_d}'^c \neq \emptyset$.

Conversely, suppose $v_0, \dots, v_d \in \mathcal{BT}_0$ be vertices such that $\cap V_{v_i}'^c \neq \emptyset$ and y be a point in this intersection. Let Φ be a chamber containing y , so that $y \in |\Phi|$. Then a similar argument as that in the previous step of the proof (showing admissibility), shows $V_x'^c \cap |\Phi| \neq \emptyset \iff x \in \Phi$. Hence $v_i \in \Phi$ for all $0 \leq i \leq d$, or equivalently they form a simplex.

Hence the nerve of $\{V_x'^c\}$ is $\mathcal{BT}_\bullet$, in particular also that of $\{W_x^c\}$. □

Since the nerve of the cover $\{W_x^c\}_{x \in \mathcal{BT}_0}$ is $\mathcal{BT}_\bullet$, the following definition makes sense

Definition 8.1.14. Let $\Delta \in \mathcal{BT}_\bullet$ be a simplex. Then define

$$W_\Delta^c := \bigcap_{x \in \Delta} W_x^c.$$

It is an affinoid subdomain of Ω^r .

8.1.3 Rigid analytic structure on arithmetic quotient

Here we will understand the rigid analytic K_∞ -structure on quotients of the form $\Gamma \backslash \Omega^r$ for certain principal congruence subgroups $\Gamma \subset \mathrm{GL}_r(K)$. We begin this subsection by describing the principal congruence subgroups Γ we will be concerned with.

Given a closed point v of C , denote by K_v the completion of K at v and $O_v \subset K_v$ the subring of integral elements. Let $\hat{A}$ denote the profinite ring $\prod_{v \neq \infty} O_v$. Let $\mathbb{A}_K^f$ denote the ring of finite adeles of K , that is, the restricted topological product $\prod'_{v \neq \infty} K_v$ with respect to the family of compact open subgroups $O_v \subset K_v$ for all $v \in \mathrm{Spec}(A)$. Similarly one denotes the restricted topological product $\prod'_{v \neq \infty} \mathrm{GL}_r(K_v)$ with respect to the family of compact opens $\mathrm{GL}_r(O_v) \subset \mathrm{GL}_r(K_v)$, for all $v \in \mathrm{Spec}(A)$, by $\mathrm{GL}_r(\mathbb{A}^f)$. Note that $\mathrm{GL}_r(\hat{A})$ is a subgroup of $\mathrm{GL}_r(\mathbb{A}_K^f)$.

Definition 8.1.15. Define $\mathcal{K}(I) := \mathrm{Ker}(\mathrm{GL}_r(\hat{A}) \rightarrow \mathrm{GL}_r(\hat{A}/I\hat{A}))$.

Let $g \in \mathrm{GL}_r(\mathbb{A}_K^f)$. Then define $\Gamma_g(I) := \mathrm{GL}_r(K) \cap g\mathcal{K}(I)g^{-1}$.

Remark 8.1.16. Note that, if we define the projective A -module $\Lambda_g := K^r \cap \hat{A}^r g^{-1}$ viewing $\hat{A}^r \subset (\mathbb{A}_K^f)^r$ as row vectors, then $\Gamma_g(I)$ is the principal congruence subgroup $\Gamma_{\Lambda_g}(I)$.

For the next result, note that the left action of $\mathrm{GL}_r(K_\infty)$ on Ω^r induces a left action of Γ as well.

Lemma 8.1.17. The elements of finite order in $\mathcal{K}(I)$ have order a power of p .

Proof. This is covered in [Boe02, Lemma 1.13]. □

Let P be a rank r projective A -submodule inside K^r and $I \subsetneq A$ be a non-zero proper ideal of A . For the rest of this subsection we set $\Gamma := \Gamma_P(I)$. One can put a locally G -ringed structure on the quotient $\Gamma \backslash \Omega^r$ via the quotient map $\pi : \Omega^r \rightarrow \Gamma \backslash \Omega^r$ as follows. The underlying G -topological space is precisely the quotient space $\Gamma \backslash \Omega^r$, with $U \subset \Gamma \backslash \Omega^r$ admissible open if and only if $\pi^{-1}U$ is admissible open in Ω^r and admissible coverings of admissible opens in $\Gamma \backslash \Omega^r$ are analogously defined. The structure sheaf $\mathcal{O}_{\Gamma \backslash \Omega^r}$ is defined by sending any admissible open $U \subset \Gamma \backslash \Omega^r$ to the ring of invariants $(\mathcal{O}_{\Omega^r}(\pi^{-1}U))^\Gamma$.

Below we will see that in fact the $\Gamma \backslash \Omega^r$ is a rigid analytic K_∞ -space. From the definition of its G -space structure, it remains to show 8.1.3(ii) for $X = \Gamma \backslash \Omega^r$, that is, it admits an admissible affinoid covering. We will achieve this using the finite quotients of the covering $\{W_x^c\}_{x \in \mathcal{BT}_0}$ from Proposition 8.1.13. We follow the treatment of the rigid analytic structure on the arithmetic quotient in [Dri74, §6] as well as its elaboration in [SS91, §5, Thm. 2], where an admissible affinoid covering is provided using Proposition 8.1.12. We follow their proof but using Proposition 8.1.13 instead and only provide more details to some of their assertions.

Let B be an affinoid algebra over K_∞ , equipped with an action of a finite group G and let B^G be the set of elements of B invariant under the action of G . In [Dri74, Prop.6.3], Drinfeld shows that B^G is also an affinoid algebra and that $B^G \rightarrow B$ is a finite map. We will use this result in what follows.

Proposition 8.1.18. There is a canonical rigid analytic K_∞ -structure on $\Gamma \backslash \Omega^r$ via the quotient $\pi : \Omega^r \rightarrow \Gamma \backslash \Omega^r$, such that it admits an admissible covering $\{\overline{W}_x^c\}_{x \in \Gamma \backslash \mathcal{BT}_0}$, equipped with a canonical isomorphism of affinoid spaces $\Gamma_{\tilde{x}} \backslash W_{\tilde{x}}^c \simeq \overline{W}_x^c$ where $\tilde{x} \in \mathcal{BT}_0$ is a representative of $x \in \Gamma \backslash \mathcal{BT}_0$.

Proof. Let $\tilde{x} \in \mathcal{BT}_0$ be a vertex, $\Gamma_{\tilde{x}}$ be its stabilizer group which is a finite set from Proposition 3.2.7 and x denote its class in $\Gamma \backslash \mathcal{BT}_0$. Define the set

$$W_{\Gamma,x}^c := \dot{\bigcup}_{g \in \Gamma/\Gamma_{\tilde{x}}} W_{g\tilde{x}}^c = \dot{\bigcup}_{g \in \Gamma/\Gamma_{\tilde{x}}} gW_{\tilde{x}}^c.$$

Then $W_{\Gamma,x}^c$ is an admissible subset of Ω^r since the intersection $W_{\Gamma,x}^c \cap W_y^c = \dot{\bigcup}_i W_{\tilde{x}g_i}^c \cap W_y^c$, where $\{g_i\}$ is a finite subset of representatives of Γ/Γ_x (since by previous proposition $\{W_{\tilde{x}}^c\}_{\tilde{x} \in \mathcal{BT}_0}$ has its nerve $\mathcal{BT}_\bullet$ and the latter is locally finite).

Hence $W_{\Gamma,x}^c$ is an admissible subset, or $\overline{W}_x^c := \pi(W_{\Gamma,x}^c)$ is an admissible subset of $\Gamma \backslash \Omega^r$. By abuse of notation, we denote the restriction of π to $W_{\Gamma,x}^c$ also by π . Then the surjection $\pi : W_{\Gamma,x}^c \rightarrow \overline{W}_x^c$ induces an isomorphism of rigid spaces $\Gamma_{\tilde{x}} \backslash W_{\tilde{x}}^c \rightarrow \overline{W}_x^c$. Denote by $\pi_{\tilde{x}} : W_{\tilde{x}}^c \rightarrow \overline{W}_x^c$ the corresponding canonical finite quotient map.

Since $\Gamma_{\tilde{x}}$ is a finite group, by [Dri74, Prop. 6.3] $\overline{W}_x^c$ is an affinoid space. By definition of rigid analytic structure on $\Gamma \backslash \Omega^r$, the covering $\{\overline{W}_x^c\}_{x \in \Gamma \backslash \mathcal{BT}_0}$ becomes an admissible covering of $\Gamma \backslash \Omega^r$. $\square$

Remark 8.1.19. Let $\tilde{\Delta} \in \mathcal{BT}_\bullet$ be a simplex and $\Delta \in \Gamma \backslash \mathcal{BT}_\bullet$ denote the corresponding class. Then the above proof applied verbatim to the affinoid $W_{\tilde{\Delta}}^c$ (see 8.1.14) allows us to construct an affinoid open $\overline{W}_\Delta^c \subset \Gamma \backslash \Omega^r$, which is seen to be $\bigcap_{x \in \Delta} \overline{W}_x^c$, together with a canonical isomorphism $\Gamma_{\tilde{\Delta}} \backslash W_{\tilde{\Delta}}^c \simeq \overline{W}_\Delta^c$. Here by “ $x \in \Delta$ ” we mean the class x in $\Gamma \backslash \mathcal{BT}_0$ of vertices $\tilde{x} \in \tilde{\Delta}$. Compare with the discussion following [Dri74, §6, Definition].

8.2 Brief recap of the theory of Drinfeld modules

The function field analogues of elliptic curves are rank 2 Drinfeld modules and were developed in the seminal work [Dri74] with the principal aim to understand Langlands conjecture for $\mathrm{GL}_{2,K}$. The analogy between elliptic curves and rank 2 Drinfeld modules goes quite deep, for example the main results of [Boe02] are motivated from the classical Eichler-Shimura isomorphism as explained

briefly in the Introduction. Drinfeld modules also admit a rich and developing theory of higher ranks. Since our goal in the part 2 of the dissertation is to emulate the arguments from [Boe02] to construct an Eichler-Shimura map in the higher rank setting, it would be prudent to recall some of the definitions and properties concerning Drinfeld modules in higher rank, their moduli spaces and its relation to arithmetic quotients of Ω^r from previous section. We will mainly follow the treatment contained in [Pin12; BBP24; Boe02; Leh09] and refer the reader there for more details. For checking claims on Drinfeld modules, one could also refer to any standard textbook on Drinfeld modules, for instance [Pap23].

8.2.1 Drinfeld modules and their moduli spaces

Let S be a scheme over K and $r \geq 2$ an integer. An invertible sheaf $\mathcal{L}$ on S determines a *line bundle* on S , given by

$$\mathbb{G}_{a,\mathcal{L}} := \text{Spec}\left(\bigoplus_{i \geq 0} \mathcal{L}^{-i}\right).$$

See [Liu02, Chap.5, Ex.1.17, 1.18] for details on this construction. Then $\mathbb{G}_{a,\mathcal{L}}$ is endowed with a canonical structure of a group scheme over S such that Zariski locally it is isomorphic to $\mathbb{G}_{a,S}$. In fact, if $S = \text{Spec}(R)$ and $\mathcal{L} \simeq \mathcal{O}_S$, then $\mathbb{G}_{a,\mathcal{L}} \simeq \text{Spec}(R[X])$. Moreover one has for any S -scheme T ,

$$\mathbb{G}_{a,\mathcal{L}}(T) = H^0(T, \mathcal{L} \otimes_{\mathcal{O}_S} \mathcal{O}_T)^+.$$

Let τ_p denote the relative Frobenius morphism on $\mathbb{G}_{a,\mathcal{L}}$, then the endomorphism ring of the group scheme $\mathbb{G}_{a,\mathcal{L}}$ is described as ([Leh09, Chap.1, Prop 2.3, Rmk.2.4])

$$\text{End}(\mathbb{G}_{a,\mathcal{L}}) = \bigoplus_{i \geq 0} H^0(S, \mathcal{L}^{1-p^i}) \tau_p^i.$$

.

Remark 8.2.1. For example, when $(S, \mathcal{L}) = (\text{Spec}(R), \mathcal{O}_S)$, then the above endomorphism ring is given by the non-commutative ring $R[\tau]$ such that $\tau a = a^p \tau$ for all $a \in R$ ([Leh09, Prop.2.1]).

Denote by $\partial_0 : \text{End}(\mathbb{G}_{a,\mathcal{L}}) \rightarrow \mathcal{O}_S(S)$ the map sending $\sum_{i \geq 0} a_i \tau_p^i \mapsto a_0$. Suppose $q = p^m$ for some integer $m > 0$ and denote by $\tau_q := \tau_p^m$. Then the subalgebra of $\text{End}(\mathbb{G}_{a,\mathcal{L}})$ consisting of $\mathbb{F}_q$ -linear endomorphisms, $\bigoplus_{i \geq 0} H^0(S, \mathcal{L}^{1-q^i}) \tau_q^i$ will be denoted by $\text{End}_{\mathbb{F}_q}(\mathbb{G}_{a,\mathcal{L}})$.

Definition 8.2.2. ([Pin12, Defn.3.1])

- (i) Let L be a field containing K . A *Drinfeld A -module* of rank r over L is a ring homomorphism $\phi : A \rightarrow L[\tau_q]$, denoted by $a \mapsto \phi_a := \sum_{i \geq 0} \phi_{i,a} \tau_q^i$, such that $\partial_0 \circ \phi$ is the structural map $A \rightarrow L$ and for all $a \in A \setminus \mathbb{F}_q$, $\phi_{i,a} = 0 \forall i > r \cdot \deg(a)$ and $\phi_{r \cdot \deg(a), a} \neq 0$.

- (ii) A Drinfeld A -module $\mathbb{E}$ over S of rank r is a pair $\mathbb{E} := (\mathcal{L}, \varphi)$ of an invertible sheaf $\mathcal{L}$ on S and a ring homomorphism

$$\varphi : A \rightarrow \text{End}_{\mathbb{F}_q}(\mathbb{G}_{a, \mathcal{L}}),$$

denoted $a \mapsto \varphi_a := \sum_{i \geq 0} \varphi_{i,a} \tau_q^i$, such that

1. $\forall a \in A$, $\partial \circ \varphi_a$ is the composition of $A \hookrightarrow K$ with the structure morphism $K \rightarrow \mathcal{O}_S(S)$;
2. for $s \in S$ the canonical map $\varphi_s : A \rightarrow \text{End}_{\mathbb{F}_q}(\mathbb{G}_{a, \mathcal{L}_s})$ induced from φ denoting the fiber at s , is a Drinfeld A -module of rank r over $k(s)$ (the residue field of the local ring at s). Denote the resulting Drinfeld A -module by $\mathbb{E}_s$;
3. $\varphi_{i,a} = 0$ for all $i > r \cdot \deg(a)$, $a \in A$.

If $\mathbb{E}' := (\mathcal{L}', \varphi')$ is another Drinfeld A -module over S , then a morphism $\mathbb{E} \rightarrow \mathbb{E}'$ is defined to be a map of line bundle $f : \mathbb{G}_{a, \mathcal{L}} \rightarrow \mathbb{G}_{a, \mathcal{L}'}$ such that $f \circ \varphi_a = \varphi'_a \circ f$ for all $a \in A$.

- (iii) Let S and the Drinfeld A -module $\mathbb{E}$ be as in (ii). Given a nonzero proper ideal $I \subset A$, a (full)level I structure on $\mathbb{E}$ is an isomorphism of S -group schemes

$$\psi : \underline{(I^{-1}/A)^r} := S \times ((I^{-1}/A)^r) \xrightarrow{\sim} \mathbb{E}[I] := \bigcap_{a \in I} \ker(\varphi_a).$$

A morphism between pairs of Drinfeld modules with level I structures, $(\mathbb{E}, \psi) \rightarrow (\mathbb{E}', \psi')$, is an isomorphism $\mathbb{E} \rightarrow \mathbb{E}'$ which is compatible in a canonical sense with the level structures ψ and ψ' .

Remark 8.2.3. (i) Note that the space of sections $\mathbb{E}(S)$ for a Drinfeld A -module $\mathbb{E}$ over a K -scheme S is equipped with a left A -module structure.

- (ii) Given a non-zero proper ideal $I \subset A$, the closed subscheme $\mathbb{E}[I] \subset \mathbb{E}$ is a relative Cartier divisor over S . Similarly given any $x \in \mathbb{E}(S)$, $x(S) \subset \mathbb{E}$ is also a relative Cartier divisor over S , which we simply denote by $[x]$. From [Leh09, Chap.2, Defn.4.2, Rem.4.3(2)], there is equivalence between, giving a level I structure $\psi : S \times ((I^{-1}/A)^r) \xrightarrow{\sim} \mathbb{E}[I]$, and providing a homomorphism of A -modules

$$i : I^{-1}/A \rightarrow \mathbb{E}(S)$$

such that, for any prime ideal $\mathfrak{p} \in V(I)$, there is an equality of relative Cartier divisors

$$\mathbb{E}[\mathfrak{p}] = \sum_{a \in \mathfrak{p}^{-1}/A} [i(a)].$$

- (iii) For a point $s \in S$, a level I structure $S \times ((I^{-1}/A)^r) \xrightarrow{\sim} \mathbb{E}[I]$ also induces a level I structure on the Drinfeld A -module $\mathbb{E}_s$ on the fiber at s . As a consequence of this observation as well as (ii), we can conclude that for $a \in I^{-1}/A \setminus \{0\}$ the section $i(a)$ (i as in (ii)) provides a trivialization of the line bundle underlying $\mathbb{E}$. In particular, if $\mathbb{E}$ admits a level I structure, then its underlying line bundle is trivial.

Let $I \subset A$ be a non-zero proper ideal. Consider the moduli functor associated with rank r Drinfeld A -modules over K -schemes with a level I structure,

$$\mathbb{M}_I^r : \mathbf{Sch}_K \rightarrow \mathbf{Sets},$$

sending

$$S \mapsto \{\text{Isomorphism classes of rank } r \text{ Drinfeld } A\text{-modules over } S \text{ with a level } I \text{ structure}\}.$$

Before stating the next result, let us recall the notion of a fine moduli space. For any K -scheme X denote by $h_X : \mathbf{Sch}_K \rightarrow \mathbf{Sets}$ the functor sending $S \mapsto \text{Hom}_{K\text{-Sch}}(S, X)$. Then a functor $F : \mathbf{Sch}_K \rightarrow \mathbf{Sets}$ is said to be represented by a K -scheme X if there is an isomorphism of functors, $F \simeq h_X$. In such a case, X is determined uniquely upto a unique isomorphism and is said to be *the fine moduli space representing F* .

Proposition 8.2.4. $\mathbb{M}_I^r$ is representable by a smooth, finite type affine scheme M_I^r of dimension $r - 1$ over K .

Proof. In [Dri74, Prop.5.3], the moduli functor for Drinfeld A -modules with a level I structure over an A -scheme is shown to be fine assuming that $\#V(I) > 1$. But for the functor $\mathbb{M}_I^r$ over K -schemes, the proof there goes through even in the case $\#V(I) \geq 1$, to furnish the existence of a fine moduli space. In fact, in op.cit., choosing a maximal ideal $\mathfrak{m} \in V(I)$ and setting $B := H^0(\text{Spec}(A) \setminus \{\mathfrak{m}\}, \mathcal{O}_C)$, the functor $\mathbf{Sch}_B \rightarrow \mathbf{Sets}$ sending a B -scheme S to isomorphism classes of Drinfeld A -modules with level I structure over S is shown to be represented by affine, finite type scheme $\widetilde{M}_I^r$ over B . Then its generic fiber $M_I^r := \widetilde{M}_I^r|_{\text{Spec}(K)}$ is the affine, finite type scheme over K representing $\mathbb{M}_I^r$. The claim on smoothness and dimension follow from [Dri74, §5, Cor.]. For more details, one could also check [Leh09, Chap.2, Prop.5.1] and [Leh09, Chap.3, Prop.4.1]. $\square$

Consequently there is a natural isomorphism $\mathbb{M}_I^r \simeq h_{M_I^r}$.

Definition 8.2.5. Denote by $(\mathcal{L}^{un}, \varphi^{un}, \psi^{un})$, the Drinfeld A -module with level I structure over M_I^r corresponding to the identity morphism in $h_{M_I^r}(M_I^r)$. This will be called the *universal Drinfeld A -module over M_I^r* .

Remark 8.2.6. From Remark 8.2.3(iii) it follows that the underlying line bundle $\mathcal{L}^{un}$ of the universal Drinfeld module is trivial.

8.2.2 Drinfeld modules over rigid spaces

Let L be a complete field between K_∞ and $\mathbb{C}_\infty$, and X be a rigid L -space of finite type. In this subsection, we will recapitulate the results from [Boe02, Chap.4]. First we will provide a definition of a Drinfeld A -module on X and construct the prime examples of such Drinfeld modules we will be concerned with later. This section partly reuses some material from [GV25, §2.8].

For defining a Drinfeld A -module over X , we follow the treatment in [Leh09, Chap.2, §6] regarding *pseudo Drinfeld modules*. Since we are only concerned with Drinfeld modules over rigid spaces, we will restrict our attention to the ringed space in op.cit. being a rigid analytic L -space. Consider a locally free sheaf $\mathcal{L}$ of rank one on X . Analogous to the situation of line bundles over schemes, one can construct a group object $\mathbb{G}_{a,\mathcal{L}} := \mathrm{Sym}_{\mathcal{O}}(\mathcal{L}^{-1}) := \bigoplus_{i \geq 0} \mathcal{L}^{-i}$. Analogous considerations as before gives us the non-commutative ring of endomorphisms $\mathrm{End}^{\mathrm{bdl}}(\mathbb{G}_{a,\mathcal{L}})$ (see [Leh09, Chap.2, Defn.6.1] and discussion above it for details).

Definition 8.2.7. (i) A *Drinfeld A -module of rank r over X* is a pair $(\mathcal{L}, \phi)$ such that $\phi : A \rightarrow \mathrm{End}^{\mathrm{bdl}}(\mathbb{G}_{a,\mathcal{L}})$ is a ring homomorphism such that, for all admissible open $U \subset X$ trivializing $\mathcal{L}$, that is $\mathcal{L}|_U \simeq \mathcal{O}_U$, the induced homomorphism obtained by restriction, $\phi_U : A \rightarrow \mathrm{End}_{\mathbb{F}_q}(\mathbb{G}_{a,\widetilde{\mathcal{L}(U)}})$, is a Drinfeld A -module of rank r over $\mathrm{Spec}(\mathcal{O}_X(U))$, in the sense of Definition 8.2.2(ii), where $\widetilde{}$ is the quasi-coherent sheaf construction of $\mathcal{O}_X(U)$ -modules.

Morphisms between two Drinfeld A -modules over X is defined in a natural manner.

(ii) Let $(\mathcal{L}, \phi)$ be a Drinfeld A -module of rank r over X and $I \subset A$ be a nonzero proper ideal. Then a *level I structure* is a homomorphism of A -modules

$$i : (I^{-1}/A)^r \rightarrow H^0(X, \mathcal{L}),$$

such that the restriction to an admissible open $U \subset X$ trivializing $\mathcal{L}$, induces a level I structure on $\mathrm{Spec}(\mathcal{O}_X(U))$ in the sense of Remark 8.2.3(ii). A morphism between two Drinfeld A -modules over X with a level I structures is defined analogous to that in the previous subsection.

Similar to $\mathbb{M}_I^r$, we can define the moduli functor

$$\mathbb{RM}_I^r : \mathbf{Rig}_{K_\infty} \rightarrow \mathbf{Sets},$$

sending X to the isomorphism classes of Drinfeld A -modules of rank r over X with a level I structure. Let $\mathbb{RM}^r$ denote the functor sending X to the isomorphism classes of Drinfeld A -modules of rank r over X .

We next present the definition of an *A-lattices over X*, which are used to construct important instances of Drinfeld modules over X .

Definition 8.2.8. ([Boe02, Defn.4.1])

- (i) Let B be an affinoid algebra over L and $|\cdot|_B$ denote a residue norm on B . Let $i_B : A \rightarrow B$ be the canonical map. Then an A -module $\Lambda \subset B$ is said to be an *A-lattice of rank r in B* if it satisfies the following:
- (1) Λ is a projective A -module of rank r ,
 - (2) the elements of $\Lambda \setminus \{0\}$ are units,
 - (3) for all $c \in \mathbb{R}_{>0}$, the set $\{x \in \Lambda \setminus \{0\} \mid |x^{-1}|_B \geq c\}$ is finite.
- (ii) An *A-lattice of rank r over X* is a triple $(\mathcal{L}, \underline{\Lambda}, s)$, consisting of
- 1. a locally free sheaf $\mathcal{L}$ of rank one over X ,
 - 2. a sheaf $\underline{\Lambda}$ of projective A -modules of rank r over the rigid analytic site of X ,
 - 3. a monomorphism $s : \underline{\Lambda} \rightarrow \mathcal{L}$ such that, for any affinoid open set U of X , over which $\mathcal{L}$ can be trivialized, $\underline{\Lambda}(U)$ is an A -lattice of rank r in $\mathcal{M}(U)$ in the sense of part (i).

A morphism between two A -lattices of rank r over X , $(\Lambda, \mathcal{L}, s) \rightarrow (\Lambda', \mathcal{L}', s')$ is defined to be a pair of maps $\Lambda \rightarrow \Lambda'$ and $\mathcal{L} \rightarrow \mathcal{L}'$ so that they commute with s and s' .

- (iii) Let $I \subset A$ be a non-zero proper ideal and $(\mathcal{L}, \underline{\Lambda}, s)$ be a rank r A -lattice over X . Moreover denote by $(I^{-1}/A)_X^r$ the constant A -module sheaf with stalk I^{-1}/A over points of X . Then a *level I -structure of $(\mathcal{L}, \underline{\Lambda}, s)$* is an A -module isomorphism of sheaves $i : (I^{-1}/A)_X^r \simeq I^{-1}\underline{\Lambda}/\underline{\Lambda}$. Morphisms between A -lattices over X with a level I structure is defined in the canonical manner.

Denote by $\mathbb{L}^r : \mathbf{Rig}_{K_\infty} \rightarrow \mathbf{Sets}$ the functor sending X to the isomorphism classes of rank r A -lattices over X . Similarly define $\mathbb{L}_I^r$ to be the functor sending X to the isomorphism classes of rank r A -lattices over X equipped with a level I structure.

The above definition is motivated from the correspondence between A -lattices in $\mathbb{C}_\infty$ and Drinfeld A -modules over $\mathbb{C}_\infty$ (cf. [Dri74, Prop.3.1], [Pap23, Thm.5.2.8]), and was generalized to the following in [Boe02, Prop. 4.2],

Proposition 8.2.9. ([Boe02, Prop.4.7]) There is a natural transformation $\mathbb{L}^r \rightarrow \mathbb{RM}^r$ such that it is an equivalence on $\mathrm{Sp}(\mathbb{C}_\infty)$. Moreover it also restricts to a natural transformation $\mathbb{L}_I^r \rightarrow \mathbb{RM}_I^r$.

Proof. We only sketch the proof of associating to an A -lattice over X a Drinfeld A -module, and refer the reader to op.cit. for more details. Firstly, passing to a trivializing cover of $\mathcal{M}$, we assume that $X = \mathrm{Sp}(B)$. Given an A -lattice Λ in B , consider the exponential function $e_\Lambda : B \rightarrow B$ given by

$$e_\Lambda(x) = x \prod_{\lambda \in \Lambda \setminus \{0\}} \left(1 - \frac{x}{\lambda}\right).$$

This is an everywhere convergent power series and hence defines a rigid analytic function on B , although unlike in the $\mathrm{Sp}(\mathbb{C}_\infty)$ case, e_Λ may no longer be surjective. Then, for each $a \in A$, the polynomial $\phi_a(z) \in B[z]$ given by

$$\phi_a(z) := az \prod_{h \in a^{-1}\Lambda/\Lambda \setminus \{0\}} \left(1 - \frac{z}{e_\Lambda(h)}\right)$$

is the unique function satisfying $\phi_a(e_\Lambda(x)) = e_\Lambda(ax)$. Hence $\phi : A \rightarrow B[\tau]$ is a Drinfeld A -module over $\mathrm{Sp}(B)$.

Now let X be an arbitrary rigid analytic space over K_∞ and $(\mathcal{M}, \underline{\Lambda}, s)$ be an A -lattice of rank r over X . We choose an admissible affinoid covering $\mathcal{U} := \{U_i\}_{i \in \mathcal{Q}}$ of X so that $\mathcal{M}$ is trivialized by $\mathcal{U}$. Each triple $(\mathcal{M}|_{U_i}, \underline{\Lambda}|_{U_i}, s|_{U_i})$ as described above determines a Drinfeld A -module $\phi_i : A \rightarrow \mathcal{M}(U_i)[\tau]$. Consequently, we obtain a Drinfeld A -module $\phi : A \rightarrow \Gamma_{X, \mathcal{M}}[\tau]$ such that $\phi|_{U_i} = \phi_i$ and it is independent of the choice of $\mathcal{U}$. Thus ϕ is a Drinfeld A module over X in the sense of Definition 8.2.7 obtained from the A -lattice $(\mathcal{M}, \underline{\Lambda}, s)$. $\square$

Example - A -lattices on Ω^r

Here we will describe the construction of certain Drinfeld A -modules on Ω^r associated to certain A -lattices over Ω^r , in the light of [Boe02, §4.3] and [BBP24, §7].

Let $g \in \mathrm{GL}_r(\mathbb{A}^f)$ and recall the projective A -module $\Lambda_g := \widehat{A}^r g^{-1} \cap K^r$ from Remark 8.1.16. Denote by $\underline{\Lambda}_g$ the constant sheaf on Ω^r with stalks Λ_g . Then there is an associated injection

$$s_g : \underline{\Lambda}_g \rightarrow \mathcal{O}_{\Omega^r},$$

obtained from the map $\Lambda_g \rightarrow \mathcal{O}_{\mathbb{P}^{r-1}}(\mathbb{P}^{r-1} \setminus \{X_r = 0\})$ sending

$$(a_1, \dots, a_r) \mapsto a_1 \frac{X_1}{X_r} + \dots + a_{r-1} \frac{X_{r-1}}{X_r} + a_r.$$

It is not difficult to check that the triple $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g)$ forms an A -lattice on Ω^r . One can also introduce a canonical level I structure on this A -lattice due to the following isomorphism of A -modules

$$I^{-1}A^r/A^r \simeq I^{-1}\widehat{A}^r/\widehat{A}^r \simeq I^{-1}\widehat{A}^r g^{-1}/\widehat{A}^r g^{-1} \simeq I^{-1}\Lambda_g/\Lambda_g.$$

The second isomorphism is multiplication by g^{-1} and the last isomorphism amounts to checking locally that the canonical map $I^{-1}\Lambda_g/\Lambda_g \rightarrow I^{-1}\widehat{A}^r g^{-1}/\widehat{A}^r g^{-1}$ is an isomorphism. The induced level structure on $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g)$ will be denoted by ψ^g .

Definition 8.2.10. (cf. [BBP24, §7]) Denote by $(\mathbb{G}_{a,\Omega^r}, \varphi^g, \psi^g)$ the Drinfeld A -module with level I structure on Ω^r associated to $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g, \psi^g)$ via the functor in Proposition 8.2.9.

Now we describe the translation of $(\mathbb{G}_{a,\Omega^r}, \varphi^g, \psi^g)$ to another Drinfeld A -module by $\gamma \in \mathrm{GL}_r(K)$. We will describe the action on the pair $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g)$ and this induces an action on the tuple $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g, \psi^g)$.

Consider the morphism of sheaves

$$\mathcal{O}_{\Omega^r} \rightarrow \gamma^* \mathcal{O}_{\Omega^r} = \gamma_*^{-1} \mathcal{O}_{\Omega^r}$$

defined by sending, for an admissible open $U \subset \Omega^r$,

$$f \in \mathcal{O}_{\Omega^r}(U) \mapsto j(\gamma^{-1}, -)f^{\gamma^{-1}} \in \mathcal{O}_{\Omega^r}(\gamma(U)),$$

where $f^{\gamma^{-1}}$ is the image of f under the canonical isomorphism $\mathcal{O}_{\Omega^r}(U) \simeq \mathcal{O}_{\Omega^r}(\gamma(U))$. Since we are regarding the elements of K^r as row vectors hence we have a canonical map of projective A -modules

$$\Lambda_g \rightarrow \Lambda_g \gamma^{-1} = \Lambda_{\gamma g}.$$

Thus we obtain an induced action on A -lattices over Ω^r , consequently inducing an action on Drinfeld A -modules over Ω^r

$$(\mathbb{G}_{a,\Omega^r}, \varphi^g) \rightarrow (\gamma^* \mathbb{G}_{a,\Omega^r}, \varphi^{\gamma g}),$$

where the RHS denotes the Drinfeld module corresponding to the action by γ on the triple $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g)$. Moreover if $\gamma \in \Gamma_g(I)$, then the above action sends level I structure ψ^g to a level I structure $\psi^{\gamma g}$ (see [BBP24, Eq.7.8]). As a result, we obtain a canonical Drinfeld A -module with a level I structure on the quotient $\Gamma_g(I) \backslash \Omega^r$ using the covering $\{\overline{W}_x^c\}_{x \in \mathcal{BT}_0}$ described in Proposition 8.1.18 for a rational number $0 < c < 1$.

Definition 8.2.11. Denote by $(\mathbb{G}_{a,\Gamma_g(I) \backslash \Omega^r}, \overline{\varphi}^g, \overline{\psi}^g)$ the Drinfeld A -module on $\Gamma_g(I) \backslash \Omega^r$ obtained by descending $(\mathbb{G}_{a,\Omega^r}, \varphi^g, \psi^g)$ via the action of $\Gamma_g(I)$.

Remark 8.2.12. Analogous to [Boe02, Rmk.4.11] we note that the A -lattice $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g, \psi^g)$ on Ω^r does not descend to an A -lattice on $\Gamma_g(I) \backslash \Omega^r$. More precisely, if $c \in \Gamma_g(I) \backslash \mathcal{BT}_0$ is in the class of the vertex $\tilde{x}$ such that $\Gamma_g(I)\tilde{x} \neq 1$, the A -lattice doesn't descend to $\overline{W}_x^c$ since, for certain $y \in \Lambda_g$ the restriction of $s_g(y) \in \mathcal{O}_{\Omega^r}(\Omega^r)$ to $W_{\tilde{x}}^c$ is not $\Gamma_g(I)_{\tilde{x}}$ invariant. Clearly the A -lattice descends to $\overline{W}_x^c$ when $\tilde{x}$ is $\Gamma_g(I)$ stable.

Denote by π_g the quotient map $\Omega^r \rightarrow \Gamma_g(I) \backslash \Omega^r$. Also for any morphism of rigid spaces $f : X \rightarrow Y$ over K_∞ , denote by f^* the map of sets $\mathbb{R}\mathbb{M}_I^r(Y) \rightarrow \mathbb{R}\mathbb{M}_I^r(X)$ associated with the pull back of line bundles via f . For example, we have $\pi_g^*(\mathbb{G}_{a, \Gamma_g(I) \backslash \Omega^r}, \overline{\varphi^g}, \overline{\psi^g}) = (\mathbb{G}_{a, \Omega^r}, \varphi^g, \psi^g)$

For the following, let $\text{Cl}_I := \text{GL}_r(K) \backslash \text{GL}_r(\mathbb{A}_K^f) / \mathcal{K}(I)$. By [BBP24, Prop.8.7], Cl_I is in bijection with the set $K^\times \backslash (\mathbb{A}_K^f)^\times$, which is a finite group by class field theory. We choose a set of representatives $g_1, \dots, g_h \in \text{Cl}_I$ and below by the phrase “ $g \in \text{Cl}_I$ ” we mean the indexing with respect to this choice of representatives.

Theorem 8.2.13. $\mathbb{R}\mathbb{M}_I^r$ is a fine moduli functor representable by $(M_I^r \times_K K_\infty)^{an}$ with the universal Drinfeld A -module induced from $(\mathcal{L}^{un}, \varphi^{un}, \psi^{un})$ and denoted here by $(\mathcal{L}^{un}, \varphi^{un}, \psi^{un})^{an}$. Moreover, there is a natural isomorphism of rigid spaces

$$\bigsqcup_{g \in \text{Cl}_I} \Gamma_g(I) \backslash \Omega^r \simeq (M_I^r \times_K K_\infty)^{an},$$

such that if we denote the embedding of the connected component $\Gamma_g(I) \backslash \Omega^r$ inside $(M_I^r \times_K K_\infty)^{an}$ by i_g , for $g \in \text{Cl}_I$, then

$$i_g^*(\mathcal{L}^{un}, \varphi^{un}, \psi^{un})^{an} = (\mathbb{G}_{a, \Gamma_g(I) \backslash \Omega^r}, \overline{\varphi^g}, \overline{\psi^g}).$$

Proof. The first half of the statement is covered in [Boe02, Thm.4.8]. The content of the latter half is in [Dri74, Prop.6.6]. The reader can also have a look at the discussion surrounding [Boe02, Thm.4.15] in the $r = 2$ case and [BBP24, §8] for the corresponding results on the $\mathbb{C}_\infty$ points of the associated spaces. $\square$

Chapter 9

τ -sheaves, crystals and A -motives

Here we will recollect some results from the theory of τ -sheaves and function field crystals, discussed more elaborately in [Boe02, Chap.7, 8, 9]. Our objective is only to setup enough prerequisites to define certain τ -sheaves constructed from the universal Drinfeld modules defined in the previous chapter as well as discuss some of its properties. With this in view, we first recap the algebraic theory of τ -sheaves and crystals, that is, those defined over a base scheme. Then following Chapter 8 of op.cit., the important notions/results about the corresponding objects over a rigid analytic base are described. Important examples of τ -sheaves are A -motives, in particular those arising from Drinfeld modules over a scheme or a rigid analytic base. It turns out that the τ -sheaves associated to the Drinfeld modules introduced in the previous chapter have certain “uniformizability” properties. This follows directly from the results in [Boe02, Chap.9] and we conclude this chapter by restating them.

9.1 Algebraic theory

Here we mainly follow the discussions in [Boe02, §7.1, 7.2]. Let X, Y etc. denote noetherian schemes over $\mathrm{Spec}(\mathbb{F}_q)$ and fix a $\mathbb{F}_q$ -algebra B locally of finite type. Examples include the Dedekind domain A , its fraction field K and finite fields. We will denote the fiber product over $\mathbb{F}_q$ by $X_B := X \times \mathrm{Spec}(B)$. Also denote by $\sigma_X : X \rightarrow X$ the Frobenius relative to $\mathbb{F}_q$, that is, the morphism of ringed spaces induced from the map $a \mapsto a^q$ on the structure sheaf $\mathcal{O}_X$. We will suppress the subscript X and simply write σ , as the scheme underlying the definition of σ would be clear from the context.

Definition 9.1.1. A pair $\underline{\mathcal{F}} := (\mathcal{F}, \tau)$ is said to be a τ -sheaf over B (or with coefficients in B) on a

scheme X , if $\mathcal{F}$ is a quasi-coherent sheaf on $X \times \text{Spec}(B)$ and τ is a morphism of $\mathcal{O}_{X \times B}$ -sheaves

$$\tau : (\sigma \times 1)^* \mathcal{F} \rightarrow \mathcal{F}.$$

We will adopt the terminology that given a property P of quasi-coherent sheaves on $X \times \text{Spec}(B)$, $\underline{\mathcal{F}}$ is said to have property P if the underlying sheaf $\mathcal{F}$ has the property P . We will mainly be concerned about P being the property “coherent” or “locally free”.

When the algebra B and scheme X are clear from the context, we simply call $\underline{\mathcal{F}}$ a τ -sheaf.

Remark 9.1.2. If $X = \text{Spec}(R)$ for an $\mathbb{F}_q$ -algebra R , then giving a τ -sheaf $\underline{\mathcal{F}}$ is equivalent to giving a $R \otimes_{\mathbb{F}_q} B$ -module M equipped with a homomorphism $\tau_M : M \rightarrow M$ which is q -linear in the R component,

$$\tau_M((r \otimes b).m) = (r^q \otimes b).\tau_M(m).$$

We will call such an M as a τ -module.

Morphisms of τ sheaves are defined to be the morphisms of underlying quasi-coherent sheaves that are compatible with the τ -structure. Denote the resulting category of τ -sheaves by $\mathbf{Quot}_\tau(X, B)$. Its full subcategory consisting of coherent τ -sheaves will be denoted by $\mathbf{Coh}_\tau(X, B)$.

A coherent τ -sheaf $\underline{\mathcal{F}}$ is said to be *nilpotent* if for an integer $n \gg 0$, $\tau^n : (\sigma^n \times 1)^* \mathcal{F} \rightarrow \mathcal{F}$ is zero, where we define iteratively $\tau^{i+1} := \tau \circ (\sigma \times 1)^* \tau^i \forall i \geq 1$ and $\tau^1 := \tau$. We denote the full subcategory of such sheaves by $\mathbf{Nil}_\tau(X, B)$.

It will follow from the definition of pullback of τ -sheaves given in Proposition 9.1.7(i) that τ induces a map of τ -sheaves $(\sigma \times 1)^* \underline{\mathcal{F}} \rightarrow \underline{\mathcal{F}}$ which is again denoted by τ below. Similar remark applies to τ^n as well, for $n \geq 0$.

We summarize from [BP09] some important properties related to $\mathbf{Nil}_\tau(X, B)$ below. For the definitions of Serre subcategories and saturated multiplicative systems, we refer the reader to the relevant portion of op.cit. and any reference contained therein.

Proposition 9.1.3. The following statements hold true:-

- (i) $\mathbf{Nil}_\tau(X, B)$ is a Serre subcategory of $\mathbf{Coh}_\tau(X, B)$.
- (ii) The collection of morphisms in $\mathbf{Coh}_\tau(X, B)$ such that both the cokernel and kernel lie in $\mathbf{Nil}_\tau(X, B)$, form a saturated multiplicative system. Such a morphism will be called a *nil-isomorphism* and we will denote them by $\Rightarrow$ to distinguish them from ordinary morphisms.
- (iii) A morphism of τ -sheaves $\underline{\mathcal{F}} \xrightarrow{\varphi} \underline{\mathcal{G}}$ is a nil-isomorphism if and only if there exists an integer $n \geq 0$ and a morphism of τ -sheaves $\alpha : (\sigma^n \times 1)^* \underline{\mathcal{G}} \rightarrow \underline{\mathcal{F}}$ such that the following diagram commutes

$$\begin{array}{ccc} (\sigma^n \times 1)^* \underline{\mathcal{F}} & \xrightarrow{\tau^n} & \underline{\mathcal{F}} \\ (\sigma^n \times 1)^* \varphi \downarrow & \nearrow \alpha & \downarrow \varphi \\ (\sigma^n \times 1)^* \underline{\mathcal{G}} & \xrightarrow{\tau^n} & \underline{\mathcal{G}} \end{array}$$

Proof. (i) This is [BP09, Prop.3.3.5].

(ii) The nil-isomorphisms are precisely the saturated multiplicative systems of morphisms that correspond to nilpotent τ -sheaves under the bijection in [BP09, Prop.2.3.2].

(iii) See [BP09, Prop.3.3.9]

□

It is clear from using 9.1.3(iii) above or otherwise that, for any integer $n \geq 0$, $\tau^n : (\sigma^n \times 1)^* \underline{\mathcal{F}} \rightarrow \underline{\mathcal{F}}$ is a nil-isomorphism.

Due to 9.1.3(ii) we have the following

Definition 9.1.4. Denote by $\mathbf{Crys}(X, B)$ the localization of $\mathbf{Coh}_\tau(X, B)$ with respect to the collection of nil-isomorphisms. Objects of this category will be called B -crystals on X or simply crystals on X if B is clear from the context.

Remark 9.1.5. (i) Following [Boe02], we consider the morphisms in $\mathbf{Crys}(X, B)$ as *left fractions*. More precisely, given two τ -sheaves $\underline{\mathcal{F}}, \underline{\mathcal{G}}$, a morphism between them regarded as crystals in $\mathbf{Crys}(X, B)$, $\underline{\mathcal{F}} \dashrightarrow \underline{\mathcal{G}}$, is represented by a diagram of the type $\underline{\mathcal{F}} \leftarrow \underline{\mathcal{H}} \rightarrow \underline{\mathcal{G}}$, for some τ -sheaf $\underline{\mathcal{H}}$. By Proposition 9.1.3(iii), any morphism $\underline{\mathcal{F}} \dashrightarrow \underline{\mathcal{G}}$ can be represented by a diagram of the kind $\underline{\mathcal{F}} \xleftarrow{\tau^n} (\sigma^n \times 1)^* \underline{\mathcal{F}} \rightarrow \underline{\mathcal{G}}$ for some integer $n \gg 0$.

(ii) Let $\underline{\mathcal{F}}$ be a τ -sheaf. Define the B -module of τ -invariants of $\underline{\mathcal{F}}$, to be

$$\underline{\mathcal{F}}^\tau(X) := H^0(X \times \mathrm{Spec}(B), \underline{\mathcal{F}})^\tau.$$

This defines a functor $\mathbf{Coh}_\tau(X, B) \rightarrow B\text{-mod}$ sending $\underline{\mathcal{F}} \mapsto \underline{\mathcal{F}}^\tau$. Clearly nil-isomorphic sheaves have isomorphic module of τ -invariants. Hence by abstract nonsense (eg. [BP09, Thm.2.2.3(b)]), the above functor factorizes via a functor $\mathbf{Crys}(X, B) \rightarrow B\text{-mod}$ which we again denote by $\underline{\mathcal{F}} \mapsto \underline{\mathcal{F}}^\tau(X)$.

Example 9.1.6. (i) There is a canonical structure of a τ -sheaf on the structure sheaf $\mathcal{O}_X \otimes B$ of X_B given by the isomorphism induced by σ

$$\tau : (\sigma \times 1)^*(\mathcal{O}_X \otimes B) \simeq \mathcal{O}_X \otimes B$$

We call the pair $(\mathcal{O}_X \otimes B, \tau)$ as the *unit crystal* denoted $\underline{\mathbb{1}}_{X,B}$.

- (ii) Let X be a K -scheme and $\mathbb{E} = (\mathcal{L}, \varphi)$ be a Drinfeld A -module of rank r on X . Consider the quasi coherent sheaf of $\mathcal{O}_X$ modules

$$\mathcal{F}(\mathbb{E}) := \mathrm{Hom}_{\mathbb{F}_q}(\mathbb{G}_{a,\mathcal{L}}, \mathbb{G}_{a,X})$$

where $\mathrm{Hom}_{\mathbb{F}_q}(-, -)$ means the $\mathbb{F}_q$ -homomorphisms between group schemes (see [Leh09, Chap.1, Prop.2.3]). It inherits a $\mathcal{O}_X \otimes A$ module structure by considering the right action of $a \in A$ by right multiplication by φ_a . Left composition by $\tau_q \in \mathrm{End}_{\mathbb{F}_q}(\mathbb{G}_{a,X})$ we obtain a morphism of $\mathcal{O}_X \otimes A$ modules $\mathcal{F}(\mathbb{E}) \rightarrow (\sigma \times 1)_* \mathcal{F}(\mathbb{E})$, which due to adjunction induces a homomorphism

$$\tau_{\mathbb{E}} : (\sigma \times 1)^* \mathcal{F}(\mathbb{E}) \rightarrow \mathcal{F}(\mathbb{E}).$$

We denote by $\underline{\mathcal{F}}(\mathbb{E})$ the τ -sheaf $(\mathcal{F}(\mathbb{E}), \tau_{\mathbb{E}})$ in $\mathbf{Coh}_{\tau}(X, A)$ and call it the A -motive associated to the Drinfeld A -module $\mathbb{E}$. It has the following properties

- (a) $\underline{\mathcal{F}}(\mathbb{E})$ is a locally free τ -sheaf sheaf of rank r on X_A ;
- (b) the cokernel of $\tau_{\mathbb{E}}$ is supported on graph of the structure map $\mathrm{ch} : X \rightarrow \mathrm{Spec}(K) \hookrightarrow \mathrm{Spec}(A)$.

Compare with [BS97, Thm.3.2.1] and [Mum78, §3]. $\underline{\mathcal{F}}(\mathbb{E})$ is a τ -sheaf avatar of Anderson's A -motive, see [And86] and [BP09, Defn.3.5.4].

Let $I \subset A$ be a non-zero proper ideal. We will denote by $\underline{\mathcal{F}}_I^{un}$ the A -motive on the Drinfeld moduli space M_I^r corresponding to the universal Drinfeld A -module $\mathbb{E}_I^{un}$ (see Definition 8.2.5).

We now recall the various functors between the categories of τ -sheaves as well as crystals induced from morphisms between the underlying schemes. For more details, the reader is referred to the relevant portion of [BP09, Chap.4].

Proposition 9.1.7. Let $f : Y \rightarrow X$ denote a morphism of schemes, $j : U \hookrightarrow X$ an open immersion and $i : Z := X \setminus U \hookrightarrow X$ the induced reduced closed subscheme.

- (i) (*Pullback*) The functor $(f \times 1)^* : \mathbf{Coh}(X \times B) \rightarrow \mathbf{Coh}(Y \times B)$ induces naturally a functor on the category of τ -sheaves, denoted $f^* : \mathbf{Coh}_{\tau}(X, B) \rightarrow \mathbf{Coh}_{\tau}(Y, B)$. This induces a functor also on the category of crystals, also denoted by f^* . On crystals f^* is exact.
- (ii) (*Derived pushforward*) Suppose f is proper. Then for $i \geq 0$, the functor $R^i(f \times 1)_* : \mathbf{Coh}(Y \times B) \rightarrow \mathbf{Coh}(X \times B)$ induces naturally a functor on the category of τ -sheaves, denoted $R^i f_* : \mathbf{Coh}_{\tau}(Y, B) \rightarrow \mathbf{Coh}_{\tau}(X, B)$. This also induces a functor on the corresponding category of crystals also denoted by $R^i f_*$.

- (iii) (*Tensor product*) Let $\underline{\mathcal{F}} = (\mathcal{F}, \tau_{\mathcal{F}}), \underline{\mathcal{G}} = (\mathcal{G}, \tau_{\mathcal{G}})$ be two τ -sheaves in $\mathbf{Coh}_{\tau}(X, B)$. Then the pair $(\mathcal{F} \otimes_{\mathcal{O}_{X \times B}} \mathcal{G}, \tau_{\mathcal{F}} \otimes \tau_{\mathcal{G}})$ defines a τ sheaf and induces a B -linear functor

$$\otimes : \mathbf{Coh}_{\tau}(X, B) \times \mathbf{Coh}_{\tau}(X, B) \rightarrow \mathbf{Coh}_{\tau}(X, B).$$

It further induuces a B -linear functor on the level of crystals as well. For $n \geq 0$ one canonically defines the τ -sheaves, and consequently also crystals, $\mathrm{Sym}^n \underline{\mathcal{F}}$ and $\wedge^n \underline{\mathcal{F}}$ as appropriate quotient of $\underline{\mathcal{F}}^{\otimes n}$.

- (iv) (*Extension by zero*) There exists a functor $j_! : \mathbf{Crys}(U, B) \rightarrow \mathbf{Crys}(X, B)$ which is uniquely characterized by the property that

- $j^* j_! = \mathrm{id}_{\mathbf{Crys}(U, B)}$;
- $i^* j_! = 0$.

- (v) (*Cohomology with compact supports*) Suppose $f : X \rightarrow Y$ be compactifiable, that is there exists a proper morphism $\bar{f} : \bar{X} \rightarrow Y$ such that there is an open immersion $j : X \hookrightarrow \bar{X}$. Then the functor

$$R^i \bar{f}_* \circ j_! : \mathbf{Crys}(X, B) \rightarrow \mathbf{Crys}(Y, B)$$

is independent of the chosen compactification.

Proof. See relevant section in [BP09, Chap.4]. Alternately one can also refer to [Boe02, §7.2]. $\square$

Remark 9.1.8. 9.1.7(iv) provides us with a blueprint to construct a representative τ -sheaf $\widetilde{\mathcal{F}}$ of the crystal $j_! \underline{\mathcal{F}}$:- if $j^* \widetilde{\mathcal{F}} = \underline{\mathcal{F}}$ and $i^* \widetilde{\mathcal{F}}$ is nilpotent, then $\widetilde{\mathcal{F}}$ represents $j_! \underline{\mathcal{F}}$.

9.2 Analytic theory

This section is a summary some main results in [Boe02, Chap.8]. One can develop analogous to the previous section a theory of τ -sheaves and crystals on rigid analytic spaces. Such a theory is necessary to show “uniformizability” of A -motives (to be defined below) associated to Drinfeld modules on Ω^r , to be discussed in the last section of this chapter.

We will denote by $\mathfrak{X}$ a rigid analytic space over a complete non-Archimedean field L such that $K_{\infty} \subset L \subset \mathbb{C}_{\infty}$. Let $\mathfrak{B}$ denote a rigid analytic Stein domain over L and set $\mathcal{B} := H^0(\mathfrak{B}, \mathcal{O}_{\mathfrak{B}})$. Denote by $\sigma_{\mathfrak{X}} : \mathfrak{X} \rightarrow \mathfrak{X}$ the map induced from the q -power map on affinoid subdomains and by $\sigma_{L, \mathfrak{B}}$ the pull back of the q -power Frobenius on L to $\mathfrak{B}$.

Definition 9.2.1. A rigid τ -sheaf on $\mathfrak{X}$ with coefficients in $\mathcal{B}$ is a pair $(\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}})$ consisting of a coherent $\mathcal{O}_{\mathfrak{X} \times \mathfrak{B}}$ -modules equipped with a $\mathcal{O}_{\mathfrak{X} \times \mathfrak{B}}$ -linear homomorphism

$$\tau_{\tilde{\mathcal{F}}} : (\sigma_{\mathfrak{X}} \times \sigma_{L, \mathfrak{B}})^* \tilde{\mathcal{F}} \rightarrow \tilde{\mathcal{F}}.$$

Remark 9.2.2. Analogous to 9.1.2, if $\mathfrak{X} = \mathrm{Sp}(\mathcal{R})$ for an affinoid algebra $\mathcal{R}$ over L and suppose $\mathfrak{B}$ is also an affinoid L -space, then giving a rigid τ -sheaf $\tilde{\mathcal{F}}$ on $\mathfrak{X}$ with coefficients in $\mathcal{B}$ is equivalent to providing a finitely generated $\mathcal{R} \hat{\otimes}_L \mathcal{B}$ -module $\mathcal{M}$ equipped with an additive map

$$\tau : \mathcal{M} \rightarrow \mathcal{M}$$

such that for $r \otimes b \in \mathcal{R} \otimes_L \mathcal{B}$ and $m \in \mathcal{M}$, we have $\tau((r \otimes b)m) = (r^q \otimes \sigma_{L, \mathfrak{B}}(b))\tau(m)$ where by slight abuse of notation we also denote by $\sigma_{L, \mathfrak{B}}$ the pullback of the q -power map on L to $\mathcal{B}$. We will call $\mathcal{M}$ a τ -module on $\mathcal{R}$ with coefficients in $\mathcal{B}$.

Morphisms between rigid τ -sheaves are defined canonically and the resulting category of such sheaves will be denoted by $\widetilde{\mathbf{Coh}}_{\tau}(\mathfrak{X}, \mathcal{B})$. If we denote by $\mathcal{B}^{\sigma}$ by elements of $\mathcal{B}$ fixed with respect to the Frobenius $x \mapsto x^q$, then $\widetilde{\mathbf{Coh}}_{\tau}(\mathfrak{X}, \mathcal{B})$ is an abelian $\mathcal{B}^{\sigma}$ -linear category.

Given a τ -sheaf $\tilde{\mathcal{F}} = (\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}})$ we can iteratively define the morphism of rigid τ -sheaves $\tau_{\tilde{\mathcal{F}}}^n : (\sigma_{\mathfrak{X}}^n \times \sigma_{L, \mathfrak{B}}^n)^* \tilde{\mathcal{F}} \rightarrow \tilde{\mathcal{F}}$ analogous to that in the previous section. If $j : \mathfrak{U} \rightarrow \mathfrak{X}$ is an admissible open subdomain then it can be check that the restriction $\tilde{\mathcal{F}}|_{\mathfrak{U}} := (j \times 1_{\mathfrak{B}})^* \tilde{\mathcal{F}}$ is a well defined rigid τ -sheaf on $\mathfrak{U} \times \mathfrak{B}$.

Definition 9.2.3. A rigid τ -sheaf $(\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}})$ is said to be *nilpotent* if and only if for each affinoid subdomain $\mathfrak{U}$ of $\mathfrak{X}$, there exists $n \in \mathbb{N}$ such that $\tau_{\tilde{\mathcal{F}}|_{\mathfrak{U}}}^n = 0$.

A morphism of rigid τ -sheaves $(\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}}) \rightarrow (\tilde{\mathcal{G}}, \tau_{\tilde{\mathcal{G}}})$ is said to be a *nil-isomorphism* if its kernel and cokernel are nilpotent.

Denote by $\mathbb{1}_{\mathfrak{X}, \mathcal{B}}$ the rigid τ -sheaf on the structure sheaf $\mathcal{O}_{\mathfrak{X} \times \mathfrak{B}}$ similar to 9.1.6(i).

Analogous to Proposition 9.1.3 it can be shown that the subcategory of nilpotent rigid τ -sheaves form a Serre subcategory and hence nil-isomorphisms form a saturated multiplicative system.

Definition 9.2.4. Denote by $\widetilde{\mathbf{Crys}}(\mathfrak{X}, \mathfrak{B})$ the localization of $\widetilde{\mathbf{Coh}}_{\tau}(\mathfrak{X}, \mathcal{B})$ with respect to the multiplicative system of nil-isomorphisms. Objects of $\widetilde{\mathbf{Crys}}(\mathfrak{X}, \mathfrak{B})$ will be called rigid crystals over $\mathfrak{X}$ with coefficients in $\mathcal{B}$ or simply rigid crystals over $\mathfrak{X}$ if $\mathcal{B}$ is clear from the context.

Now we will describe functors on the categories of rigid τ -sheaves as well as crystals similar to Proposition 9.1.7. All the functors there except, extension by zero and cohomology with compact supports, have a straightforward analogue in the rigid setting.

For a rigid space $\mathfrak{X}$, let $\mathbf{RCoh}(\mathfrak{X})$ denote the category of coherent sheaves on $\mathfrak{X}$.

Proposition 9.2.5. Let $\tilde{f} : \mathfrak{Y} \rightarrow \mathfrak{X}$ denote a morphism of rigid analytic L spaces.

- (i) (*Pullback*) The functor $(\tilde{f} \times 1)^* : \mathbf{RCoh}(\mathfrak{X} \times \mathfrak{B}) \rightarrow \mathbf{RCoh}(\mathfrak{Y} \times \mathfrak{B})$ induces naturally a functor on the category of rigid τ -sheaves, denoted $\tilde{f}^* : \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(\mathfrak{Y}, \mathcal{B})$. This induces a functor also on the corresponding category of crystals, also denoted by $\tilde{f}^*$, which is $\mathcal{B}^\sigma$ linear and exact.
- (ii) (*Direct image*) Suppose $\tilde{f} : \mathfrak{Y} \rightarrow \mathfrak{X}$ is proper morphism (see [Bos14, Chap.6.3, Defn.8]). Then for $i \geq 0$, the functor $R^i(\tilde{f} \times 1)_* : \mathbf{RCoh}(\mathfrak{Y} \times \mathfrak{B}) \rightarrow \mathbf{RCoh}(\mathfrak{X} \times \mathfrak{B})$ induces naturally a functor on the category of rigid τ -sheaves, denoted $R^i \tilde{f}_* : \widetilde{\mathbf{Coh}}_\tau(\mathfrak{Y}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B})$. This also induces a functor on the corresponding category of rigid crystals also denoted by $R^i \tilde{f}_*$ which is $\mathcal{B}^\sigma$ linear.
- (iii) (*Tensor product*) Let $(\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}})$ and $(\tilde{\mathcal{G}}, \tau_{\tilde{\mathcal{G}}})$ be rigid τ sheaves on $\mathfrak{X}$ with coefficients in $\mathcal{B}$. Then $(\tilde{\mathcal{F}} \otimes \tilde{\mathcal{G}}, \tau_{\tilde{\mathcal{F}}} \otimes \tau_{\tilde{\mathcal{G}}})$ is also in $\widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B})$ and this induces a $\mathcal{B}^\sigma$ linear functor

$$\otimes : \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B}) \times \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B})$$

which is right exact in each factor. Furthermore it also induces a $\mathcal{B}^\sigma$ -linear functor on the level of rigid crystals. Consequently one can also form $\mathrm{Sym}^n(\underline{\tilde{\mathcal{F}}})$ and $\wedge^n \tilde{\mathcal{F}}$.

- (iv) (*Change of coefficients*) Let $\mathfrak{B}'$ be a Stein domain over L with the entire functions on it denoted by $\mathcal{B}'$, $h : \mathfrak{B}' \rightarrow \mathfrak{B}$ a map of rigid spaces and $(\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}}) \in \mathbf{RCoh}(\mathfrak{X} \times \mathfrak{B})$. Then the pair $(\tilde{\mathcal{F}} \otimes_{\mathcal{B}} \mathcal{B}', \tau_{\tilde{\mathcal{F}}} \otimes 1)$ is a rigid τ sheaf with coefficients in $\mathcal{B}'$ inducing a right exact $\mathcal{B}^\sigma$ linear functor between the corresponding categories of crystals

$$\otimes_{\mathcal{B}}^{\mathcal{B}'} : \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B}').$$

It is compatible with taking pullback and transitive with respect to changing coefficients. Analogous to (iii), there is also an induced functor on the level of rigid crystals compatible with pullback.

Proof. See the relevant portions of [Boe02, §8.2]. □

Remark 9.2.6. Given a rigid τ -sheaf $\underline{\tilde{\mathcal{F}}}$ on $\mathfrak{X}$ and $\mathfrak{U}$ an admissible open subset of $\mathfrak{X}$, below we will denote by $\tilde{\mathcal{F}}|_{\mathfrak{U}}$ the pull back $\underline{\tilde{\mathcal{F}}}$ along the open immersion $\mathfrak{U} \rightarrow \mathfrak{X}$.

9.2.1 Extension by zero and direct image with compact support

Let $\mathfrak{U}$ be an admissible open subset of $\mathfrak{X}$ denoted by an open immersion $j : \mathfrak{U} \rightarrow \mathfrak{X}$ and $i : \mathfrak{Z} \rightarrow \mathfrak{X}$ denote the inclusion of the complement. Unlike the functors described so far in this section which were a straightforward generalization of their analogues from the algebraic theory in 9.1, it is apriori unclear whether a functor $j_!$ as in 9.1.7(iv) exists. The following observations showcase that one cannot hope to obtain such a functor from $\widetilde{\mathbf{Coh}}_\tau(\mathfrak{U}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{B})$:-

- (a) In general, there exist coherent sheaves on $\mathfrak{U}$ which admit no coherent extension to $\mathfrak{X}$.
- (b) If a coherent extension of coherent sheaf on $\mathfrak{U}$ exists, one could define a τ map which has essential singularities at $\mathfrak{Z}$.

For instance, one could take $\mathfrak{X} := \{z \in \mathbb{C}_\infty \mid |z| \leq 1\}$ and $\mathfrak{U}$ to be the punctured disk $\{z \in \mathbb{C}_\infty \mid 0 < |z| \leq 1\}$ and construct examples where (a) and (b) hold.

To rectify this situation one restricts to the setting where all the rigid spaces resp. maps above are analytifications of schemes of finite type resp. morphisms between schemes. More precisely, there exist schemes of finite type U and X over L such that $\mathfrak{U} = U^{an}$, $\mathfrak{X} = X^{an}$ and j is the analytification of an open immersion $U \hookrightarrow X$. Moreover $\mathfrak{Z}$ is the analytification Z^{an} of a closed subscheme $Z \hookrightarrow X$ defined by a sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$. $\mathcal{I}$ is far from being unique but we fix a choice, which suffices for our purposes. We will call a morphism of rigid spaces $\mathfrak{X} \rightarrow \mathfrak{Y}$ *algebraic* if both $\mathfrak{X}, \mathfrak{Y}$ are analytification of schemes over L and that f arises as analytification of a morphism of the underlying schemes.

In this setup one further narrows down the domain category for the prospective functor $j_!$, which reveals itself due to the following observation (see [Boe02, Lemmma 8.27]). Suppose $j_1 : \mathfrak{U} \rightarrow \mathfrak{Y}_1$ and $j_2 : \mathfrak{U} \rightarrow \mathfrak{Y}_2$ be two *algebraic compactifications*, that is, for $i = 1, 2$ there exist proper, finite type schemes Y_i such that $\mathfrak{Y}_i = Y_i^{an}$ and j_i is the analytification of an open immersion $U \hookrightarrow Y_i$. Suppose there exists a morphism $g : \mathfrak{Y}_1 \rightarrow \mathfrak{Y}_2$ so that $g \circ j_1 = j_2$. Then $j_1^* \widetilde{\mathbf{Coh}}_\tau(\mathfrak{Y}_1, \mathcal{B}) = j_2^* \widetilde{\mathbf{Coh}}_\tau(\mathfrak{Y}_2, \mathcal{B}) \subset \widetilde{\mathbf{Coh}}_\tau(\mathfrak{U}, \mathcal{B})$. This motivates the following definition.

Definition 9.2.7. Let $j' : \mathfrak{U} \rightarrow \mathfrak{Y}'$ be an algebraic compactification. Then define

$$\widetilde{\mathbf{Coh}}_\tau^e(\mathfrak{U}, \mathcal{B}) := (j')^* \widetilde{\mathbf{Coh}}_\tau(\mathfrak{Y}', \mathcal{B}).$$

The objects of $\widetilde{\mathbf{Coh}}_\tau^e(\mathfrak{U}, \mathcal{B})$ will be called the *extendible rigid τ -sheaves*.

Denote by $\widetilde{\mathbf{Crys}}_\tau^e(\mathfrak{U}, \mathcal{B})$ the essential image of $\widetilde{\mathbf{Coh}}_\tau^e(\mathfrak{U}, \mathcal{B})$ under the localization functor $\widetilde{\mathbf{Coh}}_\tau(\mathfrak{U}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Crys}}_\tau(\mathfrak{U}, \mathcal{B})$ whose objects will be called the *extendible rigid crystals*.

We now have all the necessary ingredients to define an extension by zero functor

Theorem-Definition 9.2.8. ([Boe02, Thm.8.35, Prop.8.36])

- (i) For every extendible rigid crystal $\tilde{\mathcal{F}}$ on $\mathfrak{U}$ there exists an extendible rigid crystal $\tilde{\mathcal{G}}$ on $\mathfrak{X}$ such that $\tilde{\mathcal{F}} \simeq j^* \tilde{\mathcal{G}}$ and $i^* \tilde{\mathcal{G}} = 0$. This defines a functor $j_! : \widetilde{\mathbf{Crys}}^e(\mathfrak{U}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Crys}}^e(\mathfrak{X}, \mathcal{B})$ uniquely characterized by the property that there exists a natural isomorphism of functors $j^* j_! \simeq \text{id}$ and $i^* j_! = 0$.
- (ii) $j_!$ is exact and the following is a short exact sequence of functors on $\widetilde{\mathbf{Crys}}^e(\mathfrak{X}, \mathcal{B})$

$$0 \rightarrow j_! j^* \rightarrow \text{id} \rightarrow i_* i^* \rightarrow 0.$$

Moreover $j_!$ is compatible with the change of coefficients functor.

Now let $\mathfrak{V}$ be the analytification of a scheme of finite type V over L . An algebraic morphism $f : \mathfrak{V} \rightarrow \mathfrak{X}$ is said to be *algebraically compactifiable* if there exists an algebraic proper morphism $\bar{f} : \mathfrak{W} \rightarrow \mathfrak{X}$ and an algebraic open immersion $j : \mathfrak{V} \rightarrow \mathfrak{W}$ such that obvious triangle commutes.

Definition 9.2.9. Let $f : \mathfrak{V} \rightarrow \mathfrak{X}$ be algebraic compactifiable with respect to $\mathfrak{W}$ as above. Then define

$$R^i f_! := R^i \bar{f}_* \circ j_! : \widetilde{\mathbf{Crys}}^e(\mathfrak{V}, \mathcal{B}) \rightarrow \widetilde{\mathbf{Crys}}^e(\mathfrak{X}, \mathcal{B})$$

for $i \geq 0$ as the *i -th right derived direct image with compact supports*.

Analogous to their classical analogues, it turns out that $R^i f_!$ is independent of the chosen compactification of f and is a well defined functor (see [Boe02, Prop.8.40]).

9.2.2 Rigid τ sheaves with algebraic coefficients

Now let B be a finite type $\mathbb{F}_q$ algebra as in section 9.1. Here we will briefly discuss the definition of rigid τ -sheaves with B coefficients, which arise naturally when considering the analytification of A -motives to a rigid analytic base from 9.1.6(ii) ([Boe02, §8.5]).

Definition 9.2.10. (i) A sheaf $\mathcal{F}$ of $\mathcal{O}_{\mathfrak{X}}$ -modules is said to be *quasi coherent*, if it is a direct limit of coherent sheaves on $\mathfrak{X}$.

- (ii) Denote by $B_{\mathfrak{X}}$ the quasi coherent sheaf, which is defined on admissible affinoid open $\mathfrak{U} \subset \mathfrak{X}$ by $B_{\mathfrak{X}}(\mathfrak{U}) := \mathcal{O}_{\mathfrak{X}}(\mathfrak{U}) \otimes_{\mathbb{F}_q} B$, equipped with the obvious restriction maps.

A quasi coherent sheaf $\mathcal{F}$ on $\mathfrak{X}$ is said to be a *coherent $B_{\mathfrak{X}}$ -module* if for every admissible affinoid open $\mathfrak{U}$ of $\mathfrak{X}$, $\mathcal{F}(\mathfrak{U})$ is finitely generated as a $B_{\mathfrak{X}}(\mathfrak{U})$ module, compatible with the restriction maps. Analogously a coherent $B_{\mathfrak{X}}$ -module $\mathcal{F}$ is called a *locally free sheaf of rank r* if for every admissible affinoid open $\mathfrak{U}$ of $\mathfrak{X}$, $\mathcal{F}(\mathfrak{U})$ is projective $B_{\mathfrak{X}}(\mathfrak{U})$ -module of rank r .

It turns out that kernels, cokernels, images and coimages of a map of coherent $B_{\mathfrak{X}}$ modules is again a coherent $B_{\mathfrak{X}}$ module. In fact, the category of coherent $B_{\mathfrak{X}}$ modules is abelian ([Boe02, Prop.8.47]).

Denote by $\sigma_{\mathfrak{X}/B}$ the map $\sigma_{\mathfrak{X}} \otimes 1$ on $\mathcal{O}_{\mathfrak{X}} \otimes B$. Then $\sigma_{\mathfrak{X}/B}^*$ preserves coherence of $B_{\mathfrak{X}}$ modules.

Definition 9.2.11. A pair $\tilde{\mathcal{F}} := (\tilde{\mathcal{F}}, \tau_{\tilde{\mathcal{F}}})$ is called a *rigid τ sheaf on $\mathfrak{X}$ with coefficients in B* if $\tilde{\mathcal{F}}$ is a coherent sheaf of $B_{\mathfrak{X}}$ modules and

$$\tau_{\tilde{\mathcal{F}}} : (\sigma_{\mathfrak{X}/B})^* \tilde{\mathcal{F}} \rightarrow \tilde{\mathcal{F}}$$

is a morphism of $B_{\mathfrak{X}}$ modules.

Denote the category of such sheaves by $\widetilde{\mathbf{Coh}}_{\tau}(\mathfrak{X}, B)$ which is a B -linear abelian category.

Example 9.2.12. (cf. 9.4) Let $(\mathcal{L}, \phi)$ be a Drinfeld A -module on $\mathfrak{X}$ of rank r . Let $\mathfrak{G}/\mathfrak{X}$ denote the category of rigid locally free sheaves on $\mathfrak{X}$ with $\mathbb{F}_q$ -linear homomorphisms $\alpha : \mathcal{V} \rightarrow \mathcal{W}$, that is locally on a affinoid $\mathfrak{U} = \mathrm{Sp}(\mathcal{R})$, α is given by maps $\mathcal{R}^n \rightarrow \mathcal{R}^m$ of the form $\sum_{0 \leq i \leq N} A_i \sigma_{\mathfrak{U}}^i$, where $A_i \in M_{m \times n}(\mathcal{R})$. Analogous to 9.1.6(ii), consider the following quasi coherent sheaf on $\mathfrak{X}$

$$\tilde{\mathcal{F}}(\mathcal{L}, \phi) := \mathrm{Hom}_{\mathfrak{G}/\mathfrak{X}}(\mathcal{L}, \mathcal{O}_{\mathfrak{X}}).$$

Considering the multiplication of A via ϕ , it turns out that $\tilde{\mathcal{F}}(\mathcal{L}, \phi)$ is a coherent sheaf of $A_{\mathfrak{X}}$ modules, which is locally free of rank r . The left composition by $\sigma_{\mathfrak{X}}$ induces a canonical morphism $\tilde{\tau} : (\sigma_{\mathfrak{X}/A})^* \tilde{\mathcal{F}}(\mathcal{L}, \phi) \rightarrow \tilde{\mathcal{F}}(\mathcal{L}, \phi)$. The pair $\tilde{\mathcal{F}}(\mathcal{L}, \phi) := (\tilde{\mathcal{F}}(\mathcal{L}, \phi), \tilde{\tau})$ is rigid τ sheaf with coefficients in A , that will be called a *rigid A -motive on $\mathfrak{X}$* .

Let $I \subset A$ be a non-zero proper ideal and $g \in \mathrm{GL}_r(\mathbb{A}_K^f)$ be given. We will denote by $\tilde{\mathcal{F}}^g$ resp. $\tilde{\mathcal{F}}_I^g$ the rigid A -motive on Ω^r resp. $\Gamma_g(I) \backslash \Omega^r$ corresponding to the rank r Drinfeld A -modules $(\mathbb{G}_{a, \Omega^r}, \varphi^g)$ resp. $(\mathbb{G}_{a, \Gamma_g(I) \backslash \Omega^r}, \overline{\varphi^g})$ (cf. Definition 8.2.11).

Remark 9.2.13. For a rational number $0 < c < 1$ and $\Delta \in \Gamma_g(I) \backslash \mathcal{BT}_{\bullet}$, denote by $\mathfrak{U}_{g, \Delta}^c$ the affinoid subdomain $\overline{W}_{\Delta}^c$ of $\Gamma_g(I) \backslash \Omega^r$ described in Remark 8.1.19 for $\Gamma = \Gamma_g(I)$. It has the following properties:-

- (a) Let $\tilde{\Delta}$ be an i -simplex representing Δ . Suppose $\tilde{\Delta} = \{\tilde{x}_0, \dots, \tilde{x}_i\}$ and let $x_j \in \Gamma_g(I) \backslash \mathcal{BT}_0$ denote the class of $\tilde{x}_j$ for $0 \leq j \leq i$. From the same remark 8.1.19, we see that $\mathfrak{U}_{g, \Delta}^c = \bigcap_{0 \leq j \leq i} \mathfrak{U}_{g, x_j}^c$.
- (b) Suppose that $\Delta \in \Gamma_g(I) \backslash \mathcal{BT}_{\bullet}^{\Gamma\text{-st}}$ so that for any simplex $\tilde{\Delta}$ representing Δ , there is an isomorphism $W_{\tilde{\Delta}}^c \simeq \mathfrak{U}_{g, \Delta}^c$ from the same remark as before.

- (c) Continuing the notation in (b), there is an isomorphism of the Drinfeld modules $(\mathbb{G}_{a, W_{\Delta}^c}, \varphi_{|W_{\Delta}^c}^g)$ and $(\mathbb{G}_{a, \mathfrak{U}_{g, \Delta}^c}, \overline{\varphi}_{|\mathfrak{U}_{g, \Delta}^c}^g)$ with respect to the isomorphism in (b) (cf. the observation in Remark 8.2.12). Consequently we obtain an induced isomorphism of rigid A -motives, $\tilde{\mathcal{F}}_{|W_{\Delta}^c}^g$ and $\tilde{\mathcal{F}}_{|\mathfrak{U}_{g, \Delta}^c}^g$ under the isomorphism $W_{\Delta}^c \simeq \mathfrak{U}_{g, \Delta}^c$.

9.3 Comparison

Let X be a finite type scheme over L as before. Here we will compare the categories $\mathbf{Coh}_{\tau}(X, A)$ with $\widetilde{\mathbf{Coh}}_{\tau}(X^{an}, \mathcal{B})$, for certain coefficients $\mathcal{B}$, as well as their crystal counterparts following [Boe02, §8.6]. Of particular importance for us is the compatibility of the various functors described in the previous two sections with the comparison functors.

Let us begin by defining the coefficient rings $\mathcal{B}$ and their associated rigid spaces that we will be concerned with below.

Definition 9.3.1. Denote by $\mathfrak{A}$ the rigid space $\mathrm{Spec}(L \otimes_{\mathbb{F}_q} A)^{an}$ and by $\mathcal{A} := \mathcal{O}(\mathfrak{A})$ the ring of entire function on it.

Fix a finite flat morphism $\mathbb{F}_q[T] \rightarrow A$ and define

$$\mathfrak{D}_A := \mathrm{Sp}((L\langle T \rangle) \otimes_{\mathbb{F}_q[T]} A), \quad \mathcal{D}_A := \mathcal{O}_{\mathfrak{D}_A}(\mathfrak{D}_A).$$

Remark 9.3.2. (i) $\mathfrak{D}_A$ is an affinoid subdomain of $\mathfrak{A}$ and independent of the choice of a finite flat morphism $\mathbb{F}_q[T] \rightarrow A$. Indeed, the former can be shown using the description of the rigid analytification of a finite scheme, see 8.1.6 and the references there. For the latter, it is shown in [Boe02, §8.6] that $\mathfrak{D}_A$ can also realized as the points x of $\mathfrak{A}$ such that $\sigma_{\mathfrak{A}}^n(x)$ is “bounded” as $n \rightarrow \infty$. See also the description given in [BH07, above Defn.1.2] where it is denoted by $\mathfrak{A}(1)$.

(ii) If $A = \mathbb{F}_q[T]$ then $\mathfrak{D}_A$ is the unit disc $\mathrm{Sp}(L\langle T \rangle)$.

Recall the analytification morphism $X^{an} \rightarrow X$. The pull back along the analytification of a coherent sheaf on X is a coherent sheaf on X^{an} . With this in view, in order to compare the categories $\mathbf{Coh}_{\tau}(X, A)$ and $\widetilde{\mathbf{Coh}}_{\tau}(X^{an}, A)$ we can emulate the definition in 9.2.10 and analogously define *coherent sheaf of A_X -modules*, which are in turn defined using quasi coherent sheaves on X which are direct limit of coherent sheaves on X , and can also equivalently be described in the category of coherent sheaves of $\mathcal{O}_{X \times A}$ -modules. Similarly one could define the category of coherent sheaves of $\mathcal{O}_{X^{an} \times \bullet}$ modules as coherent sheaves of $\bullet'_{X^{an}}$ modules, for $(\bullet, \bullet') \in \{(\mathfrak{A}, \mathcal{A}), (\mathfrak{D}_A, \mathcal{D}_A)\}$.

In particular, one is able to then construct a well defined functor

$$(\cdot)^{A-an} : \mathbf{Coh}_\tau(X, A) \rightarrow \widetilde{\mathbf{Coh}}_\tau(X^{an}, A),$$

as well as the functor

$$(\cdot)^{\mathfrak{A}-an} : \widetilde{\mathbf{Coh}}_\tau(X^{an}, A) \rightarrow \mathbf{Coh}_\tau(X^{an}, \mathcal{A}).$$

Further the change of coefficients functor with respect to $\mathcal{A} \rightarrow \mathcal{D}_A$ yields

$$(\cdot)^{\mathfrak{D}_A-an} : \widetilde{\mathbf{Coh}}_\tau(X^{an}, \mathcal{A}) \rightarrow \widetilde{\mathbf{Coh}}_\tau(X^{an}, \mathcal{D}_A).$$

By a slight abuse of notation, we will also denote the composition of the three functors above also by $(\cdot)^{\mathfrak{D}_A-an} : \mathbf{Coh}_\tau(X, A) \rightarrow \widetilde{\mathbf{Coh}}_\tau(X^{an}, \mathcal{D}_A)$. In [Boe02, Prop.8.49] it is shown that this induces a functor $\mathbf{Crys}(X, A) \rightarrow \widetilde{\mathbf{Crys}}(X^{an}, \mathcal{D}_A)$, which we also denote by $(\cdot)^{\mathfrak{D}_A-an}$, and that the image lies in the subcategory $\widetilde{\mathbf{Crys}}^e(X^{an}, \mathcal{D}_A)$ of extendible rigid crystals.

Remark 9.3.3. Analogous to 9.1.5(ii), given a rigid τ -sheaf $\tilde{\mathcal{F}}$ with $\mathcal{D}_A$ coefficients on $\mathfrak{X}$, one has the A -module of τ -invariants of $\tilde{\mathcal{F}}$ given by

$$\tilde{\mathcal{F}}^\tau(\mathfrak{X}) := \text{Ker}(\tilde{\mathcal{F}}(\mathfrak{X} \times \mathfrak{D}_A) \xrightarrow{1-\tau_{\tilde{\mathcal{F}}}} \tilde{\mathcal{F}}(\mathfrak{X} \times \mathfrak{D}_A)).$$

This induces a left exact functor $\widetilde{\mathbf{Coh}}_\tau(\mathfrak{X}, \mathcal{D}_A) \rightarrow A\text{-mod}$ and factors through $\widetilde{\mathbf{Crys}}(\mathfrak{X}, \mathcal{D}_A)$. Given an admissible open $\mathfrak{U} \subset \mathfrak{X}$ we will denote by $(\tilde{\mathcal{F}})^\tau(\mathfrak{U})$ the τ -invariants of the restriction $(\tilde{\mathcal{F}}|_{\mathfrak{U}})^\tau(\mathfrak{U})$.

Similarly given a τ -module $\mathcal{M}$ over an affinoid algebra $\mathcal{R}$ with coefficients in $\mathcal{D}_A$ (see 9.2.2), one can define the A -module of τ -invariants $\mathcal{M}^\tau := \widetilde{\mathcal{M}}^\tau(\text{Sp}(\mathcal{R}))$.

In [BH07, Defn.1.6] a more general *étale sheaf* $\tilde{\mathcal{F}}$ of $\underline{A}_{\mathfrak{X}}$ -modules is defined, whose set of global sections is precisely what is described above.

Theorem 9.3.4. ([Boe02, Thm.8.50]) The functors $(\cdot)^{A-an}, (\cdot)^{\mathfrak{D}_A-an}$ on coherent τ sheaves on X are compatible with the following functors:- pullback, higher direct image under proper morphisms and tensor products. Moreover the induced functor $(\cdot)^{\mathfrak{D}_A}$ on crystals is compatible with the functors listed before as well as the functors, extension by zero and direct image with compact support.

Remark 9.3.5. (i) Suppose X is a finite type scheme over L , $\overline{X}$ a proper scheme over L containing X as an open scheme and $\mathcal{F}$ a τ -sheaf on $\overline{X}$ over B . Let $\bar{f} : \overline{X} \rightarrow \text{Spec}(L)$ denote the structure map. Then $R^i \bar{f}_*(\mathcal{F})$ is the τ -sheaf corresponding to the cohomology group $H^i(\overline{X} \times \text{Spec}(B), \mathcal{F})$, which as a τ -module (9.1.2) we denote by $H^i(\overline{X}, \mathcal{F})$. From standard theory we know that for a (finite) affine cover $\{U_i\}_{i \in I}$ of $\overline{X}$, the above cohomology group can also be realized as a τ -module $\check{H}^i(\{(U_i)_A\}_{i \in I}, \mathcal{F})$ defined using Čech cohomology. Now

consider the $\mathcal{D}_A$ -analytification $R^i \bar{f}_*(\mathcal{F})^{\mathcal{D}_{A-an}}$, which is again similarly a rigid τ -sheaf corresponding to $H^i(\bar{X}^{an} \times \mathcal{D}_A, \mathcal{F}^{\mathcal{D}_{A-an}})$, which again as an analytic τ -module (9.2.2), on L with coefficients in $\mathcal{D}_A$, we denote by $H^i(\bar{X}^{an}, \mathcal{F}^{\mathcal{D}_{A-an}})$. As in previous paragraph, one can also define τ -modules corresponding to Čech covers. In the next chapter we will use the following Čech cohomological description of these groups. Let $\mathcal{U} := \{U_{i'}^{an}, \mathcal{U}_j\}_{i' \in I', j \in J}$ be a cover of $\bar{X}^{an}$ where $I \subset I', I, J$ are finite and $\mathcal{U}_j \subset X^{an}$ are admissible affinoid subsets such that $\{U_{i'}\}_{i' \in I'}$ covers the closed subscheme $\bar{X} \setminus X$. Then by [Bos14, Chap.6.2, Thm.5], since intersection of sets in $\mathcal{U}$ is again a Stein domain there is a natural isomorphism of τ -modules with $\mathcal{D}_A$ coefficients

$$\check{H}^i(\mathcal{U}, \mathcal{F}) \simeq H^i(\bar{X}^{an}, \mathcal{F}^{\mathcal{D}_{A-an}}), \quad i \geq 0.$$

- (ii) Recall the A -motive $\mathcal{F}_I^{un} \in \mathbf{Coh}_\tau(M_I^r, A)$ from 9.1.6(ii). Denote by $\mathcal{F}_{I, K_\infty}^{un}$ the pull back along the base change morphism $M_I^r \times_K K_\infty \rightarrow M_I^r$. Then it follows from Theorem 8.2.13 that

$$i_g^*(\mathcal{F}_{I, K_\infty}^{un})^{A-an} \simeq \tilde{\mathcal{F}}_I^g,$$

where the map i_g denotes the embedding of the component $\Gamma_g(I) \setminus \Omega^r \hookrightarrow (M_I^r \times_K K_\infty)^{an}$. Denoting $\pi_g : \Omega^r \rightarrow (M_I^r \times_K K_\infty)^{an}$ to be the composition of i_g with the quotient $\Omega^r \rightarrow \Gamma_g(I) \setminus \Omega^r$, we have

$$\pi_g^*(\mathcal{F}_{I, K_\infty}^{un})^{A-an} \simeq \tilde{\mathcal{F}}^g.$$

9.4 Uniformizable A -modules

Let $\phi : A \rightarrow \mathbb{C}_\infty[\tau]$ be a Drinfeld A -module of rank r . As mentioned in the previous chapter, in [Dri74, Prop.3.1] a *uniformization* of ϕ with respect to a rank r A -lattice Λ inside $\mathbb{C}_\infty$ is shown, similar to the classical uniformization of elliptic curves over $\mathbb{C}$ by a rank 2 lattices. The role of Weierstrass' $\wp$ function in the classical setting is played by the exponential function $\exp_\Lambda : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ (with A -action on RHS given by ϕ), which is a surjective A homomorphism with kernel Λ and hence provides the required uniformization of ϕ . We briefly described this in a more general context in 8.2.9. As was observed in the proof there the exponential function is not expected to be surjective over a general rigid analytic base. In [And86], Anderson more generally also constructed an exponential function for t -modules and in [And86, Thm.4] provided three equivalent conditions for a t -module to be uniformizable. Following the proof strategy of [And86, Thm.4], Bökke generalized Anderson's criteria for uniformizability to a family of rigid analytic A -modules. Primarily we are concerned with the description of the A -motive $(\tilde{\mathcal{F}}^g)^{\mathcal{D}_{A-an}}$ given

similar to [Boe02, Cor.9.20], which follows almost verbatim from its proof and is a consequence of the criteria for uniformizability. We will discuss the bare minimum of the theory in [Boe02, §9.2] required for stating this result and refer the reader there for details. One could also refer to [BH07] for more elaborate treatment of the contents of this section in the setting of étale topology on rigid analytic spaces.

Let L be a field as in the previous section, $\mathfrak{X}$ a rigid analytic L space and $\mathcal{E}$ be a locally free sheaf over it. One can generalize the discussion in [Leh09, Chap.2, §6] regarding the *vector bundle* associated to locally free sheaves of rank 1 on ringed spaces, in a straightforward manner to any locally free sheaf. More precisely one can define the sheaf of commutative algebras

$$S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) := \bigoplus_{n \geq 0} (\mathcal{E}^\vee)^{\otimes n}$$

which has an induced map of $\mathcal{O}_{\mathfrak{X}}$ -algebras

$$\Delta_{\mathcal{E}} : S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) \rightarrow S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) \otimes_{\mathcal{O}_{\mathfrak{X}}} S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}).$$

Definition 9.4.1. Let $\mathfrak{G}/\mathfrak{X}$ denote the category, called *rigid analytic vector bundles on $\mathfrak{X}$* , whose objects consist of locally free sheaves on $\mathfrak{X}$ and morphisms between two locally free sheaves $\mathcal{E}, \mathcal{F}$ on $\mathfrak{X}$, denoted by

$$\mathrm{Hom}_{\mathfrak{G}/\mathfrak{X}}(\mathcal{E}, \mathcal{F})$$

is the set of all $\mathcal{O}_{\mathfrak{X}}$ -algebra homomorphisms $f : S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) \rightarrow S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{F})$ together with a commutative diagram

$$\begin{array}{ccc} S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) & \xrightarrow{\Delta_{\mathcal{E}}} & S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) \otimes_{\mathcal{O}_{\mathfrak{X}}} S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}) \\ \downarrow f & & \downarrow f \otimes f \\ S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{F}) & \xrightarrow{\Delta_{\mathcal{F}}} & S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{F}) \otimes_{\mathcal{O}_{\mathfrak{X}}} S_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{F}) \end{array} .$$

Remark 9.4.2. (i) Suppose $\mathcal{E}$ resp. $\mathcal{F}$ are locally free sheaves on $\mathfrak{X}$ of rank n resp. n' . Let $\mathfrak{U} = \mathrm{Sp}(\mathcal{B}) \subset \mathfrak{X}$ be an admissible affinoid open such that both $\mathcal{E}, \mathcal{F}$ admit trivializations over $\mathfrak{U}$, so that $\mathcal{E}(\mathfrak{U}) = \mathcal{B}^n$ and $\mathcal{F}(\mathfrak{U}) = \mathcal{B}^{n'}$. In particular,

$$\mathrm{Hom}_{\mathfrak{G}/\mathfrak{U}}(\mathcal{E}|_{\mathfrak{U}}, \mathcal{F}|_{\mathfrak{U}}) = \mathrm{Hom}_{\mathrm{GrpSch}/\mathcal{B}}(\mathbb{G}_{a,\mathcal{B}}^n, \mathbb{G}_{a,\mathcal{B}}^{n'}) = \bigoplus_{i \geq 0} M_{n',n}(\mathcal{B}) \tau_{\mathcal{B}^n}^i$$

where $\mathbb{G}_{a,\mathcal{B}}$ is the line bundle $\mathrm{Spec}(\mathcal{B}[T])$ for an indeterminate T and $\tau_{\mathcal{B}^n}$ denotes the relative Frobenius on $\mathbb{G}_{a,\mathcal{B}}^n$. The description on the right can be proved using [Leh09, Chap.1, Prop.2.1, 2.3] (see also §7.2.1). If $q = p^m$, then setting $\tau_{\mathcal{B}^n,q} := \tau_{\mathcal{B}^n}^m$, the subalgebra of $\mathrm{Hom}_{\mathfrak{G}/\mathfrak{X}}(\mathcal{E}, \mathcal{F})$, consisting of homomorphisms, which when restricted to $\mathfrak{U}$ trivializing both

$\mathcal{E}$ and $\mathcal{F}$ lie in $\bigoplus_{i \geq 0} M_{n',n}(\mathcal{B})\tau_{\mathcal{B}^{n,q}}^i$, will be denoted by $\mathrm{Hom}_{\mathbb{F}_q-\mathfrak{G}/\mathfrak{X}}(\mathcal{E}, \mathcal{F})$. $\mathrm{End}_{\mathbb{F}_q-\mathfrak{G}/\mathfrak{X}}(\mathcal{E})$ is similarly defined.

- (ii) Denote by $\mathbf{Loc}/\mathfrak{X}$ the category of locally free sheaves on $\mathfrak{X}$. From (i) it is clear that there is an additive functor, denoted

$$\mathrm{Lie} : \mathfrak{G}/\mathfrak{X} \rightarrow \mathbf{Loc}/\mathfrak{X},$$

such that $\mathrm{Lie}(\mathcal{E}) := \mathcal{E}$ and $\mathrm{Hom}_{\mathbf{Loc}/\mathfrak{X}}(\mathrm{Lie}(\mathcal{E}), \mathrm{Lie}(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_{\mathfrak{X}}}(\mathcal{E}, \mathcal{F})$.

For $f \in \mathrm{End}_{\mathfrak{G}/\mathfrak{X}}(\mathcal{E})$ denote by f' the induced map in $\mathrm{End}(\mathrm{Lie}(\mathcal{E}))$.

Definition 9.4.3. A family of A -modules $\underline{\mathcal{E}}$ of rank r on the rigid space $\mathfrak{X}/L$ consists of a rank r locally free sheaf $\mathcal{E}$ on $\mathfrak{X}$ together with a ring homomorphism

$$\varphi : A \rightarrow \mathrm{End}_{\mathbb{F}_q-\mathfrak{G}/\mathfrak{X}}(\mathcal{E})$$

satisfying the following conditions:-

- (i) $\forall a \in A$, there exists $n_a \in \mathbb{N}$ such that $(a - \varphi'_a)^{n_a} = 0$ on $\mathrm{Lie}(\mathcal{E})$.
- (ii) The sheaf $\mathcal{M}(\mathcal{E}) := \mathrm{Hom}_{\mathfrak{G}/\mathfrak{X}}(\mathcal{E}, \mathcal{O}_{\mathfrak{X}})$ with the A -action given by right composition by φ is a locally free sheaf of $A_{\mathfrak{X}}$ modules of rank r , that is there exists an affinoid admissible covering $\{\mathfrak{U}_i := \mathrm{Sp}(\mathcal{R}_i)\}_{i \in I}$ such that $\mathcal{M}(\mathcal{E})(\mathfrak{U}_i) = P_i$ for a rank r projective $\mathcal{R}_i \otimes A$ -module P_i .

Left composition by $\sigma_{\mathfrak{X}}$ on $\mathcal{M}(\mathcal{E})$ endows the latter with the structure of coherent τ -sheaf on $\mathfrak{X}$ with coefficients in A , which we denote by $\mathcal{M}(\underline{\mathcal{E}})$ called a *family of rigid A -motives on $\mathfrak{X}$* .

Remark 9.4.4. (i) It is worth looking at [BH07, Defn.8.1] where an equivalent definition of a family of A -modules as described above is discussed.

- (ii) Note that when $\mathfrak{X} = \mathrm{Sp}(L')$ for a complete field extension of L , the above notion of A -module coincides with that of Anderson's [And86, §1.1]. In particular, given a point $x \in \mathfrak{X}$, the fiber $\mathcal{E}_x$ forms a well defined A -module of rank r over $\mathrm{Sp}(k(x))$ which we denote by $\underline{\mathcal{E}}_x$.
- (iii) Analogous to [And86, §4.1], a Drinfeld A -module of rank r $(\mathcal{L}, \phi)$ over $\mathfrak{X}$ (8.2.7(i)) forms a family of A -module in the above sense, with $n'_a = 1$ for all $a \in A$. Moreover the corresponding family of rigid A -motives is $\tilde{\mathcal{F}}(\mathcal{L}, \phi)$ (9.2.12).

In the following we have a generalization of the exponential function of Anderson [And86, Thm./Defn.3] to the setting of a family of A -modules as above. We assume henceforth that $\mathfrak{X}$ is reduced so that affinoid algebras defining affinoid subdomains of $\mathfrak{X}$ become Banach algebras with respect to the spectral norm. For such an affinoid algebra $\mathcal{R}$ and a matrix $(a_{ij})_{1 \leq i, j \leq d} \in M_d(\mathcal{R})$ denote by $|(a_{ij})_{1 \leq i, j \leq d}| := \max_{1 \leq i, j \leq d} |a_{ij}|$.

Theorem-Definition 9.4.5. ([Boe02, Lemma 9.10, Prop.9.12], [BH07, Prop.8.7, Cor.8.8]) There exists a unique $\mathbb{F}_q$ -linear morphism of sheaves $\exp_{\underline{\mathcal{E}}} : \text{Lie}(\underline{\mathcal{E}}) \rightarrow \underline{\mathcal{E}}$ such that

$$\exp_{\underline{\mathcal{E}}} \circ \varphi'_a = \varphi_a \circ \exp_{\underline{\mathcal{E}}}$$

$\forall a \in A$. Moreover let $\mathcal{U} = \text{Sp}(\mathcal{R})$ be an affinoid open in $\mathfrak{X}$ such there is an isomorphism $\mathcal{E}(\mathcal{U}) \simeq \mathcal{R}^r$ and an induced isomorphism $\text{Lie}(\underline{\mathcal{E}})(\mathcal{U}) \simeq \mathcal{R}^r$. Then there exists a unique series of matrices $(e_d)_{d \geq 0}$ in $M_d(\mathcal{R})$ such that $e_0 = \text{id}$, $\lim_{j \rightarrow \infty} (q^{-j}|e_j|) = -\infty$ and

$$\exp_{\underline{\mathcal{E}}|_{\mathcal{U}}}((z_i)_{1 \leq i \leq r}) = \sum_{j \geq 0} e_j(z_i^{q^j})_{1 \leq i \leq r}$$

for $(z_i)_{1 \leq i \leq r} \in \mathcal{R}^r$.

Denote by $H_*(\underline{\mathcal{E}}) := \text{Ker}(\exp_{\underline{\mathcal{E}}})$, which is a subsheaf of $\text{Lie}(\underline{\mathcal{E}})$.

Lemma 9.4.6. ([Boe02, Lemma 9.16]) Let $\mathfrak{X}, \underline{\mathcal{E}}$ be as above. For any connected affinoid subdomain $\mathcal{U}$ of $\mathfrak{X}$, the group of sections $H_*(\underline{\mathcal{E}})(\mathcal{U})$ form a projective A -module of rank at most r . Moreover for all point $x \in \mathcal{U}$, the specialization map $H_*(\underline{\mathcal{E}})(\mathcal{U}) \rightarrow H_*(\underline{\mathcal{E}}_x)(k(x))$ is injective.

Definition 9.4.7. (cf. [And86, §2.3])

- (i) Let $\underline{\mathcal{E}}$ be a family of A -modules on $\mathfrak{X}$ of rank r . $\underline{\mathcal{E}}$ is said to be *globally uniformizable* if $H_*(\underline{\mathcal{E}})(\mathfrak{X})$ is a projective A -module of rank r and for all points $x \in \mathfrak{X}$, the specialization maps $H_*(\underline{\mathcal{E}})(\mathfrak{X}) \rightarrow H_*(\underline{\mathcal{E}}_x)(k(x))$ is an isomorphism.
- (ii) A rigid τ -sheaf $\tilde{\mathcal{F}}$ on $\mathfrak{X}$ with coefficients in $\mathcal{D}_A$ is said to be *rigid analytically trivial* if on each connected component $\mathfrak{X}'$ of $\mathfrak{X}$, there exists a projective A -module $P_{\mathfrak{X}'}$ such that $\tilde{\mathcal{F}} \simeq \mathbb{1}_{\mathfrak{X}', \mathcal{D}_A} \otimes_A P_{\mathfrak{X}'}$.

Example 9.4.8. Suppose that $\mathfrak{X}$ be connected rigid space over L . Let $(\mathcal{L}, \underline{\Lambda}, s)$ be an A -lattice of rank r over $\mathfrak{X}$ (see Definition 8.2.8) such that $\underline{\Lambda}$ is the constant sheaf associated to a projective A -module Λ . Let $(\mathcal{L}, \phi)$ be the associated Drinfeld A -module over $\mathfrak{X}$ to this lattice by Proposition 8.2.9 and $\underline{\mathcal{E}}(\mathcal{L}, \phi)$ denote the associated family of A -modules á la Remark 9.4.4(ii). Then it is clear from the construction of $(\mathcal{L}, \phi)$ that $\underline{\mathcal{E}}(\mathcal{L}, \phi)$ defines a family of globally uniformizable A -modules on $\mathfrak{X}$. In particular, $H_*(\underline{\mathcal{E}}(\mathcal{L}, \phi))(\mathfrak{X}) = \Lambda$.

Theorem 9.4.9. ([Boe02, Thm.9.19]) Let $\underline{\mathcal{E}}$ be a family modules on $\mathfrak{X}$ of rank r . Then the following statements are equivalent:-

- (i) $\underline{\mathcal{E}}$ is globally uniformizable;

(ii) The rigid τ -sheaf $\mathcal{M}(\underline{\mathcal{E}})^{\mathcal{D}_A - an}$ with coefficients in $\mathcal{D}_A$ is rigid analytically trivial.

Remark 9.4.10. The previous theorem is also presented in [BH07, Thm.8.15] in a slightly more detailed manner, with more equivalence conditions. In fact, one of the equivalent conditions is the existence of a short exact sequence of étale sheaves on $\mathfrak{X}$

$$0 \rightarrow H_*(\underline{\mathcal{E}}) \rightarrow \mathrm{Lie}(\mathcal{E}) \xrightarrow{\exp_{\underline{\mathcal{E}}}} \mathcal{E} \rightarrow 0.$$

In particular, $\exp_{\underline{\mathcal{E}}}$ is surjective as expected from Drinfeld's uniformization discussed in the beginning of this section.

Recall the rigid τ sheaf with A coefficients $\tilde{\mathcal{F}}^g$, for $g \in \mathrm{GL}_r(\mathbb{A}_K^f)$, from Example 9.2.12. It is the τ -sheaf associated to the rank r Drinfeld A -module $(\mathbb{G}_{a,\Omega^r}, \varphi^g)$ on Ω^r , which in turn corresponds to the A lattice $(\mathcal{O}_{\Omega^r}, \underline{\Lambda}_g, s_g)$ (see 8.2.10 and discussion above it). In particular by Remark 9.4.8 the family of A -modules associated with $(\mathbb{G}_{a,\Omega^r}, \varphi^g)$ is globally uniformizable and hence by Theorem 9.4.9, $\tilde{\mathcal{F}}^{g,\mathcal{D}_A - an}$ is rigid analytically trivial. The proof of [Boe02, Cor.9.20] verbatim gives the following

Theorem 9.4.11. There is a canonical isomorphism of rigid τ -sheaves with coefficients in $\mathcal{D}_A$

$$\tilde{\mathcal{F}}^{g,\mathcal{D}_A - an} \simeq \mathbb{1}_{\Omega^r, \mathcal{D}_A} \otimes_A \mathrm{Hom}_A(\Lambda_g, \Omega_A),$$

where Ω_A denotes the module of $\mathbb{F}_q$ -linear Kähler differentials on A .

For the following consequence in light of [Boe02, Cor.9.21], recall the affinoid subsets of $\mathfrak{U}_{g,\Delta} \subset \Gamma_g(I) \setminus \Omega^r$ described in Remark 9.2.13.

Corollary 9.4.12. Let $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_{\bullet}^{\Gamma_g(I) - \mathrm{st}}$ denote the class of a simplex. Then there is a canonical isomorphism

$$(\tilde{\mathcal{F}}_I^{g,\mathcal{D}_A - an})|_{\mathfrak{U}_{g,\Delta}} \simeq \mathbb{1}_{\mathfrak{U}_{g,\Delta}, \mathcal{D}_A} \otimes_A \mathrm{Hom}_A(\Lambda_g, \Omega_A).$$

Proof. Let $\tilde{\Delta} \in \mathcal{BT}_{\bullet}^{\Gamma_g(I) - \mathrm{st}}$ be a simplex representing Δ . From Remark 8.1.19 and 9.2.13, it follows that the isomorphism $W_{\tilde{\Delta}}^c \simeq \mathfrak{U}_{g,\Delta}$ induces an isomorphism of rigid τ -sheaves with $\mathcal{D}_A$ coefficients, $\tilde{\mathcal{F}}_{|W_{\tilde{\Delta}}^c}^g$ and $\tilde{\mathcal{F}}_{| \mathfrak{U}_{g,\Delta}}^{g,\mathcal{D}_A - an}$. Restricting the isomorphism in 9.4.11 to $W_{\tilde{\Delta}}^c$ concludes the proof. $\square$

Chapter 10

An Eichler-Shimura map

We now have almost all the ingredients to construct a canonical map from the dual of the space of harmonic cocycles taking values in a certain finite dimensional K vector space (to be defined below) and the τ -invariants of an appropriate function field crystal associated to the universal Drinfeld module on the moduli space M_f^r for $r \geq 2$. This is motivated from, and a direct generalization after a change of coefficients by K , of a similar map of A -modules in the rank 2 case done in [Boe02, Chap.10].

10.1 Preparations

We begin by defining the space of coefficients over which the harmonic cocycles below are defined. Let $n \geq 0$ be an integer and consider an r dimensional K -vector space $V := K.e_1 \oplus \dots \oplus K.e_r$ for a fixed basis $\{e_1, \dots, e_r\}$. V has a natural left $\mathrm{GL}_r(K)$ action. Note that tensor products over K as well as K -dual's of K -vector spaces with left $\mathrm{GL}_r(K)$ action also inherit an induced left $\mathrm{GL}_r(K)$ action. In particular this holds for symmetric powers and determinants.

Definition 10.1.1. (cf. [Tei91, pg.494, (2)], [Boe02, pg.50]) Define V_n to be the K -vector space

$$\det(V)^* \otimes_K (\mathrm{Sym}^n(V^*))^*,$$

where we denote by $(.)^*$ the K -dual. It is equipped with structure of left $K[\mathrm{GL}_r(K)]$ -module induced from the action of $\mathrm{GL}_r(K)$ on V .

Remark 10.1.2. (i) From standard combinatorics we see that V_n is a finite dimensional K -vector space of dimension $\binom{n+r-1}{r-1}$.

- (ii) We will only be concerned with V_n as a representation of principal congruence subgroups of $\mathrm{GL}_r(K)$, hence we may omit the $\det(V)$ “twist” from our considerations below. However it should be noted that possible considerations of action by Hecke operators on the space of harmonic cocycles, similar to that in [Boe02, Chap.6], would require that we take the aforementioned $\det(V)$ twist into account.

Now recall the finite I class group $\mathrm{Cl}_I := \mathrm{GL}_r(K) \backslash \mathrm{GL}_r(\mathbb{A}_K^f) / \mathcal{K}(I)$ from the example in 8.2.2. We fix representatives $g_1, \dots, g_h \in \mathrm{GL}_r(\mathbb{A}_K^f)$ of Cl_I for $h := \#\mathrm{Cl}_I$. In what follows, whenever we write “ $g \in \mathrm{Cl}_I$ ” we mean the choice of these representatives.

For the following recall the Steinberg module $\mathrm{St}(W)$ resp. $\mathrm{St}^I(P)$ for a k -vector space W resp. a projective A -submodule $P \subset W$ from Definition’s 4.1.3 resp. 5.3.5.

Definition 10.1.3. (i) The *adelic space of harmonic cocycles of weight $n + r$ of level $\mathcal{K}(I)$* is defined by

$$\mathbf{C}_{har}(\mathcal{K}(I), n) := \bigoplus_{g \in \mathrm{Cl}_I} \mathbf{C}_{har}(V_n)^{\Gamma_g(I)}.$$

- (ii) Also define the *adelic stable Steinberg module*

$$\mathbf{C}_n^{st}(\mathcal{K}(I)) := \bigoplus_{g \in \mathrm{Cl}_I} \mathrm{St}^I(\Lambda_g) \otimes_{\Gamma_g(I)} V_n$$

as well as the *adelic Steinberg module*

$$\mathbf{St}_n(\mathcal{K}(I)) := \bigoplus_{g \in \mathrm{Cl}_I} \mathrm{St}(K^r) \otimes_{\Gamma_g(I)} V_n$$

All the above modules inherit the structure of a K -vector space via V_n .

Remark 10.1.4. Note that the definition above is given with respect to a fixed choice of representatives in $\mathrm{GL}_r(\mathbb{A}_K^f)$ of Cl_I . Changing the representatives, transforms them to isomorphic copies of K -vector spaces. Indeed suppose $g' = \alpha g \gamma$ for $\alpha \in \mathrm{GL}_r(K)$ and $\gamma \in \mathcal{K}(I)$. Then $\Gamma_{g'}(I) = \alpha \Gamma_g(I) \alpha^{-1}$. In particular we get an isomorphism $\mathbf{C}_{har}(V_n)^{\Gamma_g(I)} \xrightarrow{\sim} \mathbf{C}_{har}(V_n)^{\Gamma_{g'}(I)}$ by sending $f \mapsto f \circ \alpha$. Similarly there are isomorphisms $\mathrm{St}^I(\Lambda_g) \otimes_{\Gamma_g(I)} V_n \xrightarrow{\sim} \mathrm{St}^I(\Lambda_{g'}) \otimes_{\Gamma_{g'}(I)} V_n$ respectively $\mathrm{St}(K^r) \otimes_{\Gamma_g(I)} V_n \xrightarrow{\sim} \mathrm{St}(K^r) \otimes_{\Gamma_{g'}(I)} V_n$ defined by sending $s \otimes v \mapsto \alpha s \otimes \alpha v$.

Recall the maps $\mathrm{AR}_-, \mathrm{GQ}_-$ resp. Te_- from 4.2.8, 5.3.5 resp. 6.2.1 where $-$ runs over principal congruence subgroups. Given $g \in \mathrm{GL}_r(\mathbb{A}_K^f)$, we abbreviate

$$\mathrm{AR}_g := \mathrm{AR}_{\Gamma_g(I)}, \quad \mathrm{GQ}_g := \mathrm{GQ}_{\Gamma_g(I)}, \quad \mathrm{Te}_g := \mathrm{Te}_{\Gamma_g(I)}.$$

Further define

$$\mathrm{AR}_{\mathcal{K}(I)} : \mathbf{St}_n(\mathcal{K}(I)) \rightarrow \mathbf{C}_{har}(\mathcal{K}(I), n)$$

by $(\mathrm{AR}_g)_{g \in \mathrm{Cl}_I}$. Similarly define $\mathrm{GQ}_{\mathcal{K}(I)} \otimes 1 : \mathbf{C}_n^{st}(\mathcal{K}(I)) \rightarrow \mathbf{St}_n(\mathcal{K}(I))$ and $\mathrm{Te}_{\mathcal{K}(I)} : \mathbf{C}_{har}(\mathcal{K}(I), n) \rightarrow \mathbf{C}_n^{st}(\mathcal{K}(I))$.

The main results of Part I can be summarized in the present setting as follows

Theorem 10.1.5. $\mathrm{GQ}_{\mathcal{K}(I)} \otimes 1 : \mathbf{C}_n^{st}(\mathcal{K}(I)) \rightarrow \mathbf{St}_n(\mathcal{K}(I))$ is an isomorphism of vector spaces such that it induces a commutative triangle

$$\begin{array}{ccc} & \mathbf{C}_{har}(\mathcal{K}(I), n) & \\ \swarrow \mathrm{Te}_{\mathcal{K}(I)} & & \nwarrow \mathrm{AR}_{\mathcal{K}(I)} \\ \mathbf{C}_n^{st}(\mathcal{K}(I)) & \xleftarrow[\mathrm{GQ}_{\mathcal{K}(I)}^{-1} \otimes 1]{\sim} & \mathbf{St}_n(\mathcal{K}(I)) \end{array}$$

Moreover when $A = \mathbb{F}_q[t]$ all the maps above are isomorphisms.

Proof. 5.3, which follows from Theorem 5.2.2, shows that $\mathrm{GQ}_{\mathcal{K}(I)} \otimes 1$ is an isomorphism and Proposition 6.2.3 shows commutativity of the triangle. Theorem 7.3.8 combined with this commutativity shows the last statement. $\square$

10.2 Pink's Satake compactification and a Stein cover of the analytification

In [Pin12], Pink constructs a *Satake compactification* $\overline{M}_I^r$ of the Drinfeld moduli space M_I^r which is characterized by the following properties ([Pin12, Defn.4.1, Thm.4.2]):-

- (a) M_I^r is an open dense subscheme of $\overline{M}_I^r$ and $\overline{M}_I^r$ is a normal, integral projective algebraic variety over K ;
- (b) the universal Drinfeld A -module $\mathbb{E}^{un} := (\mathcal{L}^{un}, \varphi^{un})$ extends to a *weakly separating generalized Drinfeld A -module with level I structure* on $\overline{M}_I^r$.

Remark 10.2.1. (i) We provided the condition (b) for the sake of completeness. We refrain from defining the terms in it as we do not use it directly in what follows. For the definition of a *generalized* as well as *weakly separating* Drinfeld modules the reader is referred to the relevant portion of op.cit.

- (ii) In fact the result in [Pin12, Thm.4.2] holds true for Drinfeld moduli spaces $M_{\mathcal{K}}^r$ where $\mathcal{K} \subset \mathrm{GL}_r(\widehat{A})$ is a *fine* open compact subgroup. The moduli space M_I^r is precisely $M_{\mathcal{K}(I)}^r$ and $\mathcal{K}(I)$ is fine by definition (see [Pin12, Defn.1.4]).

Recall that in Theorem 8.2.13 we have

$$\bigsqcup_{g \in \mathrm{Cl}_I} \Gamma_g(I) \backslash \Omega^r \simeq (M_I^r \times_K K_\infty)^{an}.$$

From [Con99, Thm.A.1.3(1)] we can conclude that for each $g \in \mathrm{Cl}_I$, the connected component of the analytification $(M_I^r \times_K K_\infty)^{an}$ associated with $\Gamma_g(I) \backslash \Omega^r$ corresponds to a connected component $M_g \subset M_I^r \times_K K_\infty$ such that the above isomorphism restricts to an isomorphism $i_g : \Gamma_g(I) \backslash \Omega^r \simeq (M_g)^{an}$ and

$$\bigsqcup_{g \in \mathrm{Cl}_I} M_g = M_I^r \times_K K_\infty.$$

Proposition 10.2.2. Let $\overline{M}_g$ denote the Zariski closure of M_g inside the compactification $\overline{M}_I^r \times_K K_\infty$ for $g \in \mathrm{Cl}_I$. Then each $\overline{M}_g$ is a connected component of $\overline{M}_I^r \times_K K_\infty$ and we have

$$\bigsqcup_{g \in \mathrm{Cl}_I} \overline{M}_g = \overline{M}_I^r \times_K K_\infty.$$

Proof. We use the standard fact from algebraic geometry that, for a quasi compact morphism of schemes, its set theoretic image is Zariski dense in its scheme theoretic image (see [Sta21, Lemma 01R8(4)]) and moreover the scheme theoretic image commutes with flat base change (see [Sta21, Tag 081I]). Since M_I^r is dense in $\overline{M}_I^r$, the base change $M_I^r \times_K K_\infty$ is also dense in $\overline{M}_I^r \times_K K_\infty$. Consequently we obtain

$$\bigcup_{g \in \mathrm{Cl}_I} \overline{M}_g = \overline{M}_I^r \times_K K_\infty.$$

In particular, $\{\overline{M}_g\}_{g \in \mathrm{Cl}_I}$ are the irreducible components of $\overline{M}_I^r \times_K K_\infty$. Since K_∞ is a separable extension over K , from [Sta21, Lemma 0C3M] we see that $\overline{M}_I^r \times_K K_\infty$ is normal and hence its connected components are irreducible and this concludes the proof. $\square$

Remark 10.2.3. Note that M_g as well as $\overline{M}_g$ are independent of the choice of representatives $g_1, \dots, g_h$ in Cl_I fixed before.

Now we will define and fix a finite Stein cover of $(\overline{M}_g)^{an}$, for $g \in \mathrm{Cl}_I$, which we will use with regards to the considerations on Čech cohomology in the next section.

$\overline{M}_g \setminus M_g$ is a closed subset of the projective variety $\overline{M}_g$ and hence admits a finite cover by Zariski affine opens in $\overline{M}_g$, say $\{U_{g,j}\}_{j \in J_g}$ for a finite set J_g . Then $\{(U_{g,j})^{an}\}_{j \in J_g}$ is a finite Stein cover of $(\overline{M}_g)^{an} \setminus (M_g)^{an}$. Since $\Gamma_g(I) \backslash \Omega^r \simeq (M_g)^{an}$ recall that in 9.2.13(i), we denoted by $\mathfrak{U}_{g,\Delta}^c$ of $(M_g)^{an}$ the affinoid subdomain $\Gamma_g(I) \backslash \Omega^r$ corresponding to $\Delta \in \Gamma_g(I) \backslash \mathcal{BT}_\bullet$ described in

Proposition 8.1.18 as well as Remark 8.1.19 for a rational number $0 < c < 1$. Fix a choice of such a rational number c . Due to the compactness of $(\overline{M}_g)^{an}$, the admissible cover $\{\mathcal{U}_{g,x}^c\}_{x \in \Gamma_g(I) \setminus \mathcal{BT}_0} \cup \{\mathcal{U}_{g,j}^{an}\}_{j \in J_g}$ has a finite subcover say $\mathcal{U}_g$. We posit that the following requirement be satisfied by $\mathcal{U}_g$:-

- (*) If $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_d$ denotes the class of a $\Gamma_g(I)$ stable d -simplex $\tilde{\Delta}$ then $\mathcal{U}_{g,x}^c \in \mathcal{U}_g$ for all $x \in \Gamma_g(I) \setminus \mathcal{BT}_0$ representing class of vertices $\tilde{x} \in \tilde{\Delta}$;

If $\mathcal{U}_g$ does not already satisfy *, we can simply append the open affinoid subsets described above to $\mathcal{U}_g$. Note by Lemma 5.3.6 that this maintains the finiteness of $\mathcal{U}_g$.

For the rest of this chapter, for $g \in Cl_I$ we fix a choice of a finite Stein admissible cover $\mathcal{U}_g$ of $(\overline{M}_g)^{an}$ satisfying the condition *.

10.3 Construction of the map

Recall the A -motive $\mathcal{F}_I^{un}$ on M_I^r from 9.1.6(ii) and denote by $\mathcal{F}_I^{(n),un} := \text{Sym}^n \mathcal{F}_I^{un}$. Let $f : M_I^r \rightarrow \text{Spec}(K)$ be the structure map.

Definition 10.3.1. Regarding $\mathcal{F}_I^{un}$ as a crystal over M_I^r with coefficients in A , define

$$\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I)) := R^{r-1} f_! \mathcal{F}_I^{(n),un} \in \mathbf{Crys}(\text{Spec}(K), A).$$

Let $b : \text{Spec}(K_\infty) \rightarrow \text{Spec}(K)$ denote the obvious map and by slight abuse of notation denote $b^*(\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I)))^{\mathfrak{D}_{A-an}} \in \widetilde{\mathbf{Crys}}(\text{Sp}(K_\infty), \mathcal{D}_A)$ also by $\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))^{\mathfrak{D}_{A-an}}$.

Remark 10.3.2. (i) Let $j : M_I^r \hookrightarrow \overline{M}_I^r$ denote the open immersion. From the definition of $f_!$ in 9.1.7 as well as Remark 9.1.8, we see that for a coherent extension $\mathcal{H}_I \subset j_* \mathcal{F}_I^{(n),un}$ of $\mathcal{F}_I^{(n),un}$, the τ -modules $H^j(\overline{M}_I^r \times_K \text{Spec}(A), \mathcal{I}^m \mathcal{H}_I)$ represent the crystals $R^j f_! \mathcal{F}_I^{(n),un}$, for $0 \leq j \leq r-1$, $\mathcal{I} \subset \mathcal{O}_{\overline{M}_I^r}$ an ideal sheaf having the zero locus the Zariski closed set $\overline{M}_I^r \setminus M_I^r$ and any $m \geq 1$. Using [Pin12, Prop.4.10, Thm.5.3] a choice for such an $\mathcal{I}$ arises from the ideal sheaf determined by the top coefficient of the universal family of generalized Drinfeld A -modules on $\overline{M}_I^r$, which moreover is anti-ample, that is $\mathcal{I}^\vee$ is an ample invertible sheaf. In particular, for $j < r-1$ one should expect $H^j(\overline{M}_I^r \times_K \text{Spec}(A), \mathcal{I}^m \mathcal{H}_I)$ to vanish or equivalently in such cases $R^j f_! \mathcal{F}_I^{(n),un}$ should be the zero crystal.

- (ii) From observations in (i) as well as Theorem 9.3.4 we see that, if we denote by $b' : \overline{M}_I^r \times_K K_\infty \rightarrow \overline{M}_I^r$ the base change morphism and $\mathcal{H}_I$ is as in (i), then

$$\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))^{\mathfrak{D}_{A-an}} = H^{r-1}((\overline{M}_I^r \times_K K_\infty)^{an} \times \mathcal{D}_A, (b')^* \mathcal{H}_I^{\mathfrak{D}_{A-an}}).$$

Using the result 9.3.4 again together with Proposition 10.2.2, the RHS can be decomposed as

$$\bigoplus_{g \in \text{Cl}_I} H^{r-1}((\overline{M}_g)^{an} \times \mathfrak{D}_A, ((b')^* \underline{\mathcal{H}}_I)^{\mathfrak{D}_{A-an}}|_{\overline{M}_g}),$$

where, for $(\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})^{(n)} := \text{Sym}^n(\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})$, $((b')^* \underline{\mathcal{H}}_I)^{\mathfrak{D}_{A-an}}|_{\overline{M}_g}$ is a coherent τ -sheaf representing $(j_g)_!(\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})^{(n)}$, for $j_g : (M_g)^{an} \hookrightarrow (\overline{M}_g)^{an}$ being the open immersion (see above Remark 9.2.13 for definition of $\tilde{\mathcal{F}}_I^g$).

The previous definition is motivated from the analogous crystals described in the rank 2 case in [Boe02, Defn.10.1] and can be regarded as higher rank generalization of them. Though it should be pointed out that $\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))$ is the base change to $\text{Spec}(K)$ of the corresponding crystal defined over $\text{Spec}(A) \setminus V(I)$ in op.cit.. The main result of this chapter relates the τ -invariants of the above analytic crystal to the previously defined space of adelic stable Steinberg module $\mathbf{C}_n^{st}(\mathcal{K}(I))$ via an *Eichler-Shimura map*. It coincides with the analogous map defined in [Boe02, Defn.10.7] for $\mathcal{K}(I)$, albeit after a tensor product $\otimes_A K$. For convenience while presenting its proof, let us first introduce some terminology regarding cohomological complexes and note some basic observations.

We make the following convention about denoting classes of simplices $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i$. We write

$$\Delta = (x_0, \dots, x_i)$$

for $x_i \in \Gamma_g(I) \setminus \mathcal{BT}_0$ $0 \leq j \leq i$, to mean that, given an i -simplex $\tilde{\Delta}$ representing Δ , we have $\tilde{\Delta} = \{\tilde{x}_0, \dots, \tilde{x}_i\}$ for vertices $\tilde{x}_j$ representing x_i satisfying $\text{type}(\tilde{x}_0) < \dots < \text{type}(\tilde{x}_i)$. See 3.1.14 for definition of *type* and from Remark 3.1.15(i) we see that the above convention is independent of the representative $\tilde{\Delta}$. Similarly, we write $x \in \Delta$ whenever $x \in \Gamma_g(I) \setminus \mathcal{BT}_0$ is the class of a vertex $\tilde{x} \in \tilde{\Delta}$ and denote $\Delta' \supset \Delta$ for $\Delta' \in \Gamma_g(I) \setminus \mathcal{BT}_\bullet$ the class of a simplex $\tilde{\Delta}' \supset \tilde{\Delta}$.

Given $\Delta = (x_0, \dots, x_i) \in \Gamma_g(I) \setminus \mathcal{BT}_i$, by Remark 9.2.13(i,a), we have $\mathfrak{U}_{g, \Delta}^c = \bigcap_{0 \leq j \leq i} \mathfrak{U}_{g, x_j}^c$.

Definition 10.3.3. (i) Let $\Delta, \Delta' \in \Gamma_g(I) \setminus \mathcal{BT}_\bullet$ with $\Delta' \supset \Delta$. By discussion above $\mathfrak{U}_{g, \Delta'}^c$ is an affinoid subdomain of $\mathfrak{U}_{g, \Delta}^c$. Denote the resulting restriction map by

$$r_{\Delta'}^\Delta : (\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})^{(n)}(\mathfrak{U}_{g, \Delta}^c) \rightarrow (\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})^{(n)}(\mathfrak{U}_{g, \Delta'}^c)$$

(ii) Define the cohomological complex, denoted by $\check{C}_n^{st, \bullet}(g, I)$, as

$$\check{C}_n^{st, i}(g, I) := \bigoplus_{\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)-st}} (\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_{A-an}})^{(n)}(\mathfrak{U}_{g, \Delta}^c), \quad 0 \leq i \leq r-1$$

equipped with co-differentials $d^{st,i} : \check{C}_n^{st,i}(g, I) \rightarrow \check{C}_n^{st,i+1}(g, I)$ given by sending

$$f \in (\tilde{\mathcal{F}}_I^{g, \mathcal{D}^{A-an}})^{(n)}(\mathfrak{Y}_{g,\Delta}^c) \mapsto \sum_{\substack{\Delta'=(x_0,\dots,x_{i+1}) \in \Gamma_g(I) \setminus \mathcal{BT}_{i+1}^{\Gamma_g(I)-st} \\ \Delta' \supset \Delta=(x_0,\dots,\hat{x}_j,\dots,x_{i+1})}} (-1)^j r_{\Delta'}^\Delta(f).$$

We omit the dependence of g in the notation for $r_{\Delta'}^\Delta$ and $d^{st,\bullet}$ for convenience and this should not cause confusion in what follows.

Remark 10.3.4. As $\mathcal{BT}_\bullet^{\Gamma_g(I)-st}$ is not a simplicial subcomplex $\check{C}_n^{st,\bullet}(g, I)$ should not be confused with the Čech complex of an admissible open cover.

Remark 10.3.5. For each $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_\bullet^{\Gamma_g(I)-st}$, $(\tilde{\mathcal{F}}_I^{g, \mathcal{D}^{A-an}})^{(n)}(\mathfrak{Y}_{g,\Delta}^c)$ is a τ -module on $\mathcal{O}(\mathfrak{Y}_{g,\Delta}^c)$ with coefficients in $\mathcal{D}_A$ (see Remark 9.2.2) and the co-differential $d^{st,\bullet}$ is compatible with the corresponding τ -structure. In particular, denoting for $0 \leq i \leq r-1$

$$(\check{C}_n^{st,i}(g, I))^\tau := \bigoplus_{\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)-st}} ((\tilde{\mathcal{F}}_I^{g, \mathcal{D}^{A-an}})^{(n)})^\tau(\mathfrak{Y}_{g,\Delta}^c),$$

$d^{st,i}$ restricts to a well defined map of A -modules

$$(d^{st,i})^\tau : (\check{C}_n^{st,i}(g, I))^\tau \rightarrow (\check{C}_n^{st,i+1}(g, I))^\tau.$$

The resulting cohomological complex is denoted $(\check{C}_n^{st,\bullet}(g, I))^\tau$. For $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_\bullet^{\Gamma_g(I)-st}$, the module of τ -invariants $((\tilde{\mathcal{F}}_I^{g, \mathcal{D}^{A-an}})^{(n)})^\tau(\mathfrak{Y}_{g,\Delta}^c)$ is a projective A -module of rank equal to the rank of V_n (see Corollary 9.4.12).

Recall the homological complex $C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)-st}$ from 5.3.3. Since it is a complex of finite free $\mathbb{Z}[\Gamma_g(I)]$ modules, we have that $C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)-st} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n$ is a complex of finite dimensional K -vector spaces and hence one could form the cohomological complex of its K -dual $(C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)-st} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$. It is not hard to check that the K -rank of this complex matches the A -rank of $(\check{C}_n^{st,\bullet}(g, I))^\tau$ on each component. More precisely they are related to each other as follows

Lemma 10.3.6. There is a canonical isomorphism of cohomological complexes of K -vector spaces

$$(C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)-st} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^* \simeq (\check{C}_n^{st,\bullet}(g, I))^\tau \otimes_A K.$$

Proof. We have

$$(C_i(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)-st} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^* = \bigoplus_{\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)-st}} (V_n)^*$$

($0 \leq i \leq r-1$) and the co-differentials are given by

$$d^i : (C_i(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^* \rightarrow (C_{i+1}(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$$

sending

$$(v_\Delta)_{\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)\text{-st}}} \mapsto \left(\sum_{\substack{0 \leq j \leq i+1 \\ \Delta = (x_0, \dots, \hat{x}_j, \dots, x_{i+1})}} (-1)^j [v_\Delta]_{st} \right)_{\Delta' = (x_0, \dots, x_{i+1}) \in \Gamma_g(I) \setminus \mathcal{BT}_{i+1}^{\Gamma_g(I)\text{-st}}},$$

where similar to Remark 5.3.4 $[v_\Delta]_{st} = v_\Delta$ if $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)\text{-st}}$, else $[v_\Delta]_{st} = 0$. On the other hand, by Remark 9.2.13(i) combined with Corollary 9.4.12, for $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I)\text{-st}}$, the module of τ -invariants $((\tilde{\mathcal{F}}_I^{g, \mathcal{D}_A - an})^{(n)})^\tau(\mathcal{U}_{g, \Delta}^c)$ is canonically isomorphic to $\text{Sym}^n(\text{Hom}_A(\Lambda_g, \Omega_A))$. We can identify $\text{Sym}^n(\text{Hom}_A(\Lambda_g, \Omega_A)) \otimes_A K$ canonically with $\text{Sym}^n(V^*) \simeq (V_n)^*$. Consequently this yields the required isomorphism of cohomological complexes from the statement of the lemma. $\square$

Consequently the top cohomologies of the two complexes are isomorphic. We get

Corollary 10.3.7. There is a canonical isomorphism of K -vector spaces

$$(\text{St}^I(\Lambda_g) \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^* \simeq H^{r-1}((\check{C}_n^{st, \bullet}(g, I))^\tau) \otimes_A K.$$

Proof. Since $- \otimes_A K$ is exact, we have $H^{r-1}((\check{C}_n^{st, \bullet}(g, I))^\tau \otimes_A K) \simeq H^{r-1}((\check{C}_n^{st, \bullet}(g, I))^\tau) \otimes_A K$. Hence it suffices to show that the left side of the isomorphism in the statement of the Corollary is the top cohomology of $(C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$. Indeed $\text{St}^I(\Lambda_g)$ is the top homology of $C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}}$ by Definition 5.3.5. Since the latter is a complex of finite free $\mathbb{Z}[\Gamma_g(I)]$ -modules, $\text{St}^I(\Lambda_g) \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n$ is the top homology of $C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n$ and due to the exactness of the K -dual $(-)^*$, $(\text{St}^I(\Lambda_g) \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$ is the top cohomology of $(C_\bullet(\mathcal{BT}_\bullet(K^r), \mathbb{Z})^{\Gamma_g(I)\text{-st}} \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$. $\square$

Continuing the notation from Remark 10.3.2, let us choose a rigid τ -sheaf $\underline{\mathcal{H}}_g \in \widetilde{\mathbf{Coh}}_\tau(\overline{M}_g^{an}, \mathcal{D}_A)$ representing the crystal $(j_g)_!(\tilde{\mathcal{F}}_I^{g, \mathcal{D}_A - an})^{(n)} \in \widetilde{\mathbf{Crys}}(\overline{M}_g^{an}, \mathcal{D}_A)$ obtained as the $\mathcal{D}_A$ -analytification of a suitable coherent τ -sheaf on $\overline{M}_g$. Fix this choice. For the following, recall the finite admissible cover $\mathcal{U}_g$ of $\overline{M}_g^{an}$ fixed at the end of the previous section.

Definition 10.3.8. (i) Denote by $\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g)$ the Čech cohomological complex

$$\check{C}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g) := \prod_{\mathfrak{V}_0, \dots, \mathfrak{V}_i \in \mathcal{U}_g} \underline{\mathcal{H}}_g((\mathfrak{V}_0 \cap \dots \cap \mathfrak{V}_i) \times \mathcal{D}_A), \quad 0 \leq i \leq r-1$$

equipped with co-differential d^i for $0 \leq i \leq r-2$ defined on $f \in \check{C}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)$, by describing its component in $\underline{\mathcal{H}}_g((\mathfrak{V}'_0 \cap \dots \cap \mathfrak{V}'_{i+1}) \times \mathfrak{D}_A)$ as

$$\sum_{0 \leq j \leq i+1} (-1)^j f_{|\mathfrak{V}'_0 \cap \dots \cap \mathfrak{V}'_{i+1}}^{\mathfrak{V}'_0, \dots, \hat{\mathfrak{V}}'_j, \dots, \mathfrak{V}'_{i+1}},$$

where $f_{|\mathfrak{V}'_0 \cap \dots \cap \mathfrak{V}'_{i+1}}^{\mathfrak{V}'_0, \dots, \hat{\mathfrak{V}}'_j, \dots, \mathfrak{V}'_{i+1}}$ denotes the component of f in $\underline{\mathcal{H}}_g((\mathfrak{V}'_0 \cap \dots \cap \hat{\mathfrak{V}}'_j \cap \dots, \mathfrak{V}'_{i+1}) \times \mathfrak{D}_A)$.

Denote the cohomology of this complex by $\check{H}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)$ for $i \geq 0$.

(ii) Define the complex of A -modules $\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau$ by

$$\check{C}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau := \underline{\mathcal{H}}_g^\tau(\mathfrak{V}_0 \cap \dots \cap \mathfrak{V}_i), \quad 0 \leq i \leq r-1$$

(cf. 9.3.3) with the induced co-differentials from (i) denoted by $(d^i)^\tau : (\check{C}^i)^\tau \rightarrow (\check{C}^{i+1})^\tau$ for $0 \leq i \leq r-2$.

For any rigid τ -sheaf or crystal $\tilde{\mathcal{F}}$ on $\mathrm{Sp}(K_\infty)$ with coefficients in $\mathcal{D}_A$, we will denote by $\tilde{\mathcal{F}}^\tau$ the A -module of invariants $\tilde{\mathcal{F}}^\tau(\mathrm{Sp}(K_\infty))$ from Remark 9.3.3.

Remark 10.3.9. (i) Note that each group $\check{H}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)$ is a τ -module over K_∞ with coefficients in $\mathcal{D}_A$ and we observed in Remark 9.3.5 that $\check{H}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)$ is isomorphic to $\check{H}^i((\overline{M}_g)^{an}, \underline{\mathcal{H}}_g)$ as τ -modules over K_∞ with coefficients in $\mathcal{D}_A$. In particular, the space of τ -invariants $\check{H}^{r-1}(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau$ is independent of the choice of representative $\underline{\mathcal{H}}_g$ and is equal to

$$(R^i(f_g)_!)(\tilde{\mathcal{F}}_I^{g, \mathcal{D}_A - an})^{(n)\tau},$$

where $f_g : M_g \rightarrow \mathrm{Spec}(K_\infty)$ is the structure map.

(ii) Due to the left exactness of the operation of taking τ -invariants, there is a canonical injection of A -modules $(\mathrm{Ker}(d^i))^\tau / (\mathrm{Im}(d^{i-1}))^\tau \hookrightarrow (\mathrm{Ker}(d^i) / \mathrm{Im}(d^{i-1}))^\tau = \check{H}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau$, for $i \geq 1$. Since there is also a canonical map (not necessarily injective), $H^i(\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau) = \mathrm{Ker}((d^i)^\tau) / \mathrm{Im}((d^{i-1})^\tau) \rightarrow (\mathrm{Ker}(d^i))^\tau / (\mathrm{Im}(d^{i-1}))^\tau$, we consequently get by composition

$$H^i(\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau) \rightarrow \check{H}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)^\tau.$$

Let $f_g : M_g \rightarrow \mathrm{Spec}(K_\infty)$ is the structure map as before.

Lemma 10.3.10. There exists a canonical map of A -modules

$$\beta_i : H^i((\check{C}_n^{st, \bullet}(g, I))^\tau) \rightarrow (R^i(f_g)_!)(\tilde{\mathcal{F}}_I^{g, \mathcal{D}_A - an})^{(n)\tau}$$

for all $i \geq 0$.

Proof. By definition of $(j_g)_!$, we have that $(j_g)^*\underline{\mathcal{H}}_g = (\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_A - an})^{(n)}$. Hence for $\Delta \in \Gamma_g(I) \setminus \mathcal{BT}_i^{\Gamma_g(I) - \text{st}}$,

$$\underline{\mathcal{H}}_g(\mathfrak{U}_{g, \Delta} \times \mathfrak{D}_A) = (\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_A - an})^{(n)}(\mathfrak{U}_{g, \Delta} \times \mathfrak{D}_A)$$

appears as a direct summand in $\check{C}^i(\mathcal{U}_g, \underline{\mathcal{H}}_g)$ by the property of the cover $\mathcal{U}_g$ and the fact that if $\Delta = (x_0, \dots, x_i) \in \Gamma_g(I) \setminus \mathcal{BT}_i$ then $\mathfrak{U}_{g, \Delta}^c = \bigcap_{0 \leq j \leq i} \mathfrak{U}_{g, x_j}^c$. This gives rise to the natural embedding of complexes

$$\check{C}_n^{st, \bullet}(g, I) \hookrightarrow \check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g).$$

Due to left exactness of the operation of τ -invariants, we further obtain an embedding

$$(\check{C}_n^{st, \bullet}(g, I))^\tau \hookrightarrow (\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g))^\tau.$$

Consequently we obtain a map on the i -th cohomology

$$H^i(\check{C}_n^{st, \bullet}(g, I)^\tau) \rightarrow H^i((\check{C}^\bullet(\mathcal{U}_g, \underline{\mathcal{H}}_g))^\tau).$$

Composing this with the map described in 10.3.9(ii) and the observation in (i) of the same remark, we get the desired β_i . $\square$

Theorem 10.3.11. (cf. [Boe02, Thm.10.3, Cor.10.6]) There exists the following canonical map of finite dimensional K -vector spaces

$$\text{ES}_{r, I}^{(n)} : \mathbf{C}_n^{st}(\mathcal{K}(I))^* \rightarrow (\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))^{\mathfrak{D}_A - an})^\tau \otimes_A K$$

where $(\cdot)^*$ denotes the K -dual.

Proof. By Remark 10.3.2 and the definition of $\mathbf{C}_n^{st}(\mathcal{K}(I))$ it suffices to construct a map

$$(\text{St}^I(\Lambda_g) \otimes_{\Gamma_g(I)} V_n)^* \rightarrow (R^{r-1}(f_g)_!(\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_A - an})^{(n)})^\tau \otimes_A K,$$

for all $g \in \text{Cl}_I$. By Lemma 10.3.10 we have a map

$$\beta_{r-1} \otimes 1 : H^{r-1}((\check{C}_n^{st, \bullet}(g, I))^\tau) \otimes_A K \rightarrow (R^{r-1}(f_g)_!(\tilde{\mathcal{F}}_I^{g, \mathfrak{D}_A - an})^{(n)})^\tau \otimes_A K.$$

Corollary 10.3.7 shows that the LHS is canonically isomorphic to $(\text{St}^I(\Lambda_g) \otimes_{\mathbb{Z}[\Gamma_g(I)]} V_n)^*$, hence composing $\beta_{r-1} \otimes 1$ with this isomorphism gives us the required map. $\square$

Remark 10.3.12. Composing $\text{ES}_{r, I}^{(n)}$ with dual of the map $\text{AR}_{\mathcal{K}(I)} \circ (\text{GQ}_{\mathcal{K}(I)} \otimes 1) : \mathbf{C}_n^{st}(\mathcal{K}(I)) \rightarrow \mathbf{C}_{har}(\mathcal{K}(I), n)$ from 10.1, gives a map of K -vector spaces

$$\mathbf{C}_{har}(\mathcal{K}(I), n)^* \rightarrow (\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))^{\mathfrak{D}_A - an})^\tau \otimes_A K.$$

10.4 Some contemplations

We conclude our discussions by placing the Eichler-Shimura map $\text{ES}_{r,I}^{(n)}$ in a broader context, analogous to 1.12, with the help of results in [Grä21]. We hope that this gives credence to our expectation of a higher rank Eichler-Shimura isomorphism and also end by posing few questions with the prospect for future research.

10.4.1 Relation to Gräf's work

Recall from section 1.1.2 that in the rank 2 case, from the work of Böckle [Boe02], combining the Eichler-Shimura map $\text{ES}_2^{(n)}$ with residue isomorphism of Teitelbaum 1.8 relates (via an isomorphism) the space of Drinfeld cusp forms $S_{n+2}(\mathcal{K}(I)) := \bigoplus_{g \in \text{Cl}_I} S_{n+2}(\Gamma_g(I))$ and the τ -invariants of $\underline{\mathcal{S}}_{n+2}(\mathcal{K}(I))^{\mathcal{D}_{A-an}}$ for non-zero proper ideal $I \subset A$, see 1.11. For $r \geq 3$, over the past decade, a theory of higher rank Drinfeld modular forms has been developed due to the efforts of Basson, Breuer and Pink [BBP24] and also independently by Gekeler in [Gek25b]. A theory of Drinfeld cusp forms in higher ranks also arise naturally and one can define analogous to the rank 2 case, the space $S_k(\Gamma)$ of *Drinfeld cusp forms of weight n for Γ* a congruence subgroup of $\text{GL}_r(K)$. We refer the reader to op.cit. for details about this definition. One can set, for I as before

$$S_k(\mathcal{K}(I)) := \bigoplus_{g \in \text{Cl}_I} S_k(\Gamma_g(I)), \quad k \geq 0.$$

In light of the rank 2 theory it is natural to expect a relation between $(\underline{\mathcal{S}}_{n+r}(\mathcal{K}(I))^{\mathcal{D}_{A-an}})^{\tau}$ with the space of Drinfeld cusp forms $S_{n+r}(\mathcal{K}(I))$. This is possible in the case $r = 3$ and $A = \mathbb{F}_q[t]$, thanks to the doctoral work of Gräf in [Grä21] and the Eichler-Shimura map constructed in the previous section. From discussions in [Grä21, §8.4] gives a residue map

$$S_{k+3}(\mathcal{K}(I)) \xrightarrow{\text{Res}_I} \mathbf{C}_{har}(\mathcal{K}(I), n) \otimes_K \mathbb{C}_{\infty},$$

which is conjecturally an isomorphism. With this map as well as the discussions from previous section and Part I affirm, at least in the present setting, the schemata for a higher rank Eichler-Shimura isomorphism expected in 1.12

$$\begin{aligned}
S_{n+3}(\mathcal{K}(I)) &\xrightarrow[\text{[Grä21]}]{\text{Res}_I} \mathbf{C}_{har}(\mathcal{K}(I), n) \otimes_K \mathbb{C}_\infty \\
&\xrightarrow[\sim]{\text{Te}_I(10.1.5)} \mathbf{C}_n^{st}(\mathcal{K}(I)) \\
&\xrightarrow[10.3.11]{\text{ES}_{3,I}^{(n)}} (\underline{\mathcal{S}}_{n+3}(\mathcal{K}(I))^{\mathfrak{D}_A - an})^\tau \otimes_A K
\end{aligned} \tag{10.1}$$

In [Grä21] it also is shown that Res_I is a conjectural isomorphism for $I = (t)$. Let us conclude this section by briefly discussing it and also some of its consequences.

Define $V_{n, \mathbb{C}_\infty} := V_n \otimes_K \mathbb{C}_\infty$. Note that for $I = (t)$, Cl_I is a singleton and we have

$$\mathbf{C}_{har}(\mathcal{K}(t), n) \otimes_K \mathbb{C}_\infty = \mathbf{C}_{har}(V_{n, \mathbb{C}_\infty})^{\Gamma(t)}$$

where $\Gamma(t) := \Gamma_{A \oplus 3}(t)$. As alluded to below 1.8 to provide an inverse to Res_I in the rank 2 case, Teitelbaum has to pass through the space of *measures* on the projective line $\mathbb{P}_{K_\infty}^1$ and he was motivated to do so following his own work in the p -adic context in [Tei90]. Extending this idea, Gräf conjecturally constructs a $\text{GL}_3(K_\infty)$ -equivariant map ([Grä21, Thm.4.29])

$$\mathbf{C}_{har}(V_{n, \mathbb{C}_\infty})^b \rightarrow \text{St}_3^{an}(n)', \tag{10.2}$$

where

- $\mathbf{C}_{har}(V_{n, \mathbb{C}_\infty})^b$ denotes the space of *bounded* harmonic cocycles in $\mathbf{C}_{har}(V_{n, \mathbb{C}_\infty})$, see [Grä21, Defn.3.5]. It turns out that harmonic cocycles invariant with respect to a congruence subgroup are bounded [Grä21, Cor.8.3],
- $\text{St}_3^{an}(n)'$ is the continuous dual of the *locally analytic Steinberg representation* $\text{St}_3^{an}(n)$, see [Grä21, Defn.4.5].

Gräf's construction holds true more generally for congruence subgroups containing $\Gamma(t)$.

Admitting the existence of the above map, he proves an isomorphism similar to [Tei91, Thm.16]:-

$$S_{n+3}(\mathcal{K}(t)) \xrightarrow{\text{Res}_{(t)}} \mathbf{C}_{har}(\mathcal{K}(t), V_n) \otimes_K \mathbb{C}_\infty.$$

Composing the dual of $\text{Res}_{(t)}$ with the following map induced from Remark 10.3.12

$$\mathbf{C}_{har}(\mathcal{K}(t), n)^* \otimes_K \mathbb{C}_\infty \rightarrow (\underline{\mathcal{S}}_{n+r}(\mathcal{K}(t))^{\mathfrak{D}_A - an})^\tau \otimes_K \mathbb{C}_\infty$$

we obtain a map, albeit conjecturally, analogous to 1.11

$$S_{n+3}(\mathcal{K}(t))^* \xrightarrow[\text{conj.}]{} (\underline{\mathcal{S}}_{n+3}(\mathcal{K}(t))^{\mathcal{D}_{A-an}})^{\tau} \otimes_A \mathbb{C}_{\infty}.$$

10.4.2 Future directions

Here we list three problems for further research which arise immediately from our discussions so far as well as in keeping with our expectations from rank 2.

- (a) **Case of $A = \mathbb{F}_q[t]$.** In [Boe02; Pin12], for a fine open compact subgroups $\mathcal{K} \subset \widehat{\mathrm{GL}}_r(\mathbb{F}_q[t])$ a Drinfeld moduli space with level $\mathcal{K}$ -structure, $M_{\mathcal{K}}^r$ is defined and as briefly mentioned before it admits a Satake compactification $\overline{M}_{\mathcal{K}}^r$. Consequently one can form a crystal $\underline{\mathcal{S}}_{n+r}(\mathcal{K})$ analogous to 10.3.1. It is straightforward to also extend the definition of $\mathbf{C}_n^{st}(\mathcal{K}(I))$ to $\mathbf{C}_n^{st}(\mathcal{K})$. This prods the question whether the Eichler-Shimura map in 10.3.11 can also be extended to get $\mathrm{ES}_{r,\mathcal{K}}^{(n)}$. If true, then more generally is $\mathrm{ES}_{r,\mathcal{K}}^{(n)}$ injective/surjective/isomorphism? We are primarily motivated to ask this question due to the work of Pink [Pin12], where the Satake compactification of the Drinfeld moduli space $M_{\mathcal{K}}^r$ has been explicitly worked out for $\mathcal{K} = \mathcal{K}(t), \mathcal{K}_1(t)$. For instance $M_{\mathcal{K}_1(t)}^r$ is $\mathbb{P}_{\mathbb{F}_q(t)}^{r-1}$ minus the r standard coordinate hyperplanes and $\overline{M}_{\mathcal{K}_1(t)}^r = \mathbb{P}_{\mathbb{F}_q(t)}^{r-1}$. We expect that this should enable one to handle the construction of $\mathrm{ES}_{r,\mathcal{K}}^{(n)}$ with more precision. As briefly mentioned in 1.1.2, to show the rank 2 Eichler-Shimura map is an isomorphism, Böckle uses a formula of Pink about Euler-Poincare characteristic of étale constructible sheaves on the compactified Drinfeld modular curve, from [Pin00]. It is natural to wonder whether such a formula exists for the Pink's Satake compactification $\overline{M}_{\mathcal{K}}^r$ in the case $\mathcal{K} = \mathcal{K}(t), \mathcal{K}_1(t)$ and if so, does it relate to the dimension of $\mathbf{C}_n^{st}(\mathcal{K})$. When $r = 3$, for analyzing the dimension of the latter, the results from [BGK25] should be relevant.
- (b) **Hecke operators.** In [Boe02, Chap.6], an appropriate action by *Hecke operators* is defined on all the objects in an analogous diagram 10.1 for rank 2 and the corresponding maps are shown to be Hecke equivariant. Does this generalize in particular to 10.1 and more generally to higher ranks as well? A theory of Hecke operators and their action on Drinfeld modular forms has been developed in [BBP24; Pin12] and also by Gekeler [Gek25a]. In fact the residue isomorphism of Gräf from previous subsection is shown to be Hecke equivariant in [Grä21, Cor.8.12].
- (c) **Automatic cuspidality of harmonic cocycles.** As mentioned in the previous subsection in the $r = 3, A = \mathbb{F}_q[t]$ case, Gräf proves that harmonic cocycles that are invariant under a congruence subgroup Γ are bounded and a key result required to prove this is [Grä21, Thm.8.2]

which says that every such harmonic cocycle is supported on a finite set of chambers in $\Gamma \backslash \mathcal{BT}_2$. It seems natural to ask whether the latter result also holds true for higher rank buildings and the general A case. Due to ideas of G. Böckle this seems within reach now.

...

Bibliography

- [AB10] P. Abramenko and K.S. Brown. *Buildings: Theory and Applications*. Graduate Texts in Mathematics. Springer New York, 2010. ISBN: 9781441927019. URL: <https://books.google.de/books?id=X1RtcgAACAAJ>.
- [AGM12] Avner Ash, Paul E. Gunnells, and Mark McConnell. “Resolutions of the Steinberg module for $GL(n)$ ”. In: *Journal of Algebra* 349.1 (2012), pp. 380–390. ISSN: 0021-8693. DOI: <https://doi.org/10.1016/j.jalgebra.2011.09.018>. URL: <https://www.sciencedirect.com/science/article/pii/S0021869311005357>.
- [And86] Greg W. Anderson. “t-Motives”. In: *Duke Mathematical Journal* 53.2 (1986), pp. 457–502. DOI: 10.1215/S0012-7094-86-05328-7. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-53/issue-2/t-Motives/10.1215/S0012-7094-86-05328-7.full>.
- [AR79] Avner Ash and Lee Rudolph. “The Modular Symbol and Continued Fractions in Higher Dimensions.” In: *Inventiones mathematicae* 55 (1979), pp. 241–250. URL: <http://eudml.org/doc/186135>.
- [Bas14] Dirk J. Basson. “On the Coefficients of Drinfeld Modular Forms of Higher Rank”. PhD thesis. Stellenbosch, South Africa: Stellenbosch University, 2014. URL: <http://sun.ac.za>.
- [BBP24] Dirk Basson, Florian Breuer, and Richard Pink. “Drinfeld Modular Forms of Arbitrary Rank”. In: *Memoirs of the American Mathematical Society* 304 (Dec. 2024). DOI: 10.1090/memo/1531.
- [BGK25] Gebhard Boeckle, Peter Mathias Graef, and Theresa Kaiser. *U-Operators Acting on Harmonic Cocycles for GL_3 and Their Slopes*. 2025. arXiv: 2503.00141 [math.NT]. URL: <https://arxiv.org/abs/2503.00141>.

- [BH07] Gebhard Böckle and Urs Hartl. “Uniformizable families of t -motives”. In: *Transactions of the American Mathematical Society* 359.8 (2007), pp. 3933–3972. DOI: 10.1090/S0002-9947-07-04136-2.
- [BKW13] Kai-Uwe Bux, Ralf Köhl, and Stefan Witzel. “Higher finiteness properties of reductive arithmetic groups in positive characteristic: The Rank Theorem”. In: *Annals of Mathematics* 177.1 (Jan. 2013), pp. 311–366. ISSN: 0003-486X. DOI: 10.4007/annals.2013.177.1.6. URL: <http://dx.doi.org/10.4007/annals.2013.177.1.6>.
- [BN81] Pilar Bayer and Jürgen Neukirch. “On automorphic forms and Hodge theory”. In: *Mathematische Annalen* 257.2 (1981), pp. 137–155. DOI: 10.1007/BF01450621. URL: <https://doi.org>.
- [Boe02] G. Boeckle. *An Eichler Shimura isomorphism over function fields between Drinfeld modular forms and cohomology classes of crystals*. 2002. URL: <https://typo.iwr.uni-heidelberg.de/groups/arith-geom/members/gebhard-boeckle/publications>.
- [Bos14] Siegfried Bosch. *Lectures on Formal and Rigid Geometry*. Vol. 2105. Lecture Notes in Mathematics. Cham: Springer-Verlag, 2014. ISBN: 978-3-319-04417-0. DOI: 10.1007/978-3-319-04417-0.
- [BP09] Gebhard Böckle and Richard Pink. *Cohomological theory of crystals over function fields*. Vol. 9. Zürich: European Mathematical Society, 2009. ISBN: 978-3-03719-074-6.
- [BS97] Arno Blum and Ulrich Stuhler. “Drinfeld modules and elliptic sheaves”. In: *Vector Bundles on Curves — New Directions*. Vol. 1649. Lecture Notes in Mathematics. Springer, 1997, pp. 110–188. DOI: 10.1007/BFb0094426.
- [BV26] Gebhard Böckle and Sriram Chinthalagiri Venkata. *The unstable complex in Bruhat-Tits buildings for arithmetic groups over function fields*. 2026. arXiv: 2603.09754 [math.NT]. URL: <https://arxiv.org/abs/2603.09754>.
- [Cak18] B. Cakir. “Algorithmic aspects of Hecke eigensystems on spaces of harmonic cocycles”. MA thesis. Universität Heidelberg, 2018.
- [Con99] Brian Conrad. “Irreducible components of rigid spaces”. eng. In: *Annales de l’institut Fourier* 49.2 (1999), pp. 473–541. URL: <http://eudml.org/doc/75345>.

- [Del71] Pierre Deligne. “Formes modulaires et représentations ℓ -adiques”. In: *Séminaire Bourbaki : vol. 1968/69, exposés 347-363*. Vol. 179. Lecture Notes in Mathematics 355. Springer-Verlag, 1971, pp. 139–172. DOI: 10.1007/BFb0058810. URL: https://www.numdam.org/item/SB_1968-1969__11__139_0/.
- [deS01] Ehud de Shalit. “Residues on buildings and de Rham cohomology of p -adic symmetric domains”. In: *Duke Mathematical Journal* 106.1 (2001), pp. 123–191. DOI: 10.1215/S0012-7094-01-10615-7. URL: <https://doi.org/10.1215/S0012-7094-01-10615-7>.
- [DH05] Anton Deitmar and Joachim Hilgert. “Cohomology of arithmetic groups with infinite dimensional coefficient spaces”. In: *Documenta Mathematica* 10 (2005), pp. 199–216. DOI: 10.4171/DM/186.
- [DH87] P Deligne and D Husemoeller. *Survey of Drinfeld modules*. Providence, Rhode Island: Contemporary Math, 1987.
- [Dri74] V. G. Drinfeld. “Elliptic modules”. In: *Mat. Sb. (N.S.)* 94(136).4(8) (1974), pp. 594–627. URL: <http://mi.mathnet.ru/sm3735>.
- [Eic57] Martin Eichler. “Eine Verallgemeinerung der Abelschen Integrale”. In: *Mathematische Zeitschrift* 67.1 (1957), pp. 267–298. DOI: 10.1007/BF01258863. URL: <https://link.springer.com/article/10.1007/BF01258863>.
- [Eme76] Ralph Waldo Emerson. *Letters and Social Aims*. Boston, MA: James R. Osgood and Company, 1876.
- [FP04] Jean Fresnel and Marius van der Put. *Rigid Analytic Geometry and Its Applications*. Vol. 218. Progress in Mathematics. Boston, MA: Birkhäuser, 2004. ISBN: 978-1-4612-6488-0.
- [Geb96] M. Gebhardt. “Operationen arithmetischer Untergruppen von $GL(3)$ auf Bruhat-Tits-Gebäuden”. MA thesis. Universität Saarbrücken, 1996.
- [Gek25a] Ernst-Ulrich Gekeler. “Modular forms for $GL(r, \mathbb{F}_q[T])$: Hecke operators and growth of expansion coefficients”. In: *arXiv preprint arXiv:2511.01712* (2025). arXiv: 2511.01712 [math.NT]. URL: <https://arxiv.org/abs/2511.01712>.
- [Gek25b] Ernst-Ulrich Gekeler. “On Drinfeld modular forms of higher rank VII: Expansions at the boundary”. In: *Journal of Number Theory* 269 (2025), pp. 240–280. DOI: 10.1016/j.jnt.2024.07.005.
- [Gek86] E. U. Gekeler. *Drinfeld modular curves*. Springer Berlin, Heidelberg, 1986.

- [Gek89] Ernst-Ulrich Gekeler. “On the de Rham isomorphism for Drinfeld modules”. In: *Journal für die reine und angewandte Mathematik* 1989.401 (1989), pp. 188–208. DOI: 10.1515/crll.1989.401.188.
- [Gek90a] Ernst-Ulrich Gekeler. “De Rham cohomology and the Gauss-Manin connection for Drinfeld modules”. In: *p-adic Analysis*. Ed. by Francesco Baldassarri, Siegfried Bosch, and Bernard Dwork. Vol. 1454. Lecture Notes in Mathematics. Berlin, Heidelberg: Springer, 1990, pp. 223–255. DOI: 10.1007/BFb0091142.
- [Gek90b] Ernst-Ulrich Gekeler. “De Rham cohomology for Drinfeld modules”. In: *Séminaire de Théorie des Nombres, Paris 1988–1989*. Ed. by David Sinnou. Vol. 91. Progress in Mathematics. Boston, MA: Birkhäuser, 1990, pp. 57–85.
- [GI63] Oscar Goldman and Nagayoshi Iwahori. “The space of p-adic norms”. In: *Algebraic groups and number theory* (1963).
- [Gos80] David Goss. “ p -adic Eisenstein series for function fields”. In: *Compositio Mathematica* 41 (1980), pp. 3–38. URL: <https://api.semanticscholar.org/CorpusID:122400355>.
- [Gos96] David Goss. *Basic Structures of Function Field Arithmetic*. Vol. 35. Ergebnisse der Mathematik und ihrer Grenzgebiete. Berlin, Heidelberg: Springer-Verlag, 1996. ISBN: 978-3-540-63541-3.
- [Grä21] Peter Mathias Gräf. “Boundary Distributions for GL_3 over a Local Field and Symmetric Power Coefficients”. English. Available at heiDOK: <https://archiv.ub.uni-heidelberg.de/volltextserver/29302/>. Inaugural-Dissertation. Heidelberg, Germany: Ruprecht-Karls-Universität Heidelberg, 2021.
- [Gra82] Daniel R. Grayson. “Finite generation of K-groups of a curve over a finite field (after D. G. Quillen)”. In: *Algebraic K-Theory*. Ed. by R. Keith Dennis. Berlin, Heidelberg: Springer Berlin Heidelberg, 1982, pp. 69–90. ISBN: 978-3-540-39553-9.
- [GV25] Oğuz Gezmiş and Sriram Chinthalagiri Venkata. “On nearly holomorphic Drinfeld modular forms for admissible coefficient rings”. In: *Research in Number Theory* 11.1 (2025), p. 87. DOI: 10.1007/s40993-025-00666-2.
- [Hat02] Allen Hatcher. *Algebraic topology*. Cambridge: Cambridge University Press, 2002, pp. xii+544. ISBN: 0-521-79160-X; 0-521-79540-0.

- [Hid93] Haruzo Hida. *Elementary Theory of L-functions and Eisenstein Series*. Vol. 26. London Mathematical Society Student Texts. Cambridge: Cambridge University Press, 1993. ISBN: 9780521435697.
- [HN74] G. Harder and M.S. Narasimhan. “On the Cohomology Groups of Moduli Spaces of Vector Bundles on Curves.” In: *Mathematische Annalen* 212 (1974), pp. 215–248. URL: <http://eudml.org/doc/182746>.
- [KY21] Satoshi Kondo and Seidai Yasuda. *Arithmetic quotients of the Bruhat-Tits building for projective general linear group in positive characteristic*. 2021. arXiv: 2101.01424 [math.NT]. URL: <https://arxiv.org/abs/2101.01424>.
- [KY25] Satoshi Kondo and Seidai Yasuda. “Arithmetic Quotients of the Bruhat-Tits Building for Projective General Linear Group in Positive Characteristic”. In: *Memoirs of the American Mathematical Society* 306 (Jan. 2025). DOI: 10.1090/memo/1547.
- [Leh09] Thomas Lehmkuhl. *Compactification of the Drinfeld Modular Surfaces*. Vol. 197. Memoirs of the American Mathematical Society 921. Providence, R.I.: American Mathematical Society, 2009, pp. xi+94. ISBN: 978-0-8218-4244-7.
- [Liu02] Q. Liu. *Algebraic Geometry and Arithmetic Curves*. Oxford graduate texts in mathematics. Oxford University Press, 2002. ISBN: 9780198502845. URL: <https://books.google.de/books?id=n9uQvKVwj8gC>.
- [LRS93] G. Laumon, M. Rapoport, and U. Stuhler. “D-elliptic sheaves and the Langlands correspondence.” In: *Inventiones mathematicae* 113.2 (1993), pp. 217–338. URL: <http://eudml.org/doc/144128>.
- [Lö23] Clara Löh. “Applied Algebraic Topology”. Lecture notes. 2023. URL: https://loeh.app.uni-regensburg.de/teaching/aat_ws2223/lecture_notes.pdf.
- [Mis03] Guido Mislin. “Equivariant K-Homology of the Classifying Space for Proper Actions”. In: *Proper Group Actions and the Baum-Connes Conjecture*. Basel: Birkhäuser Basel, 2003, pp. 1–78. ISBN: 978-3-0348-8089-3. DOI: 10.1007/978-3-0348-8089-3_1. URL: https://doi.org/10.1007/978-3-0348-8089-3_1.
- [MPC96] J.P. May, R.J. Piacenza, and M. Cole. *Equivariant Homotopy and Cohomology Theory: Dedicated to the Memory of Robert J. Piacenza*. Regional conference series in mathematics. American Mathematical Society, 1996. ISBN: 9780821803196. URL: <https://books.google.de/books?id=K0cZYVxkQ08C>.

- [Mum78] David Mumford. “An algebro-geometric construction of commuting operators and of solutions to the Toda lattice equation, Korteweg deVries equation and related non-linear equations”. In: *Proceedings of the International Symposium on Algebraic Geometry (Kyoto Univ., Kyoto, 1977)*. Ed. by Masayoshi Nagata. Tokyo: Kinokuniya Book-Store Co., Ltd., 1978, pp. 115–153.
- [Mun84] James R. Munkres. *Elements of Algebraic Topology*. Addison Wesley Publishing Company, 1984. ISBN: 0201045869. URL: <http://www.worldcat.org/isbn/0201045869>.
- [Mü15] Jennylee Müller. “Modulsymbole und Bruhat-Tits-Gebäude der $\mathrm{PGL}(3)$ über Funktionenkörpern”. PhD thesis. Kassel, Universität Kassel, Fachbereich 10 Mathematik und Naturwissenschaften, Institut für Mathematik, 2015.
- [Pap23] Mihran Papikian. *Drinfeld Modules*. Vol. 296. Graduate Texts in Mathematics. Cham, Switzerland: Springer, 2023. ISBN: 978-3-031-19706-2. DOI: 10.1007/978-3-031-19707-9.
- [Pin00] Richard Pink. “Euler–Poincaré formula in equal characteristic under ordinariness assumptions”. In: *Manuscripta Mathematica* 102.1 (2000), pp. 1–24. DOI: 10.1007/PL00005849.
- [Pin12] Richard Pink. “Compactification of Drinfeld modular varieties and Drinfeld modular forms of arbitrary rank”. In: *Manuscripta Mathematica* 140.3-4 (2012), pp. 333–361. DOI: 10.1007/s00229-012-0544-3. URL: <https://link.springer.com/article/10.1007/s00229-012-0544-3>.
- [Qui73] Daniel Quillen. “Finite generation of the groups K_i of rings of algebraic integers”. In: *Higher K-Theories*. Ed. by H. Bass. Berlin, Heidelberg: Springer Berlin Heidelberg, 1973, pp. 179–198. ISBN: 978-3-540-37767-2.
- [Qui78] Daniel G. Quillen. “Homotopy properties of the poset of nontrivial p -subgroups of a group”. In: *Advances in Mathematics* 28 (1978), pp. 101–128. URL: <https://api.semanticscholar.org/urn:37065862>.
- [Ser80] J. P. Serre. *Trees*. Springer-Verlag GmbH Germany: Springer Berlin, Heidelberg, 1980.
- [Ses82] C. S. Seshadri. *Fibrés vectoriels sur les courbes algébriques (Vector bundles on algebraic curves)*. Ed. by J.-M. Drezet. Vol. 96. Astérisque. Société Mathématique de France, 1982, p. 209. URL: <https://emath.fr>.

- [Shi71] Goro Shimura. *Introduction to the Arithmetic Theory of Automorphic Functions*. Vol. 11. Publications of the Mathematical Society of Japan. Tokyo / Princeton, NJ: Iwanami Shoten and Princeton University Press, 1971. ISBN: 9780691080925.
- [Sou79] Christophe Soulé. “Chevalley groups over polynomial rings”. In: *Homological Group Theory*. London Mathematical Society Lecture Note Series. Cambridge University Press, 1979, pp. 359–368.
- [SS91] P. Schneider and U. Stuhler. “The cohomology of p-adic symmetric spaces”. In: *Invent. Math.* 105 (1991), pp. 47–122. URL: <https://doi.org/10.1007/BF01232257>.
- [ST97] Peter Schneider and Jeremy Teitelbaum. “An integral transform for p-adic symmetric spaces”. In: *Duke Mathematical Journal* 86.3 (1997), pp. 391–433. DOI: 10.1215/S0012-7094-97-08612-9.
- [Sta21] The Stacks project authors. *The Stacks project*. 2021. URL: <https://stacks.math.columbia.edu>.
- [Tei90] Jeremy T. Teitelbaum. “Values of p-adic L-functions and a p-adic Poisson kernel”. In: *Inventiones mathematicae* 101.1 (1990), pp. 395–410. DOI: 10.1007/BF01231508. URL: <https://doi.org/10.1007/BF01231508>.
- [Tei91] Jeremy Teitelbaum. “The Poisson kernel for Drinfeld modular curves”. In: *Journal of the American Mathematical Society* 4 (1991), pp. 491–511. URL: <https://api.semanticscholar.org/CorpusID:43579969>.
- [TW91] J Thévenaz and P.J Webb. “Homotopy equivalence of posets with a group action”. In: *Journal of Combinatorial Theory, Series A* 56.2 (1991), pp. 173–181. ISSN: 0097-3165. DOI: [https://doi.org/10.1016/0097-3165\(91\)90030-K](https://doi.org/10.1016/0097-3165(91)90030-K). URL: <https://www.sciencedirect.com/science/article/pii/009731659190030K>.
- [Urb14] Eric Urban. “Nearly overconvergent modular forms”. In: *Iwasawa Theory 2012: State of the Art and Recent Advances*. Vol. 7. Contributions in Mathematical and Computational Sciences. Springer, 2014, pp. 401–441. DOI: 10.1007/978-3-642-55245-8_14.
- [Wie05] S. Wiedmann. “Drinfeld modules and elliptic sheaves”. MA thesis. Universität Goettingen, 2005.